

Algebraic Methods in the Theory of Codes, Designs, and Association Schemes

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Chapter 1

Coding Theory

1.1 Introduction to codes

We begin by defining what is meant by a code.

Definition 1.1. For any positive integer n , we define a **code of length n** over a finite non-empty set S to be a subset of S^n . The elements of a code are called **codewords**.

Essentially, it is a collection of n -tuples with entries coming from some set S . This is a general definition and in order to incorporate further structure, we need to have certain operations on codewords. This in turn, translates to imposing certain structure in the set S .

Definition 1.2. A **linear code of length n** is a code of length n over a field $\mathbb{F}$ such that the code forms a subspace of $\mathbb{F}^n$. Here, all vector spaces are viewed over the base field $\mathbb{F}$.

Generally, the field is picked to be a finite field. We shall only be dealing with linear codes over finite fields. Henceforth, by a code we shall mean a linear code over a finite field.

Definition 1.3. Let $\mathbb{F}$ be a field with q elements (that is, $\mathbb{F}_q$) and C be a code of length n over $\mathbb{F}$. By the **size of C** , we mean the number of elements of C and by the **dimension of C** we mean the dimension of C as a vector space over $\mathbb{F}$.

We use the following notation, altogether, for codes. If C is a q -ary (that is, over $\mathbb{F}_q$) linear code of length n with dimension k , we say that C is an $[n, k]_q$ code. We abuse notation and sometimes write $[n, k]_q$ to be the set of all such codes.

We also would like to have a notion of distance between codewords, or the measure of the codewords in the code themselves. This is achieved by the Hamming distance and Hamming weight.

Definition 1.4. Given a codeword $x = (x_1, \dots, x_n)$ in some code $C \in [n, k]_q$, we define the **Hamming weight** of x to be $w(x) = |\{i \in [n] \mid x_i \neq 0\}|$, that is, the number of places it is non-zero at. The **Hamming distance** between two codewords $x, y \in C$ is defined to be $d_H(x, y) = w(x - y)$.

Proposition 1.5. The Hamming distance, d_H , is a metric on a code C of length n .

Proof. We use d henceforth for simplicity. The non-negativity and symmetry is clear. It is also clear that $d(x, y) = 0$ if and only if $x = y$. We only need to verify the triangle inequality. We define

$$A(x, y) = \{i \in [n] \mid x_i \neq 0\} \quad \forall \ x, y \in C$$

so that the Hamming distance is given by $d(x, y) = |A(x, y)|$. Now observe that for any three codewords $x, y, z \in C$, we have

$$A(x, z) \subseteq A(x, y) \cup A(y, z)$$

This is because if x and z differ in the i -th coordinate and y is arbitrary, then y_i cannot be equal to both x_i and z_i since they differ. Thus y_i must differ from at least one of these. Thus taking cardinalities we get

$$d(x, z) \leq d(x, y) + d(y, z)$$

which is the triangle inequality. ■

Proposition 1.6. *The Hamming distance satisfies the translation property given by $d(x + z, y + z) = d(x, y)$ for every $x, y, z \in C$.*

Proof. It is clear that $A(x + z, y + z) = A(x, y)$ since $x_i + z_i = y_i + z_i$ holds if and only if $x_i = y_i$. The proof follows by taking cardinalities. ■

Definition 1.7. Given a code C , the **minimum distance** of the code C is defined to be

$$d(C) := \min\{d_H(x, y) \mid x, y \in C, x \neq y\}$$

Since C is a vector space, it is immediate that $d(C) = \min\{w(x) \mid x \in C, x \neq 0\}$.

Proposition 1.8. *The Hamming weight function, w , for a code C satisfies*

- (i) $w(x) \geq 0$ for every $x \in C$ with equality if and only if $x = 0$
- (ii) $w(x + y) \leq w(x) + w(y)$ for every $x, y \in C$

Proof. The first item follows from the non-negativity of the Hamming distance. Further if $w(x) = d(x, 0) = 0$, then we must have $x = 0$ since d is a metric.

For the second item we use the translation property

$$w(x + y) = d(x + y, 0) \leq d(x + y, y) + d(y, 0) = w(x) + w(y)$$

■

Proposition 1.9 (Singleton bound). *For a code $C \in [n, k]_q$, the minimum distance d satisfies*

$$d \leq n - k + 1$$

or equivalently, $|C| \leq q^{n-d+1}$.

Proof. Consider a code $C = \{v_1, \dots, v_m\}$ of length n of dimension k and minimum distance d . Delete the first $d - 1$ entries of every v_i to obtain $\tilde{C} = \{\tilde{v}_1, \dots, \tilde{v}_m\}$.

If $\tilde{v}_i = \tilde{v}_j$ then v_i and v_j can differ in at most the $d - 1$ deleted coordinates. However, they must differ in at least d coordinates. Thus $|\tilde{C}| = m$. Now $\tilde{C}$ consists of sequences of length $n - d + 1$ over $\mathbb{F}_q$ and hence its maximum size can be q^{n-d+1} which is also an upper bound for $|C|$ completing the proof. ■

Definition 1.10. A code C is said to be a **maximum distance separable code** or an **MDS code** if the minimum distance achieves equality in the Singleton bound above.

Question. Find the total number of $[n, k]_q$ codes possible.

Ans. This is simply the number of k -dimensional subspaces of $\mathbb{F}_q^n$ as vector spaces over $\mathbb{F}_q$.

We construct a basis and count the number of bases. The first basis vector has $q^n - 1$ non-zero choices. The next one will have $q^n - q$ choices since we want to avoid making it linearly dependent with the first one. The third one will have $q^n - q^2$ since it should not be a linear combination of the previous two.

Thus the number of ordered linearly independent k -tuples in $\mathbb{F}_q^n$ is given by

$$(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})$$

Now we fix a k -dimensional subspace V of $\mathbb{F}_q^n$ and count how many of these ordered bases produce the same space V . The first vector will have $q^k - 1$ choices, the next one has $q^k - q$ choices and so on. This is similar to the previous approach and we get

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})$$

many bases produce the same vector space V . Since the choice of V is arbitrary, we can group up these ‘similar’ bases and get that the number of possible V is

$$\frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})} = \binom{n}{k}_q$$

□

Question. Find the total number of $[n, k]_q$ MDS codes possible for the two cases of $k = 1$ and $k = 2$.

Ans. For $k = 1$ we simply want the number of 1-dimensional subspaces of $\mathbb{F}_q^n$ such that $d(C) = n$. Now if $C = \text{span}(x)$, then $d(C) = w(x)$. Thus we want the number of x such that $w(x) = n$. Clearly there are $(q - 1)^n$ many of these. However, given an x of this form, all its $q - 1$ non-zero multiples will span the same vector space. Thus the required answer is

$$(q - 1)^{n-1}$$

We now solve the $k = 2$ case. We are looking for 2-dimensional subspaces $C = \text{span}\{v, w\}$ such that $d(C) = n - 1$, that is, non-zero codewords are allowed to have at most 1 zero.

Without loss of generality, we may assume that

$$v = (1, 0, a_3, \dots, a_n)$$

$$w = (0, 1, b_3, \dots, b_n)$$

By assuming the generator matrix of the subspace is in reduced row echelon form, we guarantee a unique representation for each 2-dimensional subspace, circumventing the need to account for multiple bases

Now $\alpha v + \beta w$ must have at most 1 zero for any choice of α and $\beta \in \mathbb{F}_q$. Choosing $\alpha = 0$ and then $\beta = 0$ we see that all a_i and b_i are non-zero. Suppose both α and β are non-zero. Consider the set

$$\mathcal{A} = \left\{ \begin{bmatrix} a_3 \\ b_3 \end{bmatrix}, \begin{bmatrix} a_4 \\ b_4 \end{bmatrix}, \dots, \begin{bmatrix} a_n \\ b_n \end{bmatrix} \right\}$$

We claim that $\mathcal{A}$ is pairwise linearly independent if and only if $\alpha v + \beta w$ has at most 1 zero. Suppose $a_i = \lambda a_j$ and $b_i = \lambda b_j$ for some distinct $i, j \in \{3, \dots, n\}$. Then choosing $\beta = -a_j, \alpha = b_j$ gives a zero in both i -th and j -th positions of $\alpha v + \beta w$ which is not possible. Conversely, if there are α and β such that we get two zeroes in positions i and j , then $\alpha a_i + \beta b_i = 0 = \alpha a_j + \beta b_j$. Choosing $\lambda = a_i/a_j$ or b_i/b_j whichever possible gives linear dependence of $\mathcal{A}$. This proves our claim.

Thus we would now want to calculate the total number of ordered sets $\mathcal{A}$ with all a_i and b_i non-zero and vectors of $\mathcal{A}$ pairwise linearly independent. In a 2-dimensional setting, we have $q + 1$ total lines through the origin: q from $y = mx$ and $x = 0$. However, we cannot choose $x = 0$ or $y = 0$ since these will correspond to 0 coordinates which is not allowed. Thus we must choose $n - 2$ directions out of the available $q - 1$ directions. Clearly, the directions will be linearly independent pairwise.

At each direction we can choose any of the $q - 1$ non-zero vectors on it to form $[a_i b_i]^\top$. Thus the required result is

$$\binom{q-1}{n-2} (n-2)! (q-1)^{n-2}$$

□

1.2 Matrices of a code

We will use linear algebraic techniques in this section and the convention followed in classical coding theory is to treat codewords as row vectors. Thus, elements of $\mathbb{F}_q^n$ will be treated as row vectors.

Definition 1.11. A **generator matrix** of a code $C \in [n, k]_q$ is a $k \times n$ matrix G over $\mathbb{F}_q$ whose rows constitute a basis for C .

Since $k \leq n$, $\text{rank}(G) = k$. Conversely, every rank k , $k \times n$ matrix over $\mathbb{F}_q$ is a generator matrix for some code.

We now justify the name of this matrix.

Proposition 1.12. If G is a generator matrix of an $[n, k]_q$ code C , then G generates C :

$$C = \{vG \mid v \in \mathbb{F}_q^k\}$$

Proof. Follows from the definition of a basis of a vector space. ■

Clearly generator matrices are not unique. However, they are closely related as can be seen from the proposition below.

Proposition 1.13. Let $C \in [n, k]_q$. Two matrices G and G' are generator matrices of C if and only if they are row equivalent to each other if and only if there exists a non-singular $k \times k$ matrix E over $\mathbb{F}_q$ such that $G' = EG$.

Proof. The two generator matrices G and G' generate the same code C if and only if their sets of rows form a basis if and only if there is an invertible base change matrix that relates the two bases. We know this happens if and only if the rows of G' are obtained through elementary row operations on G which happens if and only if $G' = EG$ for some invertible matrix E . ■

Thus, it is generally preferred to talk of generator matrices in their reduced row echelon form which is necessarily unique.

Definition 1.14. A generator matrix G for a code $C \in [n, k]_q$ is said to be in **standard form** if it has the form

$$G = [I_{k \times k} \mid A_{k \times (n-k)}]$$

Definition 1.15. Let $C \in [n, k]_q$ be a code. A **parity check matrix** of C is an $(n - k) \times n$ matrix H over $\mathbb{F}_q$ such that C is the null space of H .

Since C has dimension k , H will have rank $n - k$. Consequently, H may serve as a generator matrix for some other code, which we define to be a dual code later on in Definition 1.19.

Proposition 1.16. If an $[n, k]_q$ code C has its generator matrix $G = [I_k \mid A]$ in standard form, then the choice of

$$H = [-A^\top \mid I_{n-k}]$$

serves as a parity check matrix of C .

Proof. Every codeword in C is of the form $uG \in \mathbb{F}_q^n$ for some $u \in \mathbb{F}_q^k$. Using the form of G , we get that every codeword will necessarily be of the form

$$[u \mid uA] = [u \mid p]$$

for some $u \in \mathbb{F}_q^k$ and $p = uA \in \mathbb{F}_q^{n-k}$.

Let us find the null space of H . It is given by

$$\begin{aligned}\mathcal{N}(H) &= \{x \in \mathbb{F}_q^n \mid Hx^\top = 0 \in \mathbb{F}_q^{n-k}\} \\ &= \{(x, y) \in \mathbb{F}_q^k \times \mathbb{F}_q^{n-k} \mid -A^\top x^\top + y^\top = 0 \in \mathbb{F}_q^{n-k}\} \\ &= \{(u, p) \in \mathbb{F}_q^k \times \mathbb{F}_q^{n-k} \mid p = uA\} \\ &= C\end{aligned}$$

Thus the given H is indeed a parity check matrix of the code C . ■

Proposition 1.17. *Let H be a parity check matrix for $C \in [n, k]_q$. Let t be a positive integer such that $w(x_0) = t$ for some $x_0 \in C$. Suppose $\mathcal{J} = \{j_1, \dots, j_t\}$ correspond to the non-zero places of this codeword x_0 , then the columns of H corresponding to $\mathcal{J}$ are linearly dependent. Conversely, if t columns of H are linearly dependent and every proper subset of these columns is linearly independent, then there is a codeword x_0 with weight t .*

Proof. Let $x = (a_1, \dots, a_n)$ be a codeword of weight t with the only non-zero entries being $a_{j_1}, \dots, a_{j_t}$. Since H is a parity check matrix we have $Hx^\top = 0$ which gives

$$a_{j_1}h_{j_1} + \dots + a_{j_t}h_{j_t} = 0$$

where h_j denotes column j of the matrix H . Thus these t columns are linearly dependent.

Conversely, suppose $h_{j_1}, \dots, h_{j_t}$ are linearly dependent. Then

$$a_{j_1}h_{j_1} + \dots + a_{j_t}h_{j_t} = 0$$

for some $a_{j_1}, \dots, a_{j_t} \in \mathbb{F}_q$ not all 0. Since no smaller subset is linearly dependent, none of these coefficients can be 0 and hence the codeword $x = (x_i)_{i=1}^n$ given by

$$x_i = \begin{cases} a_{j_k} & \text{if } i = j_k \text{ for some } k \\ 0 & \text{otherwise} \end{cases}$$

serves as a codeword of weight t . ■

The above proposition gives a very good characterisation of minimum distances.

Corollary 1.18. *The minimum distance of a code C is d if and only if some d columns of H are linearly dependent and every set of $d - 1$ columns in H is linearly independent.*

Proof. This directly follows from the above proposition. ■

Note that we can also obtain the Singleton bound (Proposition 1.9) from this corollary since the rank of H is $n - k$ and hence the maximum number of columns that can be linearly independent is $n - k$. Any number of columns more than this number will necessarily be linearly dependent.

1.3 The dual code

Definition 1.19. Let C be an $[n, k]_q$ code. The **dual code** to C is defined to be the set

$$C^\perp = \{x = (x_1, \dots, x_n) \in \mathbb{F}_q^n \mid \sum_{i=1}^n x_i c_i = 0 \ \forall c = (c_1, \dots, c_n) \in C\}$$

We usually write $\langle x, y \rangle$ to mean $\sum_{i=1}^n x_i y_i$. However, one must note that this is NOT an inner product since we are working on a finite field $\mathbb{F}_q$ and it lacks positive-definiteness.

Also, the dual is not a dual in the linear algebraic sense. The instances of cases when $C \cap C^\perp \neq \{0\}$ and $C \subseteq C^\perp$ are plentiful.

Proposition 1.20. *If $C \in [n, k]_q$, then $C^\perp$ is an $[n, n - k]_q$ code. Further, $(C^\perp)^\perp = C$.*

Proof. It is clear from the linearity of $\langle \cdot, \cdot \rangle$ that if $x, y \in C^\perp$, then $x - y \in C^\perp$. It is also clear that $0 \in C^\perp$. Furthermore, for any $\lambda \in \mathbb{F}_q$ and $x \in C^\perp$, we have $\lambda x \in C^\perp$. Thus $C^\perp$ is a linear code of length n over $\mathbb{F}_q$. We need to find its dimension.

Suppose $\{v_1, \dots, v_k\}$ is a basis of C . Consider the map $\phi : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^k$ given by

$$\phi(v) = (\langle v_1, v \rangle, \langle v_2, v \rangle, \dots, \langle v_k, v \rangle)$$

Note that $\ker(\phi) = \{v \in \mathbb{F}_q^n \mid \langle v, v_i \rangle = 0 \ \forall \ i = 1, 2, \dots, k\} = \{v \in \mathbb{F}_q^n \mid \langle v, w \rangle = 0 \ \forall \ w \in C\} = C^\perp$.

Let $u = (u_1, \dots, u_k) \in \mathbb{F}_q^k$ be arbitrary. Consider the system

$$\{\langle v, v_i \rangle = u_i \mid i = 1, \dots, k\}$$

In matrix form we have $Av^\top = u^\top$ where A is the $k \times n$ matrix with the j -th row given by v_j . Since the rows are linearly independent, the rank of A is k . Thus the column span of A is $\mathbb{F}_q^k$. Thus a solution v exists since $u \in \mathbb{F}_q^k = \text{colspace}(A)$.

Thus the map ϕ is surjective and by the rank nullity theorem for the linear map ϕ , we get

$$\dim(C^\perp) + k = n$$

This shows that $C^\perp \in [n, n - k]_q$.

Now let $x = (x_1, \dots, x_n) \in C$. Pick any arbitrary $y = (y_1, \dots, y_n) \in C^\perp$. By definition of $C^\perp$ we have $\langle x, y \rangle = 0$ and hence we have $x \in (C^\perp)^\perp$ since x is arbitrary and so is y . This shows that $C \subseteq (C^\perp)^\perp$ and both have the same dimension which proves the equality. ■

Proposition 1.21. *If H is a parity check matrix of C , then it is a generator matrix of $C^\perp$ and vice versa.*

Proof. Suppose H is a parity check matrix of C . Then by definition we have

$$C = \{x \in \mathbb{F}_q^n \mid Hx^\top = 0 \in \mathbb{F}_q^{n-k}\}$$

Let the rows of H be $r_1, \dots, r_{n-k}$. Firstly note that each row of H lies in $C^\perp$ by definition of C , since $\langle r_i, x \rangle = 0$ for every $x \in C$. Thus the entire row space of H lies in $C^\perp$.

However, the dimension of this row space is $n - k$ since H is full rank and by Proposition 1.20 we have that the dimension of the dual $C^\perp$ is also $n - k$ which shows that the row space of H is the dual code and the rows of H are linearly independent making it the generator matrix of the dual code.

Conversely let G be a generator matrix of a code C . Let the rows be given by $r_1, \dots, r_k$. Now,

$$\begin{aligned} \mathcal{N}(G) &= \{x \in \mathbb{F}_q^n \mid Gx^\top = 0 \in \mathbb{F}_q^k\} \\ &= \{x \in \mathbb{F}_q^n \mid \langle r_i, x \rangle = 0 \ \forall \ i = 1, \dots, k\} \\ &= \{x \in \mathbb{F}_q^n \mid \langle c, x \rangle = 0 \ \forall \ c \in C\} \\ &= C^\perp \end{aligned}$$

Thus G is a parity check matrix of $C^\perp$. ■

Lemma 1.22. *Let C be an $[n, k]_q$ code. Then C is MDS if and only if every set of k columns of a generator matrix of C are linearly independent.*

Proof. Suppose C is MDS. Let G be a generator matrix of C . Consider any set $S = \{j_1, \dots, j_k\}$ of indices of the k columns $\{g_{j_1}, \dots, g_{j_k}\}$.

Suppose these are linearly dependent for the sake of contradiction. Consider the $k \times k$ submatrix $\tilde{G}$ given by these k columns (there are only k rows and hence all are included). Since these columns are linearly dependent, there are $\alpha_i \in \mathbb{F}_q$, which are not all zero, ($i = 1, \dots, k$) such that

$$\sum_{i=1}^k \alpha_i g_{j_i} = 0$$

Thus we can say that $\tilde{G}\alpha^\top = 0$ where $\alpha = (\alpha_1, \dots, \alpha_k) \in \mathbb{F}_q^k \setminus \{0\}$. Since the right null space is non-trivial, the left null space must also be non-trivial (since column rank equals row rank) and hence there must be some $\beta \in \mathbb{F}_q^k \setminus \{0\}$ with $\beta\tilde{G} = 0$.

Thus βG has at least k zeroes and note that $\beta G \neq 0$ since $\beta \neq 0$. Thus the minimum distance is at most $n - k$ which is a contradiction since the minimum distance is exactly $n - k + 1$.

Conversely let any k columns be linearly independent. Let $c \in C$ be a non-zero code word with weight t . Suppose $c = uG$ for some $u \in \mathbb{F}_q^k$. Let $[n] \setminus \text{supp}(c) = \{j_1, \dots, j_{n-t}\}$. Let $\tilde{G}$ be the $k \times (n-t)$ matrix obtained by considering columns indexed by this set $[n] \setminus \text{supp}(c)$.

Thus $u\tilde{G} = 0 \in \mathbb{F}_q^{n-t}$. This shows that the rows of $\tilde{G}$ are linearly dependent which gives that $\text{rank}(\tilde{G}) < k$. If $n - t \geq k$, then from the hypothesis we have that k of these chosen columns will also be linearly independent showing that $\text{rank}(\tilde{G}) \geq k$ which is not possible. Thus we must have $n - t < k$, that is, $n - t \leq k - 1$.

This gives that the minimum distance is at least $n - k + 1$. But by the Singleton bound from Proposition 1.9 we have that it can be at most $n - k + 1$ showing that it is an MDS code. ■

Proposition 1.23. *The dual of an MDS code is also MDS.*

Proof. Suppose C is MDS. We want to show that $d(C^\perp) = k + 1$. That is, for any given parity check matrix of $C^\perp$, which will be a generator matrix G of C by Proposition 1.21, we want to show that every set of k columns of G is linearly independent and there is some set of linearly dependent $k + 1$ columns.

The former fact follows from Lemma 1.22. The latter follows from the fact that the rank of G is k . Thus, in fact, every set of $k + 1$ columns will be linearly dependent. ■

Definition 1.24. A code C is called **self-orthogonal** if $C \subseteq C^\perp$ and it is called **self-dual** if $C = C^\perp$.

1.4 Classical examples of codes

Definition 1.25 (Hamming code and Simplex code). Let q be a prime power and $r > 0$ be an integer. Let $n = (q^r - 1) / (q - 1)$. Let $H_r(q)$ denote the $r \times n$ matrix whose columns are the non-zero vectors of $\mathbb{F}_q^r$ such that they are pair-wise linearly independent (there are indeed n such vectors). The **Hamming code**, $\mathcal{H}_r(q)$, is the code with parity check matrix given by $H_r(q)$ while the **Simplex code**, $\mathcal{S}_r(q)$, is the code whose generator matrix is $H_r(q)$.

One must also note that $r \leq 1 + q + \dots + q^{r-1} = n$. This r gives us the dimension of the Hamming code (which is $n - r$) as can be predicted.

Proposition 1.26. $\text{rank}(H_r(q)) = r$ and consequently $\mathcal{H}_r(q)$ is an $[n, n - r]_q$ code while $\mathcal{S}_r(q)$ is the dual code which is in $[n, r]_q$.

Proof. Observe that we are allowing all possible non-zero vectors up to scalar multiplication, to be the columns of our matrix $H_r(q)$. Thus, there must also exist columns corresponding to the r directions given by $(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1) \in \mathbb{F}_q^r$. These will clearly be linearly independent and hence the dimension $\dim(H_r(q)) = r$ since it has only r rows. ■

Proposition 1.27. *The Hamming code has minimum distance 3 provided $r \geq 2$.*

Proof. Clearly, by construction of the parity check matrix, any two columns are linearly independent. However, the columns $\{e_1, e_2, e_1 + e_2\}$ are linearly dependent and by Corollary 1.18 we see that the minimum distance is 3.

This works as long as there are at least 3 columns. But the number of columns is $n = 1 + q + q^2 + \dots + q^{r-1}$. Also, q is a prime power and hence at least 2. Thus for any $r \geq 2$, we are through. The only case we need to worry about is $r = 1$ in which case we have $n = 1$ and the parity check matrix is simply the 1×1 matrix $[1]$. The code produced is $C = \{0\}$ which is not an interesting case. ■

Proposition 1.28. *The Simplex code, $\mathcal{S}_r(q)$ has a constant weight for every non-zero codeword which is given by q^{r-1} .*

Proof. Let the columns of $H_r(q)$ be denoted as $h_1, \dots, h_n$. Then any non-zero codeword $c = (c_1, \dots, c_n) \in \mathcal{S}_r(q)$ is of the form

$$c_j = \langle x, h_j \rangle$$

for some $x \in \mathbb{F}_q^r \setminus \{0\}$.

We find the number of zero components of c . Fix some $x \in \mathbb{F}_q^r \setminus \{0\}$. Define a map $\phi : \mathbb{F}_q^r \rightarrow \mathbb{F}_q$ given by $\phi(v) = \langle x, v \rangle$. We analyse the kernel of this map. We have that the image of ϕ has dimension 1 since ϕ is surjective. Thus by the rank nullity theorem, the kernel will have dimension $r - 1$.

Thus the kernel has $q^{r-1} - 1$ non-zero vectors. Note that a vector lies in the kernel if and only if all of its non-zero scalar multiples lie in the kernel. Thus, exactly

$$\frac{q^{r-1} - 1}{q - 1}$$

number of columns of $H_r(q)$ lie in the kernel and this is precisely the number of zero entries of c . This is independent of the choice of C and hence the Simplex code is a constant weight code. Further, this constant weight is given by

$$\frac{q^r - 1}{q - 1} - \frac{q^{r-1} - 1}{q - 1} = q^{r-1}$$

■

Definition 1.29 (Reed-Solomon code). Let n, k be positive integers and $q \geq n$ be a prime power. Fix n distinct elements $a_1, \dots, a_n \in \mathbb{F}_q$. The **Reed-Solomon code** $RS_{q,k}(a_1, \dots, a_n)$ is defined to be

$$RS_{q,k}(a_1, \dots, a_n) = \{c_f = (f(a_1), \dots, f(a_n)) \mid f \in \mathbb{F}_q[x], \deg(f) < k\}$$

Question. Analyse what happens when $k \geq n$. Conclude that the only relevant cases occur when $k \leq n$.

Ans. We claim that whenever $k \geq n$ the code is the full space $\mathbb{F}_q^n$. Suppose $b = (b_1, \dots, b_n) \in \mathbb{F}_q^n$. We want a polynomial of degree $< k$ ($k \geq n$) with coefficients in $\mathbb{F}_q$ whose evaluations are the b_i . This is indeed possible and in fact we can obtain a polynomial of degree at most $n - 1 \leq k$ through the method of Lagrange interpolation. Thus every b lies in the code. □

Proposition 1.30. *The Reed-Solomon code $RS_{q,k}(a_1, \dots, a_n)$ is an $[n, k]_q$ code. Further, it is MDS.*

Proof. It is trivial to check that it is indeed a linear subspace of $\mathbb{F}_q^n$ and hence a code. We claim that the set $\mathcal{B} = \{c_1, c_x, c_{x^2}, \dots, c_{x^{k-1}}\}$ forms a basis of the code. Firstly, it is clear that $\mathcal{B} \subseteq C$. Suppose for some $\alpha_0, \dots, \alpha_{k-1} \in \mathbb{F}_q$ we have

$$\alpha_0 c_1 + \dots + \alpha_{k-1} c_{x^{k-1}} = 0$$

Then we have $c_f = 0$ where $f(x) = \alpha_0 + \alpha_1 x + \dots + \alpha_{k-1} x^{k-1}$. Thus $f(a_i) = 0$ for all $i = 1, \dots, n$. But $\deg(f) \leq k \leq n$ and hence f must be the zero polynomial (since a polynomial of degree k over $\mathbb{F}_q$ cannot have more than k roots) giving that each α_i is 0 showing that $\mathcal{B}$ is linearly independent.

It is also easy to see that $\text{span}(\mathcal{B}) = C$. Thus C has dimension k . Now let $c_f \in C$ be arbitrary. We count the number of zeroes. This is clearly given by

$$|\{a_1, \dots, a_n\} \cap \{\text{roots of } f \text{ in } \overline{\mathbb{F}}_q\}|$$

But the maximum number of zeroes f can have is $\deg(f) \leq k - 1$ and hence

$$\text{wt}(c_f) \geq n - \deg(f) \geq n - (k - 1) = n - k + 1$$

Hence this is also a bound on the minimum distance since c_f is arbitrary. But the Singleton bound gives the other inequality which proves the equality. In fact $f(x) = (x - a_1) \cdots (x - a_{k-1})$ also proves the equality. ■

Exercise 1.31. The generalised Reed-Solomon code of length n and dimension k is defined to be

$$GRS_{q,k}(a_1, \dots, a_n, \lambda_1, \dots, \lambda_n) = \{c_f = (\lambda_1 f(a_1), \dots, \lambda_n f(a_n)) \mid f \in \mathbb{F}_q[x], \deg(f) < k\}$$

where a_i are distinct elements of $\mathbb{F}_q$ (q is some prime power greater than n) and λ_i are some elements, not necessarily distinct, of $\mathbb{F}_q$. Show that this is an MDS code.

Question. Comment about the generator and parity check matrices of the Reed-Solomon code.

Ans. Since we already have the basis, the generator matrix is given by

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{k-1} & a_2^{k-1} & a_3^{k-1} & \cdots & a_n^{k-1} \end{bmatrix}$$

Taking the first $k(\leq n)$ columns of this matrix will give a square matrix, which is simply a Vandermonde matrix. The determinant of this Vandermonde matrix is simply $\prod_{1 \leq i < j \leq k} (a_i - a_j)$ which is non-zero since the a_i are distinct. Thus, one may use row operations to reduce the generator matrix to standard form and then apply Proposition 1.16 to obtain a parity check matrix. □

Definition 1.32 (Generalised Reed-Muller code). Let q be a prime power and $m > 0, r \geq 0$ be integers. Enumerate all elements of $\mathbb{F}_q^m$ as $p_1, \dots, p_n$. The **generalised Reed-Muller code** is defined to be

$$GRM_q(r, m) = \{c_f = (f(p_1), \dots, f(p_n)) \mid f \in \mathbb{F}_q[x_1, \dots, x_m], \deg(f) \leq r\}$$

Proposition 1.33. $GRM_q(r, m)$ is an $[n, k]_q$ code with the dimension

$$k = \sum_{j=0}^m (-1)^j \binom{m}{j} \binom{m+r-jq}{m}$$

Proof. It is clear that this is a code of length n over $\mathbb{F}_q$. We need to find the dimension. The basis is simply given by

$$\mathcal{B} = \{x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m} \mid 0 \leq \alpha_i \leq q-1, \alpha_1 + \cdots + \alpha_m \leq r\}$$

Clearly any polynomial evaluation of a polynomial of degree at most r can be written in terms of monomial evaluations with monomials from $\mathcal{B}$. Note that x^q and x may be different as polynomials but they have the same polynomial evaluations.

Dropping the bounded by degree r condition, we see that the bigger set is linearly independent by induction on m and since $\mathcal{B}$ is a subset of it, so is $\mathcal{B}$. All we need to do is find out the cardinality of this set.

Let us define the sets

$$B = \{(\alpha_1, \dots, \alpha_m) \in \mathbb{F}_q^m \mid \alpha_i \geq 0, \alpha_1 + \dots + \alpha_m \leq r\}$$

$$B_j = \{(\alpha_1, \dots, \alpha_m) \in B \mid \alpha_j \geq q\} \quad \forall j = 1, 2, \dots, m$$

Via stars and bars it is clear that $|B| = \binom{m+r}{m}$ and $|B_j| = \binom{m+r-q}{m}$. Furthermore, it is easy to observe that $|B_{j_1} \cap \dots \cap B_{j_t}| = \binom{m+r-tq}{m}$ for distinct $j_i \in [1, m]$.

By the principle of inclusion and exclusion,

$$\begin{aligned} |\mathcal{B}| &= |B| + \sum_{t=1}^m (-1)^{t+1} \sum_{1 \leq j_1 < \dots < j_t \leq m} |B_{j_1} \cap \dots \cap B_{j_t}| \\ &= \sum_{i=0}^m \binom{m}{i} \binom{m+r-iq}{m} \end{aligned}$$

■

Note that if $r \geq q(m-1)$ there is no real restriction on the degree being at most r since this will always be true. And in this case, it is clear the basis will have all possible monomials and will hence span the full space $\mathbb{F}_q^n$.

Theorem 1.34. Consider $GRM_q(r, m)$ and write $r = t(q-1) + s$ for $0 \leq s \leq q-2$ and $t \geq 0$ using the division algorithm. The minimum distance of the code is given by

$$d(GRM_q(r, m)) = \begin{cases} (q-s)q^{m-t-1} & \text{if } r \leq m(q-1) \\ 1 & \text{if } r \geq m(q-1) \end{cases}$$

We shall not prove this theorem here. It requires more machinery to be proven. It is proved later as a consequence of Theorem 1.127 (just after the theorem).

1.5 Polynomials of a code

Definition 1.35. Let $C \in [n, k]_q$. For $0 \leq j \leq n$, let A_j denote the number of codewords of C having weight j . The set $\{A_j \mid 0 \leq j \leq n\}$ is often called the **spectrum of the code** C . By the **weight enumerator polynomial** of the code C , we mean the bivariate homogeneous polynomial

$$W_C(x, y) = \sum_{j=0}^n A_j x^{n-j} y^j$$

Some trivial observations are that for $C = \mathbb{F}_q^n$, $W_C(x, y) = (x + (q-1)y)^n$. Also, for every code C , $W_C(1, 1) = |C| = q^k$.

The main goal of this section is to prove the identity relating W_C and $W_{C^\perp}$. To this end, we require some additional results and tools.

Definition 1.36. Let $C \in [n, k]_q$. Given a codeword $x \in C$ and a subset $P \subseteq [n]$, we define x^P to be the tuple obtained from x by deleting information at the positions indexed by P . Thus, the new tuple x^P has a length of $n - |P|$. We may now define the sets

$$C^P = \{x^P \mid x \in C\}$$

$$C_P = \{x^P \mid x \in C \text{ with } x_i = 0 \ \forall i \in P\}$$

These are called the **puncturing of C** and the **shortening of C**, respectively.

Essentially, puncturing unconditionally deletes coordinates, whereas shortening selectively retains only the codewords known to be zero at the specified coordinates, before deleting those positions. However, note that if for some codewords of C , there is data present at locations of P , then shortening will not include these codewords post-shortening.

It is easy to observe that these are still linear codes of length $n - |P|$ and $C_P \subseteq C^P$.

Proposition 1.37. *We have for a code $C \in [n, k]_q$ and $P \subseteq [n]$,*

$$(C^P)^\perp = (C^\perp)_P$$

Proof. Let $z \in (C^\perp)_P$. Thus, $z = y^P$ for some $y \in C^\perp$ with $y_i = 0$ for every $i \in P$. Let $x \in C^P$ be arbitrary. It is obtained by puncturing some $\tilde{x} \in C$. We want to show that $\langle x, z \rangle = 0$. But observe that

$$\langle x, z \rangle = \langle y, \tilde{x} \rangle = 0$$

This follows because $y_i = 0$ on P and we are simply adding a bunch of zeroes.

Conversely let $z \in (C^P)^\perp$. Thus the length of z is $n - |P|$. Extend it to $\tilde{z}$ by padding zeroes in coordinates indexed by P . Let $x \in C$ be arbitrary. Thus,

$$\langle \tilde{z}, x \rangle = \langle z, x^P \rangle = 0$$

The second equality follows since $z \in (C^P)^\perp$ while the first one has the same logic as in the forward direction due to the extra summands being zero. Since x is arbitrary, we have $\tilde{z} \in C^\perp$. However $\tilde{z}$ is zero on P by construction and hence $\tilde{z}^P = z \in (C^\perp)_P$. ■

Corollary 1.38. *Dually, we have for a code $C \in [n, k]_q$ and $P \subseteq [n]$,*

$$(C_P)^\perp = (C^\perp)^P$$

Proof. Use $C^\perp$ in place of C and apply Proposition 1.20. ■

We now arrive at a central combinatorial result.

Proposition 1.39. *Let $C \in [n, k]_q$. Suppose $\{A_j \mid 0 \leq j \leq n\}$ and $\{B_j \mid 0 \leq j \leq n\}$ denote the spectra of C and $C^\perp$, respectively. We have*

$$\sum_{j=0}^n \binom{n-j}{t} A_j = q^{k-t} \sum_{j=0}^n \binom{n-j}{n-t} B_j$$

for every $t = 0, 1, \dots, n$.

Proof. Fix some $t \in \{0, 1, \dots, n\}$. Consider the set

$$S = \{(x, I) \mid x \in C, I \subseteq [n], x_i = 0 \text{ on } I, |I| = t\}$$

Let its cardinality be denoted by N_t . We count it in two ways

$$\begin{aligned}
N_t &= \sum_{x \in C} \sum_{\substack{I \subseteq [n] \\ |I|=t \\ \text{supp}(x) \subseteq [n] \setminus I}} 1 \\
&= \sum_{j=0}^n \sum_{\substack{x \in C \\ \text{wt}(x)=j}} \sum_{\substack{I \subseteq [n] \\ |I|=t \\ I \subseteq (\text{supp}(x))^c}} 1 \\
&= \sum_{j=0}^n \sum_{\substack{x \in C \\ \text{wt}(x)=j}} \binom{n-j}{t} \\
&= \sum_{j=0}^n \binom{n-j}{t} A_j
\end{aligned}$$

Also, counting it another way we have

$$\begin{aligned}
N_t &= \sum_{\substack{I \subseteq [n] \\ |I|=t}} \sum_{\substack{x \in C \\ x_i=0 \text{ on } I}} 1 \\
&= \sum_{\substack{I \subseteq [n] \\ |I|=t}} |\{x \in C \mid x_i = 0 \ \forall \ i \in I\}|
\end{aligned}$$

Consider the projection $\pi : x \mapsto x^{[n] \setminus I}$ as a map $C \rightarrow C^{[n] \setminus I}$. It is clearly surjective with

$$\ker(\pi) = \{x \in C \mid x_i = 0 \ \forall \ i \in I\}$$

By the rank nullity theorem the dimension of the kernel is

$$\dim(\ker(\pi)) = k - \dim(C^{[n] \setminus I}) = k - k^{[n] \setminus I} \quad (\text{say})$$

Thus, the size of the kernel is $q^{k-k^{[n] \setminus I}}$. Thus our original cardinality is

$$\begin{aligned}
N_t &= \sum_{\substack{I \subseteq [n] \\ |I|=t}} q^{k-k^{[n] \setminus I}} \\
&= \frac{1}{q^{t-k}} \sum_{\substack{I \subseteq [n] \\ |I|=t}} q^{t-k^{[n] \setminus I}} \\
&= \frac{1}{q^{t-k}} \sum_{\substack{I \subseteq [n] \\ |I|=t}} |(C^{[n] \setminus I})^\perp| \\
&= \frac{1}{q^{t-k}} \sum_{\substack{I \subseteq [n] \\ |I|=t}} |(C^\perp)_{[n] \setminus I}|
\end{aligned}$$

Note that we have made use of Proposition 1.37. Thus we have

$$q^{t-k} N_t = \sum_{\substack{I \subseteq [n] \\ |I|=t}} \sum_{\substack{y \in C^\perp \\ \text{supp}(y) \subseteq I}} 1$$

Interchanging the sums and repeating the process as in the first calculation of N_t but with C replaced by $C^\perp$ we get that the RHS is

$$\sum_{j=0}^n \binom{n-j}{n-t} B_j$$

This proves the proposition. ■

This helps us derive the theorem we were after.

Theorem 1.40 (MacWilliams identity). *Let $C \in [n, k]_q$. The weight enumerator polynomial of the dual code is given by*

$$W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x + (q-1)y, x-y)$$

Proof. We use the notation $A_j = A_j(C)$ and $B_j = A_j(C^\perp)$ throughout. Let $x = y + z$ so that the identity becomes

$$W_{C^\perp}(y + z, y) = \frac{1}{|C|} W_C(z + qy, z)$$

We start with the RHS

$$\begin{aligned} \frac{1}{|C|} W_C(z + qy, z) &= \frac{1}{q^k} \sum_{j=0}^n A_j (z + qy)^{n-j} z^j \\ &= \frac{1}{q^k} \sum_{j=0}^n A_j z^j \sum_{t=0}^{n-j} \binom{n-j}{t} z^{n-j-t} (qy)^t \\ &= \frac{1}{q^k} \sum_{t=0}^n \left(\sum_{j=0}^{n-t} A_j \binom{n-j}{t} \right) z^{n-t} q^t y^t \end{aligned}$$

Using Proposition 1.39 the bracketed sum may be re-written as follows.

$$\begin{aligned} \frac{1}{|C|} W_C(z + qy, z) &= \frac{1}{q^k} \sum_{t=0}^n z^{n-t} q^t y^t \left(q^{k-t} \sum_{j=0}^t B_j \binom{n-j}{n-t} \right) \\ &= \sum_{t=0}^n z^{n-t} y^t \sum_{j=0}^t B_j \binom{n-j}{n-t} \\ &= \sum_{j=0}^n \sum_{t=j}^n B_j \binom{n-j}{n-t} z^{n-t} y^t \\ &= \sum_{j=0}^n \sum_{\mu=0}^{n-j} B_j \binom{n-j}{\mu} z^\mu y^{n-\mu} \\ &= \sum_{j=0}^n B_j y^j \sum_{\mu=0}^{n-j} \binom{n-j}{\mu} z^\mu y^{n-j-\mu} \\ &= \sum_{j=0}^n B_j y^j (y + z)^{n-j} \\ &= W_{C^\perp}(y + z, y) \end{aligned}$$

This completes the proof. ■

This leads eventually to the definition of the Kravchuk polynomials.

Definition 1.41. For any prime power q and positive integer n we define the **k -th Kravchuk polynomial** as

$$K_k(x; n, q) = \sum_{j=0}^k (-1)^j (q-1)^{k-j} \binom{x}{j} \binom{n-x}{k-j}$$

for $k = 0, 1, \dots, n$.

Proposition 1.42. With A_j and B_j as in Proposition 1.39, we have

$$B_j = \frac{1}{|C|} \sum_{i=0}^n K_j(i; n, q) A_i$$

Proof. This follows from Theorem 1.40. We have

$$\sum_{j=0}^n B_j x^{n-j} y^j = \frac{1}{|C|} W_C(x + (q-1)y, x-y)$$

We want to find the coefficient of $x^{n-j} y^j$ on the RHS. The expanded RHS looks like

$$\frac{1}{|C|} \sum_{i=0}^n A_i (x + (q-1)y)^{n-i} (x-y)^i$$

It is not difficult to find this coefficient and arrive at the required result. In fact one may show that

$$RHS = \frac{1}{|C|} \sum_{i=0}^n A_i \left[\sum_{j=0}^n K_j(i; n, q) x^{n-j} y^j \right]$$

■

To express A in terms of B , we have the orthogonality relations for these polynomials.

Proposition 1.43. For $K_k(x)$ denoting the Kravchuk polynomial $K_k(x; n, q)$, we have the following two orthogonality relations:

$$\begin{aligned} \sum_{i=0}^n K_j(i) K_i(k) &= q^n \delta_{j,k} \\ \sum_{i=0}^n \mu(i) K_j(i) K_k(i) &= q^n \mu(j) \delta_{j,k} \end{aligned}$$

where $\mu(i) = \binom{n}{i} (q-1)^i$

Proof. Doing everything with $C^\perp$ will dualise the MacWilliams identity to

$$W_C(x, y) = \frac{1}{|C^\perp|} W_{C^\perp}(x + (q-1)y, x-y)$$

This also changes Proposition 1.42 to the following

$$A_k = \frac{1}{|C^\perp|} \sum_{j=0}^n B_j K_k(j; n, q)$$

The first orthogonality relation now follows by applying the two identities relating A_j and B_j and noting that $|C| |C^\perp| = q^n$.

For the second one we use generating functions. Define the function

$$G(z, i; n, q) = \sum_{j=0}^n K_j(i; n, q) z^j$$

and consider the function

$$S(z, w) = \sum_{i=0}^n \mu(i) G(z, i; n, q) G(w, i; n, q)$$

Note that we have an explicit formula for G as

$$G(z, i; n, q) = (1 + (q-1)z)^{n-i} (1-z)^i$$

Using this and doing some simple algebraic manipulations we arrive at

$$S(z, w) = q^n \sum_{j=0}^n \binom{n}{j} (q-1)^j z^j w^j = \sum_{j=0}^n q^n \mu(j) z^j w^j$$

However by the straightforward definition we also have

$$S(z, w) = \sum_{j=0}^n \sum_{k=0}^n \left[\mu(i) K_j(i; n, q) K_k(i; n, q) \right] z^j w^k$$

Comparing the two yields the result immediately. ■

We now have a slight variation of Proposition 1.39 below.

Proposition 1.44. *With notation as in Proposition 1.39 we have*

$$\sum_{j=0}^n \binom{j}{t} A_j = q^{k-t} \sum_{j=0}^n (-1)^j \binom{n-j}{n-t} (q-1)^{t-j} B_j$$

for every $t = 0, 1, \dots, n$.

Proof. We recall the identity

$$A_i = \frac{1}{|C^\perp|} \sum_{j=0}^n B_j K_i(j; n, q) = q^{k-n} \sum_{j=0}^n B_j K_i(j; n, q)$$

We now analyse the LHS:

$$\begin{aligned} \text{LHS} &= \sum_{i=0}^n \binom{i}{t} q^{k-n} \sum_{j=0}^n B_j K_i(j; n, q) \\ &= q^{k-n} \sum_{j=0}^n B_j \sum_{i=0}^n \binom{i}{t} K_i(j; n, q) \end{aligned}$$

We now analyse the inner sum $S_j(t) = \sum_{i=0}^n \binom{i}{t} K_i(j; n, q)$ carefully. Recall the generating function from before, but written with respect to the index j :

$$G(z, j; n, q) = \sum_{i=0}^n K_i(j; n, q) z^i = (1 + (q-1)z)^{n-j} (1-z)^j$$

Taking the derivative t times with respect to z yields

$$\frac{d^t}{dz^t} (G(z, j; n, q)) = \sum_{i=t}^n K_i(j; n, q) \frac{i!}{(i-t)!} z^{i-t}$$

This is precisely $t!S_j(t)$ when evaluated at $z = 1$. Differentiating the closed form of G and substituting $z = 1$ gives us

$$\left. \frac{d^t}{dz^t} [(1 + (q-1)z)^{n-j}(1-z)^j] \right|_{z=1} = t!q^{n-t}(-1)^j(q-1)^{t-j} \binom{n-j}{n-t}$$

This gives that

$$S_j(t) = q^{n-t}(-1)^j(q-1)^{t-j} \binom{n-j}{n-t}$$

Putting this back into our expression for the LHS gives the desired RHS, completing the proof. $\blacksquare$

1.6 Equivalence of codes

Definition 1.45. Let C_1, C_2 be two codes of length n over $\mathbb{F}_q$. We say C_1 is **permutation equivalent** to C_2 and write $C_1 \sim C_2$ if there exists a permutation matrix $P \in GL_n(\mathbb{F}_q)$ such that

$$C_2 = \{xP \mid x \in C_1\}$$

It is clear that this defines an equivalence relation on the codes since the set of permutation matrices in $GL_n(\mathbb{F}_q)$ forms a group.

We have a generalisation of the above. A **monomial matrix** is a matrix whose support is a permutation matrix, that is, every row and every column has a single non-zero entry. If all these non-zero entries happen to be 1, we say it is a permutation matrix.

One may show that every monomial matrix M is of the form $M = PD$ for some permutation matrix P and some non-singular diagonal matrix D .

Definition 1.46. Let C_1, C_2 be two codes of length n over $\mathbb{F}_q$. We say C_1 is **monomial equivalent** to C_2 and write $C_1 \approx C_2$ if there exists a monomial matrix $M \in GL_n(\mathbb{F}_q)$ such that

$$C_2 = \{xM \mid x \in C_1\}$$

It may be a good exercise to show that such matrices form a group and hence this relation also is an equivalence relation.

Proposition 1.47. Let $C_1 \approx C_2$ through a monomial matrix M . Consider $f : C_1 \rightarrow C_2$ given by $f(x) = xM$. Then f is a linear bijection which preserves Hamming weights, minimum distance and dimensions of codes.

Proof. Seeing that $x \mapsto xM$ is a linear bijection is trivial.

Let $x \in C_1$. Then $f(x)$ is simply a permutation on the entries of x followed by individually scaling them by some non-zero factors. Both these operations keep the number of zeroes and hence non-zeroes constant, thus preserving weight. Since weights are preserved, so is the minimum distance.

If $\{v_1, \dots, v_k\}$ is a basis of C_1 , then we claim that $\{f(v_1), \dots, f(v_k)\}$ is a basis of C_2 . It is obvious that it is a spanning set due to linearity and surjectivity of f . It is also linearly independent due to the injectivity of f . This proves that the dimension is preserved. $\blacksquare$

We also have the converse of the above proposition.

Proposition 1.48. Let $f : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ be a linear isometry (that is, a Hamming weight preserving linear isomorphism). Then f must have the form $f(x) = xM$ for some monomial matrix $M \in GL_n(\mathbb{F}_q)$.

Proof. Consider the action of the linear isometry f on the standard basis $e_1, \dots, e_n \in \mathbb{F}_q^n$. Since it preserves weight, each e_i is mapped to a vector with unit weight and hence must be a non-zero multiple of some e_j .

Further, e_i and e_j for distinct i and j cannot be mapped to λe_k and μe_k for some $\lambda, \mu \in \mathbb{F}_q \setminus \{0\}$ and some k else it contradicts the injectivity of f .

Thus, each e_i gets mapped to a unique direction e_j and is scaled by a non-zero factor. This is the same as the action of a monomial matrix M . Since the basis action determined the whole action we are done. ■

We can now define automorphism groups of a code.

Definition 1.49. Let C be a q -ary code of length n . We define the *permutation automorphism group* to be

$$PAut(C) = \{\sigma \in S_n \mid \sigma(C) \subseteq C\}$$

where $\sigma(x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.

Proposition 1.50. $PAut(C)$ is a group and is isomorphic to the group

$$\{P \in GL_n(\mathbb{F}_q) \mid P \text{ is a perm matrix such that } xP \in C \ \forall x \in C\}$$

Proof. Trivial observation. ■

Motivated by the above we can define the monomial automorphism group as follows.

Definition 1.51. Let C be a q -ary code of length n . We define the *monomial automorphism group* to be

$$MAut(C) = \{M \in GL_n(\mathbb{F}_q) \mid M \text{ is a monomial matrix such that } xM \in C \ \forall x \in C\}$$

Proposition 1.52. $MAut(C)$ is a group containing $PAut(C)$ as a subgroup.

Proof. Trivial observation. ■

We now have a further generalisation of the above discussion.

Definition 1.53. The *semi-linear automorphism group* of a q -ary code C of length n is defined to be

$$\Gamma Aut(C) = \{M\mu \mid M \in GL_n(\mathbb{F}_q) \text{ is monomial, } \mu : \mathbb{F}_q \rightarrow \mathbb{F}_q \text{ is a field automorphism, } xM\mu \in C \ \forall x \in C\}$$

where $xM\mu$ is to be evaluated by calculating xM first and then applying μ entry-wise.

Note that we have now lost our linearity due to the field automorphism.

Proposition 1.54. $\Gamma Aut(C)$ is a group.

Proof. Let $T_{M,\mu} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ denote the map $T_{M,\mu}(x) = (xM)^\mu$, that is, take the product xM first and then apply μ entry-wise. Firstly we show that all the $T_{M,\mu}$ form a group. This group happens to be denoted by $\Gamma L_n(\mathbb{F}_q)$ and is indeed a generalisation of the general linear group.

First we make an important observation:

$$(a^\nu)A = (aA^{\nu^{-1}})^\nu$$

This is proved easily by comparing entries on both sides.

Let us look at compositions to understand the product structure.

$$T_{M,\mu} \circ T_{N,\nu}(x) = (((xN)^\nu)M)^\mu = (((xN)M^{\nu^{-1}})^\nu)^\mu = (xNM^{\nu^{-1}})^{\mu \circ \nu} = T_{NM^{\nu^{-1}}, \mu \circ \nu}$$

Thus the set is closed under the operation since $NM^{\nu^{-1}}$ is also a monomial matrix and $\mu \circ \nu$ is a field automorphism.

The identity is clearly corresponding to $M = I, \mu = \text{id}$. Associativity needs a manual check by comparing entries of matrices element wise and will follow or one may use some algebraic manipulations on the definition of the group product directly.

One may check that the inverse of $T_{M,\mu}$ is given by $T_{(M^{-1})^\mu, \mu^{-1}}$. Thus $\Gamma L_n(\mathbb{F}_q)$ is indeed a group and when we consider its action on a code C , the stabiliser group is given by

$$\Gamma \text{Aut}(C) = \{T_{M,\mu} \mid M \text{ is monomial, } \mu : \mathbb{F}_q \rightarrow \mathbb{F}_q \text{ is a field automorphism, } T_{M,\mu}(C) = C\}$$

This completes the proof. ■

In general, we define the notion of semi-linearity.

Definition 1.55. Let V_1, V_2 be vector spaces over some field F . Let $\mu : F \rightarrow F$ be a field automorphism. We say $f : V_1 \rightarrow V_2$ is μ -**semi-linear** if for every $u, v, w \in V_1$ and $\lambda \in F$ the following hold

$$\begin{aligned} f(u + v) &= f(u) + f(v) \\ f(\lambda w) &= \mu(\lambda)f(w) \end{aligned}$$

A classic example of μ on a finite field $\mathbb{F}$ of characteristic p is the map $x \mapsto x^p$.

Proposition 1.56. *Given a fixed automorphism $\mu : F \rightarrow F$, the set of all μ -semi-linear maps forms an abelian group under addition.*

Proof. Trivial to check. Note that since V_1 and V_2 are separately mentioned, the natural operation to think of is addition and not composition. ■

We also have a characterisation as we did for linear maps in Proposition 1.48. This is due to MacWilliams.

Theorem 1.57. *Let C_1, C_2 be q -ary codes of length n and dimension at least 3. Suppose $f : C_1 \rightarrow C_2$ is a map. Then f is a weight preserving bijection that maps r -dimensional subspaces of C_1 to r -dimensional subspaces of C_2 if and only if it is a semi-linear isomorphism with respect to some μ .*

We do not prove this here as it is a non-trivial result following from projective geometry.

1.7 Cyclic codes

Definition 1.58. An $[n, k]_q$ code C is said to be a **cyclic code** if for every codeword $x \in C$, $\sigma * x \in C$ where $\sigma = (n, n-1, \dots, 1) \in S_n$. Equivalently, $\sigma(C) = C$.

Note that one may also equivalently use the left shift $\sigma = (1, 2, \dots, n)$ but we adopt the right shift convention so that

$$\sigma * x = (n, n-1, \dots, 1) * (x_1, x_2, \dots, x_n) = (x_n, x_1, \dots, x_{n-1})$$

Proposition 1.59. *Cyclic codes over $\mathbb{F}_q$ of length n are in bijection with the ideals of $R_n = \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ which are in one-to-one correspondence with ideals of $\mathbb{F}_q[x]$ containing the ideal $\langle x^n - 1 \rangle$ which are in one-to-one correspondence with monic polynomials $g(x)$ which divide $x^n - 1$.*

Proof. Associate an element $a = (a_0, a_1, \dots, a_{n-1}) \in \mathbb{F}_q^n$ to the polynomial $a_0 + a_1x + \dots + a_{n-1}x^{n-1} \in \mathbb{F}_q[x]$. Conversely given any polynomial of degree at most $n-1$, we can retrieve an element of $\mathbb{F}_q^n$ by reading off the coefficients.

This gives a bijection $\pi : \mathbb{F}_q^n \rightarrow R_n$ given by

$$\pi(a_0, a_1, \dots, a_{n-1}) = a_0 + a_1x + \dots + a_{n-1}x^{n-1}$$

It is also seen to be an $\mathbb{F}_q$ -linear map.

Thus, in particular if C is a subspace, its image under the linear map π will be a subspace of R_n . Now if C is a cyclic code, then note that we have the property for $\pi(C)$ that

$$a_0 + a_1x + \cdots + a_{n-1}x^{n-1} \in \pi(C) \implies a_n + a_0x + \cdots + a_{n-2}x^{n-1} \in \pi(C)$$

That is, $f(x) \in C$ implies $xf(x) \in \pi(C)$. Conversely any subspace of R_n which is closed under multiplication by x corresponds to a linear code (one may easily verify that this is well-defined).

This is simply the definition of an ideal. Thus, C is cyclic if and only if $\pi(C)$ is an ideal of R_n . Further $\pi|_C$ is a bijection between C and $\pi(C)$ which proves the proposition.

Now we want to find the ideals of $\mathbb{F}_q[x]$, a PID, and only consider those which contain $\langle x^n - 1 \rangle$. The ideals are of the form $\langle g(x) \rangle$ for monic polynomials $g(x) \in \mathbb{F}_q[x]$. Thus the ideals of R_n are of the form $\langle g(x) \rangle$ where $g(x) \in \mathbb{F}_q[x]$ divides $x^n - 1$. ■

Definition 1.60. Given a cyclic code C of length n over $\mathbb{F}_q$ we obtain the unique monic polynomial $g \in \mathbb{F}_q[x]$ which divides $x^n - 1$ corresponding to the ideal $\pi(C)$. We call this the **generator polynomial** of the code C . Notation has been borrowed from the above Proposition 1.59.

Proposition 1.61. A code is cyclic if and only if its dual is cyclic.

Proof. Let C be a cyclic code of length n over $\mathbb{F}_q$. Consider the dual

$$C^\perp = \{x \in \mathbb{F}_q^n \mid \langle x, c \rangle = 0 \ \forall \ c \in C\}$$

Let $v = (v_1, \dots, v_n) \in C^\perp$. We show that $(v_n, v_1, \dots, v_{n-1}) \in C^\perp$, too. Let $c \in C$ be arbitrary. Since $v \in C^\perp$, we have

$$\langle v, \sigma^{-1} * c \rangle = v_1c_2 + v_2c_3 + \cdots + v_nc_1 = 0 = \langle \sigma * v, c \rangle$$

Thus $\sigma * v = (v_n, v_1, \dots, v_{n-1}) \in C^\perp$. ■

Proposition 1.62. If C is an $[n, k]_q$ cyclic code, then the degree of the generator polynomial is $n - k$.

Proof. Consider the associated ideal, $\pi(C) = I$ of R_n , of the code C . Let $g(x)h(x) = x^n - 1$ with degree of g being r . We show that $k = n - r$. Define the map $\phi : \mathbb{F}_q[x] \rightarrow I$ given by

$$\phi(a(x)) = a(x)g(x) \pmod{x^n - 1}$$

The map is clearly surjective. The kernel is given by

$$\ker(\phi) = \{a(x) \in \mathbb{F}_q[x] \mid a(x)g(x) = b(x)(x^n - 1) \text{ for some } b \in \mathbb{F}_q[x]\} = \langle h(x) \rangle$$

Now observe that ϕ is an $\mathbb{F}_q$ -module homomorphism since $\phi(b(x)a(x)) = b(x)\phi(a(x))$ holds for every scalar $b(x) \in \mathbb{F}_q[x]$ coming from the ring. By the first isomorphism theorem we have

$$\frac{\mathbb{F}_q[x]}{\langle h(x) \rangle} \cong \text{Im}(\phi) = I \quad (\text{as } \mathbb{F}_q[x]\text{-modules})$$

The structure can be brought down to $\mathbb{F}_q$ and thus the isomorphism also holds as $\mathbb{F}_q$ -modules. Since $\mathbb{F}_q$ is a ring, the word module may be replaced by vector space. Comparing dimensions of the $\mathbb{F}_q$ -vector spaces on both sides we get

$$k = \dim_{\mathbb{F}_q}(I) = \deg(h) = n - r$$

Note that $\{g(x), xg(x), \dots, x^{n-r-1}g(x)\}$ is an $\mathbb{F}_q$ -basis of I and hence the dimension is $n - r = k$. ■

Proposition 1.63. Let C be an $[n, k]_q$ cyclic code. If $g(x) = g_0 + \cdots + g_{n-k}x^{n-k}$ is a generator polynomial, then the following is a generator matrix of C :

$$\begin{bmatrix} g_0 & g_1 & g_2 & \cdots & g_{n-k} & 0 & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & \cdots & g_{n-k-1} & g_{n-k} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & g_0 & g_1 & g_2 & \cdots & g_{n-k} \end{bmatrix}_{k \times n}$$

One may note that $g_{n-k} = 1$ since g is monic.

Proof. As in the above proof, let $I = \pi(C)$ be the ideal of R_n corresponding to the cyclic code C . Then a basis of I as a vector space over $\mathbb{F}_q$ is given by

$$\mathcal{B} = \{g(x), xg(x), \dots, x^{k-1}g(x)\}$$

The pre-images under the $\mathbb{F}_q$ -linear isomorphism π will form a basis of C and these are simply given by

$$\begin{aligned} v_0 &= \pi^{-1}(g(x)) = (g_0, g_1, g_2, \dots, g_{n-k}, 0, 0, \dots, 0) \\ v_1 &= \pi^{-1}(xg(x)) = (0, g_0, g_1, \dots, g_{n-k-1}, g_{n-k}, 0, \dots, 0) \\ &\vdots \\ v_{k-1} &= \pi^{-1}(x^{k-1}g(x)) = (0, 0, 0, \dots, g_0, g_1, g_2, \dots, g_{n-k}) \end{aligned}$$

This gives the desired generator matrix. ■

Proposition 1.64. Let C be an $[n, k]_q$ cyclic code. Let $g(x)$ be a generator polynomial of C . Obtain the polynomial h given by $g(x)h(x) = x^n - 1$. Then the polynomial $h_0^{-1}x^k h(1/x)$ is a generator polynomial of the dual code $C^\perp$.

Proof. A vector $v = (v_0, \dots, v_{n-1})$, with associated polynomial $\pi(v) = v(x)$, lies in the dual $C^\perp$ if and only if it is orthogonal to every code in C , that is,

$$\langle v, g \rangle = 0 = \langle v, \text{shift}^1(g) \rangle = \langle v, \text{shift}^2(g) \rangle = \cdots = \langle v, \text{shift}^{n-1}(g) \rangle$$

where $g = (g_0, g_1, \dots, g_{n-1})$, $\text{shift}^1(g) = (g_{n-1}, g_1, \dots, g_{n-2})$ and so on. In terms of polynomials we have

$$v(x)g(1/x) \equiv 0 \pmod{x^n - 1}$$

Using the fact that $g(x)h(x) = x^n - 1$ we arrive at

$$v(x) = x^n q(x)h(1/x)$$

for some polynomial $q(x) \in \mathbb{F}_q[x]$. Now note that $x^k h(1/x)$ is a polynomial in $\mathbb{F}_q[x]$.

Thus v lies in the dual code if and only if the associated $v(x)$ is a multiple of $x^k h(1/x)$. Hence, the dual code is generated by the polynomial $x^k h(1/x)$ and we can make it monic by multiplying with h_0^{-1} . ■

Corollary 1.65. Let C be an $[n, k]_q$ cyclic code. Let $g(x)$ be a generator polynomial of C . Obtain the polynomial h given by $g(x)h(x) = x^n - 1$. Then the following is a parity check matrix of C .

$$\begin{bmatrix} h_k & h_{k-1} & h_{k-2} & \cdots & h_0 & 0 & 0 & \cdots & 0 \\ 0 & h_k & h_{k-1} & \cdots & h_1 & h_0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & h_k & h_{k-1} & h_{k-2} & \cdots & h_0 \end{bmatrix}_{(n-k) \times n}$$

Proof. Apply Proposition 1.21. ■

We now analyse the divisors of $x^n - 1$ in detail. We need some knowledge of cyclotomic polynomials and primitive roots.

Recall that if we set $\zeta = e^{2\pi i/n}$ then $1, \zeta, \zeta^2, \dots, \zeta^{n-1}$ is the full set of roots of $x^n - 1 = 0$ over $\mathbb{C}$. The primitive roots among these are precisely ζ^k for those k which are coprime to n .

Definition 1.66. Set $\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} (x - \zeta^k)$. We call this the n -th *cyclotomic polynomial*.

Theorem 1.67. The n -th cyclotomic polynomial is a monic polynomial of degree $\varphi(n)$ having integer coefficients and is irreducible in $\mathbb{Z}[x]$. Further,

$$x^n - 1 = \prod_{\substack{1 \leq d \leq n \\ d|n}} \Phi_d(x)$$

Proof. It is clear that the polynomial is monic and its degree is the Euler totient function $\varphi(n)$. We now prove the identity of these polynomials multiplying out to $x^n - 1$. We have the following grouping by divisibility

$$\begin{aligned} x^n - 1 &= \prod_{0 \leq k \leq n-1} (x - \zeta^k) \\ &= \prod_{\substack{1 \leq d \leq n \\ d|n}} \prod_{\substack{0 \leq k < n \\ \text{order}(\zeta^k) = d}} (x - \zeta^k) \\ &= \prod_{\substack{1 \leq d \leq n \\ d|n}} \prod_{\substack{0 \leq k < n \\ \gcd(k, n) = n/d}} (x - \zeta^k) \end{aligned}$$

Now note $\{0 \leq k \leq n-1 \mid \gcd(k, n) = n/d\} = \{nt/d \mid t = 1, 2, \dots, d\}$ and hence we get

$$\begin{aligned} x^n - 1 &= \prod_{\substack{1 \leq d \leq n \\ d|n}} \prod_{\substack{1 \leq t \leq d \\ \gcd(t, d) = 1}} (x - \zeta^{nt/d}) \\ &= \prod_{\substack{1 \leq d \leq n \\ d|n}} \prod_{\substack{1 \leq t \leq d \\ \gcd(t, d) = 1}} (x - \Omega^t) \quad (\text{where } \Omega = \zeta^{n/d}) \\ &= \prod_{\substack{1 \leq d \leq n \\ d|n}} \Phi_d(x) \end{aligned}$$

We now show that all the coefficients are integers. This is a simple proof by induction. Clearly $\Phi_1(x) = x - 1$ is in $\mathbb{Z}[x]$. Assume that for every $k < n$, $\Phi_k(x) \in \mathbb{Z}[x]$. Then we have

$$x^n - 1 = \Phi_n(x) \left(\prod_{\substack{1 \leq d \leq n-1 \\ d|n}} \Phi_d(x) \right)$$

The bracketed term, which we will call as $P(x)$ lies in $\mathbb{Z}[x]$ because of the induction hypothesis. Thus we have

$$\Phi_n(x) = \frac{x^n - 1}{P(x)}$$

Note that $P(x) \mid x^n - 1$ and hence by the division algorithm in $\mathbb{Z}[x]$ the quotient $\Phi_n(x)$ lies in $\mathbb{Z}[x]$.

We are left to show that $\Phi_n(x)$ is irreducible over $\mathbb{Z}[x]$. This is not very trivial and requires Gauss lemma, minimal polynomials and finite fields $\mathbb{F}_p$. It is a famous result by Dedekind and can be found in almost every algebra textbook. ■

This gives us the factorisation over $\mathbb{Q}[x]$ which is incidentally the same as that over $\mathbb{Z}[x]$ because of the Gauss lemma. We now wish to factorise over $\mathbb{F}_q[x]$.

Question. Show that it suffices to understand the factorisation of $x^n - 1$ over $\mathbb{F}_q[x]$ for n such that $p \nmid n$ where p is the characteristic of $\mathbb{F}_q$, that is, $q = p^e$ for some $e \geq 1$.

Ans. Firstly observe that if $f(x) = \sum_{i=0}^d a_i x^i \in \overline{\mathbb{F}_q}[x]$ for some algebraic closure of $\mathbb{F}_q$, say, $\cup_{t \geq 1} \mathbb{F}_{q^t}$, then $f(x)^q = \sum_{i=0}^d a_i^q x^{iq}$.

Hence $f \in \mathbb{F}_q[x]$ if and only if $f(x)^q = f(x^q)$. Thus if $n = p^m n'$ for some $m \geq 1$ and $p \nmid n' \in \mathbb{N}$, we may write

$$x^n - 1 = (x^{n'} - 1)^{p^m}$$

□

Proposition 1.68. *With the notation and theory as above (also includes assuming that $p \nmid n$), there exists a least positive integer ℓ such that $q^\ell \equiv 1 \pmod{n}$*

Further, given $s \in \mathbb{Z}$ with $0 \leq s < n$, there exists a least positive integer ℓ_s such that

$$sq^{\ell_s} \equiv s \pmod{n}$$

and $1 \leq \ell_s \leq \ell$.

Proof. Since $\gcd(q, n) = 1$, the element q of $\mathbb{Z}_n$ belongs to and is invertible in the group of units and hence ℓ is simply the order of q in this group. Thus, $sq^\ell \equiv s \pmod{n}$ holds. Thus we can get the existence of some $1 \leq \ell_s \leq \ell$ by using the well ordering principle. ■

This motivates us to define the following.

Definition 1.69. For $0 \leq s < n$, let C_s denote the set of representatives in $\{0, 1, \dots, n-1\}$ of the distinct elements $s, sq, \dots, sq^{\ell_s-1} \pmod{n}$. We call C_s the **q-cyclotomic coset** of $s \pmod{n}$.

Proposition 1.70. *An two q-cyclotomic cosets are either disjoint or identical and their union gives the full set $\{0, 1, \dots, n-1\}$.*

Proof. Let $x \in C_{s_1} \cap C_{s_2}$ for $C_{s_1} \neq C_{s_2}$. Then $s_1 q^{r_1} \equiv s_2 q^{r_2} \pmod{n}$ for some $0 \leq r_1 < \ell_{s_1}$ and $0 \leq r_2 < \ell_{s_2}$.

Thus, $s_1 \equiv s_2 q^{r_2-r_1}$ since q is invertible. Now consider some $y = s_1 q^r \in C_{s_1}$ for some $0 \leq r < \ell_{s_1}$. We have, $y \equiv s_2 q^{r_2-r_1+r} \in C_{s_2}$. This shows that $C_{s_1} \subseteq C_{s_2}$ and similarly the other containment follows.

It is clear that the disjoint union will produce $\mathbb{Z}_n$ since for every $0 \leq t < n$, we have $t \in C_t$. ■

We now make an important observation below:

Proposition 1.71. *If $\alpha \in \overline{\mathbb{F}_q}$ is a primitive n -th root of unity over $\mathbb{F}_q$, that is, $\alpha^n = 1$ and $\alpha^m \neq 1$ for every $1 \leq m < n$, then $1, \alpha, \dots, \alpha^{n-1}$ are all distinct and we can write*

$$x^n - 1 = \prod_{j=0}^{n-1} (x - \alpha^j)$$

Proof. Trivial. Follows from fundamental theorem of algebra and comparing coefficients. ■

Definition 1.72. Fix some primitive n -th root $\alpha \in \overline{\mathbb{F}_q}$. For any $s = 0, 1, \dots, n-1$, let C_s denote the q -cyclotomic coset corresponding to s . We define the polynomial

$$M_s(x) = \prod_{j \in C_s} (x - \alpha^j)$$

Theorem 1.73. We have $x^n - 1 = \prod_s M_s(x)$ where the product is over representatives s of distinct q -cyclotomic cosets. Further, $M_s(x)$ is a monic polynomial in $\mathbb{F}_q[x]$ which is irreducible over $\mathbb{F}_q[x]$. Thus, this is a complete factorisation of $x^n - 1$ over $\mathbb{F}_q[x]$. Moreover, it is well-defined because $M_s(x) = M_{s'}(x)$ for s and s' in the same coset.

Proof. The first identity of $x^n - 1 = \prod_s M_s(x)$ follows directly from Proposition 1.70.

$M_s(x)$ is seen to be a monic polynomial by the definition.

Also, $M_s(x)^q = \prod_{j \in C_s} (x - \alpha^j)^q = \prod_{j \in C_s} (x^q - \alpha^{jq})$. Now since we have $\{jq \mid j \in C_s\} = C_s$, we get $M_s(x)^q = M_s(x^q)$ showing that $M_s(x) \in \mathbb{F}_q$.

If $s' \in C_s$ then $C_s = C_{s'}$ and hence $M_s(x) = M_{s'}(x)$ follows trivially.

All we need to do now is prove that $M_s(x) \in \mathbb{F}_q[x]$ is irreducible. Let $f(x)$ be the minimal polynomial of α^s over $\mathbb{F}_q$. Then $f(\alpha^s) = 0$ and $f(\alpha^{sq^j}) = (f(\alpha^s))^{q^j} = 0$ for all $1 \leq j < \ell_s$. Thus, $M_s(x) \mid f(x)$.

Conversely, $M_s(\alpha^s) = 0$ and f is the minimal polynomial of α^s . Thus, $f(x) \mid M_s(x)$ and hence M_s is in fact the minimal polynomial of α^s and is hence irreducible. ■

Question. Let $s \in \mathbb{Z}_n$ be such that $\gcd(s, n) = 1$ then show that

$$|C_s| = |C_1|$$

Further if $ss' \equiv 1$ in $\mathbb{Z}_n$, then show that

$$M_s(x) = \gcd(M_1(x^{s'}), x^n - 1)$$

Ans. The size of C_1 is simply the order of q which is ℓ . Since s is invertible, $sq^k \equiv s$ is the same as $q^k \equiv 1$ and hence $\ell_s = \ell$ showing that $|C_s| = |C_1|$.

For the second part, suppose $R(x) = \gcd(M_1(x^{s'}), x^n - 1)$. Suppose $\beta \in \overline{\mathbb{F}}_q$ is a root of $R(x)$. If it is a root of $x^n - 1$, then $\beta = \alpha^k$ for some $0 \leq k \leq n - 1$. Also if $\beta = \alpha^k$ it is a root of $M_1(x^{s'})$, then $ks' \in C_1$, implying that $sks' = kss' = k \in sC_1 = C_s$.

Thus, $\beta \in \overline{\mathbb{F}}_q$ is a root of $R(x)$ if and only if $\beta = \alpha^k$ and $k \in C_s$. These are precisely the roots of $M_s(x)$ finishing the proof. □

Definition 1.74. Given a cyclic code C of length n over $\mathbb{F}_q$, we define the **zero-set** or the **complete defining set** of C to be the subset of $\mathbb{Z}_n$ given by

$$Z(C) = \{s \in \mathbb{Z}_n \mid c(\alpha^s) = 0 \ \forall \ c(x) \in \pi(C)\}$$

Note that the condition $c(\alpha^s) = 0$ is independent of the choice of the representative c chosen in $\pi(C) = \frac{\mathbb{F}_q[x]}{x^n - 1}$ since α^s is a root of $x^n - 1$.

Proposition 1.75. Let C be a cyclic code of length n over $\mathbb{F}_q$ for $\gcd(n, q) = 1$. The generator polynomial is given by

$$g(x) = \prod_{s \in Z(C)} (x - \alpha^s)$$

Proof. Since $g(x) \mid x^n - 1$, every root of g in $\overline{\mathbb{F}}_q$ is of the form α^s . Further, since $p \nmid n$, all these roots are simple roots, that is, of multiplicity 1. Let

$$Z(g) = \{j \in \mathbb{Z}_n \mid g(\alpha^j) = 0\}$$

Suppose $j \in Z(g)$. Then $g(\alpha^j) = 0$ and hence for any $c \in C$, we have (in R_n),

$$c(x) = g(x)\tilde{g}(x)$$

showing that $c(\alpha^j) = 0$. Thus, $Z(g) \subseteq Z(C)$. Conversely, since $g(x) \in \pi(C)$, we have $j \in Z(C) \implies g(\alpha^j) = 0$ showing that $Z(C) \subseteq Z(g)$ completing the proof. ■

Corollary 1.76. *The dimension of a cyclic code C of length n over $\mathbb{F}_q$ is $n - |Z(C)|$.*

Proof. Follows from the above proposition by comparing degrees and using Proposition 1.62. ■

Proposition 1.77. *The complete defining set $Z(C)$ of a cyclic code of length n over $\mathbb{F}_q$, with $\gcd(q, n) = 1$, is a disjoint union of some q -cyclotomic cosets.*

Proof. Note that $s \in Z(g) = Z(C) \implies g(\alpha^s) = 0 \implies g(\alpha^{sq^i}) = 0$ for every $i \geq 1$. This shows that $s \in Z(C) \implies C_s \subseteq Z(C)$. The result now follows from Proposition 1.70. ■

Corollary 1.78. *The number of q -ary cyclic codes of length n (for $\gcd(q, n) = 1$) is given by 2^N where N is the number of the cyclotomic cosets.*

Proof. We can choose any combination of these cosets and we will get a code. Conversely any code must arise from some disjoint union of these subsets by the above proposition. ■

Proposition 1.79. *Let C be a q -ary cyclic code of length n where $\gcd(q, n) = 1$. Then*

$$Z(C^\perp) = \mathbb{Z}_n \setminus \{-j \mid j \in Z(C)\}$$

Proof. Let g be the generator polynomial of C . We have $g(x)h(x) = x^n - 1$ and $g^\perp(x) = h_0^{-1}x^k h(1/x) = h_0^{-1}h^*(x)$ is the generator polynomial of $C^\perp$.

Let α be a primitive n -th root of unity in $\overline{\mathbb{F}}_q$. Observe that

- (i) $j \in Z(C) \iff g(\alpha^j) = 0$
- (ii) $g(\alpha^j) = 0 \iff h(\alpha^j) \neq 0$
- (iii) $h(\alpha^j) = 0 \iff h^*(\alpha^{-j}) = 0 \iff g^\perp(\alpha^{-j}) = 0$

Items (i) and (iii) are straightforward. The reverse implication of (ii) is also trivial. Suppose both $g(\alpha^j)$ and $h(\alpha^j)$ were 0, then α^j would be a repeated root of $x^n - 1$ which is not possible. This proves the forward direction of (ii).

The result now follows since $j \in Z(C) \iff g^\perp(\alpha^{-j}) \neq 0$ holds by combining the items. ■

We work out an example in detail for better clarity. We choose a very general n and q .

Question. Analyse cyclic codes of length 12 over $\mathbb{F}_8$.

Ans. We firstly factorise $x^{12} - 1$ over $\mathbb{F}_8[x]$. Note that $\gcd(n, q) \neq 1$. We write $12 = 2^2 \cdot 3$ and $\gcd(3, 8) = 1$. Thus,

$$x^{12} - 1 = (x^3 - 1)^4$$

We now have reduced to factorising $x^3 - 1$. Since $\gcd(3, 8) = 1$, we may apply our theory of cyclotomic cosets. Let us find these cyclotomic cosets as subsets of $\mathbb{Z}_n = \mathbb{Z}_3$.

The algorithm is simple. Begin with any $x \in \mathbb{Z}_n$ and keep multiplying with q until we reach x again. This forms the coset C_x . Now we pick an element $y \in \mathbb{Z}_n \setminus C_x$ and do the same to obtain C_y and so on.

Here, we get: $\{0\}, \{1, 2\}$ as the cyclotomic cosets. These are C_0 and $C_1 = C_2$, respectively.

Now $M_0(x) = (x - 1)$ and $M_1(x) = M_2(x) = (x - \alpha)(x - \alpha^2)$. Now we need to pick an α . This must be a primitive 3rd root of unity in $\overline{\mathbb{F}}_8 = \mathbb{F}_8 \cup \mathbb{F}_{64} \cup \mathbb{F}_{512} \cup \dots$. It is obvious that α must be a root of $x^2 + x + 1 = 0$ (could be any one). Using this α , $M_1(x) = M_2(x) = x^2 + x + 1$.

Thus, we get

$$x^{12} - 1 = (x - 1)^4(x^2 + x + 1)^4$$

as a factorisation over $\mathbb{F}_q$.

Thus the number of such cyclic codes is $5 \times 5 = 25$ since we can choose any combination of these factors to generate the code. □

1.8 Bounds on codes

We first discuss the BCH bound since it is closely associated with cyclic codes.

Theorem 1.80 (BCH Bound). *Let C be a cyclic code over $\mathbb{F}_q$ of length n . Let $\alpha \in \overline{\mathbb{F}}_q$ be a primitive n -th root of unity. If the generator polynomial $g(x) \in \mathbb{F}_q[x]$ has $\delta - 1$ consecutive powers of α as its roots, then*

$$d(C) \geq \delta$$

Note that wrapping powers is allowed, that is, $\{n-1, 0, 1\}$ is considered to be a contiguous block.

Proof. Suppose the condition is satisfied. Then $Z(C) = Z(g)$ has a continuous set $\{b, b+1, \dots, b+\delta-1\}$ as a subset for some $b \in \mathbb{Z}$. That is,

$$c(\alpha^j) = 0 \quad \forall j = b, b+1, \dots, b+\delta-1 \text{ and } \forall c(x) \in \pi(C)$$

Thus, if we write $c(x) = c_0 + c_1x + \dots + c_{n-1}x^{n-1}$, then we get a system

$$\sum_{i=0}^{n-1} c_i \alpha^{i(b+j)} = 0 \quad \forall j = 0, 1, \dots, \delta-1 \quad \forall (c_0, \dots, c_{n-1}) \in C$$

In matrix form we get

$$\begin{bmatrix} 1 & \alpha^b & \alpha^{2b} & \dots & \alpha^{(n-1)b} \\ 1 & \alpha^{b+1} & \alpha^{2(b+1)} & \dots & \alpha^{(n-1)(b+1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{b+\delta-1} & \alpha^{2(b+\delta-1)} & \dots & \alpha^{(n-1)(b+\delta-1)} \end{bmatrix}_{\delta \times n} c^\top = 0 \quad \forall c \in C$$

Now let $\tilde{C}$ be the linear code of length n with parity check matrix $\tilde{H}$ as above. For this to even make sense, one has to observe that $\tilde{H}$ is indeed full rank (hint: Vandermonde).

Thus, $C \subseteq \tilde{C}$ holds. An interesting remark is that the entries of $\tilde{H}$ are not in $\mathbb{F}_q$ since α is not necessarily in this field. However, $C \subseteq \mathbb{F}_q^n$ and hence

$$C \subseteq \tilde{C} \cap \mathbb{F}_q^n \subseteq \tilde{C} \subseteq \overline{\mathbb{F}}_q^n$$

Thus since C is a subcode of $\tilde{C}$ (now treated as codes in $\overline{\mathbb{F}}_q^n$), we have

$$d(C) \geq d(\tilde{C})$$

Now any $\delta \times \delta$ minor of $\tilde{H}$ will have a non-zero determinant since it is simply a Vandermonde determinant. Also any $\delta+1$ columns are linearly dependent since the length of a column is δ . Hence, by Corollary 1.18 we have $d(\tilde{C}) = \delta$ and we are done. ■

This motivates us to define a class of codes which are known as BCH codes.

Definition 1.81. If $Z(C)$ is itself a single contiguous block, then C is called a **BCH code** with designated distance $\delta+1$. We call it a BCH code in the **narrow sense** if $b=1$ and call it **primitive** if $n = q^e - 1$ for some $e \geq 1$.

Question. How to construct a BCH code with designated distance $\delta+1$?

Ans. We choose the following generator polynomial:

$$g(x) = \text{LCM}(M_b(x), M_{b+1}(x), \dots, M_{b+\delta-1}(x))$$

The code corresponding to this will be a BCH code with designated distance $\delta+1$ for obvious reasons. □

We can improve this bound and make it much stronger.

Definition 1.82. Let $a \in \mathbb{Z}_n$ be such that $\gcd(a, n) = 1$. A subset $S \subseteq \mathbb{Z}_n$ is said to be **consecutive of period $a \pmod{n}$** if S is of the form

$$S = \{b, b + a, b + 2a, \dots, b + (\delta - 2)a\}$$

for some $\delta \leq n + 1, b \in \mathbb{Z}$.

Theorem 1.83 (Generalised BCH Bound). *Let C be a cyclic code over $\mathbb{F}_q$ of length n . Let $g(x) \in \mathbb{F}_q[x]$ be the generator polynomial. If $Z(C) = Z(g)$ contains a subset which is consecutive of period $a \pmod{n}$ and has size $\delta - 1$ (some $a, \delta \in \mathbb{Z}$) with $\gcd(a, n) = 1$ and $\delta \leq n + 1$, then*

$$d(C) \geq \delta$$

Proof. We could do it directly from scratch by modifying the BCH bound proof. We however demonstrate a proof which uses the BCH bound.

Since $\gcd(a, n) = 1$, a is invertible in the group of units. Choose c such that $ac \equiv 1 \pmod{n}$. Then choose $\beta = \alpha^a$ so that $\alpha = \beta^c$.

Note that β is also a primitive n -th root of unity in $\overline{\mathbb{F}}_q$ and with respect to β we have

$$\{bc, bc + 1, \dots, bc + \delta - 2\} \in Z(C)$$

Applying Theorem 1.80 we get our result. ■

Question. How does one spot these consecutive sets of period a , practically?

Ans. Suppose $r \in Z(C)$ with $\gcd(r, n) = 1$. Then $rs \equiv 1 \pmod{n}$ for some $s \in \mathbb{Z}_n$. Now if $sZ(C)$ has a contiguous subset, then the original $Z(C)$ will also have a contiguous subset of this same size. □

Below is a nice application of this bound.

Proposition 1.84. *The Hamming code is monomial equivalent to a cyclic code provided $\gcd(r, q - 1) = 1$.*

Proof. Since $n \mid q^r - 1 = |\mathbb{F}_{q^r}^*|$, by Cauchy's theorem, we have an element $\alpha \in \mathbb{F}_{q^r}^*$ of order n and hence this is a primitive n -th root of unity.

Consider the cyclic code C with $\pi(C) = \{c(x) \in R_n \mid c(\alpha) = 0\}$. By the BCH bound we can say that

$$d(C) \geq 2$$

Suppose $d(C) = 2$. Then there exists $c = c_i x^i + c_j x^j \in \pi(C)$ for some $c_i, c_j \in \mathbb{F}_q^*$ (WLOG assume $i < j$). But $c(\alpha) = 0$ gives

$$c_j \alpha^j = -c_i \alpha^i \implies \alpha^{j-i} = -c_i c_j^{-1} \in \mathbb{F}_q^*$$

Thus, by Lagrange's theorem $(\alpha^{j-i})^{q-1} = 1$. Hence, $n \mid (j-i)(q-1)$.

By the given hypothesis, $\gcd(r, q-1) = 1$ and hence since $n = 1 + q + \dots + q^{r-1} \equiv r \pmod{q-1}$ we have $\gcd(n, q-1) = \gcd(r, q-1) = 1$.

Thus, $n \mid j-i$ which is a contradiction since $0 \leq i < j < n$.

Hence we must have $d(C) \geq 3$. Thus if H is a parity check matrix of C , then no two columns of H are linearly dependent. However, the order of H is $n \times r$ since the dimension of C is $n - |Z(C)| = n - |\{1, q, \dots, q^{r-1}\}| = n - r$.

Thus H is forced to be $H_r(q)$ up to scaling and reordering of the columns and hence we are done. ■

We are now done with the topic of cyclic codes. They will of course come back when we obtain the minimum distance of the Generalised Reed-Muller code.

We continue exploring other bounds.

Theorem 1.85 (Griesmer bound). *Let $C \in [n, k]_q$ with minimum distance d . Then*

$$\sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil \leq n$$

Proof. We require a lemma to prove this bound.

Lemma. *Let $C \in [n, k]_q$ and $x_0 \in C$ be a minimum distance codeword with $\text{wt}(x_0) = d$. Then there exists a code $C' \in [n - d, k - 1]_q$ with minimum distance $d' \geq \lceil d/q \rceil$.*

Proof of Lemma. Choose $C' = C^P$ for $P = \text{supp}(x_0)$, where C^P denotes the code punctured on P . Clearly $\text{wt}(x_0) = d$ and hence $n' = n - d$. We want to find $\dim(C')$ and the minimum distance of C' . Since monomially equivalent codes have the same parameters, we pass to one so that we may assume

$$x_0 = (1, 1, \dots, 1, 0, 0, \dots, 0) \in \mathbb{F}_q^n$$

where the first d entries are all 1.

Now $y \mapsto y^P$ is a surjective map $C \rightarrow C^P$ and hence $\dim(C') \leq \dim(C) = k$. If $\dim(C') < k - 1$, then the kernel of this map has dimension at least 2. Thus, there exists a y in the kernel so that y is linearly independent from x_0 . Because y is in the kernel, its support is completely contained in P .

Thus we can find a coordinate $i \in P$ where $y_i \neq 0$. Setting $\mu = y_i/x_{0,i} \in \mathbb{F}_q$, the codeword $y - \mu x_0$ is non-zero (by linear independence) but its i -th coordinate is 0. Hence $0 < \text{wt}(y - \mu x_0) < d$. This is a contradiction to x_0 being a minimum weight codeword, and hence $\dim(C') = k - 1$.

We now want to show that $d' \geq \lceil d/q \rceil$. Let $z = y^P$ for some $y \in C$ such that $z \neq 0$. By the pigeonhole principle, there must exist some $\alpha \in \mathbb{F}_q$ which occurs at least $\lceil d/q \rceil$ times among the first d coordinates $y_1, \dots, y_d$.

Consider the codeword $y - \alpha x_0$. If $y - \alpha x_0 = 0$, then $y = \alpha x_0$, which implies $z = y^P = \alpha x_0^P = 0$, contradicting our assumption that $z \neq 0$. Thus $y - \alpha x_0$ is a non-zero codeword, which implies $\text{wt}(y - \alpha x_0) \geq d$.

In the first d coordinates, $y - \alpha x_0$ has at least $\lceil d/q \rceil$ zeros. In the remaining $n - d$ coordinates, x_0 is zero, so the entries exactly match z . Hence

$$d \leq \text{wt}(y - \alpha x_0) \leq d - \left\lceil \frac{d}{q} \right\rceil + \text{wt}(z)$$

This shows that $\text{wt}(z) \geq \left\lceil \frac{d}{q} \right\rceil$. Since this holds for all non-zero $z \in C'$, we conclude that the minimum distance $d' \geq \left\lceil \frac{d}{q} \right\rceil$. \\

Using the lemma we may prove our result easily via induction. The bound is clearly true for $k = 0$ and $k = 1$. Suppose $k > 1$ and the result holds for codes with dimensions less than k . Then for the code C , we use the lemma to obtain code C' with parameters $n' = n - d$ and $d' \geq \lceil d/q \rceil$. Using the induction hypothesis on C' we have

$$n' \geq \sum_{i=0}^{k-2} \left\lceil \frac{d'}{q^i} \right\rceil$$

Since $d' \geq \lceil d/q \rceil$, it follows that $\lceil d'/q^i \rceil \geq \lceil d/q^{i+1} \rceil$. Substituting $n' = n - d$, we get

$$n - d \geq \sum_{i=0}^{k-2} \left\lceil \frac{d}{q^{i+1}} \right\rceil = \sum_{j=1}^{k-1} \left\lceil \frac{d}{q^j} \right\rceil$$

Adding $d = \lceil d/q^0 \rceil$ to both sides gives

$$n \geq \sum_{j=0}^{k-1} \left\lceil \frac{d}{q^j} \right\rceil$$

This matches the desired inequality and completes the proof. ■

We now have the sphere packing bound. This holds for a non-linear code, too.

Definition 1.86. Let Q be a finite set of size q . Let $q \leq n \in \mathbb{Z}$. Let $Q^n = Q \times Q \times \cdots \times Q$ be the n -space over Q . Given any $x \in Q^n$ and an integer $t \geq 0$ we define the **ball of radius t centred at x** to be

$$\mathcal{S}_t(x) = \{y \in Q^n \mid d_H(x, y) \leq t\}$$

Proposition 1.87. *The number of points in a ball of radius $t \geq 0$ does not depend on the centre of the ball.*

Proof. We have for some $x \in Q^n$ and integer $t \geq 0$,

$$\begin{aligned} |\mathcal{S}_t(x)| &= \sum_{i=0}^t |\{y \in Q^n \mid d_H(x, y) = i\}| \\ &= \sum_{i=0}^t \binom{n}{i} (q-1)^i \end{aligned}$$

We often denote this number as $V_q(n, t)$. ■

Definition 1.88. Given any positive integers q, n and a positive integer $d \leq n$, we define the set

$$\mathcal{A}_q(n, d) = \max \{|C| \mid C \subseteq Q^n \text{ is a code of length } n \text{ with minimum distance } d\}$$

where Q is some set of alphabet of size q .

The Singleton bound is the statement that $\mathcal{A}_q(n, d) \leq q^{n-d+1}$.

Definition 1.89. Let q be a prime power and n, d be positive integers with $d \leq n$. We define the set

$$\mathcal{B}_q(n, d) = \max \{|C| \mid C \text{ is a } q\text{-ary linear code of length } n \text{ with minimum distance } d\}$$

It is obvious that $\mathcal{B}_q(n, d) \leq \mathcal{A}_q(n, d)$.

Theorem 1.90 (Sphere-Packing bound). *Let q be a prime power and n, d be positive integers with $d \leq n$. Suppose $t = \lfloor \frac{d-1}{2} \rfloor$ then*

$$\mathcal{A}_q(n, d) \leq \frac{q^n}{V_q(n, t)}$$

Proof. Suppose $x \neq y \in C$. Let $z \in \mathcal{S}_t(x) \cap \mathcal{S}_t(y)$. Then,

$$d(x, y) \leq d(x, z) + d(z, y) \leq 2t \leq d-1 < d$$

This contradicts the minimum distance. Hence, all spheres of radius t are disjoint.

Clearly their disjoint union is a subset of Q^n and hence

$$V_q(n, t) |C| \leq q^n$$

This proves the theorem. ■

Definition 1.91. Any code (non-linear, too) which achieves the SPB is called a perfect code.

Proposition 1.92. *The Hamming code is a perfect code.*

Proof. Using Proposition 1.27 we have $t = 1$. Thus

$$V_q(n, 1) = \sum_{i=0}^1 \binom{n}{i} (q-1)^i = 1 + n(q-1) = 1 + (q^r - 1) = q^r$$

Thus we have to show that $|C| q^r = q^n$. This is clearly true since the dimension is $n - r$ by Proposition 1.26. ■

Question. Is the dual of a perfect linear code, perfect?

Ans. No. The simplex code for instance is not perfect. This is a constant weight code by Proposition 1.28. Thus the minimum distance is q^{r-1} . One can check that it will not be perfect for $r = q = 2$. $\square$

Theorem 1.93 (Gilbert bound/Sphere covering bound). *Let q be a prime power and n, d be positive integers with $d \leq n$. Then*

$$\mathcal{A}_q(n, d) \geq \frac{q^n}{V_q(n, d-1)}$$

Proof. Suppose $C \subseteq Q^n$ (some Q with $|Q| = q$) is a code of minimum distance d with $\mathcal{A}_q(n, d) = |C| = M$. That is, C is a code attaining the maximum.

Let $C = \{c_1, c_2, \dots, c_M\}$. We claim that for every x we can find some $1 \leq i \leq M$ so that $d(x, c_i) < d$. Suppose not. Then x can be added to C preserving the minimum distance d which contradicts maximality of C .

Thus, we have

$$Q^n = \bigcup_{i=1}^M \mathcal{S}_{d-1}(c_i)$$

Thus taking cardinalities we get

$$q^n \leq \sum_{i=1}^M |\mathcal{S}_{d-1}(c_i)| = M \cdot V_q(n, d-1)$$

This proves the theorem. $\blacksquare$

We can do better for linear codes. We need some lemmas before we state and prove this theorem for $\mathcal{B}_q(n, d)$.

Lemma 1.94. *Let $n \geq d \geq 2$ be integers. If there is a k such that $V_q(n-1, d-2) < q^{n-k}$ then there exists a linear code $C \in [n, k]_q$ with minimum distance $d(C) \geq d$.*

Proof. We construct a matrix H and show that it is a valid parity check matrix of some code with minimum distance at least d . Let the columns be denoted as $h_1, h_2, \dots$.

Choose the columns $h_1, h_2, \dots, h_{n-k}$ to be the standard basis of $\mathbb{F}_q^{n-k}$. Having chosen $h_1, h_2, \dots, h_j$ for $n-k \leq j < n$ such that any $d-1$ columns among $h_1, \dots, h_j$ are linearly independent, choose h_{j+1} so that h_{j+1} is not a linear combination of any $d-2$ columns among $h_1, \dots, h_j$.

Now note that if $d-1 > n-k$, such a code cannot exist since it will violate the Singleton bound. Thus, the first step check of our algorithm is satisfied.

Continuing the process at any stage is possible if

$$q^{n-k} > \text{number of linear combinations of any } d-2 \text{ columns of } h_1, \dots, h_j$$

But the RHS is equal to

$$\begin{aligned} & \sum_{i=0}^{d-2} \binom{j}{i} (q-1)^i \\ &= V_q(j, d-2) \\ &\geq V_q(n-1, d-2) \end{aligned}$$

By construction the rank of H is $n-k$ and hence it is a parity check matrix of some code. Further, any set of $d-1$ columns will be linearly independent giving that $d(C) \geq d$. $\blacksquare$

Lemma 1.95. *If there is an $[n, k]_q$ linear code C with $d(C) \geq d$, then we can obtain a code C' from C with same length and dimension over $\mathbb{F}_q$ with $d(C') = d$*

Proof. Suppose $x \in C$ is a minimum weight code word. Then $|\text{supp}| = d(C) \geq d$. Choose a $(d(C) - d)$ -sized subset $P \subseteq \text{supp}(x)$ and consider the map $\phi : C \rightarrow \mathbb{F}_q^n$ given by

$$\phi(C)_i = \begin{cases} c_i & \text{if } i \notin P \\ 0 & \text{otherwise} \end{cases}$$

It is clear that ϕ is a linear map. Now suppose $c \neq 0 \in \ker(\phi)$. Then c is 0 in the positions of P , that is $d(C) - d$ number of places. Hence,

$$\text{wt}(\phi(c)) \geq \text{wt}(c) - (d(C) - d) \geq d$$

But the LHS is just $\text{wt}(0) = 0$ which is a contradiction. Thus, ϕ is injective. Moreover, by construction we have

$$\text{wt}(\phi(x)) = d$$

Thus, C' is a minimum weight d code of the same length and same dimension (since ϕ is injective). ■

Theorem 1.96 (Varshamov bound). *Let q be a prime power and n, d be positive integers with $d \leq n$. Then*

$$\mathcal{B}_q(n, d) \geq q^{n - \lceil \log_q(1 + V_q(n-1, d-2)) \rceil}$$

Proof. For $d = 1$ we choose $C = \mathbb{F}_q^n$ and we are done. Thus suppose $2 \leq d \leq n$. Let $k = n - \lceil \log_q(1 + V_q(n-1, d-2)) \rceil$. Then

$$q^{n-k} \geq 1 + V_q(n-1, d-2)$$

Using Lemma 1.94 and then Lemma 1.95 we get an $[n, k]_q$ linear code C with minimum distance d and hence

$$\mathcal{B}_q(n, d) \geq |C| = q^k = q^{n - \lceil \log_q(1 + V_q(n-1, d-2)) \rceil}$$

■

Theorem 1.97 (Plotkin bound). *Let q be an integer and n, d be positive integers with $d \leq n$. Suppose we have*

$$n \left(1 - \frac{1}{q}\right) < d$$

then the following bound holds

$$\mathcal{A}_q(n, d) \leq \left\lfloor \frac{d}{d - \left(1 - \frac{1}{q}\right)n} \right\rfloor$$

Proof. Let C be some code (non-linear possibly) of length n over an alphabet of size q . Suppose it has size M and minimum distance d . We want to show that

$$M \leq \left\lfloor \frac{d}{d - \left(1 - \frac{1}{q}\right)n} \right\rfloor$$

To this end consider

$$\begin{aligned} T &= \sum_{c \in C} \sum_{c' \in C} d(c, c') \\ &= \sum_{c \in C} \sum_{\substack{c' \in C \\ c' \neq c}} d(c, c') \\ &\geq \sum_{c \in C} \sum_{\substack{c' \in C \\ c' \neq c}} d \\ &= dM(M-1) \end{aligned}$$

On the other hand consider the $M \times n$ matrix P whose rows are codewords of C . Each row will be of the form $(c_1, \dots, c_n)$. For a given $a \in Q$, let $n_{i,a}$ denote the number of times a occurs in the i -th column of P .

Clearly, for any given i we have $M = \sum_{a \in Q} n_{i,a}$ and

$$\begin{aligned} T &= \sum_{i=1}^n \sum_{a \in Q} n_{i,a} (M - n_{i,a}) \\ &= \sum_{i=1}^n M^2 - \sum_{a \in Q} \sum_{i=1}^n n_{i,a}^2 \end{aligned}$$

Thus, $nM^2 - \sum_{a \in Q} \sum_{i=1}^n n_{i,a}^2 \geq dM^2 - dM$. By applying Cauchy-Schwarz (fix some column i) we have

$$\begin{aligned} M^2 &= \left(\sum_{a \in Q} n_{i,a} \right)^2 \\ &\leq \left(\sum_{a \in Q} n_{i,a}^2 \right) \left(\sum_{a \in Q} 1^2 \right) \end{aligned}$$

Thus, we get

$$\sum_{a \in Q} n_{i,a}^2 \geq \frac{M^2}{q}$$

Putting it together we get

$$nM^2 - n \frac{M^2}{q} \geq dM^2 - dM$$

which gives the desired bound on M . ■

1.9 Encoding and Decoding

This section deals with encoding and decoding of codes. A particular message $(m_1, \dots, m_k)$ is to be sent to a receiver. We first need to encode this in some way such that it can be decoded uniquely. That is, we need an injective map $\mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$ for some n . Note that injectivity forces $n \geq k$.

Naturally, a larger n increases the computational complexity of decoding. For linear codes, the generator matrix does this job well. The encoded word is now, say, $(c_1, \dots, c_n)$. We can now send it to the receiver. In practice, the channel through which the message is sent is noisy and can disturb our data. Thus the receiver, on his end, sees the message as some $(r_1, \dots, r_n)$.

The error introduced by the channel is $(e_1, \dots, e_n) = (r_1 - c_1, \dots, r_n - c_n)$. If this happens to be the zero vector, then we are in the ideal case and the decoding may be done easily since the encoding process is known by the receiver.

In this section we cover the Berlekamp-Welch algorithm which deals with the decoding of the Reed-Solomon code.

Recall the form of the generator matrix of the code $C = RS_{q,k}(a_1, \dots, a_n)$:

$$V_{k,n} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{k-1} & a_2^{k-1} & a_3^{k-1} & \cdots & a_n^{k-1} \end{bmatrix}$$

We had also remarked that this can be brought into standard form. We can in fact explicitly write down the standard form. One needs to pick a suitable basis for this purpose.

Consider the standard basis $\{e_1, \dots, e_k\}$ of $\mathbb{F}_q^k$ viewed as a subset of $\mathbb{F}_q^n$ by padding with $n-k$ trailing zeroes. We are looking for k polynomials $L_1, \dots, L_k$ in $\mathbb{F}_q[x]$ of degree at most $k-1$ such that $L_i(x) = \text{ev}_a^{-1}(e_i)$ and they form a basis.

Here, $\text{ev}_a : \mathbb{F}_q[x]_k \rightarrow \mathbb{F}_q^n$ is a map from the set of polynomials with degree at most $k-1$ to $\mathbb{F}_q^n$ given by evaluation of the polynomial at the points $a_1, \dots, a_n$ given by the coordinates of the vector a .

If we can do this, then with respect to this basis, the generator matrix will have the first block as the identity matrix. In particular observe that

$$L_i(a_j) = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{otherwise} \end{cases}$$

Thus by the Lagrange interpolation method we may get an explicit form as

$$L_i(x) = \prod_{\substack{1 \leq j \leq k \\ j \neq i}} \frac{x - a_j}{a_i - a_j}$$

Note that the degree of L_i is indeed $k-1$, all L_i are uniquely determined with the above formula and they also form a basis for the space of all polynomials of degree at most $k-1$, that is, $\mathbb{F}_q[x]_k$.

Thus with respect to this basis, the generator matrix takes the form

$$\left[\begin{array}{c|ccc} I_{k \times k} & L_1(a_{k+1}) & \dots & L_1(a_n) \\ & \vdots & \ddots & \vdots \\ & L_k(a_{k+1}) & \dots & L_k(a_n) \end{array} \right]_{k \times n}$$

Thus a parity-check matrix is

$$\left[\begin{array}{ccc|c} -L_1(a_{k+1}) & \dots & -L_k(a_{k+1}) & \\ \vdots & \ddots & \vdots & \\ -L_1(a_n) & \dots & -L_k(a_n) & \end{array} \middle| I_{(n-k) \times (n-k)} \right]_{(n-k) \times n}$$

Question. Let $\gamma_i = \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{1}{a_i - a_j}$ for $i = 1, 2, \dots, n$. Then show that the following matrix is a parity check matrix of $RS_{q,k}(a_1, \dots, a_n)$

$$V_{n-k,n} \begin{bmatrix} \gamma_1 & 0 & \dots & 0 \\ 0 & \gamma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \gamma_n \end{bmatrix}$$

Ans. Firstly let the diagonal matrix be denoted as D . Clearly D is invertible since none of the γ_i are zero. Thus, the rank of the product is same as the rank of $V_{n-k,n}$ which is $n-k$. Thus we only need to show that every code word is annihilated by this matrix. Equivalently, we aim to show that

$$V_{n-k,n} D V_{k,n}^\top = 0$$

since $V_{k,n}$ is a generator matrix of the code. The product is an $(n-k) \times k$ matrix. Let us find the (α, β) -th entry of the matrix. It is given by

$$\sum_{i=1}^n \sum_{j=1}^n a_i^{\alpha-1} D_{ij} a_j^{\beta-1} = \sum_{i=1}^n a_i^{\alpha+\beta-2} \gamma_i$$

Since $1 \leq \alpha \leq n - k$ and $1 \leq \beta \leq k$, we have $0 \leq \alpha + \beta - 2 \leq n - 2$. We claim that for every $0 \leq m \leq n - 2$,

$$\sum_{i=1}^n a_i^m \gamma_i = 0$$

Define the polynomials $\ell_j \in \mathbb{F}_q[x]$ for $j = 1, 2, \dots, n$ as:

$$\ell_j(x) = \prod_{\substack{1 \leq i \leq n \\ i \neq j}} \frac{x - a_i}{a_j - a_i}$$

Then each ℓ_j is a degree $n - 1$ polynomial with leading coefficient γ_j . Also, we know that the ℓ_j form a basis for the set of all polynomials over $\mathbb{F}_q$ of degree at most $n - 1$. Thus if $f \in \mathbb{F}_q[x]_n$ we can write

$$f(x) = \sum_{j=1}^n f(a_j) \ell_j(x)$$

Taking the coefficient of x^{n-1} on both sides gives

$$[x^{n-1}]f(x) = \sum_{j=1}^n f(a_j) \gamma_j$$

Choosing $f(x) = x^m$ for $0 \leq m \leq n - 2$ gives us our result. $\square$

Definition 1.98. Given an encoded message $\underline{c} = (c_1, \dots, c_n)$ and a received message $\underline{r} = (r_1, \dots, r_n)$, we define the **error vector** to be

$$\underline{e} = (e_1, \dots, e_n) = (r_1 - c_1, \dots, r_n - c_n)$$

The **error set** is defined to be

$$E = \{i \in [n] \mid e_i = r_i - c_i \neq 0\}$$

As remarked before, in the ideal case, this vector is 0. Even when this is non-zero, the Berlekamp-Welch algorithm shows us how to obtain the original message through decoding under certain conditions and assumptions.

Definition 1.99. Given an error vector, we define the **error locator polynomial** to be

$$\Lambda(x) = \prod_{i \in E} (x - a_i)$$

Throughout the decoding we will assume that the error set is not too large. Specifically,

$$|E| \leq \left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{n-k}{2} \right\rfloor = \ell \quad (\text{say})$$

Further, the message vector $(m_1, \dots, m_k)$ may be thought as a polynomial $m_1 + m_2x + \dots + m_kx^{k-1}$. Then using the generator matrix of the Reed-Solomon code, the encoded vector is

$$c_i = m(a_i) \quad \forall \quad i = 1, 2, \dots, n$$

Lemma 1.100. *With the standard notation as discussed above, consider the bivariate polynomial*

$$Q(x, y) = \Lambda(x)(y - m(x)) = Q_0(x) + yQ_1(x)$$

Then we have

$$Q(a_i, r_i) = 0 \tag{*}$$

$$\deg(Q_1(x)) \leq \ell ; \deg(Q_0(x)) \leq \ell + k - 1 \tag{**}$$

$$m(x) = -\frac{Q_0(x)}{Q_1(x)} \tag{***}$$

Proof. The proof for second item follows trivially since $Q_1(x) = \Lambda(x)$ has degree $|E| \leq \ell$ and $Q_0(x) = -\Lambda(x)m(x)$ has degree $|E| + k - 1 \leq \ell + k - 1$ and the results follows.

For the first item fix some $i = 1, 2, \dots, n$. Then $Q(a_i, r_i) = \Lambda(a_i)(r_i - m(a_i))$. Now if $m(a_i) = c_i \neq r_i$, then $i \in E$ and hence $\Lambda(a_i) = 0$. Thus this evaluation is always zero proving the first item.

The last one is trivial and follows by definition. ■

We have a result which is based on the lemma and makes the third item redundant.

Proposition 1.101. *Let $\underline{c} \in RS_{q,k}(\underline{a})$ be an encoded message vector given by $\underline{c} = ev_{\underline{a}}m((x))$ for $m \in \mathbb{F}_q[x]_k$. Let $\underline{r} \in \mathbb{F}_q^n$ be the received codeword such that $\text{wt}(\underline{e}) = \text{wt}(\underline{r} - \underline{c}) \leq \ell = \lfloor \frac{n-k}{2} \rfloor$.*

Suppose there exists a bivariate polynomial $Q(x, y)$ of the form $Q(x, y) = Q_0(X) + Q_1(x)y$ which satisfies items $(\star)$ and $(\star\star)$, then it also satisfies item $(\star\star\star)$.

Proof. Consider $h(x) = Q(x, m(x))$. Then

$$h(a_i) = Q(a_i, m(a_i)) = Q(a_i, c_i) = \begin{cases} 0 & \text{if } i \notin E \\ -Q_1(a_i)e_i & \text{if } i \in E \end{cases}$$

This is because if $i \notin E$, then $c_i = r_i$ and hence by $(\star)$ we have $Q(a_i, r_i) = 0$. If not, then we have

$$Q_0(a_i) + Q_1(a_i)c_i = Q_0(a_i) + Q_1(a_i)(r_i - e_i) = Q(a_i, r_i) - Q_1(a_i)e_i = -Q_1(a_i)e_i$$

Now $|E| = \text{wt}(\underline{e}) \leq \ell$. Thus,

$$|E^c| \geq n - \ell \geq \frac{n+k}{2}$$

Let us also look at the degree of this polynomial h

$$\deg(h) \leq \ell + k - 1 \leq \frac{n-k}{2} + k - 1 = \frac{n+k}{2} - 1$$

Since h is zero on E^c , it has more zeroes than its degree and hence must be the zero polynomial. Thus, $Q_0(x) + Q_1(x)m(x) = 0$ and we have our result. ■

By virtue of the above Proposition 1.101 all we have to do is obtain such a polynomial Q so that we may recover our message $m(x)$. The rest of this section will be devoted to getting such a polynomial.

Definition 1.102. We define some polynomials which we will regularly make use of. Here, $\underline{a} = (a_1, \dots, a_n)$ is the fixed set of distinct elements of $\mathbb{F}_q$ used in the Reed-Solomon code. We define

$$G(x) = \prod_{i=1}^n (x - a_i)$$

$$R(x) = \prod_{i=1}^n r_i \ell_i(x)$$

where

$$\ell_i(x) = \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{x - a_j}{a_i - a_j}$$

Note that $\deg(R) \leq n - 1$ and $\deg(G) = n$ and $R(a_i) = r_i$.

Lemma 1.103. *If $Q(x, y)$ is a bivariate polynomial of the form $Q_0(x, y) + yQ_1(x)$ such that $(\star)$ holds, then*

$$Q(x, y) = B_0(x)G(x) + B_1(x)(y - R(x))$$

for some $B_0, B_1 \in \mathbb{F}_q[x]$.

Proof. Set $\tilde{Q}(x) = Q_0(x) + Q_1(x)R(x)$. Note that $\tilde{Q}(a_i) = Q_0(a_i) + Q_1(a_i)r_i = 0$ by $(\star)$. This holds for every $i \in [n]$ and hence we conclude that $G \mid \tilde{Q}$. Thus if we write

$$\tilde{Q}(x) = Q_0(x) + Q_1(x)R(x) = G(x)B_0(x)$$

the result follows by setting $B_1(x) = Q_1(x)$. ■

Thus now if we can get hold of polynomials B_0 and B_1 then we are done.

Proposition 1.104. *If $Q(x, y) = Q_0(x) + Q_1(x)y$ satisfies $(\star)$ and $(\star\star)$, then the vector $[Q_0(x) \ Q_1(x)]$ lies in the row space of the matrix*

$$M = \begin{bmatrix} G(x) & 0 \\ -R(x) & 1 \end{bmatrix}$$

Further, $\max(\deg(Q_0(x)), \deg(Q_1(x)) + k - 1) \leq \ell + k - 1$.

Proof. Applying Proposition 1.103 we have

$$\begin{bmatrix} B_0(x) & B_1(x) \end{bmatrix} \begin{bmatrix} G(x) & 0 \\ -R(x) & 1 \end{bmatrix} = \begin{bmatrix} Q_0(x) & Q_1(x) \end{bmatrix}$$

The remark follows trivially by $(\star\star)$. ■

We shall use this notation of matrix M throughout the section without explicitly writing it down every time.

Definition 1.105. Given two polynomials $A, B \in \mathbb{F}_q[x]$, we define the $(\mathbf{k} - 1)$ -*degree* of the ordered pair (A, B) to be

$$\deg_{k-1}(A, B) = \max\{\deg(A), \deg(B) + k - 1\}$$

We use the convention that the zero polynomial has degree $-\infty$.

The theorem below says that we do not need to worry about specific B_0 and B_1 because every choice will work provided there is some degree condition satisfied.

Theorem 1.106. *Let $[A \ B]$ be a vector in the row space of M such that $\deg_{k-1}(A, B) \leq \ell + k - 1$. Then the polynomial $D(x, y) = A(x) + B(x)y$ satisfies $(\star)$ and $(\star\star)$.*

Proof. Write for some polynomials $C_0, C_1 \in \mathbb{F}_q[x]$,

$$A(x) = C_0(x)G(x) - C_1(x)R(x)$$

$$B(x) = C_1(x)$$

Thus $D(x, y) = C_0G(x) - C_1(x)(y - R(x))$. Since $G(a_i) = 0$ and $r_i = R(a_i)$, condition $(\star)$ follows.

From the given degree condition, $(\star\star)$ also follows directly. ■

Owing to this proposition, it suffices to find an element of the row space of this matrix which satisfies the degree condition so that by applying Proposition 1.101 to $D(x, y)$ we are done.

Definition 1.107. A polynomial matrix

$$\begin{bmatrix} P_{00}(x) & P_{01}(x) \\ P_{10}(x) & P_{11}(x) \end{bmatrix}$$

with entries $P_{ij} \in \mathbb{F}_q[x]$ is said to be $\mathbf{k}$ -*reduced* if the following hold

$$\deg(P_{00}) \geq \deg(P_{01}) + k - 1 \tag{i}$$

$$\deg(P_{10}) < \deg(P_{11}) + k - 1 \tag{ii}$$

As the reader may have guessed by observing the nomenclature, this concept is closely related to the concept of $(k-1)$ -degree.

Proposition 1.108. *Let P be a k -reduced 2×2 matrix over $\mathbb{F}_q[x]$. Suppose $[A \ B]$ lies in the row space of P . Then either one of the following holds*

$$\deg_{k-1}(A, B) \geq \deg_{k-1}(P_{00}, P_{01})$$

$$\deg_{k-1}(A, B) \geq \deg_{k-1}(P_{10}, P_{11})$$

Proof. Write for some polynomials $H_0, H_1 \in \mathbb{F}_q[x]$,

$$A(x) = H_0(x)P_{00}(x) + H_1(x)P_{10}(x)$$

$$B(x) = H_0(x)P_{01}(x) + H_1(x)P_{11}(x)$$

Suppose if possible both the given conditions do not hold. Thus we have

$$\deg_{k-1}(A, B) < \deg_{k-1}(P_{00}, P_{01}) = \deg(P_{00})$$

$$\deg_{k-1}(A, B) < \deg_{k-1}(P_{10}, P_{11}) = \deg(P_{11}) + k - 1$$

Now by definition of the $(k-1)$ -degree, we know $\deg(A) \leq \deg_{k-1}(A, B) < \deg(P_{00})$ and $\deg(B) + k - 1 \leq \deg_{k-1}(A, B) < \deg(P_{11}) + k - 1$. Using the form of A and B we get

$$\deg(H_0(x)P_{00}(x) + H_1(x)P_{10}(x)) < \deg(P_{00})$$

$$\deg(H_0(x)P_{01}(x) + H_1(x)P_{11}(x)) < \deg(P_{11})$$

This immediately shows that H_0 and H_1 are non-zero. Further, we must have some cancellations so that the degree drops which gives

$$\deg(H_0(x)P_{00}(x)) = \deg(H_1(x)P_{10}(x))$$

$$\deg(H_0(x)P_{01}(x)) = \deg(H_1(x)P_{11}(x))$$

Using $\deg(fg) = \deg(f) + \deg(g)$ we get

$$\deg(P_{00}) + \deg(P_{11}) = \deg(P_{10}) + \deg(P_{01})$$

which contradicts the hypothesis of P being k -reduced. ■

Theorem 1.109 (Berlekamp-Welch Algorithm). *Let M be the matrix as before. Then M can be transformed into a k -reduced matrix by a finite sequence of elementary row operations.*

Proof. We describe the algorithm. We begin with a matrix P over $\mathbb{F}_q[x]$ satisfying

(i) P is not k -reduced because (ii) fails but (i) holds

(ii) $\deg(P_{01}) < \deg(P_{11})$

(iii) $\deg(P_{10}) < \deg(P_{00})$

We then apply the transformations of $R_2 \leftrightarrow R_1$ (switch rows) and then apply $R_2 \mapsto R_2 - t(x)R_1$ where $t(x)$ is the quotient when P_{00} is divided by P_{10} . Thus if we write $P_{00} = t(x)P_{10}(x) + s(x)$ where $\deg(s) < \deg(P_{10})$, then the matrix P transforms into the matrix P' after the two transformations as below.

$$P \mapsto P' = \begin{bmatrix} P_{10}(x) & P_{11}(x) \\ s(x) & P_{01}(x) - t(x)P_{11}(x) \end{bmatrix}$$

This is the algorithm we apply to a matrix satisfying the three conditions. We claim that the new matrix P' obtained will continue satisfying the hypothesis and will further have the properties

- (i) $\deg(P'_{10}) < \deg(P_{10})$
- (ii) $\deg(P'_{11}) > \deg(P_{11})$

Assuming the claim is true, note that M satisfies the initial three conditions and since while checking the condition of the matrices obtained being k -reduced, condition (i) always holds while condition (ii) fails, the second condition will eventually be satisfied because every iteration bridges the gap between $\deg(P_{10})$ and $\deg(P_{11}) + k - 1$ by at least 2 and eventually after a finite number of steps, we must have

$$\deg(P_{10}) < \deg(P_{11}) + k - 1$$

We now prove the claim made about P and P' . We need to check five conditions which are as follows.

- (i) $\deg(P'_{10}) < \deg(P_{10})$, that is, $\deg(s) < \deg(P_{10})$
- (ii) $\deg(P'_{11}) > \deg(P_{11})$, that is, $\deg(P_{01} - tP_{11}) > \deg(P_{11})$
- (iii) $\deg(P'_{00}) \geq \deg(P'_{01}) + k - 1$ but $\deg(P'_{10}) \geq \deg(P'_{11}) + k - 1$
- (iv) $\deg(P'_{01}) < \deg(P'_{11})$, that is, $\deg(P_{11}) < \deg(P_{01} - tP_{11})$
- (v) $\deg(P'_{10}) < \deg(P_{00})'$, that is, $\deg(s) < \deg(P_{10})$

These are assuming that P' is not k -reduced.

Note that (i) and (v) are the same and are trivially true by construction and definition of s . Let us restate condition (iii) in terms of the original polynomials. It becomes

$$\deg(P_{10}) \geq \deg(P_{11}) + k - 1 \quad \text{and} \quad \deg(s) \geq \deg(P_{01} - tP_{11}) + k - 1$$

The first condition is just the first condition of P being k -reduced which was assumed to be true. Since P' is not k -reduced, the second condition of being k -reduced must be false which is essentially the same as

$$\deg(s) \geq \deg(P_{01} - tP_{11}) + k - 1$$

Because if P' is k -reduced, the algorithm stops and we are done. All we need to do now is check condition (ii) which is the same as (iv). Now we have

$$\deg(P_{11}) \leq \deg(tP_{11}) \quad \text{and} \quad \deg(P_{11}) > \deg(P_{01})$$

Thus, $\deg(P_{01} - tP_{11}) = \deg(tP_{11})$. Now since $\deg(P_{00}) > \deg(P_{10})$, the quotient will have to be at least linear and hence t is at least linear. Thus, $\deg(tP_{11}) > \deg(P_{11})$. This proves condition (ii) (and (iv)) and finishes the proof. ■

Now we simply begin with M , apply the algorithm to reach a k -reduced matrix P and pick the row with least k -degree. Suppose this row is $[f(x) \ g(x)]$. Since row operations preserve the row space, this vector still lies in the row space of M . Now we know by Lemma 1.100 and Proposition 1.104, that there is supposed to exist some row vector $[Q_0(x) \ Q_1(x)]$ such that $\deg_{k-1}(Q_0, Q_1) \leq \ell + k - 1$. But since we are picking the row with the lesser k -degree, we have by Proposition 1.108 that

$$\deg_{k-1}(f, g) = \min \{ \deg_{k-1}(P_{\text{row } 1}), \deg_{k-1}(P_{\text{row } 2}) \} \leq \deg_{k-1}(Q_0, Q_1) \leq \ell + k - 1$$

Thus the hypothesis of Theorem 1.106 is satisfied and the message can be determined as

$$m(x) = -\frac{f(x)}{g(x)}$$

1.10 Generalised Hamming weights

Definition 1.110. Given a linear code C and a subset $D \subseteq C$, we define the *weight of the subset* to be

$$\text{wt}(D) = |\{i \in [n] \mid \exists x \in D \text{ with } x_i \neq 0\}|$$

Note that if D is a singleton set $D = \{x\}$ then $\text{wt}(D) = \text{wt}(x)$ and if D is a one-dimensional subspace of C spanned by a non-zero x , then again, $\text{wt}(D) = \text{wt}(x)$. We can talk more generally about the support of such a subset.

Definition 1.111. Given $D \subseteq C$ for a linear code C , the *support* of D is defined to be

$$\text{supp}(D) = \{i \in [n] \mid \exists x \in D \text{ with } x_i \neq 0\}$$

Below are some trivial properties of supports and weights of subsets of a code.

Proposition 1.112. Let C be a code with subsets $A, B, D \subseteq C$.

$$(i) \quad A \subseteq B \implies \text{wt}(A) \leq \text{wt}(B)$$

$$(ii) \quad \forall x \in D, \text{wt}(D) \geq \text{wt}(x)$$

$$(iii) \quad \text{supp}(D) = \bigcup_{x \in D} \text{supp}(x)$$

Proof. All of these properties are trivial to verify. We leave this to the reader. ■

We can now define the concept of a generalised Hamming weight.

Definition 1.113. Let C be an $[n, k]_q$ linear code. For $1 \leq r \leq k$, the *r -th generalised Hamming weight of C* is defined to be

$$d_r(C) = \min \{\text{wt}(D) : D \text{ is an } r\text{-dimensional subspace of } C\}$$

Note that $d_1(C)$ is simply the minimum distance $d(C)$ of the code.

Definition 1.114. A code $C \in [n, k]_q$ is said to be *non-degenerate* if C is not contained in a coordinate hyperplane of $\mathbb{F}_q^n$, that is, for every $i \in [n]$ we have $x_i \neq 0$ for some $x \in C$.

One can understand this as a code which cannot be shortened to a code of the same cardinality.

Proposition 1.115. A code $C \in [n, k]_q$ is non-degenerate if and only if $d_k(C) = n$.

Proof. Since C is k -dimensional, the only k -dimension subspace of C is C itself and hence $d_k(C) = \text{wt}(C)$. Suppose C is non-degenerate. By definition, every $i \in [n]$ lies in the support of C and hence $\text{wt}(C) = n$. Conversely, if $\text{wt}(C) = n$, then by definition of weight every $i \in [n]$ lies in the support and hence C is non-degenerate. ■

Proposition 1.116. The Reed-Solomon, Reed-Muller, Simplex and Hamming code are all non-degenerate.

Proof. The Reed-Solomon code is non-degenerate because the evaluations of a non-zero constant polynomial give a codeword which has no zeroes and has full weight n . Thus, it is non-degenerate.

The Reed-Muller code also has the same reasoning and we can evaluate a non-zero constant polynomial to obtain a codeword of weight n .

The Simplex code is non-degenerate because no column of the generator matrix is entirely 0.

For the case of the Hamming code we note that if a code C is not non-degenerate then it must have a coordinate position i at which every code word is 0 and hence the standard basis vector e_i must lie in the dual. The converse is also clearly true. Thus, a code C is non-degenerate if and only if every non-zero codeword in the dual has weight at least 2. By Proposition 1.28 this is satisfied and hence the Hamming code is non-degenerate except for $r = 1$ when it is the zero code. ■

Theorem 1.117. *The generalised Hamming weights of a linear code $C \in [n, k]_q$ satisfy the following*

$$d(C) = d_1(C) < d_2(C) < \cdots < d_k(C) \leq n$$

Proof. Fix some $1 \leq r < k$. We show that $d_r(C) < d_{r+1}(C)$. To this end, let D be an $(r+1)$ -dimensional subcode such that $d_{r+1}(C) = \text{wt}(D)$, that is, D achieves the minimum weight among all possible $(r+1)$ -dimensional subspaces of C .

Fix some $i \in \text{supp}(D)$ and consider $D' = D \cap \{x \in \mathbb{F}_q^n \mid x_i = 0\} = D \cap H_i$.

Firstly since $i \in \text{supp}(D)$, we have $D' \subsetneq D$. Further, D' is a subspace being the intersection of subspaces of $\mathbb{F}_q^n$. The dimension formula from linear algebra gives

$$\dim(D') = \dim(D) + \dim(H_i) - \dim(D + H_i) = (r+1) + (n-1) - \dim(D + H_i) \geq r + n - n = r$$

However, $D' \subsetneq D$ implies $\text{wt}(D') < \dim(D) = r+1$. This shows that D' is a subspace of D and hence of C with dimension r .

It is clear that the weight of D' is at most $\text{wt}(D) - 1$ since we considered vectors with the i -th coordinate zero. Thus,

$$d_r(C) \leq \text{wt}(D') \leq \text{wt}(D) - 1 = d_{r+1}(C) - 1$$

This proves the result. ■

Corollary 1.118 (Generalised Singleton Bound). *For a linear code $C \in [n, k]_q$ we have*

$$d_r(C) \leq n - k + r$$

Definition 1.119. A linear code $C \in [n, k]_q$ is called r -**MDS** for some $1 \leq r \leq k$ if $d_r(C) = n - k + r$.

For $r = 1$ we recover the definition of MDS codes. It is also clear that if a code is r -MDS, then it is also s -MDS for every $s \geq r$.

Question. Is the dual of an r -MDS also r -MDS?

Ans. Since we have asked it as a question and not put it as a proposition, the answer is expected to be negative. This is indeed false. The idea follows immediately from Wei-duality which we shall see soon. Explicitly pick a binary code with generator matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

A parity check matrix is

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

One can check that the code C is a $[4, 2]_2$ code with $d_1(C) = 1$ while $d_2(C) = 4$ and $C^\perp$ is also a $[4, 2]_2$ code with $d_1(C^\perp) = 2$ and $d_2(C^\perp) = 3$.

Thus C is 2-MDS but the dual is not 2-MDS. □

Lemma 1.120. *For a linear code $C \in [n, k]_q$ and $1 \leq r \leq k$, we have*

$$d_r(C) = \min \{|X| : X \subseteq [n], \text{nullity}(H_X) \geq r\}$$

where H is a parity check matrix and H_X denotes the submatrix of H obtained by considering all rows but only those columns indexed by X .

Proof. Let $1 \leq r \leq k$ be fixed. Set $d = \min \{|X| : X \subseteq [n], \text{nullity}(H_X) \geq r\}$. Suppose $I \subseteq [n]$ achieves this minimum d . Then it is clear that $\text{nullity}(H_I) \geq r$ and in fact $\text{nullity}(H_I) = r$ since we could keep deleting one element of I to eventually drop the nullity and reach r exactly.

Now consider the set

$$D = \left\{ x \in \mathbb{F}_q^n \mid x_i = 0 \text{ outside } I \text{ and } \sum_{i \in I} x_i h_i = 0 \right\}$$

where $\{h_j\}_{j=1}^n$ denotes the set of all columns of H .

It is clear that this is a subcode of C . Further, the condition $\sum_{i \in I} x_i h_i = 0$ is equivalent to $H_I x_I = 0$ where $x_I = x^{[n] \setminus I}$. Since x is 0 on coordinates outside I , it follows that the dimension of D is the nullity of H_I which was known to be r .

Thus we have found an r -dimensional subspace of C with $\text{wt}(D) \leq |I| = d$ which implies $d_r(C) \leq d$. In short, D was constructed by taking the vectors from the null space of H_I and padding them with extra zeroes keeping in mind the coordinates of I .

Now suppose D is any optimal subcode of dimension r which has weight $d_r(C)$. Let $I = \text{supp}(D)$ be the support of size $d_r(C)$. Consider $D^{[n] \setminus I} = D_{[n] \setminus I}$. It is clear that the dimension is preserved since only the zeroes were deleted.

Now since H is a parity check matrix we must have for every $x \in D$, $Hx^\top = 0$. Thus for every $y = x^{[n] \setminus I} \in D^{[n] \setminus I} = D_{[n] \setminus I}$, we have $H_I y^\top = 0$. In particular we have

$$D^{[n] \setminus I} \subseteq \text{NullSpace}(H_I)$$

which on taking dimensions gives $r \leq \text{nullity}(H_I)$. Thus this set I satisfies the condition and hence by definition of d , we get $d \leq |I| = d_r(C)$.

Together we have $d = d_r(C)$ completing the proof. ■

We can now state and prove our Wei duality theorem.

Theorem 1.121 (Wei's duality). *Let C be an $[n, k]_q$ code. We have*

$$\{d_1(C), \dots, d_k(C)\} \cup \{n+1-d_1(C^\perp), \dots, n+1-d_{n-k}(C^\perp)\} = [n]$$

Proof. Since the sets have sizes k and $n-k$ it suffices to show that they have no element in common. That is, we cannot have $v \in [n]$ which a weight of C such that $n+1-v$ is a weight of $C^\perp$. In other words, we prove that for any $1 \leq s \leq n-k$, $n+1-d_s(C^\perp) \notin \{d_t(C) \mid 1 \leq t \leq k\}$.

Let $1 \leq s \leq n-k$ be arbitrary. Let D be an optimal subcode of $C^\perp$ of dimension s achieving the support size $d_s(C^\perp)$. Let J be the support of D so that $|J| = d_s(C^\perp)$. Let $I = [n] \setminus J$.

Suppose we have a basis of s vectors for D . Extend this to a basis of $C^\perp$ and arrange these vectors to be the rows of a matrix H . Then the first s rows of H form a basis of D and hence lie in D while the other $n-k-s$ rows are not in D . This is by definition a generator matrix for $C^\perp$ and hence a parity check matrix for the code C .

If a vector v lies in D , it is zero outside J . Thus the first s rows of H are all zero on the columns indexed by $I = J^c$. Thus the matrix H_I has first s rows all zero which gives us $\text{rank}(H_I) \leq n-k-s$. By the rank-nullity theorem, since $|I| = n - |J| = n - d_s(C^\perp)$ we get

$$n - d_s(C^\perp) - \text{nullity}(H_I) \leq n - k - s$$

That is,

$$\text{nullity}(H_I) \geq k + s - d_s(C^\perp)$$

By Lemma 1.120, since I satisfies the condition of nullity for $r_0 = k + s - d_s(C^\perp)$ we get

$$d_{r_0}(C) \leq |I| = n - d_s(C^\perp)$$

Using the monotonicity, for every $r < r_0$ we must have $d_r(C) < d_{r_0}(C) < n + 1 - d_s(C^\perp)$. Thus we only need to show that $d_{r_0+\Delta}(C) \neq n + 1 - d_s(C^\perp)$ for every $\Delta \geq 1$. Suppose the equality holds for some Δ , for the sake of contradiction, that is, $d_{r_0+\Delta}(C) = n + 1 - d_s(C^\perp)$.

Let E be an optimal subcode of C of dimension $r_0 + \Delta$ achieving this weight. Thus, vectors of E are zero on at least $d_s(C^\perp) - 1$ bits because the support size of E is $n + 1 - d_s(C^\perp)$.

Obtain a basis $\{v_1, \dots, v_{r_0+\Delta}\}$ of E and extend it to a basis of C . Arrange these basis vectors, row-wise, to create a generator matrix G of C . Then, choose Z to be the set of columns corresponding to the zero set of vectors of E so that the first $r_0 + \Delta$ rows of G_Z are all 0 giving us

$$\text{rank}(G_Z) \leq k - (r_0 + \Delta) = d_s(C^\perp) - s - \Delta$$

Thus the nullity satisfies

$$\text{nullity}(G_Z) \geq |Z| + s + \Delta - d_s(C^\perp) = d_s(C^\perp) - 1 + s + \Delta - d_s(C^\perp) = s + \Delta - 1$$

Now G is a parity check matrix for $C^\perp$ and hence by Lemma 1.120 we get that

$$d_{s+\Delta-1}(C^\perp) \leq |Z| = d_s(C^\perp) - 1$$

Since $\Delta - 1 \geq 0$, this inequality above contradicts the strict monotonicity of the Hamming weights. Thus $n + 1 - d_s(C^\perp)$ cannot be a generalised Hamming weight of C . ■

1.11 The Reed-Muller code

We cover some odds and ends related to the Generalised Reed-Muller code. First, we show that the dual of such a code is another code of the same type.

Recall Definition 1.32.

We adopt the convention that $GRM_q(-1, m) = \{0\}$.

Lemma 1.122. *For any fixed integer $\alpha > 0$ we have*

$$\sum_{a \in \mathbb{F}_q} a^\alpha = \sum_{a \in \mathbb{F}_q^*} a^\alpha = \begin{cases} 0 & \text{if } (q-1) \nmid \alpha \\ -1 & \text{if } (q-1) \mid \alpha \end{cases}$$

Proof. If $(q-1) \mid \alpha$ then $a^\alpha = 1$ which gives the result. Now suppose $(q-1) \nmid \alpha$. Let the remainder when α is divided by $q-1$ be r . Then there must be some $b \in \mathbb{F}_q^*$ such that $b^r = b^\alpha \neq 1$. If not, then we get a contradiction by considering the polynomial $x^r - 1$.

Now we get

$$b^\alpha \sum_{a \in \mathbb{F}_q^*} a^\alpha = \sum_{a \in \mathbb{F}_q^*} (ba)^\alpha = \sum_{c \in \mathbb{F}_q^*} c^\alpha \implies (b^\alpha - 1) \sum_{a \in \mathbb{F}_q^*} a^\alpha = 0$$

Now since $b^\alpha \neq 1$, the sum must be zero and we are done. ■

Corollary 1.123. *If $\mu = x_1^{\alpha_1} \dots x_m^{\alpha_m}$ is any reduced monomial in $\mathbb{F}_q[x_1, \dots, x_m]$, then,*

$$\sum_{p \in \mathbb{F}_q^m} \mu(p) = \begin{cases} (-1)^m & \text{if } \alpha_1 = \dots = \alpha_m = q-1 \\ 0 & \text{otherwise} \end{cases}$$

Theorem 1.124. *For all $-1 \leq r \leq m(q-1)$ we have*

$$GRM_q(r, m)^\perp = GRM_q(s, m)$$

where $s + r = m(q-1) - 1$.

Proof. Fix some r in the prescribed range and as required, set $s = m(q-1) - 1 - r$. Now $GRM_q(s, m)$ is generated by evaluations (at points from $\mathbb{F}_q^m$) of reduced monomials of degree at most s in m variables.

Let $\theta = x_1^{\alpha_1} \cdots x_m^{\alpha_m}$ be one such monomial of degree at most s . Then we claim that $c_\theta = (\theta(p))_{p \in \mathbb{F}_q^m}$ lies in the dual $GRM_q(r, m)^\perp$. To prove the claim, it clearly suffices to show that $\langle c_\theta, c_\eta \rangle = 0$ for every monomial $\eta \in \mathbb{F}_q[x_1, \dots, x_m]$ of degree at most r . Now,

$$\begin{aligned} \langle c_\theta, c_\eta \rangle &= \sum_{p \in \mathbb{F}_q^m} \theta(p) \eta(p) \\ &= \sum_{p \in \mathbb{F}_q^m} (\theta \eta)(p) \end{aligned}$$

Now by Corollary 1.123 this is zero unless the exponents of $\theta \eta$ in reduced form at all $q-1$. However, the degree of this product is at most $r + s = m(q-1) - 1$. Thus the inner product is always zero.

This shows that $GRM_q(s, m) \subseteq GRM_q(r, m)^\perp$. We now show that both of these have the same dimension. Given any reduced monomial θ in m variables of degree at most s we map it as

$$\theta(x_1, \dots, x_m) = x_1^{\beta_1} \cdots x_m^{\beta_m} \mapsto x_1^{q-1-\beta_1} \cdots x_m^{q-1-\beta_m}$$

Observe that the image will always be a reduced polynomial in m variables of degree

$$m(q-1) - (\beta_1 + \cdots + \beta_m) \geq m(q-1) - s = 1 + r$$

Now the set of all reduced monomials in $\mathbb{F}_q[x_1, \dots, x_m]$ is the disjoint union of monomials of degree at most r and degree at least $r+1$. Totally we have q^m reduced monomials which also happens to be the length of the code. Thus from the map above, which is clearly a bijection, we have

$$\dim(GRM_q(s, m)) = q^m - \dim(GRM_q(r, m)) = \dim(GRM_q(r, m)^\perp)$$

■

We now circle back to the minimum distance of this code. The goal now is to prove Theorem 1.34.

Recalling the remark made just before the statement of the theorem, we do not need to concern ourselves with the case of $r \geq m(q-1)$ since in this case the code is the full space and the minimum distance is consequently 1. Thus we assume for the rest of the proof that $r < m(q-1)$.

We begin by fixing a primitive $(q^m - 1)$ -th root of unity α so that

$$\mathbb{F}_{q^m} = \{0, 1, \alpha, \dots, \alpha^{q^m-2}\}$$

As a vector space over $\mathbb{F}_q$, this has a basis $\{1, \alpha, \dots, \alpha^{m-1}\}$. Write for all $0 \leq j \leq q^m - 2$,

$$\alpha^j = \sum_{i=0}^{m-1} a_{ij} \alpha^i$$

for some unique constants $a_{ij} \in \mathbb{F}_q$ for all $0 \leq i \leq m-1$. Now as the m -tuple $(a_{0j}, a_{1j}, \dots, a_{(m-1),j})$ varies over $\mathbb{F}_q^m$ by varying j , we get all possible entries in the field $\mathbb{F}_{q^m}$. Note that for no j we have $\alpha^j = 0$, the tuple $(0, 0, \dots, 0)$ corresponds to $0 \in \mathbb{F}_{q^m}$.

This sets up a bijection between $\mathbb{F}_{q^m}$ and $\mathbb{F}_q^m$ through the definition of the point

$$P_{j+1} = (a_{0j}, a_{1j}, \dots, a_{(m-1),j}) \quad \forall 0 \leq j \leq q^m - 2$$

The point P_0 is defined to be $(0, 0, \dots, 0)$.

Theorem 1.125. Set $n = q^m - 1$. For $0 \leq r < m(q-1)$, Consider the punctured Reed-Muller code given by

$$C_r = \{f(P_1), f(P_2), \dots, f(P_n) \mid f \in \mathbb{F}_q[x_1, \dots, x_m], \deg(f) \leq r, \deg_{x_i}(f) \leq q-1\}$$

Then $C_r^\perp$ is cyclic. Moreover, α^s is a root of the generator polynomial of $C_r^\perp$ if and only if $\omega_q(s) \leq r$ where for $s = \sum_{i=0}^{m-1} s_i q^i$ with $0 \leq s_i \leq q-1$, we define the q -weight of s to be $\omega_q(s) = \sum_{i=0}^{m-1} s_i$.

Proof. Firstly, note that s is an exponent in the range $0 \leq s \leq q^m - 2$ since these are all the powers of α that are considered. Now every such integer s is expressed in its q -ary representation (base q) and this representation clearly has at most m digits.

Now suppose $c = (c_0, \dots, c_{n-1}) \in C_r^\perp$. Consider the ring $R_n = \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ and the map $\pi : \mathbb{F}_q^n \rightarrow R_n$ given by

$$\pi((c_0, \dots, c_{n-1})) = c(x) := c_0 + c_1 x + \dots + c_{n-1} x^{n-1}$$

We make the following claim.

Lemma. For any $c = (c_0, \dots, c_{n-1}) \in C_r^\perp$ we have

$$c(\alpha^s) = \sum_{j=0}^{n-1} c_j (\alpha^s)^j = 0 \quad \text{for every } 0 \leq s \leq q^m - 2 \text{ with } \omega_q(s) \leq r$$

Proof of Lemma. We begin by calculating α^{sj} .

$$\begin{aligned} (\alpha^s)^j &= (\alpha^j)^s \\ &= \left(\sum_{i=0}^{m-1} a_{ij} \alpha^i \right)^{\sum_{t=0}^{m-1} s_t q^t} \\ &= \prod_{t=0}^{m-1} \left[\left(\sum_{i=0}^{m-1} a_{ij} \alpha^i \right)^{s_t q^t} \right] \\ &= \prod_{t=0}^{m-1} \left[\left(\sum_{i=0}^{m-1} a_{ij} \alpha^{i q^t} \right)^{s_t} \right] \\ &= \prod_{t=0}^{m-1} \left[\sum_{0 \leq i_1, i_2, \dots, i_{s_t} \leq m-1} (a_{i_1, j}) (a_{i_2, j}) \cdots (a_{i_{s_t}, j}) \alpha^{(i_1 + i_2 + \dots + i_{s_t}) q^t} \right] \\ &= \prod_{t=0}^{m-1} \sum_{I_t} \left[\left((a_{i_1^{(t)}, j}) (a_{i_2^{(t)}, j}) \cdots (a_{i_{s_t}^{(t)}, j}) \right) \alpha^{(i_1^{(t)} + i_2^{(t)} + \dots + i_{s_t}^{(t)}) q^t} \right] \\ &= \sum_I \left(\prod_{t=0}^{m-1} (a_{i_1^{(t)}, j}) (a_{i_2^{(t)}, j}) \cdots (a_{i_{s_t}^{(t)}, j}) \right) \alpha^{\sum_{t=0}^{m-1} (i_1^{(t)} + i_2^{(t)} + \dots + i_{s_t}^{(t)}) q^t} \end{aligned}$$

where $I_t = (i_1^{(t)}, i_2^{(t)}, \dots, i_{s_t}^{(t)})$ with $0 \leq i_1^{(t)}, i_2^{(t)}, \dots, i_{s_t}^{(t)} \leq m-1$ and $I = (I_0, \dots, I_{m-1})$, that is, it ranges over all possible indices $i_\gamma^{(\beta)}$ where $0 \leq \beta \leq m-1$, $0 \leq \gamma \leq s_\beta$ and the index itself satisfies $0 \leq i_\gamma^{(\beta)} \leq m-1$.

Now for a particular choice of $I = (i_\gamma^{(\beta)})$, let $\theta_I \in \mathbb{F}_q[x_1, \dots, x_m]$ and $\eta_I \in \mathbb{Z}$ be defined as

$$\begin{aligned} \theta(x_0, \dots, x_{m-1}) &= \prod_{t=0}^{m-1} \prod_{\gamma=1}^{s_t} x_{i_\gamma^{(t)}} \\ \eta_I &= \sum_{t=0}^{m-1} \sum_{\gamma=1}^{s_t} i_\gamma^{(t)} q^t \end{aligned}$$

Thus we get

$$(\alpha^s)^j = \sum_I \theta_I(a_{0,j}, a_{1,j}, \dots, a_{(m-1),j}) \alpha^{\eta_I}$$

Finally we have

$$\begin{aligned} c(\alpha^s) &= \sum_{j=0}^{n-1} c_j \left(\sum_I \theta_I(a_{0,j}, a_{1,j}, \dots, a_{(m-1),j}) \alpha^{\eta_I} \right) \\ &= \sum_I \alpha^{\eta_I} \left(\sum_{j=0}^{n-1} c_j \theta_I(a_{0,j}, a_{1,j}, \dots, a_{(m-1),j}) \right) \end{aligned}$$

However the inner sum is zero because $c = (c_0, \dots, c_{n-1}) \in C_r^\perp$ while $\theta_I(a_{0,j}, a_{1,j}, \dots, a_{(m-1),j}) \in C_r$. Indeed, θ_I is a polynomial of m variables of degree $\omega_q(s)$ with coefficients in $\mathbb{F}_q$. It is also given that $\omega_q(s) \leq r$. The evaluations are precisely the points $P_1, \dots, P_n$.

Note that one may reduce the polynomial further and by abusing notation denote the reduced polynomial by θ_I again since we only care about evaluations and not the polynomial. This proves the lemma. $\backslash\backslash\backslash$

Now consider

$$g(x) = \prod_{\substack{0 \leq s \leq q^m - 2 \\ \omega_q(s) \leq r}} (x - \alpha^s)$$

Note that $g(x^q) = (g(x))^q$ implying that $g \in \mathbb{F}_q[x]$. Now if D is the cyclic code, of length n over $\mathbb{F}_q$, generated by this generator polynomial g , then from the lemma proved, as vector spaces over $\mathbb{F}_q$ we have

$$C_r^\perp \subseteq D$$

We now compare the dimensions below.

$$\begin{aligned} \deg(g) &= \left| \left\{ s = \sum_{i=0}^{m-1} s_i q^i \mid 0 \leq s_i \leq q-1, \omega_q(s) \leq r \right\} \right| \\ &= \left| \left\{ s = \sum_{i=0}^{m-1} s_i q^i \mid 0 \leq s_i \leq q-1, s_0 + \dots + s_{m-1} \leq r \right\} \right| \\ &= \dim(\text{GRM}_q(r, m)) \end{aligned}$$

Now we make the following claim.

Lemma. *The dimension of the punctured code C_r is the same as the original Reed-Muller code.*

Proof of Lemma. Let $f \in \mathbb{F}_q[x_1, \dots, x_m]$ be any reduced polynomial of degree at most r . Write it in terms of reduced monomials μ of degree at most r as

$$f(x_1, \dots, x_m) = \sum_{\mu} c_{\mu} \mu(x_1, \dots, x_m)$$

Now using Corollary 1.123, since all the powers in the monomial cannot be $q-1$ at once (since $r \neq m(q-1)$) we have

$$\sum_{P \in \mathbb{F}_q^m} f(P) = \sum_{\mu} c_{\mu} \left(\sum_{P \in \mathbb{F}_q^m} \mu(P) \right) = 0$$

Thus we have

$$f(P_0) = f(0) = - \sum_{j=1}^n f(P_j)$$

Now consider the puncturing map $\pi : GRM_q(r, m) \rightarrow C_r$ as specified previously while constructing C_r , that is, puncturing the coordinate corresponding to the evaluation of the polynomials at the origin 0.

Suppose $c_f \in GRM_q(r, m) \in \ker(\pi)$. Then $f(P) = 0$ at all $P \neq 0$. We cannot comment about $f(0)$. However, from the relation derived above, we have $f(0) = -\sum_{j=1}^n f(P_j) = 0$. Thus, the kernel is trivial since f is reduced and zero everywhere implying that it is the zero polynomial (this is an easy fact to prove about reduced polynomials).

Thus the puncturing map is injective and hence preserves dimension of the spaces. \\\

Thus $\dim(D) = n - \deg(g) = n - \dim(C_r) = \dim(C_r^\perp)$. This gives $D = C_r^\perp$ showing that $C_r^\perp$ is a cyclic code. The latter assertion of the theorem statement follows by choice of g . ■

Corollary 1.126. *We borrow notation from the above theorem. Let $0 \leq r < m(q-1)$. The code C_r is cyclic and α^s is a root of the generator polynomial if and only if $\omega_q(s) \leq m(q-1) - r - 1$.*

Proof. We have the already established Theorem 1.125. We now apply Proposition 1.61 to get that C_r is cyclic. As for the roots of the generator polynomial, we make use of Proposition 1.64.

Since $s = \sum_{i=0}^{m-1} s_i q^i$ we have

$$n - s = \sum_{i=0}^{m-1} (q - 1 - s_i) q^i$$

This gives $\omega_q(n - s) = m(q - 1) - \omega_q(s)$. Thus we have $m(q - 1) - \omega_q(s) > r$ or equivalently $\omega_q(s) \leq m(q - 1) - r - 1$. ■

We may now focus on obtaining the minimum distance

Theorem 1.127. *The minimum distance of the punctured code C_r is $d_r = (R + 1)q^Q - 1$ where R, Q are such that*

$$0 \leq R < q - 1 \quad \text{and} \quad \mu + 1 = m(q - 1) - r = Q(q - 1) + R$$

Moreover C_r is a subcode of the BCH code whose generator polynomial has roots $\alpha, \alpha^2, \dots, \alpha^{d_r-1}$.

Proof. Let Q, R be as above. Set $b = (R + 1)q^Q - 1$. Let h_0 be the smallest non-negative integer such that $\omega_q(h_0) = \mu + 1$. Then using the q -ary expansion we have

$$\begin{aligned} h_0 &= Rq^Q + (q - 1)q^{Q-1} + \dots + (q - 1)q + (q - 1) \\ &= Rq^Q + (q - 1) \frac{q^Q - 1}{q - 1} \\ &= b \end{aligned}$$

By definition of h_0 , there are $b = h_0$ successive integers, particularly, $0, 1, \dots, h_0 - 1$ with $\omega_q(h) \leq \mu$. Thus from Corollary 1.126, the generator polynomial of C_r has the successive roots:

$$\{1, \alpha, \dots, \alpha^{b-1}\}$$

Thus C_r is a subcode of a BCH code with minimum distance at least b . This follows from Proposition 1.80. We need to show that there is some word of this minimum distance b in C_r to finish the proof. To this end, consider the polynomial

$$f(x_1, \dots, x_m) = \prod_{i=0}^{m-Q-2} (x_i^{q-1} - 1) \prod_{j=1}^{q-1-R} (x_{m-Q-1} - \omega_j)$$

where $\omega_1, \dots, \omega_{q-1-R}$ are any fixed set of distinct non-zero elements of $\mathbb{F}_q$. Note that for f to be non-zero, we would need all of $x_0, x_1, \dots, x_{m-Q-2}$ to be 0 and x_{m-Q-1} to be any value other than the $q - 1 - R$ values

of ω_j while the remaining Q variables are free. Also note that although at $(0, 0, \dots, 0)$ we have a non-zero f , we do not count this because we are in the punctured setting. Thus,

$$\text{wt}(c_f) = (R+1)q^Q - 1 = b$$

■

We can now finally prove Theorem 1.34. We prove it below.

Proof. Using Theorem 1.127 the minimum distance of $GRM_q(r, m)$ can either be $(R+1)q^Q - 1$ or $(R+1)q^Q$. This is because we only need to factor in the one extra coordinate which was being punctured.

Suppose the minimum distance continued to be $(R+1)q^Q - 1$. Then we can find a reduced polynomial $g \in \mathbb{F}_q[x_1, \dots, x_m]$ with degree at most r such that it has $(R+1)q^Q - 1$ non-zero evaluations and $g(0) = 0$ (If $g(0) \neq 0$, the punctured weight would be $(R+1)q^Q - 2$ which is not possible by Theorem 1.127).

Since affine translations preserve weight, the weight of the codeword corresponding to $h(x) = g(x + a)$ for any $a = (a_1, \dots, a_m)$ is the same as that of c_g . In particular choosing a point a_0 at which g is non-zero, at $x = 0$ we have that $h(0) = g(a_0) \neq 0$. Such an a_0 exists because the code is non-trivial.

This is a contradiction because when evaluated at this polynomial, we have $(R+1)q^Q - 1$ number of non-zero positions which includes the non-zero evaluation at the origin. Thus, the punctured code would have a word of weight $(R+1)q^Q - 2$ which is a contradiction.

Thus the minimum distance is $(R+1)q^Q$. Now we have the following relations where s and t are as in Theorem 1.34

$$Q = m - t - 1 \quad \text{and} \quad R = q - 1 - s \quad \text{if } s \neq 0,$$

$$Q = m - t \quad \text{and} \quad R = 0 \quad \text{if } s = 0$$

Thus we have that the minimum distance in both the cases is

$$d = (R+1)q^Q = (q-s)q^{m-t-1}$$

■

Chapter 2

Design Theory

2.1 Basic introduction

We have the general definition of a design below.

Definition 2.1. Let t, v, k, λ be integers such that $0 < t \leq k \leq v$ and $\lambda > 0$. A t – (v, k, λ) **design** is a collection $\mathcal{B}$ of k -sized subsets of the set $[v] = \{1, 2, \dots, v\}$ such that for any t -sized subset T of $[v]$, the number of subsets in $\mathcal{B}$ containing T is exactly λ .

Definition 2.2. Let $\mathcal{B}$ be a design. The elements of the parent set $[v]$ are called **points** while the elements of the design $\mathcal{B}$ are called **blocks**.

Suppose we take $\mathcal{B}$ to be the full collection of all k -sized subsets for some $v \geq k$. Then this is always a t – $(v, k, \binom{v-t}{k-t})$ design for every t .

We now see how to obtain different designs from a given design.

Proposition 2.3. Let $\mathcal{B}$ be a t – (v, k, λ) design. Then for every $0 \leq s \leq t$, it is also an s – (v, k, λ_s) design where

$$\binom{k-s}{t-s} \lambda_s = \binom{v-s}{t-s} \lambda$$

Proof. Let $\lambda(S)$ denote the number of blocks containing a particular set S of size s . Our aim is to show that this is independent of the choice of S and is further given by the formula claimed in the question. It suffices to show the latter.

We count the number of pairs (T, B) such that $S \subseteq T \subseteq B \in \mathcal{B}$. Given a fixed T , the number of B from $\mathcal{B}$ which contain it is λ . The number of T containing a fixed set S of size s is clearly $\binom{v-s}{t-s}$. Thus, this count gives the RHS.

Now, given a fixed B in $\mathcal{B}$ containing S , the number of T coming in between B and S is simply $\binom{k-s}{t-s}$. The number of such B containing S is $\lambda(S)$ by definition. Thus we obtain the LHS with λ_s replaced by $\lambda(S)$.

This gives the same formula as claimed for $\lambda(S)$ and it is clearly independent of the choice of S . ■

Corollary 2.4. If a t – (v, k, λ) design exists, then λ_s is a positive integer.

Proof. Follows from positivity of λ . ■

Corollary 2.5. The number of blocks in a t – (v, k, λ) design is given by

$$\frac{\binom{v}{t}}{\binom{k}{t}} \lambda$$

Proof. Putting $s = 0$ in Proposition 2.3 gives a $0-(v, k, \lambda_0)$ design. The first parameter being 0 indicates that the quantity λ_0 is the number of blocks containing a given set of size 0, that is, the number of blocks containing the empty set. This is just the number of blocks in the design. ■

We have some special names for particular designs of interest.

Definition 2.6. A $t-(v, k, \lambda)$ design is called a **balanced incomplete block design** if $t = 2$. It is called a **Steiner system** if $\lambda = 1$.

We now have a concept of a matrix related to the design.

Definition 2.7. Let $\mathcal{B}$ be a $t-(v, k, \lambda)$ design. The $|\mathcal{B}| \times v$ matrix M whose (B, i) -th entry is given by

$$M_{B,i} = \begin{cases} 1 & \text{if } i \in B \\ 0 & \text{otherwise} \end{cases}$$

is called the **incidence matrix** of the design

Consequently, each row has k number of 1's and each column has $\left(\binom{v-1}{t-1} / \binom{k-1}{t-1} \right) \lambda$ number of 1's (obtained by setting $s = 1$ in Proposition 2.3).

As standard notation in practice, we write

$$r = \lambda_1 = \frac{\binom{v-1}{t-1}}{\binom{k-1}{t-1}} \lambda$$

$$b = \lambda_0 = \frac{\binom{v}{t}}{\binom{k}{t}} \lambda$$

We have the complement of a design obtained by taking the complement of $\mathcal{B}$ in the parent set. It is also a design.

Proposition 2.8. Let $\mathcal{B}$ be a $t-(v, k, \lambda)$ design. Then $2^{[v]} \setminus \mathcal{B}$ is also a design with parameters $t-(v, v-k, \bar{\lambda})$ where

$$\bar{\lambda} = \frac{\binom{v-k}{t}}{\binom{k}{t}} \lambda$$

Proof. The fact that the parameters v and $v - k$ are correct is trivial. We must justify that the number of blocks in the complement design containing a given t -sized subset $T \subseteq [v]$ is a constant $\bar{\lambda}$ independent of T , and find its value.

Let T be a fixed t -sized subset. The number of blocks in the complement design containing T is exactly the number of blocks $B \in \mathcal{B}$ such that $T \cap B = \emptyset$. By the Principle of Inclusion-Exclusion, this number is given by:

$$\lambda(T) = \sum_{i=0}^t (-1)^i \sum_{\substack{S \subseteq T \\ |S|=i}} (\text{number of blocks containing } S)$$

Since $\mathcal{B}$ is a t -design, any subset S of size $i \leq t$ is contained in exactly λ_i blocks. Thus, the inner sum is simply $\binom{t}{i} \lambda_i$, giving:

$$\lambda(T) = \sum_{i=0}^t (-1)^i \binom{t}{i} \lambda_i$$

This formula depends only on the size of T , proving that $\lambda(T)$ is a constant $\bar{\lambda}$ independent of the specific choice of T .

To find the elegant expression for $\bar{\lambda}$ claimed in the proposition, we can now double count the pairs (T, B) where T is a t -sized subset, $B \in \mathcal{B}$, and $T \cap B = \emptyset$.

First, fix T in $\binom{v}{t}$ ways. The number of blocks B disjoint from it is the constant $\bar{\lambda}$. This gives a total count of $\binom{v}{t} \bar{\lambda}$.

Alternatively, fix a block $B \in \mathcal{B}$ in b ways. The number of t -sized subsets T disjoint from this k -sized block is $\binom{v-k}{t}$. This gives a count of $b \binom{v-k}{t}$.

Equating the two yields $\binom{v}{t} \bar{\lambda} = b \binom{v-k}{t}$. Substituting $b = \frac{\binom{v}{k}}{\binom{k}{t}} \lambda$ gives the required formula:

$$\bar{\lambda} = \frac{\binom{v-k}{t}}{\binom{k}{t}} \lambda$$

■

Proposition 2.9. *For a $2-(v, k, \lambda)$ design, we have*

$$M^\top M = rI + \lambda(J - I) = \begin{bmatrix} r & \lambda & \lambda & \cdots & \lambda \\ \lambda & r & \lambda & \cdots & \lambda \\ \lambda & \lambda & r & \cdots & \lambda \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda & \lambda & \lambda & \cdots & r \end{bmatrix}$$

Proof. Firstly, $M^\top M$ has size $v \times v$ and for $i \neq j$, the (i, j) -th entry is given by

$$\sum_{B \in \mathcal{B}} M_{B,i} M_{B,j}$$

This counts the number of B such that i and j are both in B . This is the number of B in the design containing the set $\{i, j\}$. Since it is a 2-design, this is λ by definition. If $i = j$, then the sum just counts the number of blocks containing a given element $i = j$ which is just $r = \lambda_1$. ■

One may now ask about the bounds on the number of blocks in a design. We resolve this with the following theorem.

Theorem 2.10 (Fisher type inequality). *For a $2-(v, k, \lambda)$ design, if $v > k$ then $b \geq v$.*

Proof. Since $t = 2$, we can write out r and b explicitly as

$$r = \frac{v-1}{k-1} \lambda$$

$$b = \frac{v(v-1)}{k(k-1)} \lambda$$

Using Proposition 2.9 we get that the eigenvalues of $M^\top M$ are

$$\{(r - \lambda) + \lambda v, r - \lambda, r - \lambda, \dots, r - \lambda\}$$

Consequently the determinant is

$$\det(M^\top M) = (r + (v-1)\lambda)(r - \lambda)^{v-1}$$

Now since $v > k$, we have $r > \lambda$ and hence the determinant is non-zero. Thus, the matrix $M^\top M$ is full rank.

Now

$$v = \text{rank}(M^\top M) \leq \text{rank}(M) \leq |\mathcal{B}| = b$$

■

2.2 Symmetric Designs

We now come to the definition of a symmetric design - an important design which has been studied extensively. Motivated by the above Theorem 2.10, we define the following in general.

Definition 2.11. A t -(v, k, λ) design is said to be a *symmetric design* if $b = v$.

Since $bk = vr$, the condition $b = v$ holds if and only if $r = k$. We elaborate this by giving some more equivalent conditions for the special case of $t = 2$.

Proposition 2.12. *The following are equivalent for a 2-(v, k, λ) design with $v > k$.*

- i. $b = v$
- ii. $r = k$
- iii. *Any two blocks have λ common points*
- iv. *Any two blocks have m common points for some constant m*

Proof. The equivalence of (i) and (ii) is clear from the fundamental relation $bk = rv$. This holds for any design.

Suppose (i) and (ii) hold. Then $b = v$, so the incidence matrix M is a square $v \times v$ matrix. We know from Proposition 2.9 that $M^\top M = (r - \lambda)I + \lambda J$. Because $v > k$, we have $r > \lambda$, making $\det(M^\top M)$ non-zero. Thus, M is invertible.

Since M is the incidence matrix of a design with uniform block size k and uniform point replication r , and because $r = k$, the all-ones matrix J satisfies $MJ = kJ$ and $JM = rJ = kJ$. Therefore, M and J commute. We can now evaluate $MM^\top$:

$$MM^\top = M(M^\top M)M^{-1} = M((r - \lambda)I + \lambda J)M^{-1} = (r - \lambda)I + \lambda MJM^{-1} = (r - \lambda)I + \lambda J$$

The (A, B) -th entry of $MM^\top$ for $A \neq B$ counts exactly the number of points common to blocks A and B . Since $MM^\top = (r - \lambda)I + \lambda J$, this off-diagonal entry is exactly λ . This proves (iii).

(iii) implies (iv) trivially by setting $m = \lambda$.

We are left to show that (iv) implies (i). Consider $M^\top$. It has r 1's in each row and k 1's in each column. Further, by (iv), any two columns of $M^\top$ have exactly m common 1's. Thus, $M^\top$ is the incidence matrix of a 2-(b, r, m) design.

Consider the fundamental relations $bk = rv$ and $r(k - 1) = (v - 1)\lambda$. Applying the latter to the dual-like design of $M^\top$, we obtain $k(r - 1) = (b - 1)m$. Since $v > k$, we have $b > r$, which implies $k > m$. Consequently, the matrix $MM^\top = (k - m)I + mJ$ has the determinant $(k + (b - 1)m)(k - m)^{b-1}$, which is strictly positive.

Thus, $MM^\top$ is full rank, giving $\text{rank}(MM^\top) = b$. Since $\text{rank}(MM^\top) \leq \text{rank}(M) \leq v$, we obtain $b \leq v$. However, by Theorem 2.10, we have the opposite inequality $b \geq v$. Therefore, $b = v$, which proves (i). ■

Motivated by the above we have the following definition.

Definition 2.13. Given a t -(v, k, λ) design and its incidence matrix M , the design whose incidence matrix is $M^\top$ is called the *dual design*.

It is clear the dual of a symmetric design is also a symmetric design. We formally establish this important result, which shows that symmetric designs are exactly those whose duals retain the 2-design property.

Proposition 2.14. *Let $\mathcal{B}$ be a symmetric 2-(v, k, λ) design. Then its dual design is also a symmetric 2-(v, k, λ) design.*

Proof. Let M be the incidence matrix of $\mathcal{B}$. By definition, the dual design has the incidence matrix $M^\top$. We must verify the parameters of this new design.

The parent set of the dual design corresponds to the rows of $M^\top$, which are the columns of M . There are exactly b such rows. Since $\mathcal{B}$ is symmetric, $b = v$, meaning the dual design has exactly v points.

The blocks of the dual design correspond to the columns of $M^\top$, of which there are $v = b$. Thus, the dual design has exactly b blocks.

The size of each block in the dual design is the number of 1's in a column of $M^\top$. This is exactly the number of 1's in a row of M , which is r . Since $\mathcal{B}$ is symmetric, $r = k$, so every block in the dual design has size k .

Finally, we must show that any pair of points in the dual design is contained in exactly λ blocks. This is equivalent to showing that the inner product of any two distinct rows of $M^\top$ is λ . The matrix of these inner products is given by $M^\top(M^\top)^\top = M^\top M$. Since $\mathcal{B}$ is a symmetric design, we know from Proposition 2.12 that M is a square matrix satisfying $MM^\top = (r - \lambda)I + \lambda J$. Since $r = k$, we have $MM^\top = (k - \lambda)I + \lambda J$.

Because $MM^\top = M^\top M$ for a symmetric design (as shown in the proof of Proposition 2.12), the off-diagonal entries of $M^\top M$ are exactly λ . Thus, any two points in the dual design occur together in exactly λ blocks.

Consequently, the dual design is a $2-(v, k, \lambda)$ design. Since its number of blocks equals its number of points, it is also symmetric. ■

From the proposition we just proved, it is clear that a symmetric design and its dual have the same parameters. However they may not be isomorphic. We clarify what it means for two designs to be isomorphic below.

Definition 2.15. Two t -designs $\mathcal{B}$ and $\mathcal{B}'$ with parent sets V and V' (usually the positive integers $[v]$, for convenience) are said to be **isomorphic** if there exists a bijection from V to V' which induces a bijection between the blocks of $\mathcal{B}$ and $\mathcal{B}'$. Further, for every point $p \in B \in \mathcal{B}$ the image of p in the bijection is required to be contained in the image of B in the bijection.

A symmetric design that is isomorphic to its dual is called **self-dual**. We will see such examples later.

Symmetric designs are incredibly useful because they naturally give rise to other designs.

Definition 2.16. Let $\mathcal{B}$ be a symmetric $2-(v, k, \lambda)$ design with parent set V and fix a block $B_0 \in \mathcal{B}$. The **derived design** of $\mathcal{B}$ with respect to B_0 has the point set B_0 . Its blocks are the intersections $B \cap B_0$ for all $B \in \mathcal{B} \setminus \{B_0\}$.

Definition 2.17. Let $\mathcal{B}$ be a symmetric $2-(v, k, \lambda)$ design with parent set V and fix a block $B_0 \in \mathcal{B}$. The **residual design** of $\mathcal{B}$ with respect to B_0 has the point set $V \setminus B_0$. Its blocks are the set differences $B \setminus B_0$ for all $B \in \mathcal{B} \setminus \{B_0\}$.

We now verify the parameters for these two new families of designs.

Proposition 2.18. *Let $\mathcal{B}$ be a symmetric $2-(v, k, \lambda)$ design.*

- *The derived design is a $2-(k, \lambda, \lambda - 1)$ design.*
- *The residual design is a $2-(v - k, k - \lambda, \lambda)$ design.*

Proof. We evaluate the parameters for both designs sequentially.

First, consider the derived design. Its point set is precisely B_0 . Since B_0 is a block in the original design, its cardinality is k . Thus, the number of points in the derived design is k . The blocks of the derived design are of the form $B \cap B_0$ for $B \neq B_0$. By Proposition 2.12, any two distinct blocks in a symmetric design intersect in exactly λ points. Therefore, every block in the derived design has a uniform size of λ .

Finally, consider a pair of distinct points $x, y \in B_0$. In the original design, there are exactly λ blocks containing both x and y . One of these blocks is B_0 itself. The remaining $\lambda - 1$ blocks each intersect B_0 to

form the blocks of the derived design containing both x and y . Hence, any pair of points appears in $\lambda - 1$ blocks, meaning the derived design is a $2-(k, \lambda, \lambda - 1)$ design.

Now, consider the residual design. Its point set is $V \setminus B_0$. The total number of points is thus $v - |B_0| = v - k$. The blocks of the residual design are of the form $B \setminus B_0$ for $B \neq B_0$. The size of each such block is $|B| - |B \cap B_0|$. Since $|B| = k$ and $|B \cap B_0| = \lambda$ (again, relying on the symmetric property from Proposition 2.12), the block size is uniformly $k - \lambda$.

Finally, take any pair of distinct points $x, y \in V \setminus B_0$. In the original design $\mathcal{B}$, there are exactly λ blocks containing both x and y . Since $x \notin B_0$ and $y \notin B_0$, the block B_0 cannot be one of these λ blocks. Thus, all λ of these blocks must belong to $\mathcal{B} \setminus \{B_0\}$. When we take the set difference $B \setminus B_0$, the points x and y are preserved. Therefore, exactly λ blocks in the residual design contain both x and y , confirming it is a $2-(v - k, k - \lambda, \lambda)$ design. ■

A major open problem in this area is the following. If we are given parameters v, k and λ and have also been given a $2-(k, \lambda, \lambda - 1)$ design and a $2-(v - k, k - \lambda, \lambda)$ design, there is no way known currently by which we can find the original parent design of which these two designs are derived and residual, respectively. There might not even exist a parent design. More formally, given a $2-(v - k, k - \lambda, \lambda)$ design, whether it can always be extended to a symmetric $2-(v, k, \lambda)$ design by appending a block and adding points is a major open problem.

We now have one of the central theorems of symmetric designs - the Bruck-Ryser-Chowla theorem. This would need its own section!

2.3 The Bruck-Ryser-Chowla Theorem

In this section, we obtain a necessary condition on the parameters of a symmetric $2-(v, k, \lambda)$ design. We first develop some classical number-theoretic results related to the Legendre symbol. We then define the Hilbert symbols, whose calculation relies on the Legendre symbol. The Hilbert symbols are subsequently used to define the Hasse invariants for symmetric matrices over the integers.

Definition 2.19. For any odd prime p and for any integer a , the **Legendre symbol** $\left(\frac{a}{p}\right)$ is defined as:

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p \text{ (i.e., } x^2 \equiv a \pmod{p} \text{ has a solution)} \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p \end{cases}$$

The following properties of the Legendre symbol can be found in standard Number Theory texts. One may refer [18].

Theorem 2.20. Let p and q be distinct odd primes and let a and b be integers not divisible by p . Then

- i. if $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$;
- ii. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$;
- iii. $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$;
- iv. $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$;
- v. [Quadratic Reciprocity Theorem] $\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}$.

Property (v) of Theorem 2.20 is the celebrated **Quadratic Reciprocity Law**. This would again require its own section to prove. There are multiple different proofs available. We refer the reader to [25] for one such proof.

Properties (i) and (ii) of Theorem 2.20 almost uniquely define the Legendre symbol, as the next proposition shows.

Proposition 2.21. *Let p be an odd prime and let a function L from the set of all integers not divisible by p to the set $\{-1, 1\}$ have the following properties:*

- i. *if $a \equiv b \pmod{p}$, then $L(a) = L(b)$;*
- ii. *$L(ab) = L(a)L(b)$ for all a and b .*

Then either $L(a) = 1$ for all a or $L(a) = \left(\frac{a}{p}\right)$ for all a .

Proof. Let $G = (\mathbb{Z}/p\mathbb{Z})^\times$ denote the multiplicative group of integers modulo p . Then G is a cyclic group of order $p - 1$ which is even.

Property (i) guarantees that L respects residue classes, allowing us to view L as a well-defined function $L : G \rightarrow \{-1, 1\}$. Property (ii) dictates that L preserves the group operation, meaning L is a group homomorphism.

The image of L , denoted $\text{Im}(L)$, must be a subgroup of the codomain $\{-1, 1\}$. There are only two such subgroups: the trivial subgroup $\{1\}$ and the entire group $\{-1, 1\}$. This gives us two cases to consider.

Case 1: $\text{Im}(L) = \{1\}$. Then $L(a) = 1$ for all $a \in G$.

Case 2: $\text{Im}(L) = \{-1, 1\}$. By the First Isomorphism Theorem, $G/\ker(L) \cong \text{Im}(L)$. Since $|\text{Im}(L)| = 2$, the kernel $\ker(L)$ must be a subgroup of G of index 2. Consequently, its order is exactly $|G|/2 = (p - 1)/2$.

Let $S = \{x^2 \pmod{p} \mid x \in G\}$ denote the set of all quadratic residues in G . Consider the squaring map $\phi : G \rightarrow G$ defined by $\phi(x) = x^2$. Since G is abelian, ϕ is a homomorphism. The kernel of ϕ consists of all elements $x \in G$ such that $x^2 \equiv 1 \pmod{p}$. The polynomial $x^2 - 1 \equiv 0 \pmod{p}$ has exactly two roots in the field $\mathbb{Z}/p\mathbb{Z}$, namely 1 and -1 . Thus, $|\ker(\phi)| = 2$.

Applying the First Isomorphism Theorem to ϕ , the image of ϕ (which is precisely the set S) has order $|G|/|\ker(\phi)| = (p - 1)/2$. Therefore, the quadratic residues form a subgroup S of order $(p - 1)/2$.

Now, for any $x \in G$, we evaluate L on the square x^2 :

$$L(x^2) = L(x)L(x) = L(x)^2 = 1$$

where the final equality holds because the only possible values for $L(x)$ are 1 and -1 , both of whose squares are 1. This demonstrates that every square is mapped to 1, meaning $S \subseteq \ker(L)$.

We have established that both S and $\ker(L)$ are finite subgroups of G possessing the exact same order $(p - 1)/2$. Since S is a subset of $\ker(L)$, they must be identical; that is, $\ker(L) = S$.

This structural equality dictates the behaviour of L : $L(a) = 1$ if and only if $a \in S$ (meaning a is a quadratic residue). Since the only other element in the codomain is -1 , we must have $L(a) = -1$ for all $a \notin S$ (meaning a is a quadratic non-residue). This is the exact definition of the Legendre symbol, proving that $L(a) = \left(\frac{a}{p}\right)$ for all $a \in G$. ■

The next theorem will allow us to define the Hilbert symbols. Here, $\mathbb{N}$ denotes the set of non-negative integers.

Theorem 2.22. *For any odd prime p , there exists a unique function $(a, b) \rightarrow (a, b)_p$ from $\mathbb{N} \times \mathbb{N}$ to $\{-1, 1\}$ that satisfies the following conditions:*

- (H1) $(a, b)_p = (b, a)_p$, for any $a, b \in \mathbb{N}$;
- (H2) $(ab, c)_p = (a, c)_p(b, c)_p$, for any $a, b, c \in \mathbb{N}$;
- (H3) $(a, b)_p = 1$, for any integers $a, b \not\equiv 0 \pmod{p}$;
- (H4) if $a \not\equiv 0 \pmod{p}$, then $(a, p)_p = \left(\frac{a}{p}\right)$;

(H5) $(-p, p)_p = 1$.

Proof. We first prove uniqueness. Suppose a function $(a, b) \rightarrow (a, b)_p$ satisfies conditions (H1) – (H5). We begin by evaluating $(p, p)_p$. By (H5), we know $(-p, p)_p = 1$. Applying the multiplicativity axiom (H2), we can split this into

$$(-1, p)_p (p, p)_p = 1$$

Since $-1 \not\equiv 0 \pmod{p}$, (H4) implies $(-1, p)_p = \left(\frac{-1}{p}\right)$. Because our codomain is $\{-1, 1\}$, we can multiply both sides by $\left(\frac{-1}{p}\right)$ to obtain $(p, p)_p = \left(\frac{-1}{p}\right)$.

Consequently, by repeated application of (H1) and (H2), for any non-negative integers s and t , we have:

$$(p^s, p^t)_p = ((p, p)_p)^{st} = \left(\frac{-1}{p}\right)^{st}$$

Now, let $a, b \in \mathbb{N}$. We can factor out the highest powers of p to write $a = p^s a_0$ and $b = p^t b_0$, where s and t are non-negative integers, and $a_0, b_0 \not\equiv 0 \pmod{p}$. Using the bilinearity provided by (H1) and (H2), we expand the symbol $(a, b)_p$:

$$(a, b)_p = (p^s a_0, p^t b_0)_p = (p^s, p^t)_p (p^s, b_0)_p (a_0, p^t)_p (a_0, b_0)_p$$

We evaluate each of the four factors on the right-hand side separately:

- As established, $(p^s, p^t)_p = \left(\frac{-1}{p}\right)^{st}$.
- By (H1) and (H2), $(p^s, b_0)_p = (b_0, p^s)_p = (b_0, p)_p^s$. Then by (H4), this equals $\left(\frac{b_0}{p}\right)^s$.
- Similarly, $(a_0, p^t)_p = (a_0, p)_p^t = \left(\frac{a_0}{p}\right)^t$.
- By (H3), since neither a_0 nor b_0 is divisible by p , $(a_0, b_0)_p = 1$.

Multiplying these evaluations together yields a strictly determined explicit formula:

$$(a, b)_p = \left(\frac{-1}{p}\right)^{st} \left(\frac{a_0}{p}\right)^t \left(\frac{b_0}{p}\right)^s \quad (2.1)$$

This proves that if such a function exists, it is unique.

Conversely, to prove existence, we simply define the function $(a, b) \rightarrow (a, b)_p$ from $\mathbb{N} \times \mathbb{N}$ to $\{-1, 1\}$ using (2.1). Relying on the multiplicative properties of the Legendre symbol from Theorem 2.20 we verify that this explicit construction satisfies all five conditions (H1) – (H5).

- **(H1):** The expression in (2.1) is perfectly symmetric with respect to exchanging the pairs (s, a_0) and (t, b_0) , immediately yielding $(a, b)_p = (b, a)_p$.
- **(H2):** Let $c \in \mathbb{Z}^*$ with $c = p^r c_0$ where $c_0 \not\equiv 0 \pmod{p}$. Then the product $ab = p^{s+t}(a_0 b_0)$. Applying (2.1) and the multiplicativity of the Legendre symbol, we compute:

$$\begin{aligned} (ab, c)_p &= \left(\frac{-1}{p}\right)^{(s+t)r} \left(\frac{a_0 b_0}{p}\right)^r \left(\frac{c_0}{p}\right)^{s+t} \\ &= \left(\frac{-1}{p}\right)^{sr} \left(\frac{-1}{p}\right)^{tr} \left(\frac{a_0}{p}\right)^r \left(\frac{b_0}{p}\right)^r \left(\frac{c_0}{p}\right)^s \left(\frac{c_0}{p}\right)^t \\ &= \left(\left(\frac{-1}{p}\right)^{sr} \left(\frac{a_0}{p}\right)^r \left(\frac{c_0}{p}\right)^s\right) \left(\left(\frac{-1}{p}\right)^{tr} \left(\frac{b_0}{p}\right)^r \left(\frac{c_0}{p}\right)^t\right) \\ &= (a, c)_p (b, c)_p \end{aligned}$$

- **(H3):** If $a, b \not\equiv 0 \pmod{p}$, then their highest dividing powers of p are trivial, meaning $s = 0$ and $t = 0$. Substituting these exponents into the formula leaves $(a, b)_p = \left(\frac{-1}{p}\right)^0 \left(\frac{a_0}{p}\right)^0 \left(\frac{b_0}{p}\right)^0 = 1$.
- **(H4):** If $a \not\equiv 0 \pmod{p}$, then $s = 0$ and $a_0 = a$. Evaluating against $b = p$ gives $t = 1$ and $b_0 = 1$. The formula then simplifies to $(a, p)_p = \left(\frac{-1}{p}\right)^0 \left(\frac{a}{p}\right)^1 \left(\frac{1}{p}\right)^0 = \left(\frac{a}{p}\right)$.
- **(H5):** For $a = -p$ and $b = p$, we have $s = 1, a_0 = -1$ and $t = 1, b_0 = 1$. The formula yields $(-p, p)_p = \left(\frac{-1}{p}\right)^{1 \cdot 1} \left(\frac{-1}{p}\right)^1 \left(\frac{1}{p}\right)^1 = \left(\frac{-1}{p}\right)^2 = 1$.

■

Definition 2.23. The functions $(a, b) \rightarrow (a, b)_p$ from $\mathbb{N} \times \mathbb{N}$ to $\{-1, 1\}$ defined, for odd primes p , by (2.1) are called the **Hilbert symbols**.

The next proposition gives further properties of Hilbert symbols.

Proposition 2.24. The Hilbert symbol $(a, b)_p$ satisfies the following properties for any non-zero integers a and b and odd prime p :

- (H6) $(a^2, b)_p = 1$;
- (H7) if $a + b$ is a square, then $(a, b)_p = 1$;
- (H8) $(a, -a)_p = 1$;
- (H9) if $a + b \neq 0$, then $(a, b)_p = (a + b, -ab)_p$.

Proof. (H6) follows immediately from (H2).

(H7) If $a \not\equiv 0 \pmod{p}$ and $b \not\equiv 0 \pmod{p}$, then $(a, b)_p = 1$ by (H3).

Suppose that $a \not\equiv 0 \pmod{p}$ and $b \equiv 0 \pmod{p}$. Let $a + b = x^2$ and $b = p^t b_0$ where $b_0 \not\equiv 0 \pmod{p}$. Then $a \equiv x^2 \pmod{p}$, so, by (H2), (H3), and (H6), we obtain:

$$(a, b)_p = (a, b_0)_p (a, p)_p^t = (x^2, p)_p^t = 1.$$

Suppose that $a \equiv b \equiv 0 \pmod{p}$. Let $a = p^s a_0, b = p^t b_0$ where $a_0, b_0 \not\equiv 0 \pmod{p}$. Then

$$(a, b)_p = (a_0, b_0)_p (a_0, p)_p^t (b_0, p)_p^s (p, p)_p^{st}. \quad (2.2)$$

If s and t are even, then $(a, b)_p = 1$. Suppose that s is even and t is odd. Since $a + b = p^s a_0 + p^t b_0$ is a square and $s \neq t$, the smaller of the exponents s, t must be even, i.e., $s < t$. Then $a + b = p^s(a_0 + p^{t-s} b_0)$, so $a_0 + p^{t-s} b_0$ is a square. Therefore, $(a_0, p)_p = 1$ and (2.2) implies that $(a, b)_p = 1$. Suppose finally that both s and t are odd. If $s \neq t$, then the highest power of p dividing $a + b$ is odd, and $a + b$ cannot be a square. Therefore, $s = t$, and we have $a + b = p^s(a_0 + b_0)$. Since $a + b$ is a square and s is odd, $a_0 + b_0 \equiv 0 \pmod{p}$. Therefore, (2.2) implies that

$$\begin{aligned} (a, b)_p &= (a_0, p)_p (b_0, p)_p (p, p)_p = (a_0, p)_p (-a_0, p)_p (-1, p)_p (-p, p)_p \\ &= (a_0, p)_p^2 (-p, p)_p = 1. \end{aligned}$$

(H8) follows from (H7).

(H9) Since $a(a + b) + b(a + b) = (a + b)^2$, we apply (H7) to obtain that $(a(a + b), b(a + b))_p = 1$. Therefore,

$$\begin{aligned} (a, b)_p (a, a + b)_p (b, a + b)_p (a + b, a + b)_p &= 1, \\ (a, b)_p (ab, a + b)_p (-1, a + b)_p (-(a + b), a + b)_p &= 1, \\ (a, b)_p (-ab, a + b)_p &= 1 \implies (a, b)_p = (-ab, a + b)_p. \end{aligned}$$

■

We next use the Hilbert symbols to define the **Hasse invariants** of symmetric matrices over the integers.

Definition 2.25. Let A be a symmetric matrix of order n with integral entries. For $i = 1, 2, \dots, n$, let $D_i(A)$ be the determinant of the submatrix formed by the first i rows and the first i columns of A . Suppose that the determinants $D_1(A), D_2(A), \dots, D_n(A)$ are not equal to zero. Let p be an odd prime. Then the product

$$c_p(A) = (-1, D_n(A))_p \prod_{i=1}^{n-1} (D_i(A), -D_{i+1}(A))_p$$

is called the **Hasse p -invariant** of A .

The following theorem is central to applications of these invariants to designs. The proof is beyond the scope of this text. We would need a heavy build of Witt rings, quadratic spaces, and local class field theory for the proof. One may refer [16].

Theorem 2.26. *If N is a non-singular matrix over the integers, then $c_p(NN^\top) = 1$, for any odd prime p .*

We now prove the BRC Theorem.

Theorem 2.27 (Bruck-Ryser-Chowla). *Suppose there exists a symmetric $2-(v, k, \lambda)$ design.*

- i. If v is even, then $k - \lambda$ is a perfect square.*
- ii. If v is odd, then the following Diophantine equation has a non-trivial solution $(x, y, z) \neq (0, 0, 0)$ in integers:*

$$z^2 = (k - \lambda)x^2 + (-1)^{\frac{v-1}{2}} \lambda y^2$$

Proof. Let M be the incidence matrix of the symmetric design. By Proposition 2.9, we know that $MM^\top = (k - \lambda)I + \lambda J$. Let $A = MM^\top$. We first establish a lemma regarding the principal minors of this matrix.

Case 1: v is even.

We evaluate the determinant of the full matrix $A = MM^\top$. Setting $i = v$ in our lemma, we have $a_v = k + (v - 1)\lambda$. Since $k(k - 1) = \lambda(v - 1)$ holds in any symmetric design, this simplifies to $a_v = k^2$. Thus,

$$\det(MM^\top) = D_v = k^2(k - \lambda)^{v-1}$$

Since M is an incidence matrix with binary integer entries, its determinant $\det(M)$ is an integer. Consequently, $\det(MM^\top) = \det(M)^2$ must be a perfect square. Because v is even, the exponent $v - 1$ is odd.

We can factor the determinant as $\left(k(k - \lambda)^{\frac{v-2}{2}}\right)^2 (k - \lambda)$. For this entire expression to be a perfect square, the remaining linear factor $k - \lambda$ must itself be a perfect square.

Lemma. *For $i = 1, 2, \dots, v$, the determinant D_i of the principal submatrix formed by the first i rows and i columns of A is given by $D_i = a_i(k - \lambda)^{i-1}$, where $a_i = k + (i - 1)\lambda$.*

Proof of Lemma. Let A_i denote the $i \times i$ principal submatrix of A . Because $A = (k - \lambda)I + \lambda J$, any $i \times i$ principal submatrix inherits this exact uniform structure. Thus, $A_i = (k - \lambda)I_i + \lambda J_i$, where I_i is the $i \times i$ identity matrix and J_i is the $i \times i$ matrix of all ones.

We determine the eigenvalues of A_i . The matrix J_i is a rank-one matrix whose rows are identical. Its column space is spanned by the all-ones vector $\mathbf{1}$, which is naturally an eigenvector with eigenvalue i . The eigenspace corresponding to the eigenvalue 0 has dimension $i - 1$, consisting of all vectors orthogonal to $\mathbf{1}$.

Since A_i is a linear combination of I_i and J_i , these matrices share the same eigenvectors. Applying the transformation $x \mapsto (k - \lambda) + \lambda x$ to the eigenvalues of J_i , we find the eigenvalues of A_i are:

- $(k - \lambda) + \lambda(i) = k + (i - 1)\lambda = a_i$, with algebraic multiplicity 1.
- $(k - \lambda) + \lambda(0) = k - \lambda$, with algebraic multiplicity $i - 1$.

The determinant is the product of these eigenvalues, yielding $D_i = a_i(k - \lambda)^{i-1}$. This completes the proof of the lemma. $\backslash\backslash\backslash$

We now proceed with the proof for the odd case.

Case 2: v is odd.

We invoke the Hasse invariant. Since M is a non-singular matrix over the integers, Theorem 2.26 dictates that $c_p(A) = 1$ for all odd primes p .

By Definition 2.25, we expand $c_p(A)$. First, note that $D_v = k^2(k - \lambda)^{v-1}$. Because v is odd, $v - 1$ is even, making D_v a perfect square. By property (H6) from Proposition 2.24, any perfect square argument reduces the Hilbert symbol to 1. Thus, $(-1, D_v)_p = 1$.

The remaining product in the Hasse invariant groups naturally into pairs since $v - 1$ is even:

$$\begin{aligned} 1 = c_p(A) &= \prod_{i=1}^{\frac{v-1}{2}} (D_{2i-1}, -D_{2i})_p (D_{2i}, -D_{2i+1})_p \\ &= \prod_{i=1}^{\frac{v-1}{2}} (a_{2i-1}(k - \lambda)^{2i-2}, -a_{2i}(k - \lambda)^{2i-1})_p (a_{2i}(k - \lambda)^{2i-1}, -a_{2i+1}(k - \lambda)^{2i})_p \end{aligned}$$

We apply (H6) to eliminate the even powers of $(k - \lambda)$, reducing the symbols to:

$$= \prod_{i=1}^{\frac{v-1}{2}} (a_{2i-1}, -a_{2i}(k - \lambda))_p (a_{2i}(k - \lambda), -a_{2i+1})_p$$

Expanding via multiplicativity (H2), we obtain:

$$= \prod_{i=1}^{\frac{v-1}{2}} (a_{2i-1}, -a_{2i})_p (a_{2i-1}, k - \lambda)_p (a_{2i}, -a_{2i+1})_p (k - \lambda, -a_{2i+1})_p$$

We group the terms containing $(k - \lambda)$. By (H2), $(a_{2i-1}, k - \lambda)_p (k - \lambda, -a_{2i+1})_p = (-a_{2i-1}a_{2i+1}, k - \lambda)_p$, which splits into $(-1, k - \lambda)_p (a_{2i-1}a_{2i+1}, k - \lambda)_p$. Factoring out the exactly $\frac{v-1}{2}$ copies of $(-1, k - \lambda)_p$ leaves:

$$= \left((-1)^{\frac{v-1}{2}}, k - \lambda \right)_p \prod_{i=1}^{\frac{v-1}{2}} (a_{2i-1}, -a_{2i})_p (a_{2i}, -a_{2i+1})_p (a_{2i-1}a_{2i+1}, k - \lambda)_p$$

We now invoke the critical additive property (H9). Observing that $a_{2i-1} - a_{2i} = -\lambda$, we apply (H9) with $a = a_{2i-1}$ and $b = -a_{2i}$ to find $(a_{2i-1}, -a_{2i})_p = (-\lambda, a_{2i-1}a_{2i})_p$. Applying this similarly to the second term gives $(a_{2i}, -a_{2i+1})_p = (-\lambda, a_{2i}a_{2i+1})_p$. Their product is $(-\lambda, a_{2i-1}a_{2i}^2a_{2i+1})_p$, which reduces to $(-\lambda, a_{2i-1}a_{2i+1})_p$ by (H6). Our master equation now simplifies to:

$$\begin{aligned} 1 &= \left((-1)^{\frac{v-1}{2}}, k - \lambda \right)_p \prod_{i=1}^{\frac{v-1}{2}} (-\lambda, a_{2i-1}a_{2i+1})_p (k - \lambda, a_{2i-1}a_{2i+1})_p \\ &= \left((-1)^{\frac{v-1}{2}}, k - \lambda \right)_p \left(-\lambda(k - \lambda), \prod_{i=1}^{\frac{v-1}{2}} a_{2i-1}a_{2i+1} \right)_p \end{aligned}$$

The product inside the symbol telescopes modulo squares (applying H6 repeatedly): $a_1a_3^2a_5^2 \dots a_{v-2}^2a_v \equiv a_1a_v$. Substituting $a_1 = k$ and $a_v = k^2$, the product becomes k^3 , which is equivalent to k modulo squares. We are left with:

$$1 = \left((-1)^{\frac{v-1}{2}}, k - \lambda \right)_p (-\lambda(k - \lambda), k)_p$$

Finally, observe that $(k - \lambda) + \lambda = k$. Using (H9) with $a = k - \lambda$ and $b = \lambda$, we have $(k - \lambda, \lambda)_p = (k, -\lambda(k - \lambda))_p$. By symmetry (H1), this implies $(-\lambda(k - \lambda), k)_p = (\lambda, k - \lambda)_p$. Substituting this into our equation yields:

$$1 = \left((-1)^{\frac{v-1}{2}}, k - \lambda \right)_p (\lambda, k - \lambda)_p = \left((-1)^{\frac{v-1}{2}} \lambda, k - \lambda \right)_p$$

This equality holds for all odd primes p . By the definition of the Hilbert symbol, $(A, B)_p = 1$ signifies that the quadratic form $Z^2 - AX^2 - BY^2 = 0$ has a non-trivial solution over the p -adic numbers. Setting $A = k - \lambda$ and $B = (-1)^{\frac{v-1}{2}} \lambda$, this gives local solutions for all p . Since $k > \lambda$, $A > 0$, guaranteeing a real solution as well. By the Hasse-Minkowski Theorem, a quadratic form that has local solutions over $\mathbb{R}$ and all p -adic fields must have a global non-trivial rational solution. Clearing denominators provides a non-trivial integer solution (x, y, z) to the Diophantine equation

$$z^2 = (k - \lambda)x^2 + (-1)^{\frac{v-1}{2}} \lambda y^2$$

thus completing the proof. ■

Example 2.28. If there exists a symmetric $2-(43, 7, 1)$ design, then $(-1, 6)_p = 1$ for any odd prime p . However, $(-1, 6)_3 = (-1, 3)_3 = \left(\frac{-1}{3}\right) = -1$. Therefore, there is no symmetric $2-(43, 7, 1)$ design.

Example 2.29. If there exists a symmetric $2-(29, 8, 2)$ design, then $(2, 6)_3 = 1$. On the other hand, $(2, 6)_3 = (2, 3)_3 = \left(\frac{2}{3}\right) = -1$. Therefore, there is no symmetric $2-(29, 8, 2)$ design.

The condition of the Bruck-Ryser-Chowla Theorem is not sufficient for the existence of symmetric designs. The only known counterexample is the parameter set $(111, 11, 1)$. It satisfies the condition of the theorem, however, it has been computationally proven that there is no symmetric $2-(111, 11, 1)$ design.

2.4 Classical Families - Finite Projective Planes

We now embark on a detailed study of one of the most prominent families of symmetric designs: finite projective planes. Instead of relying on algebraic constructions over finite fields, we can define these structures purely through an elegant axiomatisation of incidence geometry.

Definition 2.30. Let P be a finite set of elements called *points* and L be a finite set of subsets of P called *lines*. The pair (P, L) is called a *finite projective plane* if it satisfies the following three incidence axioms:

- A1.** For any two distinct points $x, y \in P$, there exists exactly one line $\ell \in L$ containing both x and y .
- A2.** For any two distinct lines $\ell_1, \ell_2 \in L$, there exists exactly one point $z \in P$ such that $z \in \ell_1 \cap \ell_2$.
- A3.** There exist four points in P , no three of which are collinear (i.e., contained in the same line).

Axiom A3 is of paramount importance. Axioms A1 and A2 alone allow for trivial, ‘degenerate’ geometries—such as a configuration where all points lie on a single line except for one external point. Axiom A3 strictly eliminates these cases, enforcing a rich, two-dimensional structure.

To establish the uniform parameters of a projective plane, we first require a fundamental lemma regarding the projection of points onto lines.

Lemma 2.31. Let $\pi = (P, L)$ be an incidence structure satisfying A1 and A2. Let $\ell \in L$ be a line and $p \in P$ be a point such that $p \notin \ell$. Then there exists a bijection between the points on ℓ and the lines passing through p . Consequently, the number of points on ℓ is equal to the number of lines passing through p .

Proof. Let S_ℓ be the set of points on ℓ , and let S_p be the set of lines containing p . We define a mapping $f : S_\ell \rightarrow S_p$ by $f(x) = px$, where px denotes the unique line passing through p and x guaranteed by axiom A1. Since $p \notin \ell$, the line px is distinct from ℓ .

To show f is injective, suppose $f(x) = f(y)$ for some $x, y \in S_\ell$. This implies $px = py$. If $x \neq y$, then the distinct points x and y are both contained in the line ℓ and the line px . By A1, the line containing x and y is unique, forcing $px = \ell$. However, this would mean $p \in \ell$, which contradicts our premise. Thus, $x = y$, and f is injective.

To show f is surjective, let $m \in S_p$ be any line containing p . By A2, the lines m and ℓ must intersect at exactly one point, say z . Thus, $z \in \ell$ and $z \in m$. Since m contains both p and z , it must be the line pz . Therefore, $f(z) = m$, proving f is surjective.

Being both injective and surjective, f is a bijection, and the result follows. $\blacksquare$

We can now establish the structural rigidity of finite projective planes. If we know the size of a single line, the entire global parameter set of the plane is strictly determined.

Theorem 2.32. *Let $\pi = (P, L)$ be a finite projective plane. For an integer $n \geq 2$, the following six conditions are equivalent. If they hold, n is called the **order** of the projective plane.*

- i. A line contains exactly $n + 1$ points.
- ii. A point is contained in exactly $n + 1$ lines.
- iii. Every line contains exactly $n + 1$ points.
- iv. Every point is contained in exactly $n + 1$ lines.
- v. There exist $n^2 + n + 1$ points in total.
- vi. There exist $n^2 + n + 1$ lines in total.

Proof. We will prove the cycle of implications (i) $\implies$ (iii) $\implies$ (iv) $\implies$ (ii) $\implies$ (i), followed by establishing (v) and (vi) from the uniform parameters. Then using (v) and separately (vi), we prove one of the uniform parameter conditions.

(i) $\implies$ (iii): Suppose there exists a line ℓ_0 with exactly $n + 1$ points. Let ℓ be any other line. We wish to show $|\ell| = n + 1$.

Suppose there exists a point $p \notin \ell_0 \cup \ell$. By Lemma 2.31, since $p \notin \ell_0$, the number of lines through p is exactly $|\ell_0| = n + 1$. Applying Lemma 2.31 again, since $p \notin \ell$, the number of points on ℓ is equal to the number of lines through p . Thus, $|\ell| = n + 1$.

We must now eliminate the possibility that no such point p exists. Suppose for the sake of contradiction that $P = \ell_0 \cup \ell$. By A3, there exist four points A, B, C, D , no three of which are collinear. They must all lie in $\ell_0 \cup \ell$. Since no three are collinear, exactly two must lie on $\ell_0 \setminus \ell$ (say, A and B) and exactly two must lie on $\ell \setminus \ell_0$ (say, C and D). Consider the line AC . It intersects ℓ_0 at A and ℓ at C . Because all points in P belong to either ℓ_0 or ℓ , any point on AC must be either the intersection $AC \cap \ell_0 = A$ or $AC \cap \ell = C$. Thus, the line AC contains exactly 2 points. However, since $|\ell_0| = n + 1 \geq 3$, there exists a point $y \in \ell_0$ distinct from A and $\ell_0 \cap \ell$. Since $D \notin \ell_0$, Lemma 2.31 dictates that the number of lines through D is exactly $|\ell_0| = n + 1$. Furthermore, since $D \notin AC$, Lemma 2.31 also dictates that the number of lines through D is equal to $|AC| = 2$. This implies $n + 1 = 2$, meaning $n = 1$, which violates our hypothesis that $n \geq 2$. Thus, $P \neq \ell_0 \cup \ell$, and we can always find a point off both lines, completing the proof that (i) implies (iii).

(iii) $\implies$ (iv): Let p be any point. By A3, not all points in the plane are collinear, so there exists at least one line ℓ not containing p . By (iii), $|\ell| = n + 1$. By Lemma 2.31, the number of lines containing p is exactly $|\ell| = n + 1$.

(iv) $\implies$ (ii): This is trivial, as the existence of the quadrangle in A3 guarantees the existence of points, any of which trivially satisfies (ii).

(ii) $\implies$ (i): Suppose there exists a point p contained in exactly $n + 1$ lines. By A3, there exist lines not containing p . For any such line ℓ , Lemma 2.31 states that $|\ell|$ equals the number of lines through p , which is $n + 1$. This yields a line with $n + 1$ points, satisfying (i).

Establishing (v) and (vi): Assume the uniform parameters from (iii) and (iv) hold. To prove (v), we select an arbitrary point $p \in P$. Every other point in the plane must lie on exactly one of the lines passing through

p by A1. By (iv), there are $n + 1$ lines passing through p . By (iii), each of these lines contains $n + 1$ points, meaning each line contains exactly n points distinct from p . Because these lines only intersect at p (by A2), they partition the remaining points of the plane. Thus, the total number of points is $1 + (n + 1)n = n^2 + n + 1$.

To prove (vi), we select an arbitrary line $\ell \in L$. Every other line in the plane must intersect ℓ at exactly one point by A2. By (iii), there are $n + 1$ points on ℓ . By (iv), there are $n + 1$ lines passing through each of these points, meaning there are exactly n lines distinct from ℓ passing through each point on ℓ . Because any two lines intersect exactly once, this accounts for all lines in the plane. Thus, the total number of lines is $1 + (n + 1)n = n^2 + n + 1$.

(v) $\implies$ (i): Assume (v) holds, so the total number of points is $n^2 + n + 1$. We first establish that there exists a line with at least 3 points. By A3, there exist four points A, B, C, D , no three of which are collinear. The lines AB and CD must intersect at exactly one point E by A2. If $E = A$, then A would lie on CD , making A, C, D collinear, a contradiction. Similar logic ensures E is distinct from B, C , and D . Thus, the line AB contains at least three distinct points: A, B , and E .

Let $m + 1 \geq 3$ be the exact number of points on this line AB . Treating this as our starting condition (i) for an integer $m \geq 2$, our established cycle of implications (i) $\implies$ (iii) $\implies$ (iv) dictates that the total number of points in the plane *must* be equal to $m^2 + m + 1$. However, we are given by (v) that the total number of points is $n^2 + n + 1$. Equating these yields $m^2 + m + 1 = n^2 + n + 1$, which implies $m = n$ (since both are positive integers). Therefore, the line AB contains exactly $n + 1$ points, satisfying (i).

(vi) $\implies$ (ii): Assume (vi) holds, meaning there are $n^2 + n + 1$ lines in total. By the inherent duality of projective planes, we can exchange the roles of points and lines to apply the exact same argument as above. Axiom A3 dually guarantees the existence of a point with at least 3 lines passing through it. Let this point have $m + 1 \geq 3$ lines passing through it. By the dual cycle of implications, the total number of lines in the plane must be $m^2 + m + 1$. Equating this to $n^2 + n + 1$ yields $m = n$. Thus, there exists a point contained in exactly $n + 1$ lines, satisfying (ii). ■

The parameters derived in Theorem 2.32 immediately reveal that axiomatic projective planes are merely symmetric designs in disguise.

Proposition 2.33. *A symmetric $2-(n^2 + n + 1, n + 1, 1)$ design exists if and only if a finite projective plane of order n exists.*

Proof. Let $\pi = (P, L)$ be a finite projective plane of order n . We view the set of points P as the parent set of the design and the set of lines L as the collection of blocks $\mathcal{B}$. By Theorem 2.32, the total number of points is $v = n^2 + n + 1$, and every block has a uniform size of $k = n + 1$. By Axiom A1, any pair of distinct points is contained in exactly one line, meaning $\lambda = 1$. Thus, it is a $2-(n^2 + n + 1, n + 1, 1)$ design. Furthermore, Theorem 2.32 establishes that the total number of blocks (lines) is $b = n^2 + n + 1$. Since $b = v$, the design is symmetric by Definition 2.11.

Conversely, let $\mathcal{B}$ be a symmetric $2-(n^2 + n + 1, n + 1, 1)$ design defined on a parent set of points P . We define our lines L to be the blocks of $\mathcal{B}$ and verify the three axioms. Since it is a 2-design with $\lambda = 1$, every pair of distinct points is contained in exactly one block. This immediately satisfies Axiom A1. Because $\mathcal{B}$ is a symmetric design, Proposition 2.12 establishes that any two distinct blocks intersect in exactly $\lambda = 1$ points. Thus, any two distinct lines meet at exactly one point, satisfying Axiom A2.

Finally, we must verify Axiom A3. Since $n \geq 2$, the block size is $k = n + 1 \geq 3$. Select an arbitrary block B_1 and two distinct points $x, y \in B_1$. The number of blocks containing a given point is $r = k \geq 3$. Select a block $B_2 \neq B_1$ containing x , and a block $B_3 \neq B_1$ containing y . By A2, B_2 and B_3 intersect at exactly one point, say u . Because $B_1 \cap B_2 = \{x\}$ and $B_1 \cap B_3 = \{y\}$, u cannot be in B_1 (otherwise $u = x$ and $u = y$, implying $x = y$).

Since the block size is at least 3, we can pick a point $z \in B_2 \setminus \{x, u\}$ and a point $w \in B_3 \setminus \{y, u\}$. We claim x, y, z, w form the required quadrangle. The points x and y lie on B_1 , but $z, w \notin B_1$, so no three lie on B_1 . The points x and z lie on B_2 . Since $B_1 \cap B_2 = \{x\}$, $y \notin B_2$; and since $B_2 \cap B_3 = \{u\}$ and $w \neq u$, $w \notin B_2$. So no three lie on B_2 . The points y and w lie on B_3 . By symmetrical logic, $x \notin B_3$ and $z \notin B_3$. So no three

lie on B_3 . Lastly, the unique block containing z and w cannot contain x (as that would be B_2 , which does not contain w) and cannot contain y (as that would be B_3 , which does not contain z). Thus, no three of the four points are collinear, satisfying Axiom A3. This concludes the proof that (P, L) is a projective plane of order n . ■

When considering the axioms of a finite projective plane, it is worth noting the inherent duality of these incidence structures. If the three axioms (A1, A2, and A3) of Definition 2.30 hold, then one can rigorously deduce the following dual condition to Axiom A3: there exists a set of four lines, no three of which are concurrent (meeting at a single point). This symmetry guarantees that the dual structure $\pi' = (L, P)$, obtained by exchanging the roles of points and lines in a projective plane $\pi = (P, L)$, inherently satisfies all the axioms and thus becomes a projective plane itself.

We have established that the parameters of a finite projective plane are entirely determined by its order n . The most natural examples of these incidence structures are constructed using the algebraic machinery of finite fields. A projective plane constructed over the finite field $\mathbb{F}_q$ is called a **Desarguesian plane**, and it necessarily has order q (which is inherently a prime power).

However, there exist several non-standard projective planes that do not arise from finite fields; these are known as **non-Desarguesian planes**. In the search for algebraic systems that coordinatise these non-Desarguesian planes, mathematicians have conducted extensive studies using some structures much weaker than fields.

Up to the present day, the orders of all known finite projective planes are prime powers. It remains an incredibly important open problem whether the order of a finite projective plane is *always* a prime power. Specialists in the field remain divided over the expected resolution of this issue.

Furthermore, many attempts have been made to fully classify non-Desarguesian planes. These are exceptionally difficult problems, and very few definitive results are known. For a given order n , there are undeniably ‘finitely many’ projective planes up to isomorphism (for $n = 9$ there are 4 different ones (counting up to isomorphisms)). However, as is often the case in combinatorics, the guarantee of ‘finitely many’ is not a practically powerful condition. One might assume that with a finite search space, computers can simply exhaust all possibilities to resolve these open problems. This is not true at all. While computational searches are useful for very small parameters, the combinatorial explosion as the number of points increases renders computers useless exceptionally quickly. Deep theoretical breakthroughs, rather than pure computational power, are required to advance the classification of these fascinating structures.

We now apply the Bruck-Ryser-Chowla Theorem directly to the parameters of finite projective planes to obtain a powerful classification tool.

Theorem 2.34. *Let $n \equiv 1 \pmod{4}$ or $n \equiv 2 \pmod{4}$. If there exists a finite projective plane of order n , then n can be expressed as the sum of two integer squares, $n = x^2 + y^2$.*

Proof. Recall from Proposition 2.33 that a finite projective plane of order n corresponds to a symmetric $2-(n^2 + n + 1, n + 1, 1)$ design. The number of points $v = n^2 + n + 1 = n(n + 1) + 1$ is always an odd integer, allowing us to invoke Case (ii) of Theorem 2.27.

We evaluate the parameters for the Diophantine equation: the coefficient $k - \lambda$ becomes $(n + 1) - 1 = n$, and the exponent for the sign evaluates to $\frac{v-1}{2} = \frac{n^2+n}{2} = \frac{n(n+1)}{2}$.

Because $n \equiv 1 \pmod{4}$ or $n \equiv 2 \pmod{4}$, the product $n(n + 1)$ leaves a remainder of 2 when divided by 4. Consequently, the exponent $\frac{n(n+1)}{2}$ is strictly an odd integer. This reduces the term $(-1)^{\frac{v-1}{2}}$ to -1 .

The Diophantine equation from the Bruck-Ryser-Chowla Theorem simplifies to:

$$z^2 = nx^2 - y^2 \implies nx^2 = y^2 + z^2$$

The existence of a non-trivial integer solution (x, y, z) to this equation implies that nx^2 can be written as the sum of two squares. By classical number theory, if the product of an integer and a perfect square is expressible as the sum of two squares, then the integer itself can be expressed as the sum of two squares. Thus, $n = x^2 + y^2$ for some integers x, y . ■

Corollary 2.35. *There exist no projective planes of order $n = 6, 14, 21, 22, \dots$*

Proof. For $n = 6$, we have $n \equiv 2 \pmod{4}$, but 6 cannot be expressed as the sum of two integer squares. The same algebraic obstruction applies to $14 \equiv 2 \pmod{4}$, $21 \equiv 1 \pmod{4}$, and $22 \equiv 2 \pmod{4}$. The non-existence follows directly from Theorem 2.34. ■

This brings us to the natural question: how many projective planes of order n exist for a given n ? Extensive studies have been conducted on this classification problem, yet numerous aspects remain unresolved. It is known that for orders $n = 2, 3, 4, 5, 7$, and 8, the projective plane is strictly unique up to isomorphism. For the case of $n = 9$, the classification is complete, concluding with exactly four non-isomorphic projective planes.

The case of order 10 holds a particularly storied history in combinatorial mathematics. Because $10 \equiv 2 \pmod{4}$ and $10 = 3^2 + 1^2$, the Bruck-Ryser-Chowla theorem is satisfied and fails to rule out its existence. At one time, there was a legend that mathematicians who attacked this problem lost their minds due to the sheer volume and overwhelming complexity of the required calculations. Building upon the foundational early works of M. Hall Jr., Parker, MacWilliams, Sloane, and Thompson, the problem was finally resolved in 1989. Lam, Thiel, and Swiercz definitively proved the non-existence of a projective plane of order 10. Their proof was essentially based on a massive brute-force search; it is recorded that they ran the fastest large computer available at the time for over 1000 hours. In this sense, the solution made a profound impact on the mathematical world regarding the role of computers in rigorous proofs, drawing parallels to the Four Colour Theorem and the Kepler Conjecture.

Following the resolution of order 10, several researchers have seriously dedicated study to the case of order 12 (and 15). As stated previously, whether the order n of a finite projective plane must *always* be a prime power remains the most important open problem in this area. In particular, if n is restricted to a prime p , whether a projective plane of order p is always unique up to isomorphism is also an incredibly interesting open problem (for which no counterexample is currently known). Furthermore, it is a well-established fact that depending on the choice of n , an extraordinarily large number of non-isomorphic projective planes can exist.

2.5 Hadamard Matrices and Paley Constructions

We now turn our attention to a fascinating class of matrices that possess deep connections with symmetric designs. These matrices are defined purely by an extremal property regarding their determinant.

Definition 2.36. An $n \times n$ matrix H is said to be a **Hadamard matrix** of order n if every entry is either 1 or -1 , and it satisfies the extremal determinant bound $|\det H| = n^{n/2}$.

This maximal determinant property is equivalent to a very clean geometric condition on the rows and columns of the matrix.

Proposition 2.37. *For an $n \times n$ matrix H with entries in $\{1, -1\}$, the following are equivalent:*

- i. $|\det H| = n^{n/2}$ (i.e., H is a Hadamard matrix).
- ii. $H^\top H = nI$.
- iii. $HH^\top = nI$.

Proof. We first prove the equivalence of (i) and (ii). Let $c_1, c_2, \dots, c_n$ be the column vectors of H . We consider the matrix $H^\top H$. The (i, j) -th entry of $H^\top H$ is given by the standard inner product $c_i^\top c_j$.

Since every entry of H is either 1 or -1 , the diagonal entries of $H^\top H$ are simply the dot product of a column with itself. For any i , $c_i^\top c_i = \sum_{k=1}^n (\pm 1)^2 = n$.

By Hadamard's Inequality, the square of the determinant of a matrix is bounded by the product of the squared Euclidean lengths of its column vectors. That is,

$$(\det H)^2 \leq \prod_{i=1}^n \|c_i\|^2 = \prod_{i=1}^n n = n^n$$

Equality holds in Hadamard's Inequality if and only if the column vectors are mutually orthogonal. Thus, $(\det H)^2 = n^n$ (which is equivalent to $|\det H| = n^{n/2}$) if and only if $c_i^\top c_j = 0$ for all $i \neq j$. This condition precisely means the off-diagonal entries of $H^\top H$ are 0. Combined with our observation about the diagonal entries, this is equivalent to the matrix equation $H^\top H = nI$.

The equivalence of (ii) and (iii) is a standard result in linear algebra. Since $H^\top H = nI$, the matrix $\frac{1}{n}H^\top$ acts as the left inverse of H . For square matrices over a field, a left inverse is necessarily a right inverse. Therefore, $H(\frac{1}{n}H^\top) = I$, which immediately implies $HH^\top = nI$. ■

The condition $H^\top H = nI$ dictates that any two distinct columns of a Hadamard matrix must be orthogonal. Since the entries are strictly ± 1 , this orthogonality places severe restrictions on the possible orders of such a matrix.

Theorem 2.38. *If a Hadamard matrix of order n exists, then $n = 1$, $n = 2$, or n is a multiple of 4.*

Proof. Let H be a Hadamard matrix of order n . The cases $n = 1$ and $n = 2$ are trivially satisfied by the matrices $[1]$ and $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

Assume $n \geq 3$. Because H is a Hadamard matrix, it satisfies $HH^\top = nI$, which strictly guarantees that any pair of distinct rows in H is mutually orthogonal. Since $n \geq 3$, the matrix contains at least three distinct rows, which are inherently orthogonal to one another.

Furthermore, multiplying any column of a Hadamard matrix by -1 preserves the orthogonality of its rows (and thus preserves the property of being a Hadamard matrix). We can therefore multiply specific columns by -1 to ensure that every entry in the first row is 1. Such a matrix is called *normalised*.

Let us look at the first three rows of this normalised matrix. The columns can be partitioned into four disjoint sets based on the entries in the second and third rows:

- Let a be the number of columns of the form $(1, 1, 1)^\top$.
- Let b be the number of columns of the form $(1, 1, -1)^\top$.
- Let c be the number of columns of the form $(1, -1, 1)^\top$.
- Let d be the number of columns of the form $(1, -1, -1)^\top$.

Since every column must fall into one of these four categories, we have

$$a + b + c + d = n$$

The inner product of the first row and the second row must be 0. The entries match in the first $a + b$ columns and differ in the last $c + d$ columns. Thus,

$$a + b - c - d = 0$$

Similarly, the inner product of the first row and the third row must be 0. The entries match in the columns counted by a and c , and differ in those counted by b and d . Thus,

$$a - b + c - d = 0$$

Finally, the inner product of the second and third rows must be 0. They match in the columns where they are both 1 (count a) or both -1 (count d), and differ elsewhere. Thus,

$$a - b - c + d = 0$$

Adding all four equations yields $4a = n$. Since a represents a count of columns, it must be a non-negative integer. Therefore, n must be a multiple of 4. ■

The necessary condition that n must be a multiple of 4 (for $n > 2$) naturally leads to the converse question, which is one of the oldest and most famous open problems in combinatorial mathematics.

Conjecture 2.39 (Hadamard Conjecture). *There exists a Hadamard matrix of order $4k$ for every positive integer k .*

While the conjecture remains unproven, massive progress has been made. For instance, the existence of a Hadamard matrix of order 428 was an open problem for decades until it was constructed by Kharaghani and Tayfeh-Rezaie in 2005. As of May 2026, the smallest order for which the existence of a Hadamard matrix remains unknown is 668.

We now explicitly demonstrate the deep connection between Hadamard matrices and symmetric designs.

Proposition 2.40. *Let H be a Hadamard matrix of order $n = 4t$. One can construct a symmetric $2-(4t - 1, 2t - 1, t - 1)$ design from H .*

Proof. Since $n \geq 4$, we can normalise H by multiplying appropriate rows and columns by -1 such that its first row and first column consist entirely of 1's. Let this normalised matrix be H . We form an $(n-1) \times (n-1)$ matrix H' by deleting the first row and the first column of H .

We construct an incidence matrix A of size $(n-1) \times (n-1)$ by setting $A_{ij} = 1$ if $H'_{ij} = 1$, and $A_{ij} = 0$ if $H'_{ij} = -1$. This algebraic substitution is equivalent to the linear transformation

$$A = \frac{1}{2}(H' + J)$$

where J is the $(n-1) \times (n-1)$ matrix of all ones. We must verify that A is the incidence matrix of a symmetric design.

Recall that $HH^\top = nI$. Because the first row of H is all 1's, the inner product of any other row of H with the first row is exactly the sum of its entries, which must be 0. Since the first entry of every row is 1, the remaining $n-1$ entries in any row of H' must sum to -1 . This implies that every row of H' contains exactly $(n/2) - 1$ ones and $n/2$ negative ones. Substituting $n = 4t$, every row of H' has $2t - 1$ ones. Consequently, every row of A contains exactly $2t - 1$ ones. This establishes uniform block size $k = 2t - 1$.

Now consider two distinct rows of H' , say row x and row y . In the full matrix H , these rows are orthogonal. Since they both begin with a 1 in the first column, the dot product of their remaining $n-1$ entries (which constitute rows x and y of H') must be -1 . Let α be the number of positions where both rows have a 1. Let β be the number of positions where row x is 1 and row y is -1 . Since row x has exactly $2t - 1$ ones in total, we have $\alpha + \beta = 2t - 1$.

The dot product of these two rows in H' contributes 1 for the α positions where they agree and are both positive, and -1 for the β positions where they differ. There are symmetrical arguments for when they are both negative. However, we can use a direct algebraic approach on the incidence matrix A .

Consider $AA^\top$. Using our linear transformation, we have

$$AA^\top = \frac{1}{4}(H' + J)(H'^\top + J) = \frac{1}{4}(H'H'^\top + H'J + JH'^\top + J^2)$$

From $HH^\top = nI$, we can deduce $H'H'^\top = nI - J$. We also know every row sum of H' is -1 , so $H'J = -J$. Similarly $JH'^\top = -J$. Finally, $J^2 = (n-1)J$. Substituting these yields:

$$AA^\top = \frac{1}{4}(nI - J - J - J + (n-1)J) = \frac{n}{4}I + \frac{n-4}{4}J$$

Substituting $n = 4t$, we obtain

$$AA^\top = tI + (t-1)J = (t)I + (t-1)(J - I)$$

This matches the fundamental equation $MM^\top = (r-\lambda)I + \lambda J$ for a symmetric design. We read off $r-\lambda = t$ and $\lambda = t-1$, giving $r = 2t-1$. Since the matrix is square ($b = v = 4t-1$) and $r = k$, this confirms A is the incidence matrix of a symmetric $2-(4t-1, 2t-1, t-1)$ design. ■

Definition 2.41. A symmetric design with parameters $2-(4t-1, 2t-1, t-1)$ is classically called a **Hadamard 2-design**.

The construction in Proposition 2.40 is completely reversible. Given a Hadamard 2-design, one can replace the 0's in its incidence matrix with -1 's, prepend a row and a column of all 1's, and recover a Hadamard matrix of order $4t$. Thus, the existence of a Hadamard matrix of order $4t$ is mathematically equivalent to the existence of a Hadamard $2-(4t-1, 2t-1, t-1)$ design.

To build larger Hadamard matrices from smaller ones, we define a standard matrix operation.

Definition 2.42. The **Kronecker product** of an $n \times n$ matrix $A = (a_{ij})$ and an $m \times m$ matrix B is the $nm \times nm$ block matrix $A \otimes B$ where the (i, j) -th block is the matrix $a_{ij}B$.

Proposition 2.43. If H_n and H_m are Hadamard matrices of order n and m respectively, then $H_n \otimes H_m$ is a Hadamard matrix of order nm .

Proof. Let $H = H_n \otimes H_m$. By the definition of the Kronecker product, every entry of H is the product of an entry from H_n and an entry from H_m . Since both matrices contain only 1's and -1 's, every entry of H is also strictly 1 or -1 .

We must show that $HH^\top = (nm)I$. A standard property of the Kronecker product is that the transpose distributes across the product, $(A \otimes B)^\top = A^\top \otimes B^\top$, and multiplication is performed component-wise, $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$. Applying these properties, we have:

$$\begin{aligned} HH^\top &= (H_n \otimes H_m)(H_n \otimes H_m)^\top \\ &= (H_n \otimes H_m)(H_n^\top \otimes H_m^\top) \\ &= (H_n H_n^\top) \otimes (H_m H_m^\top) \end{aligned}$$

Since H_n and H_m are Hadamard matrices, we substitute $H_n H_n^\top = nI_n$ and $H_m H_m^\top = mI_m$.

$$\begin{aligned} HH^\top &= (nI_n) \otimes (mI_m) \\ &= (nm)(I_n \otimes I_m) \\ &= (nm)I_{nm} \end{aligned}$$

Thus, $H_n \otimes H_m$ is a Hadamard matrix of order nm . ■

As an immediate consequence, since we know the 2×2 matrix $H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ is a Hadamard matrix, we can recursively take the Kronecker product of H_2 with itself to establish the existence of a Hadamard matrix of order 2^r for any positive integer r . This specific family of matrices is often referred to as Sylvester-Hadamard matrices.

While the Kronecker product allows us to build larger Hadamard matrices from smaller known ones, it cannot generate matrices of orders that are not multiples of the starting matrices. To establish the existence of Hadamard matrices for new, fundamental orders, we turn to the algebraic machinery of finite fields.

In 1933, Raymond Paley introduced a brilliant family of constructions relying on the multiplicative group $\mathbb{F}_q^*$ of a finite field $\mathbb{F}_q$, where $q = p^r$ for an odd prime p .

Definition 2.44. Let $\mathbb{F}_q$ be a finite field of odd order q . We define the **quadratic character** $\chi : \mathbb{F}_q \rightarrow \{0, 1, -1\}$ by:

$$\chi(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \text{ is a non-zero square (quadratic residue) in } \mathbb{F}_q \\ -1 & \text{if } x \text{ is a non-square (quadratic non-residue) in } \mathbb{F}_q \end{cases}$$

This generalises the Legendre symbol from $\mathbb{Z}/p\mathbb{Z}$ to arbitrary finite fields. Because the non-zero squares form a subgroup of index 2 in the cyclic multiplicative group $\mathbb{F}_q^*$, exactly half of the non-zero elements are

squares and half are non-squares. Furthermore, χ acts as a completely multiplicative homomorphism on $\mathbb{F}_q^*$, meaning $\chi(xy) = \chi(x)\chi(y)$ for all $x, y \in \mathbb{F}_q$.

We use this character to construct a core matrix, classically referred to as the Jacobsthal matrix, which encapsulates the field's arithmetic.

Definition 2.45. Let the elements of $\mathbb{F}_q$ be indexed as $a_1, a_2, \dots, a_q$. The **Paley core matrix** Q is a $q \times q$ matrix whose (i, j) -th entry is given by $Q_{ij} = \chi(a_i - a_j)$.

Before leveraging Q to construct a Hadamard matrix, we must establish three powerful algebraic properties regarding its structure.

Proposition 2.46. Let Q be the $q \times q$ Paley core matrix over $\mathbb{F}_q$, and let J be the $q \times q$ matrix of all ones. Then:

- i. $QJ = JQ = 0$ (all row sums and column sums are exactly zero).
- ii. $Q^\top = (-1)^{(q-1)/2}Q$.
- iii. $QQ^\top = qI - J$.

Proof. To prove (i), we evaluate the sum of the entries in an arbitrary row i . The sum is given by $\sum_{j=1}^q \chi(a_i - a_j)$. As a_j ranges over all elements of the field $\mathbb{F}_q$, the shifted value $(a_i - a_j)$ also ranges over all elements of $\mathbb{F}_q$. Thus, the sum is simply $\sum_{x \in \mathbb{F}_q} \chi(x)$. Since there are exactly $(q-1)/2$ squares yielding $+1$ and $(q-1)/2$ non-squares yielding -1 , and $\chi(0) = 0$, the sum evaluates precisely to 0. The same logic applies to column sums.

To prove (ii), we examine the symmetry of χ . We know $Q_{ji} = \chi(a_j - a_i) = \chi(-(a_i - a_j)) = \chi(-1)\chi(a_i - a_j)$. In any finite field of odd order q , -1 is a quadratic residue if and only if $q \equiv 1 \pmod{4}$. Thus, $\chi(-1) = (-1)^{(q-1)/2}$. This immediately gives $Q_{ji} = (-1)^{(q-1)/2}Q_{ij}$, completing the proof of symmetry or skew-symmetry depending on q .

To prove (iii), we compute the (i, k) -th entry of the matrix product $QQ^\top$. Let this entry be denoted by P_{ik} . We have:

$$P_{ik} = \sum_{j=1}^q Q_{ij}Q_{kj} = \sum_{j=1}^q \chi(a_i - a_j)\chi(a_k - a_j)$$

If $i = k$, then $P_{ii} = \sum_{j=1}^q \chi(a_i - a_j)^2$. For $j = i$, the term is $\chi(0)^2 = 0$. For the remaining $q-1$ terms, the character evaluates to either 1 or -1 , squaring to 1 in all cases. Thus, the diagonal entries of $QQ^\top$ are uniformly $q-1$.

If $i \neq k$, we evaluate the off-diagonal sum. Let $x = a_j$. The sum becomes $\sum_{x \in \mathbb{F}_q} \chi(a_i - x)\chi(a_k - x)$. Let us employ the substitution $y = a_i - x$. Then $a_k - x = a_k - a_i + y$. Let $c = a_k - a_i$. Since $i \neq k$, $c \neq 0$. The sum translates to $\sum_{y \in \mathbb{F}_q} \chi(y)\chi(y+c)$. Since $y = 0$ contributes 0 to the sum, we may restrict our summation to $y \in \mathbb{F}_q^*$. Because $\chi(y^{-2}) = 1$ for any non-zero y , we can multiply the summand by $\chi(y^{-2})$ without altering its value:

$$\sum_{y \in \mathbb{F}_q^*} \chi(y)\chi(y+c)\chi(y^{-2}) = \sum_{y \in \mathbb{F}_q^*} \chi(y(y+c)y^{-2}) = \sum_{y \in \mathbb{F}_q^*} \chi(1+cy^{-1})$$

As y ranges over all non-zero elements of $\mathbb{F}_q$, its inverse y^{-1} ranges over all non-zero elements, and thus $z = cy^{-1}$ also ranges over all non-zero elements of $\mathbb{F}_q$ (since $c \neq 0$). The summation therefore simplifies to:

$$\sum_{z \in \mathbb{F}_q^*} \chi(1+z)$$

Let $w = 1 + z$. As z ranges over all non-zero elements, w ranges over all elements of $\mathbb{F}_q$ except 1. We can therefore rewrite the sum as:

$$\sum_{w \in \mathbb{F}_q, w \neq 1} \chi(w) = \left(\sum_{w \in \mathbb{F}_q} \chi(w) \right) - \chi(1)$$

We established in (i) that the full sum over the field is 0. Since $1 = 1^2$, $\chi(1) = 1$. The sum evaluates strictly to -1 .

We have shown that $QQ^\top$ has $q-1$ on the main diagonal and -1 everywhere else. This is the exact algebraic matrix structure of $qI - J$, concluding the proof. ■

With these structural properties established, we can construct the first of Paley's two distinct families of Hadamard matrices.

Theorem 2.47 (Paley Type I Construction). *Let q be a prime power such that $q \equiv 3 \pmod{4}$. Then there exists a Hadamard matrix of order $n = q + 1$.*

Proof. Because $q \equiv 3 \pmod{4}$, the quantity $(q-1)/2$ is an odd integer. By Proposition 2.46(ii), the Paley core matrix Q satisfies $Q^\top = -Q$, meaning Q is skew-symmetric.

Let $S = Q - I$. Since Q has zeros strictly on its diagonal, the matrix S has -1 on its diagonal and values in $\{1, -1\}$ everywhere else. We evaluate $SS^\top$:

$$SS^\top = (Q - I)(Q^\top - I^\top) = (Q - I)(-Q - I) = -Q^2 - Q + Q + I = -Q^2 + I$$

From Proposition 2.46(iii), we know $QQ^\top = qI - J$. Substituting $Q^\top = -Q$, we obtain $-Q^2 = qI - J$. Substituting this back into our calculation for S yields:

$$SS^\top = (qI - J) + I = (q+1)I - J$$

We also require the row and column sums of S . Let e be the $q \times 1$ column vector of all ones. We know $Qe = 0$ from Proposition 2.46(i). Thus, $Se = (Q - I)e = -e$.

We now construct a bordered matrix H of order $n = q + 1$. We define H by appending a leading row and leading column of ones to S :

$$H = \begin{bmatrix} 1 & e^\top \\ e & S \end{bmatrix}$$

Every entry in H is strictly 1 or -1 . To prove H is a Hadamard matrix, we must show $HH^\top = (q+1)I_{q+1}$. We compute the block matrix product:

$$HH^\top = \begin{bmatrix} 1 & e^\top \\ e & S \end{bmatrix} \begin{bmatrix} 1 & e^\top \\ e & S^\top \end{bmatrix} = \begin{bmatrix} 1 + e^\top e & e^\top + e^\top S^\top \\ e + Se & ee^\top + SS^\top \end{bmatrix}$$

We evaluate these four blocks individually:

- The top-left entry is the scalar $1 + q$, which is exactly our target dimension n .
- The bottom-left block is $e + Se$. Since $Se = -e$, this becomes $e - e = 0$.
- The top-right block is $e^\top + e^\top S^\top = (e + Se)^\top = 0^\top = 0$.
- The bottom-right block is $ee^\top + SS^\top$. The matrix $ee^\top$ is exactly J . Substituting our earlier result for $SS^\top$, we get $J + (q+1)I - J = (q+1)I_q$.

Collapsing these blocks together yields $HH^\top = (q+1)I_{q+1}$, proving that H is indeed a Hadamard matrix of order $q + 1$. ■

Paley also introduced a Type II construction to handle the remaining odd prime powers where $q \equiv 1 \pmod{4}$. Because Q is purely symmetric in this case rather than skew-symmetric, the bordering requires a block construction, ultimately producing a Hadamard matrix of order $n = 2(q+1)$. The full proof is structurally similar but relies heavily on Kronecker products with the fundamental 2×2 Hadamard matrix. This derivation is left out of scope for this text, but one may refer [16] for an exhaustive treatment.

2.6 Difference Sets

We have seen that symmetric designs exhibit remarkable structural rigidity. Representing these structures via their incidence matrices or purely axiomatic geometry is powerful, but it is often more economical to generate them using algebraic groups. This is achieved by finding a single ‘seed’ block from which the entire symmetric design can be developed.

Definition 2.48. Let G be an additively written finite group of order v . A k -sized subset $D \subseteq G$ is called a (v, k, λ) -**difference set** if every non-zero element $g \in G \setminus \{0\}$ can be expressed as a difference $d_1 - d_2$ of elements $d_1, d_2 \in D$ in exactly λ distinct ways.

From this definition, a simple counting argument on the number of possible non-zero differences yields the fundamental parameter relation $\lambda(v-1) = k(k-1)$. If G is a cyclic or abelian group, the difference set is naturally referred to as cyclic or abelian, respectively. The true power of difference sets lies in their ability to completely generate symmetric designs through group translation.

Definition 2.49. Let G be an additive group and D a subset of G . The **development** of D , denoted $\text{dev } D$, is the incidence structure whose point set is G and whose blocks are the right translates $D + g = \{d + g : d \in D\}$ for all $g \in G$.

Theorem 2.50. Let G be a finite group of order v and D a proper, non-empty k -subset of G . Then D is a (v, k, λ) -difference set if and only if $\text{dev } D$ is a symmetric $2-(v, k, \lambda)$ design upon which G acts as a regular automorphism group.

Proof. Let x and y be any two distinct points in G . We determine the number of blocks in $\text{dev } D$ containing both x and y . A block $D + g$ contains both points if and only if there exist elements $d_1, d_2 \in D$ such that $x = d_1 + g$ and $y = d_2 + g$. Subtracting these equations demonstrates this is algebraically equivalent to the conditions $x - y = d_1 - d_2$ and $g = x - d_1$. Thus, the number of blocks containing both x and y is exactly the number of ways to express the target element $x - y$ as a difference $d_1 - d_2$ of elements residing in D .

If D is a difference set, this number of representations is strictly λ for any choice of distinct points x and y . Since every block has a uniform size k , the incidence structure $\text{dev } D$ is a 2-design. Furthermore, the number of blocks is exactly the order of the group, $b = v$. By Fisher’s Inequality (Theorem 2.10), any 2-design with $b = v$ is necessarily a symmetric design. The converse directly follows by reversing this counting argument. Finally, the group G clearly acts regularly on both the points and the blocks via right translation. ■

To study these sets with advanced algebraic tools, we transition from combinatorial set notation to the setting of group algebras. For the remainder of this section, we will assume G is written multiplicatively.

Definition 2.51. Let G be a multiplicatively written group. The integral group ring $\mathbb{Z}[G]$ is the free $\mathbb{Z}$ -module spanned by the elements of G , equipped with linear extension of the group multiplication.

By abuse of notation, we identify any subset $S \subseteq G$ with the group ring element $S = \sum_{g \in S} g$. Furthermore, for any integer t , we define $S^{(t)} = \sum_{g \in S} g^t$. We also conventionally denote the formal sum of all group elements simply as G .

Proposition 2.52. Let D be a k -subset of a multiplicative group G of order v . Then D is a (v, k, λ) -difference set if and only if it satisfies the following equation in the group ring $\mathbb{Z}[G]$:

$$DD^{(-1)} = n \cdot 1_G + \lambda G$$

where $n = k - \lambda$ is the order of the design, and 1_G is the identity element of G .

Proof. We expand the product $DD^{(-1)}$ algebraically within the group ring. By definition, this is the product of the formal sums $\sum_{d \in D} d$ and $\sum_{d' \in D} (d')^{-1}$. Expanding this yields a double summation of the form $\sum_{d, d' \in D} d(d')^{-1}$. We separate this sum into two distinct parts: the terms where $d = d'$, and the terms where $d \neq d'$.

For the terms where $d = d'$, the product is the identity element 1_G . Since there are precisely k elements in D , this contributes $k \cdot 1_G$ to the overall sum. For the terms where $d \neq d'$, the products $d(d')^{-1}$ represent exactly the differences generated by D evaluated in multiplicative notation. If D is a difference set, every non-identity element $g \in G \setminus \{1_G\}$ is generated exactly λ times. Thus, the sum of these distinct terms evaluates to $\lambda(G - 1_G)$.

Combining these two parts yields $k \cdot 1_G + \lambda(G - 1_G) = (k - \lambda)1_G + \lambda G = n \cdot 1_G + \lambda G$. Since the group ring expansions are reversible, the converse holds identically. ■

The group ring formulation allows us to apply the theory of characters to abelian difference sets. A complex character χ of an abelian group G is a homomorphism $\chi : G \rightarrow \mathbb{C}^*$. We linearly extend χ to the group ring $\mathbb{Z}[G]$ by mapping formal sums to $\mathbb{C}$.

Lemma 2.53. *Let D be a subset of an abelian group G of order v . Then D is a (v, k, λ) -difference set if and only if for every complex character χ of G , the following holds:*

$$|\chi(D)|^2 = \begin{cases} n & \text{if } \chi \neq \chi_0 \\ k^2 & \text{if } \chi = \chi_0 \end{cases}$$

where χ_0 is the principal character and $n = k - \lambda$.

Proof. We apply the linear character homomorphism χ to the fundamental group ring equation $DD^{(-1)} = n \cdot 1_G + \lambda G$. This evaluates to $\chi(D)\chi(D^{(-1)}) = n + \lambda\chi(G)$. For any finite group element g , the character evaluation $\chi(g^{-1})$ is identical to the complex conjugate $\overline{\chi(g)}$. Consequently, by linearity, $\chi(D^{(-1)}) = \overline{\chi(D)}$, rendering the left side of the equation equal to the absolute square $|\chi(D)|^2$.

For the right side of the equation, the standard orthogonality relations of group characters dictate that $\chi(G) = 0$ for any non-principal character $\chi \neq \chi_0$. Thus, for non-principal characters, the equation strictly simplifies to $|\chi(D)|^2 = n$. For the principal character χ_0 , the character maps every element to 1, so $\chi_0(D)$ is simply the cardinality k , and the value is trivially k^2 . The inversion formula for characters over splitting fields guarantees the converse. ■

The values $\chi(D)$ are algebraic integers in cyclotomic fields. Thus, Lemma 2.53 makes it possible to apply the heavy machinery of algebraic number theory to the study of hypothetical difference sets.

A central technique in the search for difference sets is the use of automorphisms that preserve the block structure.

Definition 2.54. Let D be a difference set in a group G . An automorphism α of G is called a **multiplier** of D if there exist elements $a, b \in G$ such that $D^\alpha = aDb$. If G is abelian and α is defined by the mapping $x \mapsto x^t$ for some integer t coprime to v , then t is called a **numerical multiplier**.

If t is a numerical multiplier, the block translation condition simply becomes $D^{(t)} = Dg$ in the group ring $\mathbb{Z}[G]$ for some group element g . The most profound results in the theory of abelian difference sets guarantee the existence of numerical multipliers based strictly on the parameters of the design.

Theorem 2.55 (First Multiplier Theorem). *Let D be an abelian (v, k, λ) -difference set, and let p be a prime dividing $n = k - \lambda$ but not v . If $p > \lambda$, then p is a numerical multiplier of D .*

Proof. We use the group algebra over the integers, passing to coefficients modulo p . We first establish two auxiliary lemmas regarding the incidence structure of symmetric designs.

Lemma. *Let S be a k -subset of the points of a symmetric $2-(v, k, \lambda)$ design. If S intersects every block in at least λ points, then S must be a block of the design.*

Proof of Lemma. Let $B_1, \dots, B_v$ denote the blocks of the symmetric design, and let $a_i = |S \cap B_i|$ denote the intersection size of S with the i -th block. By standard double-counting arguments for point-block flags and point pairs confined to the subset S , we obtain the algebraic relations $\sum a_i = k^2$ and $\sum a_i(a_i - 1) = k(k - 1)\lambda$.

We evaluate the sum of the squared deviations from λ . Expanding the binomial yields $\sum (a_i - \lambda)^2 = \sum a_i^2 - 2\lambda \sum a_i + v\lambda^2$. Substituting our known relations into this expansion gives $k(k-1)\lambda + k^2 - 2\lambda k^2 + v\lambda^2$. Utilising the fundamental symmetric design parameter relation $\lambda v - \lambda = k^2 - k$, this entire expression simplifies identically to $(k - \lambda)^2 = n^2$.

On the other hand, the sum of the linear deviations is strictly $\sum (a_i - \lambda) = k^2 - v\lambda = k - \lambda = n$. We have thus generated a sequence of non-negative integers $x_i = a_i - \lambda$ such that their linear sum is n and the sum of their squares is n^2 . This is only mathematically possible if exactly one of the x_i equals n and all others equal 0. Thus, S intersects exactly one block in $\lambda + n = k$ points, meaning S is identical to that specific block. \\

Lemma. Let N be the incidence matrix of a symmetric design, and let $e_1, \dots, e_v$ denote its column vectors. For any integer vector $x = (x_1, \dots, x_v)$ in the column span of N , the standard inner product satisfies $\langle x, e_j \rangle \equiv \sum_{i=1}^v x_i \pmod{n}$.

Proof of Lemma. Since the design is symmetric, any two columns of N have an inner product of k if they are identical, and λ if they are distinct. Because $k - \lambda = n$, this implies $\langle e_i, e_j \rangle \equiv k \pmod{n}$ for all valid indices i, j .

Since x is in the integer column space, we may express it as a linear combination $x = \sum c_i e_i$ for some integers c_i . The inner product is therefore $\langle x, e_j \rangle = \sum c_i \langle e_i, e_j \rangle \equiv k \sum c_i \pmod{n}$. Furthermore, the sum of the components of x is exactly the inner product of x with the all-ones vector. Since every column vector e_i sums exactly to k , the sum of the components of x evaluates to exactly $k \sum c_i$. Substituting this equivalence gives $\langle x, e_j \rangle \equiv \sum x_i \pmod{n}$. \\

We now return to the main proof. We identify the points and blocks of $\text{dev } D$ directly with the group elements G . The columns of the incidence matrix therefore correspond to the block translates Dg . In the group ring $\mathbb{Z}[G]$, the algebraic operation of taking the p -th power satisfies the modular congruence $D^p \equiv D^{(p)} \pmod{p}$, since all cross-terms generated in the multinomial expansion are divisible by the prime p .

Because p strictly divides n , this implies $D^{(p)}$ is congruentially equivalent to D^p modulo the order of the design n . Consequently, the vector corresponding to $D^{(p)}$ resides within the integer module generated by the translates of D modulo n , and thus certainly modulo p . We can therefore apply our second lemma. The sum of the coefficients of $D^{(p)}$ is simply the cardinality of the original subset, which is k . By the lemma, the intersection size of $D^{(p)}$ with any generic block Dg satisfies $|D^{(p)} \cap Dg| \equiv k \equiv \lambda \pmod{p}$.

We are given by hypothesis that $p > \lambda$. The only way the modular congruence $|D^{(p)} \cap Dg| \equiv \lambda \pmod{p}$ can hold with non-negative physical intersection sizes is if the intersection evaluates to at least λ for every $g \in G$. Our first lemma then immediately forces $D^{(p)}$ to structurally be a block of the design. Thus $D^{(p)} = Dg$ for some translating element g , which dictates that the mapping $x \mapsto x^p$ permutes the blocks of the design. Therefore, p acts as a numerical multiplier. ■

The condition $p > \lambda$ in Theorem 2.55 is a technical artefact of the proof method. The Multiplier Conjecture famously posits that any prime dividing n is inherently a multiplier, regardless of its size relative to λ . While this remains an open problem, higher-order multiplier theorems (such as the Second Multiplier Theorem and McFarland's Theorem) successfully remove the $p > \lambda$ restriction by imposing divisibility conditions on v instead. The proofs for these extensions require significantly more involved algebraic number theory and are beyond the scope of this text; one may refer [6] for a comprehensive treatment.

We now focus on specific families of difference sets, beginning with those that have the strongest geometric implications.

Definition 2.56. A difference set with the parameter $\lambda = 1$ is called a *planar difference set*.

The terminology is natural; by Theorem 2.50, the development of a planar difference set of order n yields a symmetric $2-(n^2 + n + 1, n + 1, 1)$ design. By Proposition 2.33, this incidence structure is precisely a finite projective plane of order n . Consequently, the existence of a planar difference set in a group G is mathematically equivalent to the existence of a projective plane that admits G as a Singer group.

The scarcity of known examples and the severe arithmetic constraints imposed by multipliers have led to one of the most famous unsolved conjectures in finite geometry.

Conjecture 2.57 (Prime Power Conjecture). *If there exists an abelian planar difference set of order n , then n must be a prime power.*

Extensive computational and theoretical tests have been devised to attack this conjecture. The most prominent non-existence criteria for non-square orders rely on the Mann test and multiplier theorems.

Theorem 2.58. *Let D be a planar abelian difference set of order n . Moreover, let p and q be prime divisors of n and $v = n^2 + n + 1$, respectively. If p is a quadratic non-residue modulo q , then n must be a perfect square.*

The proof of this theorem requires an extensive application of the group ring identity $DD^{(-1)} = n \cdot 1_G + G$ evaluated over the field of p -adic numbers, combined with the fact that non-residues have even order modulo q . This is beyond the scope of this text, and one may refer [6] or [20] for a comprehensive treatment.

The immediate consequences of Theorem 2.58, however, are highly practical for ruling out specific orders. For instance, if $n \equiv 1 \pmod{3}$, we may evaluate the condition at $q = 3$. The theorem dictates that every prime divisor of n must be congruent to 1 modulo 3. This immediately rules out the existence of abelian planar difference sets for even orders such as $n = 10$ or $n = 22$, as $2 \not\equiv 1 \pmod{3}$.

Furthermore, if a planar difference set admits a multiplier of order 2, the structural restrictions on the resulting projective plane force the order to be a square.

Theorem 2.59. *Let D be a planar difference set in an abelian group G . If D admits a multiplier t of order 2, then n is a perfect square, say $n = m^2$, and necessarily $t = m^3$.*

Proof. We use additive notation for the abelian group G . We define two subgroups based on the action of the involutory multiplier t . Let $A = \{g \in G : tg = -g\}$ and $B = \{g \in G : tg = g\}$. Since G is an abelian group of odd order $v = n^2 + n + 1$, the mappings $x \mapsto (x - tx)/2$ and $x \mapsto (x + tx)/2$ are well-defined homomorphisms onto A and B , respectively. It follows that $A \cap B = \{0\}$ and G is the direct sum $A \oplus B$.

The multiplier t induces a collineation of order 2 on the projective plane $\text{dev } D$. By classical projective geometry, an involutory collineation of a finite projective plane must either be an elation, a homology, or a Baer involution. A Baer involution fixes a subplane of order m , meaning it has exactly $m^2 + m + 1$ fixed points, and this can only occur if the order of the parent plane is a square, $n = m^2$.

The fixed points of the collineation induced by t correspond precisely to the elements of the subgroup B . Since the order of B must divide the order of the group G , and neither the number of fixed points of an elation ($n + 1$) nor a homology ($n + 2$) divides $n^2 + n + 1$, the collineation must be a Baer involution. Thus, $n = m^2$ is a perfect square, and the subgroup B has order $m^2 + m + 1$.

Since D is a planar difference set of order m^2 , we have $\lambda = 1$. For any prime factor p of m , we strictly have $p > 1 = \lambda$. Thus, the First Multiplier Theorem (Theorem 2.55) dictates that every prime factor of m is a numerical multiplier. Since the multipliers form a group under multiplication, their product m is also a multiplier. Consequently, the cube m^3 is a multiplier. One can verify algebraically that it acts as an involution on the group, forcing the unique involutory multiplier to be $t = m^3$. ■

We now shift our focus from $\lambda = 1$ to the parameters associated with Hadamard matrices.

Definition 2.60. A difference set with parameters $v = 4t - 1$, $k = 2t - 1$, and $\lambda = t - 1$ is called a *Paley-Hadamard difference set*.

As established in the previous section, the existence of such a difference set is equivalent to the existence of a Hadamard matrix of order $4t$. While the existence of Hadamard matrices of order $4t$ is conjectured for all positive integers t , abelian Paley-Hadamard difference sets are known to be far more restricted.

The most prominent infinite family of such sets is generated using the algebraic structure of finite fields, specifically the quadratic residues.

Theorem 2.61 (Paley Difference Sets). *Let q be a prime power such that $q \equiv 3 \pmod{4}$. The set D of non-zero quadratic residues (squares) in the finite field $F = GF(q)$ forms a $(q, (q-1)/2, (q-3)/4)$ -difference set in the additive group of F .*

Proof. Let G be the additive group of F , which has order $v = q$. Since the multiplicative group F^* is cyclic of even order $q-1$, exactly half of its elements are squares. Thus, the cardinality of D is $k = (q-1)/2$.

We evaluate the number of times a non-zero element $a \in F^*$ can be expressed as a difference $x - y$ for $x, y \in D$. Let this number of representations be denoted by $R(a)$.

Choose any square $s \in D$. Since the product of two squares is a square, multiplying the equation $a = x - y$ by s yields $sa = sx - sy$. The mapping $x \mapsto sx$ is a permutation of the set D . Therefore, the number of solutions to $a = x - y$ is identical to the number of solutions to $sa = u - v$ for $u, v \in D$. This implies $R(a) = R(sa)$. Consequently, all squares in F^* share the exact same number of difference representations, which we will call λ_1 . By identical logic, multiplying by a non-square shows that all non-squares in F^* share a uniform number of representations, say λ_2 .

We must show that $\lambda_1 = \lambda_2$. Because $q \equiv 3 \pmod{4}$, the element -1 is a quadratic non-residue in F . If we take any valid difference equation $a = x - y$ where a is a square, we can multiply by -1 to obtain $-a = y - x$. Since $x, y \in D$, this provides a valid difference representation for $-a$, which is a non-square. This establishes a one-to-one correspondence between the difference representations of the square a and the non-square $-a$. Thus, $\lambda_1 = \lambda_2$.

Since every non-zero element in the group has an identical number of representations, D is a difference set. The uniform parameter λ is easily determined by the fundamental relation $\lambda(v-1) = k(k-1)$. Substituting $v = q$ and $k = (q-1)/2$ yields $\lambda = (q-3)/4$. ■

The Paley construction is a specific instance of a much broader technique known as cyclotomy. By partitioning the multiplicative group F^* into cosets (cyclotomic classes) based on higher powers, one can search for difference sets by taking unions of these classes. For example, if $q = 4x^2 + 1$ is a prime power, the bi-quadratic residues (fourth powers) can form difference sets. Extensive computations of cyclotomic numbers have yielded several other sporadic infinite families, such as the octic residue difference sets discovered by Lehmer, though a complete classification remains out of reach.

We conclude this chapter with a monumental result that elegantly unifies the algebraic theory of difference sets with the classical families of finite geometry. In 1938, James Singer proved that the classical Desarguesian projective spaces are not just symmetric designs, but are actually generated by cyclic difference sets.

Theorem 2.62 (Singer's Theorem). *Let q be a prime power and $d \geq 2$ an integer. The points and hyperplanes of the projective geometry $PG(d, q)$ form a symmetric design that admits a cyclic Singer group. Consequently, there exists a cyclic difference set with parameters:*

$$v = \frac{q^{d+1} - 1}{q - 1}, \quad k = \frac{q^d - 1}{q - 1}, \quad \lambda = \frac{q^{d-1} - 1}{q - 1}$$

Proof. Consider the finite field $E = GF(q^{d+1})$ as a $(d+1)$ -dimensional vector space over its subfield $F = GF(q)$. The points of the projective space $PG(d, q)$ correspond precisely to the 1-dimensional vector subspaces of E over F .

The multiplicative group E^* is a cyclic group of order $q^{d+1} - 1$. Let α be a generator of E^* . The non-zero elements of the subfield F form a subgroup F^* of order $q - 1$ within E^* . The quotient group $G = E^*/F^*$ is therefore a cyclic group of order $v = (q^{d+1} - 1)/(q - 1)$.

The elements of G can be naturally identified with the points of $PG(d, q)$, since each coset xF^* is exactly the set of non-zero vectors in a 1-dimensional subspace spanned by x . The action of the generator α on the vector space E by simple field multiplication, $x \mapsto \alpha x$, is an invertible F -linear transformation. This transformation maps 1-dimensional subspaces to 1-dimensional subspaces, and thus induces a collineation on the projective space $PG(d, q)$.

Since α generates E^* , the induced cyclic group of collineations G acts transitively and regularly on the v points of the projective space.

A block of $PG(d, q)$ is a hyperplane, which corresponds to a d -dimensional subspace H of E over F . Since the linear transformation induced by α preserves subspace dimensions, the action of G maps hyperplanes to hyperplanes. Because the incidence structure is a symmetric design, the regular transitive action on the points guarantees a regular transitive action on the blocks.

We fix a specific hyperplane H_0 as our base block. Let D be the set of points (elements of G) contained within H_0 . Because G acts regularly on the points and blocks, the translates of D under the cyclic action of G will generate all the hyperplanes of $PG(d, q)$. The number of points in a hyperplane is $k = (q^d - 1)/(q - 1)$, and the intersection of any two distinct hyperplanes is a subspace of dimension $d - 1$, which contains $\lambda = (q^{d-1} - 1)/(q - 1)$ points. Thus, by Theorem 2.50, the subset D forms a cyclic (v, k, λ) -difference set. ■

The difference sets described by Theorem 2.62 are universally known as Singer difference sets. It is important to note that while Singer difference sets provide one construction for these parameters, they are not necessarily unique. Gordon, Mills, and Welch (GMW) established a recursive technique using relative difference sets to construct exponential numbers of pairwise inequivalent cyclic difference sets sharing the exact same Singer parameters. The analysis of GMW difference sets and their corresponding 2-ranks is a highly active area of modern research, deeply intertwined with the study of perfect binary sequences and cryptography.

2.7 Latin Squares

We now introduce one of the most fundamental objects in combinatorial design theory: the Latin square.

Definition 2.63. A *Latin square* of order n with entries from an n -set X is an $n \times n$ array L in which every cell contains an element of X such that every row of L is a permutation of X and every column of L is a permutation of X .

Latin squares are intimately connected to algebraic structures known as quasigroups. A pair $(X, \circ)$ is a *quasigroup* of order n if $\circ$ is a binary operation on X such that for every $x, y \in X$, the equations $x \circ z = y$ and $z \circ x = y$ each have a unique solution for $z \in X$. It follows directly from these definitions that $(X, \circ)$ is a quasigroup if and only if its operation table is a Latin square of order n .

We frequently impose additional structural symmetry on these objects.

Definition 2.64. A quasigroup $(X, \circ)$ is *idempotent* if $x \circ x = x$ for all $x \in X$, and it is *symmetric* if $x \circ y = y \circ x$ for all $x, y \in X$. The corresponding Latin square is likewise called idempotent or symmetric.

The existence of a symmetric idempotent quasigroup is governed strictly by the parity of its order.

Lemma 2.65. *If there exists a symmetric idempotent quasigroup of order n , then n is odd.*

Proof. Suppose $(X, \circ)$ is a symmetric idempotent quasigroup. Let $z \in X$ and define a set of pairs $S = \{(x, y) : x \circ y = z\}$. Because the quasigroup is idempotent, $(x, x) \in S$ if and only if $x = z$. Because the quasigroup is symmetric, $(x, y) \in S$ if and only if $(y, x) \in S$. Consequently, the collection of unordered pairs $\{\{x, y\} : x \neq y, x \circ y = z\}$ forms a perfect partition of $X \setminus \{z\}$ into disjoint sets of size two. This implies that $|X| - 1$ must be an even integer, which immediately forces $|X| = n$ to be odd. ■

Conversely, symmetric idempotent quasigroups can be explicitly constructed for every odd order. Consider the additive group $(\mathbb{Z}_n, +)$. While this forms a symmetric quasigroup, it is not idempotent. However, by relabelling the elements, we can force idempotency. Specifically, for an odd integer n , the element 2 has a valid multiplicative inverse $(n + 1)/2$ modulo n . By defining a new binary operation:

$$x \circ y = \left(\frac{n+1}{2} \right) (x + y) \pmod{n} \quad (2.3)$$

we immediately obtain a symmetric idempotent quasigroup defined on $\mathbb{Z}_n$.

We now turn to the interaction between multiple Latin squares.

Definition 2.66. Let L_1 and L_2 be two Latin squares of order n defined on symbol sets X and Y , respectively. L_1 and L_2 are **orthogonal** if for every $x \in X$ and every $y \in Y$, there exists a unique cell (i, j) such that $L_1(i, j) = x$ and $L_2(i, j) = y$.

Theorem 2.67. *If $n > 1$ is an odd integer, then there exist orthogonal Latin squares of order n .*

Proof. We define two arrays L_1 and L_2 of order n with entries from $\mathbb{Z}_n$ by the algebraic rules:

$$\begin{aligned} L_1(i, j) &= i + j \pmod{n} \\ L_2(i, j) &= i - j \pmod{n} \end{aligned}$$

Both L_1 and L_2 are trivially seen to be Latin squares for any positive integer n , as fixing either i or j produces a simple bijection over the elements of $\mathbb{Z}_n$.

We must prove that they are mutually orthogonal when n is odd. Suppose that $(x, y) \in \mathbb{Z}_n \times \mathbb{Z}_n$ is an arbitrary ordered pair of symbols. We seek a unique cell coordinate (i, j) such that $L_1(i, j) = x$ and $L_2(i, j) = y$. This is mathematically equivalent to solving the linear system of congruences:

$$\begin{aligned} i + j &\equiv x \pmod{n} \\ i - j &\equiv y \pmod{n} \end{aligned}$$

Because n is strictly an odd integer, the element 2 is coprime to n and thus possesses a unique multiplicative inverse 2^{-1} modulo n . Adding and subtracting the two congruences yields the unique solution:

$$\begin{aligned} i &\equiv (x + y)2^{-1} \pmod{n} \\ j &\equiv (x - y)2^{-1} \pmod{n} \end{aligned}$$

Since this unique solution dictates exactly one valid cell (i, j) for every possible ordered pair (x, y) , the Latin squares L_1 and L_2 are orthogonal. ■

Lemma 2.68. *If L and M are Latin squares of order m and n defined on symbol sets X and Y respectively, then their direct product $L \times M$ is a Latin square of order mn defined on the symbol set $X \times Y$.*

Proof. Consider an arbitrary row of the direct product $L \times M$, indexed by the pair (i_1, i_2) . Let $x \in X$ and let $y \in Y$ be an arbitrary target symbol (x, y) . We will demonstrate that this symbol appears exactly once in this row.

Because L is a Latin square, there exists a unique column index j_1 such that $L(i_1, j_1) = x$. Similarly, because M is a Latin square, there exists a unique column index j_2 such that $M(i_2, j_2) = y$. By the formal definition of the direct product, the entry in row (i_1, i_2) and column (j_1, j_2) evaluates to:

$$(L \times M)((i_1, i_2), (j_1, j_2)) = (x, y)$$

This proves that every symbol (x, y) appears exactly once in every row. A completely symmetric argument applying the identical logic to fixed columns guarantees that every column of $L \times M$ contains every symbol in $X \times Y$ exactly once. Therefore, the direct product $L \times M$ is a Latin square. ■

To construct larger orthogonal Latin squares from smaller ones, we rely on the direct product.

Definition 2.69. Let L and M be Latin squares of order m and n , defined on symbol sets X and Y , respectively. The **direct product** $L \times M$ is an $mn \times mn$ array defined on $X \times Y$ by the rule $(L \times M)((i_1, i_2), (j_1, j_2)) = (L(i_1, j_1), M(i_2, j_2))$.

Theorem 2.70 (Direct Product). *If there exist orthogonal Latin squares of order n_1 and orthogonal Latin squares of order n_2 , then there exist orthogonal Latin squares of order $n_1 n_2$.*

Proof. Suppose that L_1 and L_2 are orthogonal Latin squares of order n_1 defined on a symbol set X , and M_1 and M_2 are orthogonal Latin squares of order n_2 defined on a symbol set Y . We evaluate the direct products $L_1 \times M_1$ and $L_2 \times M_2$. By Lemma 2.68, both of these arrays are mathematically guaranteed to be Latin squares of order $n_1 n_2$. We must therefore only prove that they are mutually orthogonal.

Consider an arbitrary ordered pair of symbols from the combined alphabet, namely $((x_1, y_1), (x_2, y_2))$. We seek a unique cell coordinate $((i_1, i_2), (j_1, j_2))$ such that the superposition of the two direct product squares yields this pair. Algebraically, this requires:

$$\begin{aligned}(L_1 \times M_1)((i_1, i_2), (j_1, j_2)) &= (x_1, y_1) \\ (L_2 \times M_2)((i_1, i_2), (j_1, j_2)) &= (x_2, y_2)\end{aligned}$$

By expanding the definition of the direct product, this system strictly decouples into four independent equations over the base symbol sets:

$$\begin{aligned}L_1(i_1, j_1) &= x_1 \\ M_1(i_2, j_2) &= y_1 \\ L_2(i_1, j_1) &= x_2 \\ M_2(i_2, j_2) &= y_2\end{aligned}$$

The first and third equations stipulate that the superposition of L_1 and L_2 at cell (i_1, j_1) yields the pair (x_1, x_2) . Because L_1 and L_2 are given to be orthogonal, this uniquely determines the coordinate (i_1, j_1) . Identically, the second and fourth equations stipulate that the superposition of M_1 and M_2 at cell (i_2, j_2) yields (y_1, y_2) , which uniquely determines (i_2, j_2) due to the established orthogonality of M_1 and M_2 .

Since the combined coordinate cell is uniquely determined for any target pair of symbols, the direct product squares are mutually orthogonal. ■

Theorem 2.71. *There exist orthogonal Latin squares of order n for all integers $n \not\equiv 2 \pmod{4}$.*

Proof. We proceed by analysing the residue classes modulo 4. If n is an odd integer, the existence of orthogonal Latin squares is directly established by Theorem 2.67.

Next, suppose $n \geq 4$ is a strict power of two, expressed as $n = 2^i$ where $i \geq 2$. The foundational cases for $i = 2$ and $i = 3$ can be verified via explicit finite field constructions. For any $i \geq 4$, we proceed by induction on the exponent i . By setting $n_1 = 4$ and $n_2 = 2^{i-2}$, we can construct orthogonal Latin squares of order 2^i by applying the direct product construction from Theorem 2.70.

Finally, suppose that n is an even integer such that $n \not\equiv 2 \pmod{4}$, and n is not a strict power of two. This implies n is a multiple of 4. We can therefore factor the integer as $n = 2^i n'$, where the exponent $i \geq 2$ and the odd component $n' > 1$. We apply the direct product from Theorem 2.70 using the component orders $n_1 = 2^i$ and $n_2 = n'$. Since we have already constructed orthogonal Latin squares for the power of two 2^i and the odd integer n' , their direct product guarantees the existence of orthogonal squares of order n . ■

An immediate question is about $n \equiv 2 \pmod{4}$. Well, for $n = 2$ clearly orthogonal squares do not exist (there are only two and they are not orthogonal). The $n = 6$ case is the famous 36 officer problem of Euler and he believed that it was not possible. More generally he also believed that it is not possible for 10, 14, However, 2 and 6 are the only exceptions! Orthogonal Latin squares exist for every $n \geq 1$ except for $n = 2$ and $n = 6$. Proving this here at a basic level would take another 100 pages or so. We refer the reader to the breakthrough paper [8]. After all, the $n = 6$ problem stood for about 177 years and even after proving existence for $n = 10$, explicitly finding the squares was extremely difficult for a long period of time.

Definition 2.72. A collection of s Latin squares of order n is said to be **mutually orthogonal** (abbreviated **MOLS**) if every pair of squares within the collection is orthogonal.

The central problem in this area is determining the maximum number of MOLS of order n , denoted $N(n)$.

Theorem 2.73. *Let $N(n)$ denote the maximum number of mutually orthogonal Latin squares of order n . For $n > 1$, $N(n) \leq n - 1$.*

Proof. Suppose we have s mutually orthogonal Latin squares of order n , denoted $L_1, \dots, L_s$, all defined on the symbol set $X = \{1, \dots, n\}$. Because relabelling the symbols of any Latin square does not affect its orthogonality with another square, we may systematically relabel the symbols of each L_i so that the first row of every square is exactly the sequence $1, 2, \dots, n$.

Consider the cells in the second row and first column of each square, namely $L_i(2, 1)$ for $1 \leq i \leq s$. First, we observe that these s values must be strictly distinct. If there were indices $i \neq j$ such that $L_i(2, 1) = L_j(2, 1) = x$, then the superposition of L_i and L_j would contain the ordered pair (x, x) in cell $(2, 1)$. However, because the first row of both squares is naturally ordered, cell $(1, x)$ also contains the symbol x for both squares, meaning the ordered pair (x, x) also appears in cell $(1, x)$. This contradicts the orthogonality of L_i and L_j .

Furthermore, none of the values $L_i(2, 1)$ can equal 1. Since 1 already appears in cell $(1, 1)$ of every square, placing a 1 in cell $(2, 1)$ would violate the fundamental Latin square property that no symbol may repeat in a column. Thus, the s distinct values $L_1(2, 1), \dots, L_s(2, 1)$ must be drawn entirely from the subset $\{2, \dots, n\}$. As this subset contains exactly $n - 1$ elements, it immediately follows that $s \leq n - 1$. ■

Theorem 2.74 (MacNeish's Theorem). *Suppose that n has the prime power factorisation $n = p_1^{e_1} \dots p_l^{e_l}$, where the p_i are distinct primes. Let $s = \min \{p_i^{e_i} - 1 : 1 \leq i \leq l\}$. Then there exist at least s mutually orthogonal Latin squares of order n .*

Proof. For each index i from 1 to l , the term $p_i^{e_i}$ is a prime power. By the algebraic construction over finite fields established previously, there exist exactly $p_i^{e_i} - 1$ mutually orthogonal Latin squares of order $p_i^{e_i}$.

Because s is defined as the minimum of these available quantities, we are guaranteed that $s \leq p_i^{e_i} - 1$ for every i . Consequently, for each individual prime power component $p_i^{e_i}$, we can safely select a subset of exactly s mutually orthogonal Latin squares.

We now invoke the generalisation of the direct product construction from Theorem 2.70. Taking the direct product of the s mutually orthogonal Latin squares of order $p_1^{e_1}$ with the s mutually orthogonal Latin squares of order $p_2^{e_2}$ produces a new set of exactly s mutually orthogonal Latin squares of order $p_1^{e_1} p_2^{e_2}$.

By iteratively applying this direct product operation across all l prime power components, the mutual orthogonality of the s concurrent squares is perfectly preserved at each algebraic step, while their orders are multiplicatively combined. The final iteration yields exactly s mutually orthogonal Latin squares of order $p_1^{e_1} p_2^{e_2} \dots p_l^{e_l} = n$. ■

Corollary 2.75. *If $n \equiv 1, 5, 7$, or $11 \pmod{12}$, then there exist at least four mutually orthogonal Latin squares of order n . If $n \equiv 4$ or $8 \pmod{12}$, then there exist at least three mutually orthogonal Latin squares of order n .*

Proof. Suppose that n possesses the prime power factorisation $n = p_1^{e_1} \dots p_l^{e_l}$. By MacNeish's Theorem, a minimum of three mutually orthogonal Latin squares will exist if the prime power components satisfy $p_i^{e_i} \geq 4$ for all $1 \leq i \leq l$. The only arithmetic situations in which $p_i^{e_i} < 4$ occurs are when the component is strictly $(p_i, e_i) = (2, 1)$ or $(3, 1)$. In other words, if the prime power factorisation of n explicitly excludes the isolated factors 2^1 and 3^1 , then three mutually orthogonal Latin squares exist. By an identical algebraic threshold, if the prime power factorisation explicitly excludes the isolated factors $2^1, 2^2$, and 3^1 , then a minimum of four mutually orthogonal Latin squares exist.

Consider the first case where $n \equiv 1, 5, 7$, or $11 \pmod{12}$. These congruences dictate that $\gcd(n, 6) = 1$. Consequently, there are no prime factors involving 2 or 3 anywhere within the factorisation of n . It naturally follows that all $p_i^{e_i} \geq 5$, guaranteeing that at least four mutually orthogonal Latin squares of these orders exist.

Consider the second case where $n \equiv 4$ or $8 \pmod{12}$. These congruences strictly dictate that $n \equiv 0 \pmod{4}$ and $n \not\equiv 0 \pmod{3}$. Therefore, there is no prime factor involving 3 in the factorisation of n . Furthermore,

because n is divisible by 4, the prime base 2 must appear with an exponent $e \geq 2$. Since neither 2^1 nor 3^1 is present in the factorisation, we are mathematically guaranteed that at least three mutually orthogonal Latin squares of these orders exist. ■

We conclude this section with an elegant construction for Latin squares that are orthogonal to their own transposes.

Proposition 2.76. *Let n be a positive integer and let $a, b \in \mathbb{Z}_n$. Define an $n \times n$ array M with entries in $\mathbb{Z}_n$ by the rule $M_{i,j} = ai + bj \pmod{n}$. If $\gcd(a, n) = 1$, $\gcd(b, n) = 1$, and $\gcd(a^2 - b^2, n) = 1$, then M is a self-orthogonal Latin square of order n .*

Proof. We first verify that M is a Latin square. Fixing a row i , the mapping $j \mapsto ai + bj \pmod{n}$ is a bijection because $\gcd(b, n) = 1$. Thus, every symbol appears exactly once in each row. Symmetrically, fixing a column j , the mapping $i \mapsto ai + bj \pmod{n}$ is a bijection because $\gcd(a, n) = 1$, ensuring every symbol appears exactly once in each column. Hence, M is a Latin square.

To demonstrate that M is self-orthogonal, we must show it is orthogonal to its transpose, $M^\top$, whose entries are given by $M_{i,j}^\top = M_{j,i} = aj + bi \pmod{n}$. Suppose that the superposition of M and $M^\top$ yields the same ordered pair of symbols at cells (i, j) and (k, l) . This implies the system of congruences:

$$\begin{aligned} ai + bj &\equiv ak + bl \pmod{n} \\ aj + bi &\equiv al + bk \pmod{n} \end{aligned}$$

Rearranging the terms gives $a(i - k) + b(j - l) \equiv 0 \pmod{n}$ and $b(i - k) + a(j - l) \equiv 0 \pmod{n}$. Multiplying the first congruence by a and the second by b , and then subtracting the second from the first, eliminates the $(j - l)$ term, yielding:

$$(a^2 - b^2)(i - k) \equiv 0 \pmod{n}$$

By hypothesis, $\gcd(a^2 - b^2, n) = 1$, which forces $i \equiv k \pmod{n}$. Substituting $i = k$ back into either original congruence yields $b(j - l) \equiv 0 \pmod{n}$, and since $\gcd(b, n) = 1$, we must have $j \equiv l \pmod{n}$. Therefore, the cells (i, j) and (k, l) are identical, meaning every ordered pair of symbols occurs exactly once. Thus, M is orthogonal to its transpose. ■

Chapter 3

Association Schemes

3.1 Introduction to association schemes

We begin with the definition of an association scheme.

Definition 3.1. An *association scheme of class d* is a pair $(X, \{R_i\}_{0 \leq i \leq d})$ of a finite set X and subsets $R_i \subseteq X \times X$ (for $0 \leq i \leq d$) such that

- The first set $R_0 = \{(x, x) \mid x \in X\}$ is the diagonal.
- $\{R_i\}_{0 \leq i \leq d}$ is a partition of $X \times X$, that is, these sets are pairwise disjoint and $\bigcup_{0 \leq i \leq d} R_i = X \times X$.
- If $R_i^\top = \{(x, y) \mid (y, x) \in R_i\}$ denotes the transpose of R_i , then for every $0 \leq i \leq d$, we have some $0 \leq j \leq d$ such that $R_i^\top = R_j$. That is, the partition is closed under transposition.
- Let $0 \leq i, j, k \leq d$. Define $p_{i,j}^k(x, y) = |\{z \in X \mid (x, z) \in R_i, (z, y) \in R_j\}|$ for $(x, y) \in R_k$. Then this number is independent of the x and y chosen from R_k . These numbers are known as the ***Intersection numbers*** of the scheme.

Essentially, we need a partition containing a diagonal which is closed under transposition such that the ‘transitivity-property’ holds.

Definition 3.2. Let $(X, \{R_i\}_{0 \leq i \leq d})$ be an association scheme of class d . The number $p_{i,j}^k$, which is the number of $z \in X$ such that $(x, z) \in R_i$ and $(z, y) \in R_j$ (for some $(x, y) \in R_k$), is called an ***intersection number***. Here, $0 \leq i, j, k \leq d$.

Definition 3.3. We say that an association scheme of class d , $(X, \{R_i\}_{0 \leq i \leq d})$ is a ***commutative association scheme*** if $p_{i,j}^k = p_{j,i}^k$ for every $0 \leq i, j, k \leq d$.

Definition 3.4. A ***symmetric association scheme of class d*** is an association scheme, $(X, \{R_i\}_{0 \leq i \leq d})$, of class d such that $R_i^\top = R_i$ for every $0 \leq i \leq d$.

Henceforth, we shall drop the class of the association scheme since it is obvious. Further, we write $\mathcal{A}$ for the association scheme $(X, \{R_i\}_{0 \leq i \leq d})$. We shall use the above discussed standard notation throughout unless specified otherwise.

Proposition 3.5. *Let $\mathcal{A}$ be a symmetric association scheme. Then $\mathcal{A}$ is commutative.*

Proof. Let i, j, k be arbitrary. Fix some $(x, y) \in R_k$. We show that $p_{i,j}^k = p_{j,i}^k$.

Consider the sets $M = \{z \in X \mid (x, z) \in R_i, (z, y) \in R_j\}$ and $N = \{z \in X \mid (x, z) \in R_j, (z, y) \in R_i\}$. By symmetry, we have

$$\begin{aligned} M &= \{z \in X \mid (x, z) \in R_i, (z, y) \in R_j\} \\ &= \{z \in X \mid (z, x) \in R_i, (y, z) \in R_j\} \end{aligned}$$

Now consider the set $\tilde{N} = \{z \in X \mid (y, z) \in R_j, (z, x) \in R_i\}$. Note that N was defined with respect to (x, y) , whereas $\tilde{N}$ evaluates the intersection condition with respect to (y, x) . Because $\mathcal{A}$ is a symmetric association scheme, $(x, y) \in R_k$ implies that $(y, x) \in R_k^\top = R_k$. Therefore, evaluating the intersection parameters for (y, x) yields the exact same constant $p_{j,i}^k$. Thus, the cardinalities of N and $\tilde{N}$ are both equal to $p_{j,i}^k$. Since $M = \tilde{N}$, we conclude that $p_{i,j}^k = |M| = |\tilde{N}| = p_{j,i}^k$, completing the proof. ■

We now have the association scheme coming from group actions. This was the first motivation to study association schemes and define them.

Proposition 3.6. *Let G have a transitive group action on a set X . We can extend this action to $X \times X$ naturally. Let $\{R_i\}_{0 \leq i \leq d}$ be the orbits of this action on $X \times X$. Then $(X, \{R_i\})$ is an association scheme.*

Proof. Let R_0 be the orbit of an element (x_0, x_0) . Since G acts transitively, we are able to reach every (y, y) and hence $R_0 = \{(x, x) \mid x \in X\}$. The R_i by definition of group actions are disjoint.

Furthermore, if $R_i = \{(gx_0, gy_0) \mid g \in G\}$ is the orbit of (x_0, y_0) , then the transpose $R_i^\top$ is simply the orbit of (y_0, x_0) which may very well be R_i itself in some cases.

We now have to only show the independence number property. To this end, fix three orbits R_i, R_j and R_k . Let (x, y) and (x', y') be in R_k . Let them be related through a g_0 as $g_0x = x', g_0y = y'$. We have,

$$\begin{aligned} p_{i,j}^k(x, y) &= |\{z \in X \mid (x, z) \in R_i, (z, y) \in R_j\}| \\ p_{i,j}^k(x', y') &= |\{z \in X \mid (x', z) \in R_i, (z, y') \in R_j\}| \\ &= |\{z \in X \mid (g_0x, z) \in R_i, (z, g_0y) \in R_j\}| \\ &= |\{z \in X \mid (x, g_0^{-1}z) \in R_i, (g_0^{-1}z, y) \in R_j\}| \\ &= |\{t \in X \mid (x, t) \in R_i, (t, y) \in R_j\}| \\ &= p_{i,j}^k(x, y) \end{aligned}$$

This shows that $p_{i,j}^k$ is independent of the choice of the element of R_k completing the proof. ■

Question. Consider the dihedral group D_{10} of order 10. Consider its action on the regular pentagon. Determine the class of the association scheme generated through the group action. Also find the intersection numbers.

Ans. The class is simply the value of d which is one less than the number of orbits of the group action of D_{10} on $P_5 \times P_5$ where P_5 are the vertices of the pentagon.

Recall Burnside's orbit counting lemma which states that the number of orbits of X under the action of G is given by

$$\frac{1}{|G|} \sum_{g \in G} |X^g|$$

where $X^g = \{x \in X \mid gx = x\}$ is the set of all elements of X fixed by g .

Applying this result for $D_{10} = \{1, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$ and noting that $(P_5 \times P_5)^g = P_5^g \times P_5^g$ we have

$$\frac{1}{10} (25 + 0 + 0 + 0 + 0 + 1 + 1 + 1 + 1 + 1) = 3$$

Thus we have $d = 2$.

Let R_0 be the diagonal elements. To analyse the other orbits, we need some geometrical intuition. Let v and w be distinct vertices in P_5 . The shortest distance between them can either be 1 or 2. This determines the other two orbits since D_{10} preserves the distances between vertices.

Now we calculate the intersection numbers. There are 27 of them to find. When R_0 is considered, the calculations are simplified and we cover 19 of these numbers below

$$\begin{aligned}
p_{i,j}^0 &= |\{z \in X \mid (x, z) \in R_i, (z, x) \in R_j\}| \\
&= \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j = 0 \\ 2 & \text{if } i = j \in \{1, 2\} \end{cases} \\
p_{0,j}^k &= |\{z \in X \mid z = x, (z, y) \in R_j\}| \\
&= \begin{cases} 1 & \text{if } k = j \\ 0 & \text{otherwise} \end{cases} \\
p_{i,0}^k &= |\{z \in X \mid z = y, (x, z) \in R_i\}| \\
&= \begin{cases} 1 & \text{if } k = i \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

We leave it to the reader to find the other numbers. The principle is the same. To find $p_{i,j}^k$, fix two vertices x and y with $k - 1$ vertices between them and count the number of vertices at a distance of i from the first one and j from the other one. $\square$

3.2 Examples of association schemes

Definition 3.7. Consider a set F of size q and consider the set $X = F^d = F \times \cdots \times F$ (d times). Equip it with the Hamming distance $\partial(x, y) = |\{i \in [d] \mid x_i \neq y_i\}|$. Define the sets $R_i \subseteq X \times X$ ($i = 0, 1, \dots, d$) as

$$R_i = \{(x, y) \mid \partial(x, y) = i\}$$

Then $(X, \{R_i\}_{i=0}^d)$ forms a symmetric association scheme, $H(d, q)$, called the **Hamming scheme**.

Proposition 3.8. $H(d, q)$ is indeed a symmetric association scheme.

Proof. It is clear that $R_i^\top = R_i$. We need to show that it is an association scheme, now.

We define a group and its action on X which will happen to be transitive. Then we shall show that the orbits are indeed R_i and hence by Proposition 3.6 we will be done.

Consider the group $S = S_q \times \cdots \times S_q$ (d times) acting on X as

$$(\sigma_1, \dots, \sigma_d) * (x_1, \dots, x_d) := (\sigma_1(x_1), \dots, \sigma_d(x_d))$$

Also consider the group S_d acting on the indexing set $[d]$ and its action on X as

$$(\tau_1, \dots, \tau_d) * (x_1, \dots, x_d) := (x_{\tau_1}, \dots, x_{\tau_d})$$

Both of these are subgroups of S_X , the full permutation group on X .

Now for $\tau \in S_d$ and $\sigma \in S$, consider $\tau^{-1}\sigma\tau$. It is not difficult to show through the action that

$$\tau^{-1}\sigma\tau = (\sigma_{\tau(1)}, \dots, \sigma_{\tau(d)}) \in S$$

Thus, S_d normalises S and hence the product SS_d forms a subgroup of S_X . This product is denoted by $S_q \wr S_d$ and is known as the **wreath product** of S_q and S_d .

Now it is clear that S acts transitively on X and hence so does $S_q \wr S_d$ since the wreath product already has permutations of S and some more. Now we find the orbits. We claim that (x, y) and (a, b) lie in the same orbit if and only if there exists $g \in S_q \wr S_d$ such that $g * x = a$ and $g * y = b$ if and only if the $\partial(x, y) = \partial(a, b)$.

Note that although SS_d and S_dS are the same subgroup, their action on X may be defined using S_dS , that is, apply permutations at each coordinate and then shuffle the coordinates.

Clearly, if they are in the same orbit, the distance is preserved since the distance is preserved under both S and S_d , as can be checked.

Conversely, suppose the distance is preserved. Let this distance be some i . Start with (x, y) . Since the action is transitive, obtain $g_1 \in G = S_q \wr S_d$ so that for some fixed $a \in F$,

$$(x, y) \mapsto (g_1 * x, g_1 * y) = ((a, \dots, a), g_1 * y)$$

Since the distance is preserved, $g_1 * y$ must have i coordinates which are not a . Apply $g_2 \in S_d$ as

$$((a, \dots, a), g_1 * y) \mapsto ((a, \dots, a), g_2 * (g_1 * y)) = ((a, \dots, a), (b_1, b_2, \dots, b_i, a, \dots, a))$$

where the first i coordinates b_i are not equal to a . Finally apply $g_3 \in S$ which fixes a but maps all the b_i to some fixed b so that

$$((a, \dots, a), (b_1, b_2, \dots, b_i, a, \dots, a)) \mapsto ((a, \dots, a), (b, b, \dots, b, a, \dots, a))$$

Similarly (a, b) can be transformed to the same pair and hence by using the inverses of the group elements, we can reach (a, b) from (x, y) .

Thus the R_i are indeed the orbits and we are done. ■

Proposition 3.9. *The intersection numbers of the Hamming scheme $H(d, q)$ are given by*

$$p_{i,j}^k = \sum_c \frac{k!}{a!b!c!} \binom{d-k}{d'} (q-2)^c (q-1)^{d'}$$

where

$$a = \frac{k+j-i-c}{2} \quad b = \frac{k+i-j-c}{2} \quad d' = \frac{i+j-k-c}{2}$$

and the sum is over all integers $c \geq 0$ such that a, b and d' are non-negative integers.

Proof. Without loss of generality we may assume $F = \{0, 1, \dots, q-1\}$. Further, we may assume that $x = (0, \dots, 0)$ and $y = (1, \dots, 1, 0, \dots, 0)$ where the first k positions of y are 1 so that $\partial(x, y) = k$.

We want to count the number of $z \in \mathbb{F}_q^d$ with i non-zero entries such that $\partial(z, y) = j$. Let K_1 denote the first k positions and K_2 denote the next $d-k$ ones.

Let a be the number of zeroes of z in K_1 . Let b be the number of ones in K_1 and let c be the number of positions in K_1 where z is neither 1 nor 0. Let d' be the number of positions in K_2 where z is non-zero.

We have

$$\partial(x, z) = i \implies b + c + d' = i$$

$$\partial(y, z) = j \implies a + c + d' = j$$

$$|K_1| = k \implies a + b + c = k$$

This gives

$$a = \frac{k+j-i-c}{2} \quad b = \frac{k+i-j-c}{2} \quad d' = \frac{i+j-k-c}{2}$$

Now for a fixed c , and hence fixed a, b, d' , choose the c places in the first k places where z is neither 1 nor 0 in $\binom{k}{c}$ ways and fill each with $q-2$ choices. Then from the remaining $k-c$ places, choose a of them to be 0

in $\binom{k-c}{a}$ ways. The remaining b positions will be forced to be filled with 1's. In the second K_2 block, choose the d' positions and fill with $q-1$ choices at each. Thus, total count is

$$\binom{k}{c}(q-2)^c \binom{k-c}{a} \binom{d-k}{d'}(q-1)^{d'} = \frac{k!}{a!b!c!} \binom{d-k}{d'}(q-2)^c(q-1)^{d'}$$

This completes the proof. Alternate versions may be obtained by summing over the other variables instead of our choice of the variable c . ■

Definition 3.10. Let V be a set of v 'points'. Consider X to be the set of all d sized subsets of V and define the relations R_i for $0 \leq i \leq d$ as

$$R_i = \{(A, B) \in X \times X \mid |A \cap B| = d - i\}$$

Then the association scheme $J(v, d) = (X, \{R_i\}_{i=0}^d)$ is called the **Johnson association scheme**.

Proposition 3.11. $J(v, d)$ is a symmetric association scheme.

Proof. It is clear that $R_i^\top = R_i$.

Consider the action of S_v on X . Clearly the action is transitive since we can go between any two d -sized subsets of V . We now want to analyse the orbits of $X \times X$.

Without loss of generality assume that $V = [v]$. Let $(A, B), (C, D) \in X \times X$ with $|A \cap B| = |C \cap D| = d - i$ for some $0 \leq i \leq d$. Through transitivity, obtain $g \in S_v$ so that

$$g * (A, B) = (g * A, g * B) = ([d], B')$$

Since cardinality of the intersection is preserved, we have $|B' \cap [d]| = d - i$. Thus, i elements of B' are greater than d . Now apply $h \in S_v$ which fixes the first d elements, maps the elements of B' outside $[d]$ to $d+1, d+2, \dots, d+i$. Thus finally we have obtained $k = hg \in S_v$ such that

$$k * (A, B) = ([d], [d-i] \cup \{d+1, \dots, d+i\})$$

Similarly (C, D) will correspond to another k' and through kk'^{-1} we can map (A, B) to (C, D) . Conversely, if there is such a mapping, the cardinality is clearly preserved.

Thus the orbits are indeed the R_i and by applying Proposition 3.6 we are done. ■

Proposition 3.12. Let G be a finite group with conjugacy classes $C_0 = \{\text{id}\}, C_1, \dots, C_d$. Setting $R_i \subseteq G \times G$ for $0 \leq i \leq d$ to be

$$R_i = \{(x, y) \mid xy^{-1} \in C_i\}$$

we get the commutative association scheme $(G, \{R_i\}_{i=0}^d)$.

Proof. It is clear that all the R_i are disjoint and span $G \times G$. Also, R_0 is clearly the diagonal. Further, $R_i^\top = \{(x, y) \mid yx^{-1} \in C_i\}$. Since the inverse of a conjugacy class is another conjugacy class, possibly itself, we have that $R_i^\top$ must be some R_j . In fact, $C_j = (C_i)^{-1}$.

All we need to do is show the intersection number property and the commutativity after that.

To this end, fix some pair $(x, y) \in R_k$. Thus, $xy^{-1} = g \in C_k$. We want to find the cardinality of

$$\mathcal{B} = \{z \in G \mid xz^{-1} \in C_i, zy^{-1} \in C_j\}$$

Consider the set

$$\mathcal{A}_g = \{(a, b) \in R_i \times R_j \mid ab = g\}$$

The map $z \mapsto (xz^{-1}, zy^{-1})$ is clearly a bijection between the sets. However, it may seem that the cardinality of $\mathcal{A}_g$ now depends on the choice of g . This is however not the case.

Suppose $g, g' \in C_k$. Let $t \in G$ so that $tgt^{-1} = g'$. Then we get the sets

$$\mathcal{A}_g = \{(a, b) \in R_i \times R_j \mid ab = g\}$$

$$\mathcal{A}'_{g'} = \{(a, b) \in R_i \times R_j \mid ab = g'\}$$

The map $(a, b) \mapsto (tat^{-1}, tbt^{-1})$ provides a bijection and we are done.

For the commutativity, fix some $(x, y) \in R_k$ with $xy^{-1} = g$ and observe that

$$\begin{aligned} p_{i,j}^k &= |\{z \in G \mid xz^{-1} \in C_i, zy^{-1} \in C_j\}| \\ &= |C_i^{-1}x \cap C_jy| \\ &= |(C_i^{-1}xy^{-1} \cap C_j)y| \\ &= |(C_i^{-1}g \cap C_j)y| \\ &= |C_i^{-1}g \cap C_j| \\ &= |C_i^{-1}g \cap g^{-1}C_jg| \\ &= |C_i^{-1} \cap g^{-1}C_j| \\ &= |C_i \cap C_j^{-1}g| \\ &= p_{j,i}^k \end{aligned}$$

The sixth equality is justified since C_j is a conjugacy class and hence $h^{-1}C_jh = C_j$ for every $h \in G$. The eighth equality holds because the size of a subset of G and its inverse are the same.

This proves that $(G, \{R_i\}_{i=0}^d)$ is a commutative association scheme. ■

All of the above schemes arise from the transitive action of a group. Although not proved in this way in the conjugacy class scheme example, it can be shown easily through the conjugacy action. However, there are important schemes which do not arise from a transitive group action.

Definition 3.13. Define the *Shrikhande graph* on 16 vertices as shown below.

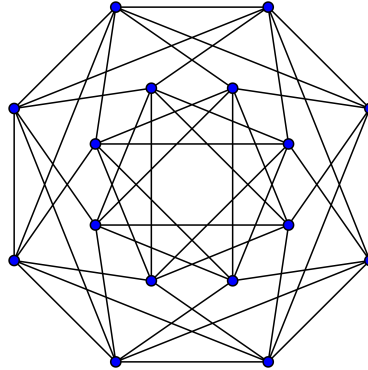


Figure 3.1: The Shrikhande graph

It is in fact a Cayley graph on $\mathbb{Z}_4 \times \mathbb{Z}_4$ with the connecting set $\{\pm(1,0), \pm(0,1), \pm(1,1)\}$. It has many interesting properties. It is a strongly regular graph with parameters $(16, 6, 2, 2)$. It also admits an integral spectrum with the characteristic polynomial being $(x - 6)(x + 2)^9(x - 2)^6$. However, the property which perhaps shines the most is that this graph is the smallest distance regular graph which is not distance transitive.

We can create an association scheme out of any given strongly regular graph in the following way.

Definition 3.14. Let $G = (V, E)$ be a strongly regular graph. Define the following relations on $V \times V$.

$$R_0 = \{(x, x) \mid x \in V\}$$

$$R_1 = \{(x, y) \mid x \text{ and } y \text{ are adjacent vertices}\}$$

$$R_2 = \{(x, y) \mid x \text{ and } y \text{ are not adjacent}\}$$

Then we define the **srg scheme with respect to G** to be $X(G) = (V, \{R_0, R_1, R_2\})$.

Proposition 3.15. *The srg scheme is a symmetric and commutative association scheme.*

Proof. All properties other than the intersection number are trivial. The intersection numbers can be verified exhaustively for all cases. A table is given below.

	(0, 0)	(0, 1)	(0, 2)	(1, 0)	(1, 1)	(1, 2)	(2, 0)	(2, 1)	(2, 2)
0	1	0	0	0	k	0	0	0	$n - k - 1$
1	0	1	0	1	λ	$k - \lambda - 1$	0	$k - \lambda - 1$	$n - 2k + \lambda$
2	0	0	1	0	μ	$k - \mu$	1	$k - \mu$	$n - 2k + \mu - 2$

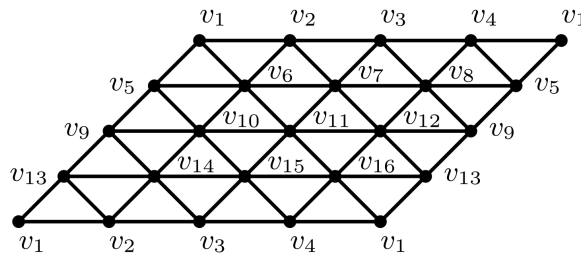
Here, the row index is k while the column index is (i, j) . The commutativity is clear from the table. ■

Proposition 3.16. *The srg scheme corresponding to the Shrikhande graph does not arise from a transitive group action.*

Proof. Suppose for the sake of contradiction, there is a group G which acts transitively on V and the orbits of its action on $V \times V$ are R_0, R_1 and R_2 . Then since R_1 and R_2 are orbits G must be a subgroup of the automorphism group of the graph since it preserves adjacency and non-adjacency.

We now claim that for every $a \in V$, the stabiliser $G_a = \{g \in G \mid g * a = a\}$ acts transitively on each of the neighbourhoods $\Gamma_i(a) = \{x \in V \mid (x, a) \in R_i\}$. Suppose (x, a) and (y, a) are in the same neighbourhood. Then by definition of orbits, there is a $g \in G$ mapping x to y and fixing a . This proves the claim. We will show that $\text{Aut}(G)_a$ is not transitive on the neighbourhood $\Gamma_2(a)$ for some $a \in V$ so that $G_a \subseteq \text{Aut}(G)_a$ will also fail to be transitive giving a contradiction.

Consider the Shrikhande graph as drawn below.



Vertices with same label are identified.

We have $\Gamma_2(v_6) = \{v_3, v_4, v_8, v_9, v_{12}, v_{13}, v_{14}, v_{15}, v_{16}\}$. Choose $b = v_3$ and $c = v_{16}$. Obtain $g \in \text{Aut}(G)_{v_6}$ such that $g(v_3) = v_{16}$. Since automorphisms preserve neighbourhoods, we must have

$$g(\Gamma_1(v_6) \cap \Gamma_1(v_3)) = \Gamma_1(g(v_6)) \cap \Gamma_1(g(v_3))$$

By inspecting the graph, we find $\Gamma_1(v_6) \cap \Gamma_1(v_3) = \{v_2, v_7\}$ and $\Gamma_1(v_6) \cap \Gamma_1(v_{16}) = \{v_1, v_{11}\}$. Therefore, any such automorphism g must induce a bijection between these sets, meaning:

$$g(\{v_2, v_7\}) = \{v_1, v_{11}\}$$

However, this is not possible since v_2 and v_7 are adjacent but v_1 and v_{11} are not. This is a contradiction and we are done. ■

Definition 3.17. Consider the finite field $\mathbb{F}_q$ for some prime power q . Let $d \mid q-1$ and $r = \frac{q-1}{d}$. Let ω be a generator of the cyclic multiplicative group $\mathbb{F}_q^*$. Consider the subgroup $H_r = \langle \omega^d \rangle$ of order r and consider the cosets $C_i = \omega^{i-1}H_r$. Thus we have the coset decomposition $\mathbb{F}_q^* = \bigcup_{i=1}^d C_i$. Set $C_0 = \{0\}$. Then if we define the relations

$$R_i = \{(x, y) \in \mathbb{F}_q \times \mathbb{F}_q \mid x - y \in C_i\} \quad \forall 0 \leq i \leq d$$

then $(\mathbb{F}_q, \{R_i\}_{i=0}^d)$ is an association scheme which we call the **Cyclotomic association scheme**

Proposition 3.18. *The scheme defined above is indeed an association scheme. It is further, commutative.*

Proof. Consider the group G to be the set of all maps σ from $\mathbb{F}_q$ to itself of the form $\sigma(x) = ax + b$ for some $a \in H_r$ and $b \in \mathbb{F}_q$. Clearly, this a group under compositions.

Further it is a transitive action since we can choose $a = 1$ and $b = \beta - \alpha$ for any $\alpha, \beta \in \mathbb{F}_q$ so that $\sigma(x) = x + (\beta - \alpha)$ is an element of the group mapping α to β .

Now we find the orbits of this action. Firstly if (x, y) are such that $x - y \in C_i$ for some $0 \leq i \leq d$, then it is clear that $(ax + b, ay + b)$ will also lie in C_i since $a \in H_r$.

Conversely let (x, y) and (f, g) be such that $x - y \in C_i$ and $f - g \in C_i$. Since both differences are in the same C_i , there must exist an element $a \in H_r$ such that $a(x - y) = (f - g)$. Thus choosing $b = f - ax = g - ay$ we have shown that the map $x \mapsto ax + b$ maps (x, y) to (f, g) and hence the orbits are indeed R_i .

Thus we have an association scheme by Proposition 3.6. Now we have to prove the commutativity. To this end, fix $(x, y) \in R_k$, that is, $x - y \in C_k$. We evaluate the intersection number $p_{i,j}^k$:

$$p_{i,j}^k = |\{z \in \mathbb{F}_q \mid x - z \in C_i, z - y \in C_j\}|$$

Consider the bijective substitution $w = x + y - z$. As z ranges over $\mathbb{F}_q$, w uniquely ranges over $\mathbb{F}_q$. We substitute $z = x + y - w$ into our membership conditions:

$$\begin{aligned} x - z &= x - (x + y - w) = w - y \\ z - y &= (x + y - w) - y = x - w \end{aligned}$$

Thus, $x - z \in C_i$ is equivalent to $w - y \in C_i$, and $z - y \in C_j$ is equivalent to $x - w \in C_j$. Substituting these conditions back into the set cardinality yields:

$$p_{i,j}^k = |\{w \in \mathbb{F}_q \mid w - y \in C_i, x - w \in C_j\}| = |\{w \in \mathbb{F}_q \mid x - w \in C_j, w - y \in C_i\}| = p_{j,i}^k$$

This proves that the cyclotomic association scheme is commutative. ■

3.3 Bose-Mesner algebras

In this section we look at the matrix representation of an association scheme. This is an equivalent way to look at and define schemes.

Definition 3.19. Given an association scheme $X = (X, \{R_i\}_{i=0}^d)$ we can define the following set of $d+1$ matrices of size $|X| \times |X|$ with entries in $\mathbb{C}$. For each $0 \leq i \leq d$, define

$$A_i(x, y) = \begin{cases} 1 & \text{if } (x, y) \in R_i \\ 0 & \text{if } (x, y) \notin R_i \end{cases}$$

These are called the **adjacency matrices of the relations R_i** .

Proposition 3.20. *The adjacency matrices A_i of the relations of an association scheme $X = (X, \{R_i\}_{i=0}^d)$ satisfy*

- (i) $A_0 = I$
- (ii) $A_0 + A_1 + \cdots + A_d = J$, the all one matrix
- (iii) For each $0 \leq i \leq d$ there is some $0 \leq i' \leq d$ such that $A_i^\top = A_{i'}$
- (iv) For every $0 \leq i, j \leq d$ there exist non-negative integers $p_{i,j}^k$ ($0 \leq k \leq d$) such that

$$A_i A_j = \sum_{k=0}^d p_{i,j}^k A_k$$

Moreover if X is commutative, then $A_i A_j = A_j A_i$ and if it is symmetric then $A_i^\top = A_i$ for every $0 \leq i \leq d$. Conversely if for a finite set X , there exist $(0,1)$ -matrices A_i for $0 \leq i \leq d$ satisfying the above four properties, then we can turn X into an association scheme by defining

$$R_i = \{(x, y) \in X \times X \mid A_i(x, y) = 1\}$$

Further, if $A_i A_j = A_j A_i$ then the association scheme is commutative and if $A_i^\top = A_i$ then it is symmetric.

Proof. All the properties directly follow from the definition of association schemes and from the basic definitions of operations on matrices. ■

Observe that property 4 of intersection numbers implies that the linear subspace spanned by the adjacency matrices is closed under matrix multiplication and hence is a $(d+1)$ -dimensional subalgebra of the algebra of $|X| \times |X|$ matrices over $\mathbb{C}$.

Definition 3.21. The $(d+1)$ -dimensional algebra $\mathfrak{A}$ generated by the adjacency matrices of an association scheme of class d is called the **adjacency algebra of the scheme**. If the association scheme is commutative, we call it the **Bose-Mesner Algebra**.

Observe that by the second condition of the matrices summing up to the all ones matrix J , we have

$$A_i \circ A_j = \delta_{i,j} A_i$$

where $\circ$ denotes the Hadamard product of matrices, that is, the naive entry-wise product. Thus, the linear space can also be regarded as an algebra under this product. Thus we have two product structures. Note that the Hadamard structure will always be commutative.

We now want to show that if an algebra of such matrices is closed under both these products, then under certain conditions it must be an adjacency algebra.

Lemma 3.22. *Let $\mathfrak{M}$ be a vector subspace of $M_X(\mathbb{C})$, the set of all $|X| \times |X|$ matrices over $\mathbb{C}$. If $\mathfrak{M}$ is closed under the Hadamard product, then there exists a basis $\{A_0, \dots, A_d\}$ of $(0,1)$ -matrices for some d such that $A_i \circ A_j = \delta_{i,j} A_i$ for every $0 \leq i, j \leq d$.*

Proof. Let $M \in \mathfrak{M}$. Let $\{\beta_1, \dots, \beta_r\}$ be the set of distinct non-zero entries of M . For each β_i we can define

$$M^{(i)} = \begin{cases} 1 & \text{if } M(x, y) = \beta_i \\ 0 & \text{otherwise} \end{cases}$$

Then it is clear that

$$M = \sum_{i=1}^r \beta_i M^{(i)}$$

In fact, taking Hadamard powers we have

$$M^{\circ j} = \sum_{i=1}^r (\beta_i)^j M^{(i)}$$

Since the β_i are distinct, the Vandermonde determinant

$$\begin{vmatrix} \beta_1 & \beta_2 & \dots & \beta_r \\ \beta_1^2 & \beta_2^2 & \dots & \beta_r^2 \\ \vdots & \vdots & & \vdots \\ \beta_1^r & \beta_2^r & \dots & \beta_r^r \end{vmatrix} \neq 0$$

Thus the system can be inverted and the matrices $M^{(i)}$ may be written in terms of $M^{\circ j}$, hence showing that $M^{(i)}$ is an element of $\mathfrak{M}$ for every $1 \leq i \leq r$.

Now let $\{B_0, \dots, B_d\}$ form a basis of $\mathfrak{M}$. Then as before, each $B_j^{(i)}$ ($1 \leq i \leq r_j$) is a $(0,1)$ -matrix in $\mathfrak{M}$, and each B_j is a linear combination of these $B_j^{(1)}, \dots, B_j^{(r_j)}$. Thus we have a new spanning set consisting entirely of $(0,1)$ -matrices: $\{B_j^{(i)} \mid 1 \leq i \leq r_j, 0 \leq j \leq d\}$. Choose a basis $\{A_0, \dots, A_{d'}\}$ from this spanning set. Since the dimension of $\mathfrak{M}$ is $d+1$, we must have $d' = d$.

Suppose the desired condition is not satisfied. Then pick $i \neq j$ so that $C_1 = A_i \circ A_j \neq 0$. Set $C_2 = A_i - C_1$ (or possibly $A_j - C_1$) so that $C_2 \neq 0$. Note that C_1 and C_2 are also $(0,1)$ -matrices. Since C_1 and C_2 belong to $\mathfrak{M}$ with $C_1 + C_2 = A_i$, the following is a valid spanning set for $\mathfrak{M}$:

$$\{A_0, A_1, \dots, A_{i-1}, C_1, C_2, A_{i+1}, \dots, A_d\}$$

Thus we can remove one element from this set to make it into a basis again. Comparing with the original basis, this new basis will have a strictly lesser total number of ones. This is because both C_1 and C_2 individually have a strictly lesser number of ones as compared to A_i . In fact, the number of ones in A_i is equal to the total number of ones in C_1 and C_2 . We continue repeating this operation until we reach a basis of $(0,1)$ -matrices satisfying the desired condition. ■

Proposition 3.23. *Let $\mathfrak{M}$ be a vector subspace of $M_X(\mathbb{C})$ which is closed under normal multiplication and Hadamard product. Suppose the following are satisfied*

(i) *if $M \in \mathfrak{M}$ then $M^\top \in \mathfrak{M}$*

(ii) *$I, J \in \mathfrak{M}$*

(iii) *for every $M \in \mathfrak{M}$, there is a constant $\alpha(M)$ such that $M \circ I = \alpha(M)I$*

Then there exists a basis $\{A_0, \dots, A_d\}$ of $\mathfrak{M}$ so that these matrices are the adjacency matrices of some association scheme.

Proof. Obtain a basis $\{A_0, \dots, A_d\}$ of $\mathfrak{M}$ such that $A_i \circ A_j = \delta_{i,j} A_i$ using the previous lemma. Since $I \in \mathfrak{M}$, write $I = \sum_{i=0}^d a_i A_i$. Since $I \circ I = I$ we must have $a_i = 0$ or 1 .

For one such t where $a_t = 1$ we have

$$\alpha(A_t)I = A_t \circ I = A_t$$

Through scaling and relabelling of the basis elements, we may assume $A_0 = I$. Now let $J = \sum_{i=0}^d b_i A_i$. Then $A_j = A_j \circ J = b_j A_j$. Thus $b_0 = \dots = b_d = 1$.

Since $J^\top = J$, we have $A_0 + \dots + A_d = A_0^\top + \dots + A_d^\top$. Since $A_i \circ A_j = \delta_{i,j} A_i$ and every $A_j^\top$ is a linear combination of A_i (by property (i)), we must have $A_j^\top = A_i$ for some unique A_i .

The final transitivity property follows from the fact that $\mathfrak{M}$ is closed under the usual matrix multiplication. ■

Definition 3.24. We define a *coherent algebra* to be a matrix algebra over $\mathbb{C}$ which is closed under transposition and Hadamard product, and contains I and J .

It is clear that any Bose-Mesner algebra is a commutative coherent algebra. Conversely, it is not difficult to show using Proposition 3.23 that every commutative coherent algebra is a Bose-Mesner algebra for some association scheme.

3.4 Commutative Association Schemes

Throughout the section, we assume our association schemes to be commutative. The default notation used will be $X = (X, \{R_0, \dots, R_d\})$.

Definition 3.25. Given the scheme X and some $0 \leq i \leq d$, we define the index i' to be the *transpose of the index* i , that is, the index such that $R_i^\top = R_{i'}$.

Definition 3.26. For a scheme X , given $x \in X$ and $0 \leq i \leq d$, we define the *neighbourhood set* $\Gamma_i(x) = \{z \in X \mid (x, z) \in R_i\}$.

It follows that $p_{i,j}^k = |\Gamma_i(x) \cap \Gamma_{j'}(y)|$ for some $(x, y) \in R_k$.

Definition 3.27. Given $0 \leq i \leq d$ for a scheme X , we define the *valency of the relation* R_i to be

$$k_i = p_{i,i'}^0$$

Explicitly, $k_i = |\Gamma_i(x)|$ for any $x \in X$.

Proposition 3.28. *The following hold for a commutative association scheme:*

- (i) $k_0 = 1$
- (ii) $k_i = k_{i'}$
- (iii) $k_0 + \dots + k_d = |X|$

Proof. $R_0^\top = R_0$ proving the first one. For the second one, note that $(i')' = i$ and by commutativity, the result follows. The third one is also trivial since $k_i = |\Gamma_i(x)|$ for any fixed $x \in X$ and every $y \in X$ must satisfy $(x, y) \in R_j$ for some unique j . Thus the y 's get sorted into the neighbourhoods $\Gamma_j(x)$ of the fixed x whose cardinalities are k_j . ■

Proposition 3.29. *The following hold for a commutative association scheme:*

- (i) $p_{i,0}^k = \delta_{i,k}$
- (ii) $p_{0,j}^k = \delta_{j,k}$
- (iii) $p_{i,j}^0 = k_i \delta_{i,j'}$
- (iv) $p_{i,j}^k = p_{i',j'}^{k'}$
- (v) $\sum_{j=0}^d p_{i,j}^k = k_i$
- (vi) $k_\ell p_{i,j}^\ell = k_j p_{i',\ell}^j = k_i p_{\ell,j'}^i$
- (vii) $\sum_{\alpha=0}^d p_{i,j}^\alpha p_{k,\alpha}^\ell = \sum_{\alpha=0}^d p_{k,i}^\alpha p_{\alpha,j}^\ell$

Proof. The first four properties follow directly from the basic definitions of association schemes:

- (i) Let $(x, y) \in R_k$. For $p_{i,0}^k$, we seek the number of $z \in X$ such that $(x, z) \in R_i$ and $(z, y) \in R_0$. The latter condition forces $z = y$. Thus, we require $(x, y) \in R_i$. Since the relations are pairwise disjoint, this is possible if and only if $i = k$, yielding exactly one such z . Hence, $p_{i,0}^k = \delta_{i,k}$.
- (ii) By an identical argument to (i), $(x, z) \in R_0$ forces $z = x$, so $(x, y) \in R_j$, yielding $p_{0,j}^k = \delta_{j,k}$.
- (iii) Let $(x, x) \in R_0$. We count $z \in X$ such that $(x, z) \in R_i$ and $(z, x) \in R_j$. The second relation is equivalent to $(x, z) \in R_{j'}$. This implies z must lie in $\Gamma_i(x) \cap \Gamma_{j'}(x)$. Since the neighbourhoods are disjoint for distinct indices, this intersection is empty unless $i = j'$, in which case it is simply $\Gamma_i(x)$, which has cardinality k_i . Thus, $p_{i,j}^0 = k_i \delta_{i,j'}$.
- (iv) Let $(x, y) \in R_k$, which means $(y, x) \in R_{k'}$. A vertex z satisfies $(x, z) \in R_i$ and $(z, y) \in R_j$ if and only if $(y, z) \in R_{j'}$ and $(z, x) \in R_{i'}$. Viewed from the perspective of the pair (y, x) , this precisely counts $p_{j',i'}^{k'}$. Because our scheme is commutative, we have $p_{j',i'}^{k'} = p_{i',j'}^{k'}$.

For the fifth property, fix some $(x, y) \in R_k$. Observe that we can partition the neighbourhood $\Gamma_i(x)$ based on the relation each element has with y :

$$\Gamma_i(x) = \Gamma_i(x) \cap X = \Gamma_i(x) \cap \left(\bigcup_{j=0}^d \Gamma_{j'}(y) \right) = \bigcup_{j=0}^d \left(\Gamma_i(x) \cap \Gamma_{j'}(y) \right)$$

Since the union on the right is disjoint, taking cardinalities yields $k_i = \sum_{j=0}^d p_{i,j}^k$.

For the sixth property, consider the set of all valid paths of length three parametrised by the indices i, j , and ℓ . We construct a set of ordered triples and evaluate its cardinality in three different ways:

$$\begin{aligned} \mathcal{S} &= \{(x, y, z) \in X^3 \mid (x, y) \in R_\ell, (x, z) \in R_i, (z, y) \in R_j\} \\ &= \bigcup_{x \in X} \bigcup_{y \in \Gamma_\ell(x)} \{z \in X \mid (x, z) \in R_i, (z, y) \in R_j\} \\ &= \bigcup_{z \in X} \bigcup_{y \in \Gamma_j(z)} \{x \in X \mid (z, x) \in R_{i'}, (x, y) \in R_\ell\} \\ &= \bigcup_{x \in X} \bigcup_{z \in \Gamma_i(x)} \{y \in X \mid (x, y) \in R_\ell, (y, z) \in R_{j'}\} \end{aligned}$$

Taking the cardinality of $\mathcal{S}$ by evaluating the nested disjoint unions in the three respective expressions yields $|X| k_\ell p_{i,j}^\ell$, $|X| k_j p_{i',\ell}^{j'}$, and $|X| k_i p_{\ell,j'}^i$. Dividing by $|X|$ provides the desired identities.

The last property is a consequence of associativity in counting paths of length three between two fixed vertices. Fix $(x, y) \in R_\ell$. We count the number of pairs (z, w) satisfying a specific relation chain:

$$\begin{aligned} \{(z, w) \in X^2 \mid (x, z) \in R_k, (z, w) \in R_i, (w, y) \in R_j\} &= \bigcup_{\alpha=0}^d \bigcup_{z \in \Gamma_k(x) \cap \Gamma_{\alpha'}(y)} \{w \in X \mid (z, w) \in R_i, (w, y) \in R_j\} \\ &= \bigcup_{\alpha=0}^d \bigcup_{w \in \Gamma_\alpha(x) \cap \Gamma_{j'}(y)} \{z \in X \mid (x, z) \in R_k, (z, w) \in R_i\} \end{aligned}$$

In the first expression, we sum over the intermediate relation α between z and y , yielding $\sum_{\alpha=0}^d p_{k,\alpha}^\ell p_{i,j}^\alpha$. In the second expression, we sum over the intermediate relation α between x and w , yielding $\sum_{\alpha=0}^d p_{\alpha,j}^\ell p_{k,i}^\alpha$. Equating them completes the proof. $\blacksquare$

We now obtain a different basis for the Bose-Mesner algebra. Contrast the following theorem with Proposition 3.23.

Proposition 3.30. *Let $\mathfrak{M}$ be a vector subspace of $M_X(\mathbb{C})$ which is closed under matrix multiplication and is commutative. Assume the following:*

(i) if $M \in \mathfrak{M}$ then $M^\top \in \mathfrak{M}$ and $\overline{M} \in \mathfrak{M}$

(ii) $I, J \in \mathfrak{M}$

Then there exists a basis $\{E_0, \dots, E_d\}$ of $\mathfrak{M}$ such that

(a) $E_0 = \frac{1}{|X|}J$

(b) $E_0 + E_1 + \dots + E_d = I$

(c) $E_i E_j = \delta_{i,j} E_i$ for every $0 \leq i, j \leq d$

(d) for each $0 \leq i \leq d$ there is some $0 \leq j \leq d$ such that $E_i^\top = \overline{E_i} = E_j$

Proof. For any $M \in \mathfrak{M}$, we have $M^\top, \overline{M} \in \mathfrak{M}$ by hypothesis, which implies the conjugate transpose $M^* = \overline{M}^\top \in \mathfrak{M}$. Since $\mathfrak{M}$ is commutative, we have $MM^* = M^*M$ for all $M \in \mathfrak{M}$, meaning every matrix in $\mathfrak{M}$ is a normal matrix. Therefore, $\mathfrak{M}$ consists of mutually commuting normal matrices, which implies they are simultaneously diagonalisable by some unitary matrix U .

In other words, every matrix in the vector space $U^*\mathfrak{M}U$ is a diagonal matrix. For a set of diagonal matrices, the ordinary matrix product and the Hadamard product are identical. Since $U^*\mathfrak{M}U$ is closed under the ordinary matrix product, it is also closed under the Hadamard product. By Lemma 3.22, there exists a basis $\{\Lambda_0, \Lambda_1, \dots, \Lambda_d\}$ of $U^*\mathfrak{M}U$ consisting of $(0, 1)$ -matrices such that $\Lambda_i \Lambda_j = \Lambda_i \circ \Lambda_j = \delta_{i,j} \Lambda_i$.

If we define $E_i = U \Lambda_i U^*$ for $0 \leq i \leq d$, then $\{E_0, E_1, \dots, E_d\}$ forms a basis of $\mathfrak{M}$, and we trivially have $E_i E_j = U \Lambda_i \Lambda_j U^* = \delta_{i,j} U \Lambda_i U^* = \delta_{i,j} E_i$, proving (c).

Since $J \in \mathfrak{M}$, we can write $J = |X| \sum_{j=0}^d a_j E_j$. Since $J^2 = |X|J$, squaring the sum yields $a_j^2 = a_j$, meaning each $a_j \in \{0, 1\}$. Note that the rank of the all-ones matrix J is exactly 1. Because the matrices E_j are mutually orthogonal, the rank of their sum is the sum of their individual ranks. Since each E_j is a non-zero idempotent, $\text{rank}(E_j) \geq 1$. This implies that exactly one coefficient a_j can be 1 while all others must be 0. By relabelling the indices, we may set $a_0 = 1$, which gives $E_0 = \frac{1}{|X|}J$, proving (a).

Similarly, since $I \in \mathfrak{M}$, we can write $I = \sum_{j=0}^d b_j E_j$. As $I^2 = I$, we obtain $b_j^2 = b_j$, meaning $b_j \in \{0, 1\}$. If there were some index k for which $b_k = 0$, then multiplying by E_k would give $IE_k = 0$, contradicting $IE_k = E_k \neq 0$. Thus $b_j = 1$ for all $0 \leq j \leq d$, yielding $E_0 + E_1 + \dots + E_d = I$, proving (b).

Finally, observe that the matrices Λ_i are real diagonal matrices (their entries are 0 and 1), meaning $\Lambda_i^* = \Lambda_i$. Consequently, $E_i^* = (U \Lambda_i U^*)^* = U \Lambda_i^* U^* = U \Lambda_i U^* = E_i$, demonstrating that each E_i is Hermitian. This immediately gives $\overline{E_i}^\top = E_i$, or equivalently, $\overline{E_i} = E_i^\top$. The operation of taking the transpose preserves the algebraic relations of the basis (orthogonality, idempotency, and summing to I). By the uniqueness of the decomposition of I into mutually orthogonal primitive idempotents in $\mathfrak{M}$, the set $\{E_0^\top, \dots, E_d^\top\}$ must simply be a permutation of the basis $\{E_0, \dots, E_d\}$. Thus, for each i , there exists some j such that $E_i^\top = \overline{E_i} = E_j$, proving (d). $\blacksquare$

We summarise our efforts made for finding the two different bases of the Bose-Mesner algebra below.

Let X be a commutative association scheme and let $\mathfrak{M}$ be its Bose-Mesner Algebra. Then the hypotheses of Proposition 3.23 and Proposition 3.30 are satisfied, and we obtain two canonical bases for $\mathfrak{M}$.

The **Bose-Mesner Basis** is given by $\{A_0, A_1, \dots, A_d\}$ and satisfies:

(i) $A_0 = I$

(ii) $\sum_{i=0}^d A_i = J$

(iii) $A_i^\top = A_j$ for some j

(iv) $A_i \circ A_j = \delta_{i,j} A_i$

(v) $A_i A_j = \sum_{k=0}^d p_{i,j}^k A_k$, where $p_{i,j}^k$ are the intersection numbers.

The **Dual Basis** is given by $\{E_0, E_1, \dots, E_d\}$ and satisfies:

- (i) $E_0 = \frac{1}{|X|}J$
- (ii) $\sum_{i=0}^d E_i = I$
- (iii) $E_i^\top = E_j$ for some j
- (iv) $E_i E_j = \delta_{i,j} E_i$
- (v) $E_i \circ E_j = \frac{1}{|X|} \sum_{k=0}^d q_{i,j}^k E_k$, where $q_{i,j}^k$ are the Krein parameters.

Note that we forced the fifth property for the dual basis. Since the algebra is closed under Hadamard products, $E_i \circ E_j$ can be written as a linear combination of the E_k and the coefficients $q_{i,j}^k$ are defined to be the **Krein parameters**.

Another fact which can be deduced from the proof of Proposition 3.30 is that $E_i^\top$ is in fact $\overline{E_i}$. This follows from the definition of E_i in the proof.

3.5 Character Tables and Intersection Matrices

Let $\mathfrak{M}$ be the Bose-Mesner algebra of a commutative association scheme $X = (X, \{R_i\})$. Consider the $|X|$ -dimensional vector space $V = \mathbb{C}^X$ over $\mathbb{C}$ (indexed by the elements of X).

For each $0 \leq i \leq d$, define the subspace $V_i = \{E_i v \mid v \in V\}$.

Proposition 3.31. *Under the standard inner product $\langle x, y \rangle = x^* y$, the following hold:*

- (i) $V = V_0 \perp V_1 \perp \dots \perp V_d$ (orthogonal direct sum).
- (ii) $V_0 = \text{span}\{\mathbb{1}\}$, where $\mathbb{1}$ is the all-ones vector.
- (iii) Each V_i is a simultaneous eigenspace for every matrix $M \in \mathfrak{M}$.

Proof. From the properties of the primitive idempotents, we know that $E_0 + E_1 + \dots + E_d = I$ and $E_i E_j = \delta_{i,j} E_i$. For any $v \in V$, we can write $v = Iv = \sum_{i=0}^d E_i v$. Since $E_i v \in V_i$ by definition, this demonstrates that $V = V_0 + V_1 + \dots + V_d$. The idempotence and orthogonality of the E_i ensure this sum is a direct sum.

To see that it is an orthogonal direct sum, let $x \in V_i$ and $y \in V_j$ for $i \neq j$. There exist $u, w \in V$ such that $x = E_i u$ and $y = E_j w$. Recall from the properties of the dual basis that $E_i^\top = \overline{E_i}$, which implies that the conjugate transpose is $E_i^* = \overline{E_i}^\top = E_i$. Thus, the idempotents are Hermitian matrices. The inner product is therefore given by:

$$\langle x, y \rangle = x^* y = (E_i u)^* (E_j w) = u^* E_i^* E_j w = u^* E_i E_j w = u^* (0) w = 0$$

Thus, $V_i \perp V_j$ for all $i \neq j$, establishing (i).

For (ii), we have $E_0 = \frac{1}{|X|}J$, where J is the all-ones matrix. For any $v \in V$, the product Jv results in a vector where every component is equal to the sum of the components of v . Thus, Jv is simply a scalar multiple of the all-ones vector $\mathbb{1}$, which implies $V_0 = \{E_0 v \mid v \in V\} \subseteq \text{span}\{\mathbb{1}\}$. Conversely, since $J\mathbb{1} = |X|\mathbb{1}$, we have $E_0\mathbb{1} = \mathbb{1} \neq 0$, showing that $\mathbb{1} \in V_0$. Therefore, V_0 is precisely $\text{span}\{\mathbb{1}\}$.

For (iii), let $M \in \mathfrak{M}$ be arbitrary. Since $\{E_0, E_1, \dots, E_d\}$ forms a basis for $\mathfrak{M}$, there exist scalars $\lambda_0, \dots, \lambda_d \in \mathbb{C}$ such that $M = \sum_{j=0}^d \lambda_j E_j$. For any $v \in V_i$, we have $E_i v = v$. Applying M to v yields:

$$Mv = \left(\sum_{j=0}^d \lambda_j E_j \right) v = \sum_{j=0}^d \lambda_j (E_j E_i) v = \lambda_i E_i v = \lambda_i v$$

This shows that every vector in V_i is an eigenvector of M corresponding to the eigenvalue λ_i . Since this holds for all $M \in \mathfrak{M}$, V_i is a simultaneous eigenspace for the entire algebra $\mathfrak{M}$. ■

The upshot is the following. Since every adjacency matrix A_j belongs to $\mathfrak{M}$, it acts on each V_i as a scalar multiple. We denote this scalar by $P_j(i)$. That is,

$$A_j v = P_j(i) v \quad \forall v \in V_i$$

Proposition 3.32. *Writing one basis in terms of the other, we have:*

$$A_i = \sum_{j=0}^d P_i(j) E_j$$

where the coefficients are exactly the scalars $P_i(j)$ defined previously.

Proof. Since $\{E_0, E_1, \dots, E_d\}$ forms a basis for the Bose-Mesner algebra $\mathfrak{M}$, we can express each adjacency matrix as a linear combination $A_i = \sum_{j=0}^d c_j E_j$ for some scalars $c_j \in \mathbb{C}$. Multiplying both sides by the primitive idempotent E_k yields:

$$A_i E_k = \left(\sum_{j=0}^d c_j E_j \right) E_k = c_k E_k$$

since $E_j E_k = \delta_{j,k} E_k$. Now, for any non-zero vector $v \in V_k$, we know that $E_k v = v$. Applying both sides of our operator equation to v gives $A_i v = c_k v$. However, by the definition of the eigenspace V_k , A_i acts on V_k as the scalar $P_i(k)$, so $A_i v = P_i(k) v$. Thus, $c_k = P_i(k)$ for all $0 \leq k \leq d$, completing the proof. ■

Definition 3.33. Conversely, let us define constants $Q_i(j)$ such that:

$$E_i = \frac{1}{|X|} \sum_{j=0}^d Q_i(j) A_j$$

Proposition 3.34. *If we define the $(d+1) \times (d+1)$ matrices P and Q over $\mathbb{C}$ by their entries as $P = (P_j(i))$ and $Q = (Q_j(i))$, one can verify directly that:*

$$PQ = QP = |X| I$$

Proof. From Proposition 3.32 and Definition 3.33, we have the two base change equations:

$$A_j = \sum_{k=0}^d P_j(k) E_k \quad \text{and} \quad E_i = \frac{1}{|X|} \sum_{j=0}^d Q_i(j) A_j$$

Substituting the expression for A_j into the equation for E_i yields:

$$E_i = \frac{1}{|X|} \sum_{j=0}^d Q_i(j) \left(\sum_{k=0}^d P_j(k) E_k \right) = \sum_{k=0}^d \left(\frac{1}{|X|} \sum_{j=0}^d Q_i(j) P_j(k) \right) E_k$$

Because the primitive idempotents $\{E_0, E_1, \dots, E_d\}$ are linearly independent and form a basis, we can equate the coefficients of E_k on both sides to obtain:

$$\frac{1}{|X|} \sum_{j=0}^d Q_i(j) P_j(k) = \delta_{i,k}$$

which simplifies to $\sum_{j=0}^d P_j(k) Q_i(j) = |X| \delta_{i,k}$. Since $P_j(k)$ is the (k, j) -th entry of P and $Q_i(j)$ is the (j, i) -th entry of Q , the left-hand side precisely represents the definition of matrix multiplication for the (k, i) -th entry of the product PQ . Therefore, $PQ = |X| I$. Since P and Q are square matrices of finite dimension, having a right inverse guarantees it is also a left inverse, so $QP = |X| I$ as well. ■

Definition 3.35. The matrix $P = [P_j(i)]$ is called the **first eigenmatrix** or the **character table** of the association scheme X . Similarly, the matrix $Q = [Q_j(i)]$ is called the **second eigenmatrix** or the **dual character table** of the association scheme. These are $(d+1) \times (d+1)$ matrices over $\mathbb{C}$.

Proposition 3.36. Let X be a commutative association scheme. For every $0 \leq i \leq d$, the adjacency matrices A_i and the primitive idempotents E_i satisfy the following properties:

- (i) $JA_i = A_iJ = k_iJ$
- (ii) $\text{tr}(A_i) = |X| \delta_{i,0}$
- (iii) $\text{tr}(E_i) = m_i = \dim(V_i) = \text{rank}(E_i)$ for some positive integers m_i .

Proof. For (i), recall that the adjacency matrix A_i represents the relation R_i , meaning its row sums and column sums are precisely the valency k_i . Multiplying A_i on the right by the all-ones matrix J replaces every entry with the row sum of A_i , yielding k_iJ . Similarly, left multiplication by J replaces every entry with the column sum, yielding $JA_i = k_iJ$.

For (ii), we know $A_0 = I$, whose trace is clearly $|X|$. For $i > 0$, the relation R_i contains no diagonal elements (x, x) . Thus, the diagonal entries of A_i are all identically 0, meaning $\text{tr}(A_i) = 0$. Combining these facts gives $\text{tr}(A_i) = |X| \delta_{i,0}$.

For (iii), each E_i is a non-zero primitive idempotent matrix, meaning $E_i^2 = E_i$. The eigenvalues of an idempotent matrix are strictly 0 and 1. The trace of any matrix is the sum of its eigenvalues, which for an idempotent matrix perfectly counts the algebraic multiplicity of the eigenvalue 1. This multiplicity is exactly the rank of the matrix. Furthermore, since $V_i = \{E_i v \mid v \in V\}$ is the image of the linear transformation represented by E_i , its dimension is $\text{rank}(E_i)$. Thus, $m_i = \text{tr}(E_i) = \text{rank}(E_i) = \dim(V_i)$. Since $E_i \neq 0$, this must be a strictly positive integer. ■

Definition 3.37. The constants $m_i = \text{tr}(E_i)$ are called the **multiplicities** of the association scheme X .

Proposition 3.38. The character tables P and Q satisfy the following properties:

- (i) $P_0(i) = 1$ and $Q_0(i) = 1$
- (ii) $P_i(0) = k_i$ and $Q_i(0) = m_i$
- (iii) $P_{i'}(j) = \overline{P_i(j)}$ and $Q_i(j) = \overline{Q_{i'}(j)}$ $[A_i^\top = A_{i'}, E_i^\top = E_i]$
- (iv) $\frac{Q_j(i)}{m_j} = \frac{\overline{P_i(j)}}{k_i}$
- (v) $\sum_{v=0}^d \frac{1}{k_v} P_v(i) \overline{P_v(j)} = \delta_{i,j} \frac{|X|}{m_i}$ (First Orthogonality Relation)
- (vi) $\sum_{v=0}^d m_v P_i(v) \overline{P_j(v)} = \delta_{i,j} |X| k_i$ (Second Orthogonality Relation)
- (vii) $P_i(\ell) P_j(\ell) = \sum_{k=0}^d p_{i,j}^k P_k(\ell)$
- (viii) $Q_i(\ell) Q_j(\ell) = \sum_{k=0}^d q_{i,j}^k Q_k(\ell)$
- (ix) $P_i(j) Q_j(\ell) = \sum_{k=0}^d p_{i,k}^\ell Q_j(k)$

Proof. (i) Since $A_0 = I = \sum_{j=0}^d E_j$ and $A_0 = \sum_{j=0}^d P_0(j) E_j$, equating the coefficients of the basis matrices E_j gives $P_0(j) = 1$. Similarly, substituting $J = \sum_{j=0}^d A_j$ into $E_0 = \frac{1}{|X|} J$ and comparing it with $E_0 = \frac{1}{|X|} \sum_{j=0}^d Q_0(j) A_j$ yields $Q_0(j) = 1$.

(ii) Applying A_i to the all-ones vector $\mathbf{1} \in V_0$ yields $A_i \mathbf{1} = k_i \mathbf{1}$. Since A_i acts on V_0 as the scalar $P_i(0)$, we have $P_i(0) = k_i$. For the dual, taking the trace of $|X| E_i = \sum_{j=0}^d Q_i(j) A_j$ and applying $\text{tr}(A_j) = |X| \delta_{j,0}$ from the previous proposition gives $|X| \text{tr}(E_i) = Q_i(0) |X|$. Thus, $Q_i(0) = m_i$.

(iii) Since A_i acts on V_j as the scalar $P_i(j)$, its transpose $A_i^\top = A_{i'}$ acts as the complex conjugate, meaning $\overline{P_{i'}(j)} = \overline{P_i(j)}$. A symmetric argument on the idempotents E_i under the Hadamard product yields $Q_i(j) = \overline{Q_{i'}(j)}$.

(iv) Consider the trace of $A_{i'}E_j$. Since $A_{i'}$ acts on V_j as $P_{i'}(j) = \overline{P_i(j)}$, we have $\text{tr}(A_{i'}E_j) = \overline{P_i(j)} \text{tr}(E_j) = \overline{P_i(j)}m_j$. Alternatively, expanding E_j , we get $\text{tr}(A_{i'}E_j) = \frac{1}{|X|} \sum_{k=0}^d Q_j(k) \text{tr}(A_{i'}A_k)$. The matrix $A_{i'}A_k$ has non-zero trace only if $k = i$, where its diagonal entries are exactly k_i . Thus, $\text{tr}(A_{i'}E_j) = \frac{1}{|X|} Q_j(i)(|X| k_i) = Q_j(i)k_i$. Equating both expressions for the trace gives the result.

(v) & (vi) These are the First and Second Orthogonality Relations. They follow immediately by writing out the entries of the matrix identities $QP = |X|I$ and $PQ = |X|I$ respectively, and substituting $Q_j(i) = m_j \overline{P_i(j)}/k_i$ derived in (iv).

(vii) Apply the identity $A_iA_j = \sum_{k=0}^d p_{i,j}^k A_k$ to an arbitrary non-zero vector $v \in V_\ell$. The left side scales v by $P_i(\ell)P_j(\ell)$, whilst the right side scales it by $\sum_{k=0}^d p_{i,j}^k P_k(\ell)$. Equating these scalars yields the result.

(viii) This is the dual analogue of (vii). It is obtained by substituting the base change formulas into the Hadamard product relation $E_i \circ E_j = \frac{1}{|X|} \sum_{k=0}^d q_{i,j}^k E_k$ and comparing coefficients.

(ix) Expand the operator identity $A_iE_j = P_i(j)E_j$ by expressing E_j in terms of the adjacency matrices:

$$A_i \left(\frac{1}{|X|} \sum_{k=0}^d Q_j(k) A_k \right) = P_i(j) \left(\frac{1}{|X|} \sum_{\ell=0}^d Q_j(\ell) A_\ell \right)$$

Expanding the left side using $A_iA_k = \sum_{\ell=0}^d p_{i,k}^\ell A_\ell$ and comparing the coefficient of A_ℓ on both sides produces the desired relation. $\blacksquare$

Theorem 3.39. *The Krein parameters and intersection numbers can be explicitly expressed in terms of the entries of the first and second eigenmatrices:*

$$(i) \quad q_{i,j}^\ell = \frac{m_i m_j}{|X|} \sum_{t=0}^d \frac{1}{k_t^2} P_t(i) P_t(j) \overline{P_t(\ell)}$$

$$(ii) \quad p_{i,j}^\ell = \frac{k_i k_j}{|X|} \sum_{t=0}^d \frac{1}{m_t^2} Q_t(i) Q_t(j) \overline{Q_t(\ell)}$$

Proof. (i) Consider the identity $\frac{1}{|X|} q_{i,j}^\ell E_\ell = (E_i \circ E_j) E_\ell$, which is obtained by expanding $E_i \circ E_j$ and multiplying by E_ℓ . We calculate the trace of both sides.

The trace of the left-hand side is $\frac{1}{|X|} q_{i,j}^\ell \text{tr}(E_\ell) = \frac{1}{|X|} q_{i,j}^\ell m_\ell$.

For the right-hand side, we expand the trace explicitly:

$$\begin{aligned} \text{tr}((E_i \circ E_j) E_\ell) &= \sum_{x \in X} ((E_i \circ E_j) E_\ell)(x, x) \\ &= \sum_{x \in X} \sum_{y \in X} (E_i \circ E_j)(x, y) E_\ell(y, x) \\ &= \sum_{x \in X} \sum_{y \in X} E_i(x, y) E_j(x, y) E_\ell^\top(x, y) \\ &= \sum_{x \in X} \sum_{y \in X} (E_i \circ E_j \circ E_{\hat{\ell}})(x, y) \end{aligned}$$

Substituting the dual base change formula $E_k = \frac{1}{|X|} \sum_{t=0}^d Q_k(t) A_t$ into the Hadamard product, and recalling that $A_t \circ A_u = \delta_{t,u} A_t$, we get:

$$E_i \circ E_j \circ E_{\hat{\ell}} = \frac{1}{|X|^3} \sum_{t=0}^d Q_i(t) Q_j(t) Q_{\hat{\ell}}(t) A_t$$

The sum over all $x, y \in X$ computes the sum of all entries of the matrix. Since the sum of all entries of A_t is exactly $|X| k_t$, we obtain:

$$\text{tr}((E_i \circ E_j)E_\ell) = \frac{1}{|X|^3} \sum_{t=0}^d Q_i(t)Q_j(t)\overline{Q_\ell(t)}(|X| k_t) = \frac{1}{|X|^2} \sum_{t=0}^d Q_i(t)Q_j(t)\overline{Q_\ell(t)}k_t$$

Equating the two traces gives:

$$q_{i,j}^\ell m_\ell = \frac{1}{|X|} \sum_{t=0}^d Q_i(t)Q_j(t)\overline{Q_\ell(t)}k_t$$

From Proposition 3.38 (iv), we have $Q_r(t) = m_r \overline{P_t(r)}/k_t$. Substituting this relation into our expression yields:

$$q_{i,j}^\ell m_\ell = \frac{1}{|X|} \sum_{t=0}^d \left(\frac{m_i \overline{P_t(i)}}{k_t} \right) \left(\frac{m_j \overline{P_t(j)}}{k_t} \right) \left(\frac{m_\ell P_t(\ell)}{k_t} \right) k_t$$

Dividing by m_ℓ isolates $q_{i,j}^\ell$. Since the Krein parameters are real numbers, taking the complex conjugate of both sides preserves the value and yields the desired formula.

(ii) We employ a dual argument. Multiply the expansion $A_i A_j = \sum_{\ell=0}^d p_{i,j}^\ell A_\ell$ entry-wise by A_ℓ to obtain $p_{i,j}^\ell A_\ell = (A_i A_j) \circ A_\ell$, and calculate the sum of all entries on both sides.

For the left-hand side, the sum of the entries of A_ℓ is $|X| k_\ell$, so the total sum is $p_{i,j}^\ell |X| k_\ell$.

For the right-hand side, the sum of all entries is:

$$\begin{aligned} \sum_{x \in X} \sum_{y \in X} ((A_i A_j) \circ A_\ell)(x, y) &= \sum_{x \in X} \sum_{y \in X} (A_i A_j)(x, y) A_\ell^\top(y, x) \\ &= \text{tr}(A_i A_j A_\ell) \\ &= \text{tr} \left(\sum_{t=0}^d P_i(t) P_j(t) P_\ell(t) E_t \right) \\ &= \sum_{t=0}^d P_i(t) P_j(t) \overline{P_\ell(t)} m_t \end{aligned}$$

Equating the sums, we have:

$$p_{i,j}^\ell |X| k_\ell = \sum_{t=0}^d P_i(t) P_j(t) \overline{P_\ell(t)} m_t$$

Using Proposition 3.38 (iv) again, we substitute $P_r(t) = k_r \overline{Q_t(r)}/m_t$:

$$p_{i,j}^\ell |X| k_\ell = \sum_{t=0}^d \left(\frac{k_i \overline{Q_t(i)}}{m_t} \right) \left(\frac{k_j \overline{Q_t(j)}}{m_t} \right) \left(\frac{k_\ell Q_t(\ell)}{m_t} \right) m_t$$

Dividing by $|X| k_\ell$ isolates $p_{i,j}^\ell$. Taking the complex conjugate of both sides, which is valid since the intersection numbers are real integers, completes the proof. $\blacksquare$

We now have some properties of the Krein parameters. Contrast the following remarkable dual symmetries of the Krein parameters with the corresponding identities from Proposition 3.29.

Proposition 3.40. *The Krein parameters satisfy the following properties:*

- (i) $q_{0,j}^k = \delta_{j,k}$
- (ii) $q_{i,0}^k = \delta_{i,k}$
- (iii) $q_{i,j}^0 = m_i \delta_{i,j}$

- (iv) $q_{i,j}^k = q_{i,j}^{\hat{k}}$
- (v) $\sum_{j=0}^d q_{i,j}^k = m_i$
- (vi) $m_k q_{i,j}^k = m_j q_{i,k}^j = m_i q_{k,j}^i$
- (vii) $\sum_{\alpha=0}^d q_{i,j}^\alpha q_{k,\alpha}^\ell = \sum_{\alpha=0}^d q_{k,i}^\alpha q_{\alpha,j}^\ell$

Proof. These properties are the dual analogues of the intersection number identities, and they follow naturally from the algebraic structure of the Bose-Mesner algebra $\mathfrak{M}^\circ$ under the Hadamard product. Recall that $|X| E_i \circ E_j = \sum_{k=0}^d q_{i,j}^k E_k$, $E_0 = \frac{1}{|X|} J$, and $\text{tr}(E_i) = m_i$.

- (i) Since $E_0 = \frac{1}{|X|} J$ and J acts as the identity for the Hadamard product, $E_0 \circ E_j = \frac{1}{|X|} E_j$. Comparing this with the definition $\frac{1}{|X|} \sum_{k=0}^d q_{0,j}^k E_k$, we deduce that $q_{0,j}^k = \delta_{j,k}$.
- (ii) This follows directly from (i) due to the commutativity of the Hadamard product: $E_i \circ E_0 = E_0 \circ E_i$.
- (iii) We compute the sum of all entries of $E_i \circ E_j$ in two different ways. First, substituting the expansion gives the sum of entries of $\frac{1}{|X|} \sum_{k=0}^d q_{i,j}^k E_k$. The sum of the entries of any matrix M is given by $\text{tr}(MJ) = |X| \text{tr}(ME_0)$. Thus, the sum of entries of E_k is $|X| \text{tr}(E_k E_0) = |X| \delta_{k,0} \text{tr}(E_0) = |X| \delta_{k,0}$. Therefore, the sum of the entries of the right-hand side is precisely $q_{i,j}^0$. Alternatively, the sum of the entries of $E_i \circ E_j$ is given by $\sum_{x,y \in X} E_i(x,y) E_j(x,y) = \text{tr}(E_i E_j^\top)$. Since $E_j^\top = E_j$, this equals $\text{tr}(E_i E_j) = \delta_{i,j} \text{tr}(E_i) = m_i \delta_{i,j}$. Equating the two yields $q_{i,j}^0 = m_i \delta_{i,j}$.
- (iv) Taking the transpose of the expansion $|X| E_i \circ E_j = \sum_{k=0}^d q_{i,j}^k E_k$ yields $|X| E_i^\top \circ E_j^\top = \sum_{k=0}^d q_{i,j}^k E_k^\top$. Since $E_i^\top = E_i$, this becomes $|X| E_i \circ E_j = \sum_{k=0}^d q_{i,j}^k E_k$. However, by definition, $|X| E_i \circ E_j = \sum_{k=0}^d q_{i,j}^k E_k$. Substituting k with $\hat{k}$ gives $q_{i,j}^k = q_{i,j}^{\hat{k}}$.
- (v) Summing over j , we have $\sum_{j=0}^d E_i \circ E_j = E_i \circ (\sum_{j=0}^d E_j) = E_i \circ I$. Recall the base change $E_i = \frac{1}{|X|} \sum_{\ell=0}^d Q_i(\ell) A_\ell$. Since $A_0 = I$ and the relations are disjoint, the Hadamard product with I isolates the A_0 term: $E_i \circ I = \frac{1}{|X|} Q_i(0) I = \frac{m_i}{|X|} I$. On the other hand, substituting the Krein expansion yields $\frac{1}{|X|} \sum_{k=0}^d (\sum_{j=0}^d q_{i,j}^k) E_k$. Equating the coefficients of $I = \sum_{k=0}^d E_k$ shows that $\sum_{j=0}^d q_{i,j}^k = m_i$.
- (vi) Consider the trace of the ordinary product of $(E_i \circ E_j)$ with E_k . Using the expansion, $\text{tr}((E_i \circ E_j) E_k) = \frac{1}{|X|} \sum_{\ell=0}^d q_{i,j}^\ell \text{tr}(E_\ell E_k) = \frac{1}{|X|} q_{i,j}^k m_k$. Alternatively, writing out the trace explicitly gives:

$$\text{tr}((E_i \circ E_j) E_k) = \sum_{x,y \in X} (E_i \circ E_j)(x,y) E_k(y,x) = \sum_{x,y \in X} E_i(x,y) E_j(x,y) E_k(y,x)$$

This sum is highly symmetric. By substituting $E_i(x,y) = E_i(y,x)$ and rearranging the terms to match the trace format, we get:

$$\sum_{x,y \in X} E_i(y,x) E_k(y,x) E_j(x,y) = \text{tr}((E_i \circ E_k) E_j) = \frac{1}{|X|} q_{i,k}^j m_j$$

Similarly, cyclic permutation yields $\text{tr}((E_k \circ E_j) E_i) = \frac{1}{|X|} q_{k,j}^i m_i$. Equating these expressions and multiplying by $|X|$ produces the required dual identities.

- (vii) This property is an immediate consequence of the associativity of the Hadamard product: $(E_i \circ E_j) \circ E_k = (E_k \circ E_i) \circ E_j$. Both expressions represent $E_i \circ E_j \circ E_k$. Expanding the left side yields $\frac{1}{|X|^2} \sum_{\ell=0}^d \left(\sum_{\alpha=0}^d q_{i,j}^\alpha q_{\alpha,k}^\ell \right) E_\ell$. Expanding the right side yields $\frac{1}{|X|^2} \sum_{\ell=0}^d \left(\sum_{\alpha=0}^d q_{k,i}^\alpha q_{\alpha,j}^\ell \right) E_\ell$. Using $q_{\alpha,k}^\ell = q_{k,\alpha}^\ell$ and equating the coefficients of the linearly independent basis matrices E_ℓ completes the proof.

■

Theorem 3.41. *The numbers $q_{i,j}^k$ are non-negative real numbers.*

Proof. Recall that the primitive idempotents E_i and E_j are Hermitian matrices ($E_i^* = E_i$) and satisfy $E_i^2 = E_i$. Consequently, their eigenvalues are exclusively 0 and 1. A Hermitian matrix with non-negative eigenvalues is positive semi-definite. Thus, E_i and E_j are both positive semi-definite matrices.

We now consider their Kronecker product, $E_i \otimes E_j$. The eigenvalues of a Kronecker product of two matrices are precisely the products of their respective eigenvalues. Since the eigenvalues of E_i and E_j are non-negative, the eigenvalues of $E_i \otimes E_j$ are also non-negative, meaning $E_i \otimes E_j$ is positive semi-definite.

The Hadamard product $E_i \circ E_j$ can be naturally embedded within the Kronecker product. Specifically, the entries of $E_i \circ E_j$ are given by $(E_i \circ E_j)(x, y) = E_i(x, y)E_j(x, y)$, which correspond exactly to the entries $(x, x), (y, y)$ of the larger matrix $E_i \otimes E_j$. Thus, $E_i \circ E_j$ is a principal submatrix of $E_i \otimes E_j$.

It is a standard result in linear algebra that any principal submatrix of a positive semi-definite matrix is itself positive semi-definite. Therefore, the Hadamard product $E_i \circ E_j$ is a positive semi-definite Hermitian matrix. This implies that all of its eigenvalues are non-negative real numbers.

On the other hand, we have the algebraic expansion:

$$E_i \circ E_j = \frac{1}{|X|} \sum_{\ell=0}^d q_{i,j}^\ell E_\ell$$

Let $k \in \{0, 1, \dots, d\}$ be arbitrary, and let $v \in V_k$ be a non-zero vector in the k -th eigenspace. By the properties of the primitive idempotents, $E_\ell v = \delta_{\ell,k} v$. Applying the operator $E_i \circ E_j$ to v yields:

$$(E_i \circ E_j)v = \left(\frac{1}{|X|} \sum_{\ell=0}^d q_{i,j}^\ell E_\ell \right) v = \frac{1}{|X|} q_{i,j}^k v$$

This shows that $\frac{1}{|X|} q_{i,j}^k$ is an eigenvalue of the positive semi-definite matrix $E_i \circ E_j$. Therefore, we must have:

$$\frac{1}{|X|} q_{i,j}^k \geq 0$$

Since $|X|$ is a strictly positive integer, it immediately follows that $q_{i,j}^k$ is a non-negative real number. ■

Definition 3.42. For each $0 \leq i \leq d$, we define the *intersection matrix* B_i as a $(d+1) \times (d+1)$ matrix given by its entries:

$$(B_i)_{j,k} = p_{i,j}^k$$

Let $\mathcal{B}$ be the subalgebra of $M_{d+1}(\mathbb{C})$ generated by $B_0, \dots, B_d$ under the standard matrix product.

Theorem 3.43. *The map $A_i \mapsto B_i$ extends to an algebra isomorphism between the Bose-Mesner algebra $\mathfrak{M}$ (equipped with standard matrix multiplication) and $\mathcal{B}$.*

Proof. For each $M \in \mathfrak{M}$, define a linear map $L_M \in \text{End}(\mathfrak{M})$ given by left multiplication: $L_M(N) = MN$ for any $N \in \mathfrak{M}$. This endows $\mathfrak{M}$ with the structure of a module over itself. Let $\rho(M)$ denote the matrix representation of L_M with respect to the basis $\{A_0, A_1, \dots, A_d\}$. The mapping $\rho : \mathfrak{M} \rightarrow M_{d+1}(\mathbb{C})$ is an algebra homomorphism, known as the left regular representation of $\mathfrak{M}$.

Suppose $\rho(M) = 0$. Then $L_M(N) = 0$ for all $N \in \mathfrak{M}$. In particular, evaluating at the identity gives $L_M(I) = MI = M = 0$. Thus, ρ is a faithful representation (an injection).

Consider the action of A_i on the basis element A_j . We have:

$$L_{A_i}(A_j) = A_i A_j = \sum_{k=0}^d p_{i,j}^k A_k$$

By the definition of the matrix representation with respect to this basis, the (k, j) -entry of $\rho(A_i)$ is exactly the coefficient of A_k , which is $p_{i,j}^k$. Since $(B_i)_{j,k} = p_{i,j}^k$, we see that the entries are transposed, yielding $\rho(A_i) = B_i^\top$.

Because the Bose-Mesner algebra $\mathfrak{M}$ is commutative, the transpose mapping $M \mapsto \rho(M)^\top$ is also an algebra homomorphism. Thus, the mapping $A_i \mapsto \rho(A_i)^\top = B_i$ is an algebra isomorphism between $\mathfrak{M}$ and $\mathcal{B}$. ■

Definition 3.44. Similarly, for each $0 \leq i \leq d$, we define the *dual intersection matrix* B_i^* as a $(d+1) \times (d+1)$ matrix given by its entries:

$$(B_i^*)_{j,k} = q_{i,j}^k$$

Let $\mathcal{B}^*$ be the subalgebra generated by $B_0^*, \dots, B_d^*$ under the standard matrix product.

Theorem 3.45. *Let $\mathfrak{M}^\circ$ denote the Bose-Mesner algebra equipped with the Hadamard product (entry-wise multiplication). Then the map $|X| E_i \mapsto B_i^*$ extends to an algebra isomorphism between $\mathfrak{M}^\circ$ and $\mathcal{B}^*$.*

Proof. We proceed similarly to the previous proof, using the dual structures. For each $M \in \mathfrak{M}^\circ$, define a linear transformation $L_M^* \in \text{End}(\mathfrak{M}^\circ)$ using the Hadamard product: $L_M^*(N) = M \circ N$ for any $N \in \mathfrak{M}^\circ$. Let $\rho^*(M)$ be the matrix representation of L_M^* with respect to the basis $\{|X| E_0, |X| E_1, \dots, |X| E_d\}$ of $\mathfrak{M}^\circ$.

The mapping $\rho^* : \mathfrak{M}^\circ \rightarrow M_{d+1}(\mathbb{C})$ is an algebra homomorphism, representing the left regular representation of $\mathfrak{M}^\circ$. Since the all-ones matrix J acts as the identity for the Hadamard product and $J = |X| E_0 \in \mathfrak{M}^\circ$, if $\rho^*(M) = 0$, then $L_M^*(J) = M \circ J = M = 0$. Thus, ρ^* is a faithful representation.

Consider the action of $|X| E_i$ on the basis element $|X| E_j$:

$$L_{|X| E_i}^* (|X| E_j) = (|X| E_i) \circ (|X| E_j) = |X|^2 (E_i \circ E_j) = |X| \sum_{k=0}^d q_{i,j}^k E_k = \sum_{k=0}^d q_{i,j}^k (|X| E_k)$$

This demonstrates that the (k, j) -entry of the matrix representation $\rho^*(|X| E_i)$ is exactly $q_{i,j}^k$. Since $(B_i^*)_{j,k} = q_{i,j}^k$, we obtain $\rho^*(|X| E_i) = (B_i^*)^\top$.

Since $\mathfrak{M}^\circ$ is commutative under the Hadamard product, the transpose mapping $M \mapsto \rho^*(M)^\top$ is an algebra homomorphism. Consequently, the mapping $|X| E_i \mapsto \rho^*(|X| E_i)^\top = B_i^*$ gives an algebra isomorphism between $\mathfrak{M}^\circ$ and $\mathcal{B}^*$. ■

We also have the following very nice diagonalisation property.

Theorem 3.46. *The first and second eigenmatrices P and Q simultaneously diagonalise the transposes of the intersection and dual intersection matrices, respectively.*

Specifically, we have:

- (i) $P B_i^\top P^{-1} = \text{diag}(P_i(0), P_i(1), \dots, P_i(d))$
- (ii) $Q (B_i^*)^\top Q^{-1} = \text{diag}(Q_i(0), Q_i(1), \dots, Q_i(d))$

Proof. We shall prove (i); the proof for (ii) follows an identical dual argument.

Recall that the matrix P is defined such that its (v, k) -entry is $P_k(v)$. Consider the (v, j) -entry of the matrix product on the left-hand side before taking the inverse, namely $\text{diag}(P_i(0), P_i(1), \dots, P_i(d))P$. Left-multiplying by a diagonal matrix scales the rows, so this entry is given by:

$$(\text{diag}(P_i(0), P_i(1), \dots, P_i(d))P)_{v,j} = P_i(v)P_j(v)$$

By Proposition 3.38 (vii), we can expand this product of character table entries as:

$$P_i(v)P_j(v) = \sum_{k=0}^d p_{i,j}^k P_k(v)$$

Now, let us compute the (v, j) -entry of the product $PB_i^\top$. By the definition of matrix multiplication:

$$(PB_i^\top)_{v,j} = \sum_{k=0}^d P_{v,k}(B_i^\top)_{k,j} = \sum_{k=0}^d P_k(v)(B_i)_{j,k}$$

Substituting the definition of the intersection matrix $(B_i)_{j,k} = p_{i,j}^k$, we get:

$$(PB_i^\top)_{v,j} = \sum_{k=0}^d P_k(v)p_{i,j}^k$$

This exactly matches our expansion of $P_i(v)P_j(v)$. Therefore, we have established the matrix equality:

$$\text{diag}(P_i(0), P_i(1), \dots, P_i(d))P = PB_i^\top$$

Right-multiplying both sides by P^{-1} yields $PB_i^\top P^{-1} = \text{diag}(P_i(0), P_i(1), \dots, P_i(d))$, which proves (i).

By substituting the corresponding dual identities and applying Proposition 3.38 (viii), an exactly symmetric calculation verifies (ii). ■

3.6 The Dual Bose-Mesner Algebra

As we have seen, the Bose-Mesner algebra $\mathfrak{M}$ possesses two natural algebra structures: one with respect to the ordinary matrix product, and another with respect to the Hadamard product, denoted by $\mathfrak{M}^\circ$. In many ways, $\mathfrak{M}^\circ$ acts as the dual of $\mathfrak{M}$. To study this duality more concretely, we construct an explicit isomorphic copy of $\mathfrak{M}^\circ$ within the full matrix algebra $M_X(\mathbb{C})$. This new algebraic structure is known as the dual Bose-Mesner algebra, and it serves as a foundational building block for the Terwilliger algebra. We do not cover Terwilliger algebras in this text. One may refer the original papers [30], [31] and [32].

We begin by fixing a base point $x_0 \in X$. We identify the standard basis of the complex vector space $V = \mathbb{C}^X$ with the elements of X , meaning each $x \in X$ corresponds to a standard basis vector in V .

Recall the neighbourhood set $\Gamma_i(x_0) = \{x \in X \mid (x_0, x) \in R_i\}$. Let V_i^* be the subspace of V spanned by $\Gamma_i(x_0)$. Since the relations partition X , we have $\dim(V_i^*) = k_i$, yielding the orthogonal decomposition:

$$V = V_0^* \perp V_1^* \perp \dots \perp V_d^*$$

Definition 3.47. Let E_i^* be the matrix representation of the orthogonal projection from V onto V_i^* . By definition, E_i^* is a diagonal matrix satisfying:

$$E_i^*(x, y) = \begin{cases} 1, & \text{if } x = y \text{ and } (x_0, x) \in R_i \\ 0, & \text{otherwise} \end{cases}$$

Consequently, we have $E_i^*E_j^* = \delta_{i,j}E_i^*$ and $I = \sum_{i=0}^d E_i^*$.

These diagonal matrices form a set of mutually orthogonal idempotents that sum to the identity, perfectly mirroring the behaviour of the original primitive idempotents E_i of $\mathfrak{M}$. We now define the dual analogues of the adjacency matrices.

Definition 3.48. For each $0 \leq i \leq d$, we define the **dual adjacency matrix** A_i^* with respect to the base point x_0 as:

$$A_i^* = \sum_{j=0}^d Q_i(j)E_j^*$$

Proposition 3.49. *The dual adjacency matrices satisfy the following properties:*

- (i) $E_i^* = \frac{1}{|X|} \sum_{j=0}^d P_i(j) A_j^*$
- (ii) $A_i^* A_j^* = A_i^* \circ A_j^* = \sum_{k=0}^d q_{i,j}^k A_k^*$
- (iii) The set of eigenvalues of A_i^* is exactly $\{Q_i(0), Q_i(1), \dots, Q_i(d)\}$.

Proof. (i) From the definition of A_j^* , we can invert the relationship using the orthogonality of the character tables. Since $PQ = |X| I$, summing over the columns of P immediately gives the inverse linear combination $E_i^* = \frac{1}{|X|} \sum_{j=0}^d P_i(j) A_j^*$.

(ii) Because A_i^* and A_j^* are diagonal matrices, their ordinary matrix product perfectly coincides with their Hadamard product. Let $x \in X$ be arbitrary, and suppose $(x_0, x) \in R_a$ for some index a . The matrix E_j^* has a 1 at the (x, x) position if and only if $j = a$. Therefore, evaluating the diagonal entry of A_i^* yields:

$$A_i^*(x, x) = \sum_{j=0}^d Q_i(j) E_j^*(x, x) = Q_i(a)$$

Recall from the dual base change formula $E_i = \frac{1}{|X|} \sum_j Q_i(j) A_j$ that $Q_i(a) = |X| E_i(x_0, x)$. Multiplying the diagonal entries, we get:

$$(A_i^* A_j^*)(x, x) = A_i^*(x, x) A_j^*(x, x) = |X|^2 E_i(x_0, x) E_j(x_0, x) = |X|^2 (E_i \circ E_j)(x_0, x)$$

Substituting the expansion $|X| (E_i \circ E_j) = \sum_{k=0}^d q_{i,j}^k E_k$ into this expression yields:

$$(A_i^* A_j^*)(x, x) = |X| \sum_{k=0}^d q_{i,j}^k E_k(x_0, x) = \sum_{k=0}^d q_{i,j}^k Q_k(a) = \sum_{k=0}^d q_{i,j}^k A_k^*(x, x)$$

Since this equality holds for every diagonal entry $x \in X$, the matrix identity follows.

(iii) By definition, $A_i^* = \sum_{a=0}^d Q_i(a) E_a^*$. Since the E_a^* are projection matrices onto mutually orthogonal subspaces V_a^* , the matrix A_i^* acts as the scalar $Q_i(a)$ on the subspace V_a^* . Thus, the eigenvalues are precisely the elements of the set $\{Q_i(0), Q_i(1), \dots, Q_i(d)\}$. ■

Proposition 3.50. *The dual idempotents E_i^* satisfy the following properties:*

- (i) $A_i^* = \sum_{j=0}^d Q_i(j) E_j^*$
- (ii) $E_i^* E_j^* = E_i^* \circ E_j^* = \delta_{i,j} E_i^*$
- (iii) The eigenvalues of E_i^* are precisely 0 and 1, with the multiplicity of 1 being k_i .

Proof. (i) This is precisely Definition 3.48, restated here to emphasise the symmetry with the base change formulas of the original Bose-Mesner algebra.

(ii) Because E_i^* and E_j^* are diagonal matrices, their ordinary product equals their Hadamard product. As they are orthogonal projections onto the disjoint subspaces V_i^* and V_j^* , their product is the zero matrix when $i \neq j$, and E_i^* when $i = j$.

(iii) Being an orthogonal projection matrix, the eigenvalues of E_i^* are strictly 0 and 1. The algebraic multiplicity of the eigenvalue 1 is equal to the rank of the projection, which is exactly the dimension of the target subspace V_i^* . Since $\dim(V_i^*) = |\Gamma_i(x_0)| = k_i$, the multiplicity is k_i . ■

With these bases established, we can formally define the dual algebra.

Definition 3.51. The subalgebra $\mathfrak{M}^* = \mathfrak{M}^*(x_0) = \text{span} \{A_0^*, A_1^*, \dots, A_d^*\}$ of the full matrix algebra $M_X(\mathbb{C})$ is called the **dual Bose-Mesner algebra** with respect to x_0 .

Proposition 3.52. *The dual Bose-Mesner algebra $\mathfrak{M}^*$ has the following structural properties:*

- (i) $\mathfrak{M}^* = \text{span} \{E_0^*, E_1^*, \dots, E_d^*\}$.
- (ii) The mapping $|X| E_i \mapsto A_i^*$ is an algebra isomorphism from $\mathfrak{M}^\circ$ (the Bose-Mesner algebra equipped with the Hadamard product) to $\mathfrak{M}^*$.
- (iii) The mapping $A_i^* \mapsto B_i^*$ is an algebra isomorphism from $\mathfrak{M}^*$ to $\mathcal{B}^*$ (the algebra generated by the dual intersection matrices).

Proof. (i) The transition matrices between the set $\{A_0^*, \dots, A_d^*\}$ and $\{E_0^*, \dots, E_d^*\}$ are the character tables P and Q , up to scaling. Since P and Q are non-singular, the two sets span the exact same vector space.

(ii) In the algebra $\mathfrak{M}^\circ$, the identity element is $J = |X| E_0$, and the matrices A_i serve as the mutually orthogonal primitive idempotents since $A_i \circ A_j = \delta_{i,j} A_i$. Correspondingly, in $\mathfrak{M}^*$, the primitive idempotents are E_i^* . The base change formula gives $|X| E_i = \sum_{j=0}^d Q_i(j) A_j$, whilst we defined $A_i^* = \sum_{j=0}^d Q_i(j) E_j^*$. Thus, the linear mapping defined by extending $|X| E_i \mapsto A_i^*$ maps the basis $\{A_j\}$ of primitive idempotents in $\mathfrak{M}^\circ$ directly to the basis $\{E_j^*\}$ of primitive idempotents in $\mathfrak{M}^*$. Because it maps a complete set of orthogonal idempotents to another complete set of orthogonal idempotents, it perfectly preserves the multiplicative structure, establishing an algebra isomorphism.

(iii) We have already established in Theorem 3.45 that the mapping $|X| E_i \mapsto B_i^*$ is an algebra isomorphism from $\mathfrak{M}^\circ$ to $\mathcal{B}^*$. Composing the inverse of the isomorphism from (ii) with this mapping yields the natural isomorphism $A_i^* \mapsto B_i^*$ between $\mathfrak{M}^*$ and $\mathcal{B}^*$. $\blacksquare$

Note that since all matrices in $\mathfrak{M}^*$ are diagonal, the Hadamard product and the ordinary matrix product coincide within $\mathfrak{M}^*$. Consequently, whilst we constructed $\mathfrak{M}^*$ specifically to mirror the Hadamard structure $\mathfrak{M}^\circ$, it functions perfectly as a standard matrix algebra under standard multiplication.

3.7 Duality and Fusion

We now explore two important structural concepts in the theory of association schemes: duality, which formalises the symmetry between the ordinary and Hadamard products of the Bose-Mesner algebra, and fusion, which deals with coarse-graining the relations of a scheme to form a smaller, simpler scheme.

Let $X = (X, \{R_i\}_{0 \leq i \leq d})$ be a commutative association scheme and let $\mathfrak{M}$ be its Bose-Mesner algebra.

Definition 3.53. A linear isomorphism $\Psi : \mathfrak{M} \rightarrow \mathfrak{M}$ is called a **duality** of $\mathfrak{M}$ if the following hold for any $M, N \in \mathfrak{M}$:

- (i) $\Psi(MN) = \Psi(M) \circ \Psi(N)$
- (ii) $\Psi^2(M) = |X| M^\top$

Definition 3.54. If $\mathfrak{M}$ has a duality, we say that the Bose-Mesner algebra $\mathfrak{M}$ and the association scheme X are **self-dual**.

Proposition 3.55. The Hamming scheme $H(d, q)$ and the association scheme constructed from the conjugacy classes of a finite abelian group G are self-dual.

The proof of this result will be deferred to a later section.

Before proving the main equivalence for self-dual schemes, we establish some basic operational properties of the duality map.

Proposition 3.56. Let Ψ be a duality of the Bose-Mesner algebra $\mathfrak{M}$. Then for any $M, N \in \mathfrak{M}$, the following hold:

$$\Psi(M^\top) = \Psi(M)^\top \quad \text{and} \quad \Psi(M \circ N) = \frac{1}{|X|} \Psi(M) \Psi(N)$$

Proof. First, observe how Ψ maps the identities of the two products. For any $M \in \mathfrak{M}$, we have $\Psi(M) = \Psi(MI) = \Psi(M) \circ \Psi(I)$. Because Ψ is an isomorphism, its image spans all of $\mathfrak{M}$. The only element that acts as the identity for the Hadamard product across all of $\mathfrak{M}$ is the all-ones matrix J . Thus, $\Psi(I) = J$. Consequently, applying Ψ again yields $\Psi(J) = \Psi^2(I) = |X| I^\top = |X| I$.

To prove the property regarding the Hadamard product, we apply Ψ^2 to $M \circ N$:

$$\Psi^2(M \circ N) = |X| (M \circ N)^\top = |X| (M^\top \circ N^\top)$$

Substituting $M^\top = \frac{1}{|X|} \Psi^2(M)$ and $N^\top = \frac{1}{|X|} \Psi^2(N)$ into the right side gives:

$$|X| \left(\frac{1}{|X|} \Psi^2(M) \circ \frac{1}{|X|} \Psi^2(N) \right) = \frac{1}{|X|} (\Psi^2(M) \circ \Psi^2(N)) = \frac{1}{|X|} \Psi(\Psi(M)\Psi(N))$$

Since Ψ is injective, we can drop the outermost Ψ from both sides, yielding $\Psi(M \circ N) = \frac{1}{|X|} \Psi(M)\Psi(N)$.

For the transpose, we simply chain the properties of Ψ^2 :

$$\Psi(M^\top) = \Psi \left(\frac{1}{|X|} \Psi^2(M) \right) = \frac{1}{|X|} \Psi^2(\Psi(M)) = \frac{1}{|X|} (|X| \Psi(M)^\top) = \Psi(M)^\top$$

■

Theorem 3.57. *For a commutative association scheme X , the following are equivalent:*

- (1) X is self-dual.
- (2) We can rearrange the order of the relations $R_0, R_1, \dots, R_d$ so that the character tables P and Q satisfy $P = \overline{Q}$.

Proof. (1) $\implies$ (2): Let Ψ be a duality of $\mathfrak{M}$. We established that $\Psi(E_0) = \Psi(\frac{1}{|X|}J) = \frac{1}{|X|}(|X|I) = A_0$. Furthermore, since E_i is idempotent, $E_i = E_i^2$, which implies $\Psi(E_i) = \Psi(E_i) \circ \Psi(E_i)$. An element is idempotent under the Hadamard product if and only if its entries are exclusively 0 or 1.

Because Ψ is an isomorphism, the set $\{\Psi(E_0), \dots, \Psi(E_d)\}$ forms a basis for $\mathfrak{M}$ consisting entirely of $(0, 1)$ -matrices that sum to $\Psi(I) = J$. This basis must therefore exactly be the set of adjacency matrices $\{A_0, \dots, A_d\}$. By rearranging the indices of the relations, we can assume without loss of generality that $\Psi(E_i) = A_i$ for all i .

Now, substitute the base change expansion $E_i = \frac{1}{|X|} \sum_{j=0}^d Q_i(j) A_j$:

$$A_i = \Psi(E_i) = \frac{1}{|X|} \sum_{j=0}^d Q_i(j) \Psi(A_j)$$

Since $\Psi(E_j) = A_j$, we have $\Psi(A_j) = \Psi^2(E_j) = |X| E_j^\top = |X| \overline{E_j}$. This reduces our equation to $A_i = \sum_{j=0}^d Q_i(j) \overline{E_j}$. Because A_i is a real $(0, 1)$ -matrix, it equals its own complex conjugate, so $A_i = \overline{A_i} = \sum_{j=0}^d \overline{Q_i(j)} E_j$. Comparing this directly to the definition $A_i = \sum_{j=0}^d P_i(j) E_j$, we conclude that $P_i(j) = \overline{Q_i(j)}$, meaning $P = \overline{Q}$.

(2) $\implies$ (1): Suppose there exists an ordering of the relations such that $P = \overline{Q}$. Define a linear map $\Psi : \mathfrak{M} \rightarrow \mathfrak{M}$ by setting $\Psi(E_i) = A_i$.

Applying Ψ to A_i , we get:

$$\Psi(A_i) = \Psi \left(\sum_{j=0}^d P_i(j) E_j \right) = \sum_{j=0}^d P_i(j) A_j = \sum_{j=0}^d \overline{Q_i(j)} A_j = |X| \overline{E_i} = |X| E_i^\top$$

From this, $\Psi^2(E_i) = \Psi(A_i) = |X| E_i^\top$, which extends by linearity to $\Psi^2(M) = |X| M^\top$ for all $M \in \mathfrak{M}$. Finally, checking the product condition on the basis elements gives:

$$\Psi(E_i E_j) = \delta_{i,j} \Psi(E_i) = \delta_{i,j} A_i = A_i \circ A_j = \Psi(E_i) \circ \Psi(E_j)$$

Since $\{E_i\}$ spans the algebra, this implies $\Psi(MN) = \Psi(M) \circ \Psi(N)$ for all $M, N \in \mathfrak{M}$, confirming Ψ is indeed a duality. $\blacksquare$

We now turn to the concept of fusion schemes. A fusion scheme arises when we can ‘clump’ or union certain relations together to form a coarser structure that remains an association scheme.

Let $X = (X, \{R_i\}_{0 \leq i \leq d})$ be a commutative association scheme.

Definition 3.58. Consider a partition of the index set $\{0, 1, \dots, d\}$ given by:

$$\{0, 1, \dots, d\} = \{0\} \cup \Lambda_1 \cup \dots \cup \Lambda_{\tilde{d}}$$

Define a new set of relations by grouping the original ones according to this partition:

$$\tilde{R}_0 = R_0, \quad \tilde{R}_i = \bigcup_{j \in \Lambda_i} R_j \quad (\text{for } 1 \leq i \leq \tilde{d})$$

Definition 3.59. Using the grouped relations defined previously, consider the resulting structure on the same set of vertices:

$$\tilde{X} = (X, \{\tilde{R}_i\}_{0 \leq i \leq \tilde{d}})$$

If $\tilde{X}$ is itself an association scheme, we call it a **fusion scheme** of X .

In some of the literature, fusion schemes are referred to as subschemes. We maintain the term fusion scheme here to distinguish it from actual subschemes, which are defined by restricting relations to a subset of the vertices rather than grouping the relations themselves.

The following powerful result gives us necessary and sufficient conditions for a fusion scheme to exist, relying directly on the character table.

Theorem 3.60 (Bannai-Muzychuk Criterion). *Let X be a commutative association scheme. The structure $\tilde{X}$ is a fusion scheme of X if and only if the following conditions hold:*

- (1) *For each i , the transpose relation satisfies $\tilde{R}_i^\top \in \{\tilde{R}_0, \tilde{R}_1, \dots, \tilde{R}_{\tilde{d}}\}$.*
- (2) *There exists another partition of the index set $\{0, 1, \dots, d\} = \{0\} \cup F_1 \cup \dots \cup F_{\tilde{d}}$ such that the primitive idempotents of the fusion scheme are exactly $\tilde{E}_0, \tilde{E}_1, \dots, \tilde{E}_{\tilde{d}}$, where $\tilde{E}_0 = E_0$ and $\tilde{E}_i = \sum_{j \in F_i} E_j$.*
- (3) *Let $P|_{F_k \times \Lambda_\ell}$ be the submatrix of the character table P of X obtained by restricting the rows to F_k and the columns to Λ_ℓ .*

Then this submatrix has a constant row sum. Namely, for any $i \in F_k$, the sum

$$\sum_{j \in \Lambda_\ell} P_i(j)$$

is constant and independent of the choice of i .

Proof. Let $\tilde{A}_\ell = \sum_{j \in \Lambda_\ell} A_j$ for $0 \leq \ell \leq \tilde{d}$. These are the candidate adjacency matrices for the fusion scheme.

(*Necessity*): Suppose $\tilde{X}$ is a fusion scheme. It is an association scheme in its own right, so its relations must be closed under transposition, proving (1). The matrices $\tilde{A}_0, \dots, \tilde{A}_{\tilde{d}}$ form a basis for its Bose-Mesner algebra $\tilde{\mathfrak{M}}$. Since $\tilde{\mathfrak{M}} \subseteq \mathfrak{M}$, the primitive idempotents of $\tilde{\mathfrak{M}}$, denoted $\tilde{E}_0 = E_0, \tilde{E}_1, \dots, \tilde{E}_{\tilde{d}}$, must be sums of subsets of the primitive idempotents of $\mathfrak{M}$. Thus, they partition the idempotents E_j , proving (2).

Let $\tilde{P}$ be the first eigenmatrix of the fusion scheme $\tilde{X}$. Then $\tilde{A}_\ell$ acts on the target space of $\tilde{E}_k$ as the scalar $\tilde{P}_\ell(k)$, so $\tilde{A}_\ell \tilde{E}_k = \tilde{P}_\ell(k) \tilde{E}_k$. Expanding the definitions of both matrices in terms of the original scheme gives:

$$\left(\sum_{j \in \Lambda_\ell} A_j \right) \left(\sum_{i \in F_k} E_i \right) = \sum_{i \in F_k} \sum_{j \in \Lambda_\ell} A_j E_i = \sum_{i \in F_k} \left(\sum_{j \in \Lambda_\ell} P_j(i) \right) E_i$$

Equating this with $\tilde{P}_\ell(k) \sum_{i \in F_k} E_i$ reveals that the coefficient of each linearly independent E_i must match. Thus, for any $i \in F_k$, the sum $\sum_{j \in \Lambda_\ell} P_j(i)$ must exactly equal the scalar $\tilde{P}_\ell(k)$. Since this scalar depends only on ℓ and k , the sum is constant across all $i \in F_k$, proving (3).

(*Sufficiency*): Assume conditions (1), (2), and (3) hold. The matrices $\{\tilde{A}_0, \dots, \tilde{A}_{\tilde{d}}\}$ inherently sum to J and start with $\tilde{A}_0 = I$. Condition (1) guarantees this basis is closed under transposition. We need only show they generate an algebra closed under matrix multiplication.

Let $\tilde{P}_\ell(k)$ denote the constant row sum from condition (3) for $i \in F_k$. We expand $\tilde{A}_\ell$ using the original character table:

$$\tilde{A}_\ell = \sum_{j \in \Lambda_\ell} A_j = \sum_{j \in \Lambda_\ell} \sum_{v=0}^d P_j(v) E_v = \sum_{k=0}^{\tilde{d}} \sum_{v \in F_k} \sum_{j \in \Lambda_\ell} P_j(v) E_v$$

By condition (3), the innermost sum is the constant $\tilde{P}_\ell(k)$, allowing us to pull it out:

$$\tilde{A}_\ell = \sum_{k=0}^{\tilde{d}} \tilde{P}_\ell(k) \left(\sum_{v \in F_k} E_v \right) = \sum_{k=0}^{\tilde{d}} \tilde{P}_\ell(k) \tilde{E}_k$$

This demonstrates that the vector space spanned by $\{\tilde{A}_\ell\}$ is identical to the vector space spanned by the partitioned idempotents $\{\tilde{E}_k\}$. Because the $\tilde{E}_k$ are disjoint sums of orthogonal idempotents, they are themselves orthogonal idempotents. The span of orthogonal idempotents is trivially closed under ordinary matrix multiplication. Therefore, the span of $\{\tilde{A}_\ell\}$ is a proper matrix algebra, confirming $\tilde{X}$ satisfies all properties of an association scheme. ■

3.8 Metric Structures: P-Polynomial Schemes

Distance-regular graphs represent a beautiful fusion of algebraic combinatorics and graph theory. They strictly generalise strongly regular graphs to arbitrary diameters by demanding that the local metric structure between any two vertices depends only on the distance between them.

Definition 3.61. A connected graph $\Gamma = (V, E)$ is **distance-regular** if for any vertices $v, w \in V$ at distance i , the number of vertices at distance j from v and distance k from w is constant and independent of the specific choice of v and w .

This definition can be simplified by focusing purely on the immediate neighbourhood of one vertex.

Definition 3.62. Equivalently, a regular graph Γ of diameter d is **distance-regular** if for any integer $1 \leq i \leq d$ and any $x, y \in V$ with shortest-path distance $\partial(x, y) = i$, the following intersection sizes are constant:

- (i) $a_i = |\{z \in V \mid \partial(x, z) = i, \partial(y, z) = 1\}|$
- (ii) $b_i = |\{z \in V \mid \partial(x, z) = i + 1, \partial(y, z) = 1\}|$
- (iii) $c_i = |\{z \in V \mid \partial(x, z) = i - 1, \partial(y, z) = 1\}|$

The proof that the above two definitions are equivalent can be found in [9] or also in [13].

In plain terms, if you stand at a vertex y which is at distance i from a base vertex x , your immediate neighbours (distance 1 from y) are strictly partitioned into three groups: those that move you closer to x (c_i), those that keep you at the same distance (a_i), and those that move you further away (b_i). Since Γ is regular of valency k , it must always hold that $a_i + b_i + c_i = k$. Also, $c_0 = 0$ and $b_d = 0$ by definition.

Definition 3.63. For a distance-regular graph, the constants a_i, b_i, c_i can be arranged into an *intersection matrix* of size $(d+1) \times (d+1)$:

$$L = \begin{bmatrix} 0 & c_1 & c_2 & \cdots & c_{d-1} & c_d \\ k & a_1 & a_2 & \cdots & a_{d-1} & a_d \\ 0 & b_1 & b_2 & \cdots & b_{d-1} & 0 \end{bmatrix}$$

Theorem 3.64. *The sum of the entries in any column of the intersection matrix is exactly k , the valency of the graph.*

Proof. Because every neighbour of y must fall into exactly one of the three distance categories with respect to x , the sum $a_i + b_i + c_i$ strictly counts the total degree of y . Since the graph is distance-regular, it is inherently regular, so this degree is the constant k . ■

Proposition 3.65. *A strongly regular graph $\text{SRG}(n, k, \lambda, \mu)$ is exactly a distance-regular graph of diameter 2 with intersection matrix:*

$$\begin{bmatrix} 0 & 1 & \mu \\ k & \lambda & k - \mu \\ 0 & k - \lambda - 1 & 0 \end{bmatrix}$$

Proof. A strongly regular graph has diameter 2. If $\partial(x, y) = 1$ (adjacent), the number of common neighbours is λ , which corresponds to a_1 . The remaining neighbours of y must be at distance 2 from x , so $b_1 = k - 1 - \lambda$. Also, $c_1 = 1$ always. If $\partial(x, y) = 2$ (non-adjacent), the number of common neighbours is μ , which corresponds to c_2 . The remaining neighbours of y stay at distance 2 from x , so $a_2 = k - \mu$. By definition $b_2 = 0$. Plugging these into the intersection matrix structure gives the desired matrix. ■

We now formally abstract the metric properties of a distance-regular graph into the framework of the Bose-Mesner algebra.

Definition 3.66. Let $X = (X, \{R_i\}_{0 \leq i \leq d})$ be a symmetric association scheme. X is called a ***P-polynomial scheme*** with respect to the ordering $R_0, R_1, \dots, R_d$ if the following condition holds:

For each $0 \leq i \leq d$, there exists a polynomial $v_i(x)$ of degree i such that:

$$A_i = v_i(A_1)$$

This definition is extraordinarily powerful. It implies that the entire $(d+1)$ -dimensional Bose-Mesner algebra is generated by powers of a single matrix A_1 .

The following theorem builds the fundamental bridge between the algebraic structure of P-polynomial schemes and the combinatorial structure of distance-regular graphs.

Theorem 3.67. *We can go between DRGs and P-polynomial schemes as follows.*

- (1) *Let X be a P-polynomial scheme. The graph $\Gamma = (X, E)$ with edge set $E = R_1$ is a distance-regular graph.*
- (2) *Conversely, let $\Gamma = (X, E)$ be a distance-regular graph of diameter d . If we define relations based on the shortest-path distance as $R_i = \{(x, y) \mid \partial(x, y) = i\}$ for each $0 \leq i \leq d$, then $X = (X, \{R_i\}_{0 \leq i \leq d})$ is a P-polynomial scheme.*

Proof. (1) Let X be a P-polynomial scheme with respect to the ordering $R_0, R_1, \dots, R_d$. By definition, $A_i = v_i(A_1)$ for some polynomial v_i of degree i . Since A_i is a polynomial of degree exactly i in the adjacency matrix A_1 of the graph $\Gamma = (X, R_1)$, the entry $(A_i)_{x,y} \neq 0$ implies there is a walk of length i between x and y in Γ . Furthermore, because the set of matrices $\{A_0, A_1, \dots, A_i\}$ spans the space of polynomials up to degree i in A_1 , we have $(A_i)_{x,y} = 1$ if and only if the shortest path between x and y has exactly length i . Thus, the relation R_i coincides perfectly with the distance- i relation in Γ .

Since X is an association scheme, the intersection number $p_{1,j}^k = |\{z \mid (x, z) \in R_1, (z, y) \in R_j\}|$ is a constant for any choice of pair $(x, y) \in R_k$. Translated to the graph Γ , this means for any vertices x, y at distance k , the number of neighbours of x that are at distance j from y is constant, independent of the chosen vertices. This precisely satisfies the definition of a distance-regular graph.

(2) Let $\Gamma = (X, E)$ be a DRG of diameter d . By defining the relations $R_i = \{(x, y) \mid \partial(x, y) = i\}$, the distance-regularity constants exactly guarantee the existence of constant intersection numbers $p_{i,j}^k$, making $(X, \{R_i\}_{0 \leq i \leq d})$ a symmetric association scheme.

To see that it is P-polynomial, consider the matrix product $A_1 A_i$. From the perspective of a vertex y at distance i from a vertex x , the neighbours of y must be at distance $i-1$, i , or $i+1$ from x . The number of such neighbours are precisely the intersection array parameters c_i, a_i , and b_i . In matrix form, this yields the three-term recurrence:

$$A_1 A_i = b_{i-1} A_{i-1} + a_i A_i + c_{i+1} A_{i+1}$$

where $c_{i+1} \neq 0$ because the graph has diameter d . Starting with $A_0 = I$ and $A_1 = A_1$, we can recursively rearrange this equation to express A_{i+1} as a polynomial of degree exactly $i+1$ in A_1 . Therefore, $A_i = v_i(A_1)$ for all $0 \leq i \leq d$, confirming that X is a P-polynomial scheme. ■

Theorem 3.68. *Let X be a symmetric association scheme. The following are equivalent:*

- (1) X is a P-polynomial scheme.
- (2) The intersection matrix B_1 (where $(B_1)_{j,k} = p_{1,j}^k$) is a tridiagonal matrix with non-zero off-diagonal entries $b_i \neq 0$ and $c_i \neq 0$.
- (3) If P is the first eigenmatrix and $\theta_j = P_1(j)$ are the eigenvalues of A_1 , then for each $0 \leq i \leq d$, there exists a polynomial $v_i(x)$ of degree i such that $P_i(j) = v_i(\theta_j)$.

Proof. (1) $\implies$ (2): By definition, $A_i = v_i(A_1)$ for some degree i polynomial v_i . Since $xv_i(x)$ is a polynomial of degree $i+1$, we can express it as a linear combination $\sum_{\ell=0}^{i+1} c_\ell v_\ell(x)$ where $c_{i+1} \neq 0$. Substituting A_1 , we get $A_1 A_i = A_1 v_i(A_1) = \sum_{\ell=0}^{i+1} c_\ell A_\ell$. The left side is by definition $\sum_{\ell=0}^d p_{1,i}^\ell A_\ell$. Equating coefficients, $p_{1,i}^\ell = 0$ for all $\ell \geq i+2$, and $p_{1,i}^{i+1} = c_{i+1} \neq 0$. By the symmetry $k_\ell p_{1,i}^\ell = k_i p_{1,\ell}^i$, we also obtain $p_{1,i}^\ell = 0$ for $\ell \leq i-2$. Thus, B_1 is strictly tridiagonal.

(2) $\implies$ (1): Because B_1 is tridiagonal, the expansion of $A_1 A_i$ simplifies to a three-term recurrence: $A_1 A_i = p_{1,i}^{i-1} A_{i-1} + p_{1,i}^i A_i + p_{1,i}^{i+1} A_{i+1}$. We construct the polynomials iteratively. Set $v_0(x) = 1$ and $v_1(x) = x$. Using the recurrence $xv_i(x) = p_{1,i}^{i-1} v_{i-1}(x) + p_{1,i}^i v_i(x) + p_{1,i}^{i+1} v_{i+1}(x)$, we can uniquely define $v_{i+1}(x)$ as a polynomial of degree $i+1$, since $p_{1,i}^{i+1} = b_i \neq 0$. By induction, $A_{i+1} = v_{i+1}(A_1)$.

(1) $\iff$ (3): Substitute the spectral decomposition $A_1 = \sum_j \theta_j E_j$ into the polynomial relation $A_i = v_i(A_1)$. Because the E_j are orthogonal idempotents, evaluating the polynomial simply evaluates it on the scalar eigenvalues: $v_i(A_1) = \sum_j v_i(\theta_j) E_j$. Equating this with $A_i = \sum_j P_i(j) E_j$ gives $P_i(j) = v_i(\theta_j)$. ■

Proposition 3.69. *The valencies k_i of the relations R_i can be computed recursively from the intersection array. Specifically, $k_0 = 1$, $k_1 = k$, and for $0 \leq i \leq d-1$:*

$$k_{i+1} = \frac{k_i b_i}{c_{i+1}}$$

Proof. Fix a vertex x . We count the edges between the sets $\Gamma_i(x)$ and $\Gamma_{i+1}(x)$ in two ways. Every vertex in $\Gamma_i(x)$ has exactly b_i neighbours in $\Gamma_{i+1}(x)$, contributing $k_i b_i$ edges. Conversely, every vertex in $\Gamma_{i+1}(x)$ has

exactly c_{i+1} neighbours in $\Gamma_i(x)$, contributing $k_{i+1}c_{i+1}$ edges. Equating the two counts yields $k_{i+1}c_{i+1} = k_i b_i$. ■

Proposition 3.70. *Let X be a P -polynomial scheme. Then the eigenvalues $\theta_0, \theta_1, \dots, \theta_d$ of the adjacency matrix A_1 are mutually distinct real numbers.*

Proof. Suppose for contradiction that $\theta_j = \theta_\ell$ for some distinct indices j and ℓ . By Theorem 3.68 (3), we have $P_i(j) = v_i(\theta_j) = v_i(\theta_\ell) = P_i(\ell)$ for all relations $0 \leq i \leq d$. This implies that the j -th column and the ℓ -th column of the character table P are identical. However, the first eigenmatrix P represents a change of basis and is strictly non-singular. A non-singular matrix cannot have identical columns, establishing the contradiction. ■

3.9 The Character Table of the Hamming Scheme

Recall from classical coding theory the definition of the Kravchuk polynomials. This is essentially Definition 3.71. We recall it for convenience and possible changes of notations which may have been introduced.

Definition 3.71. For any prime power q and $d \in \mathbb{Z}^+$, we define the **k -th Kravchuk polynomial** as

$$K_k(x; d, q) = \sum_{j=0}^k (-1)^j (q-1)^{k-j} \binom{x}{j} \binom{d-x}{k-j}$$

for $k = 0, 1, \dots, d$.

The following have already been dealt with as Proposition 3.72.

Proposition 3.72. *Let $K_k(x)$ denote the Kravchuk polynomial $K_k(x; d, q)$. These polynomials satisfy the following dual orthogonality relations:*

$$(i) \sum_{i=0}^d K_j(i) K_i(k) = q^d \delta_{j,k}$$

$$(ii) \sum_{i=0}^d \mu(i) K_j(i) K_k(i) = q^d \mu(j) \delta_{j,k}$$

where the weight function is given by $\mu(i) = \binom{d}{i} (q-1)^i$.

The weight $\mu(i)$ corresponds exactly to the valency k_i of the i -th relation in the Hamming scheme $H(d, q)$.

To evaluate the character tables, we exploit the natural group structure underlying the Hamming scheme. We defined $H(d, q)$ on the set $X = \mathbb{F}_q^d$. By viewing $\mathbb{F}_q$ as an additive abelian group, X becomes an additive abelian group as well.

The weight of an element $\mathbf{x} \in X$ is the number of its non-zero coordinates: $w(\mathbf{x}) = |\{i \mid x_i \neq 0\}|$. The relations are then purely translation-invariant: $R_i = \{(\mathbf{x}, \mathbf{y}) \mid w(\mathbf{x} - \mathbf{y}) = i\}$. Let X_i denote the set of elements of weight i , so that $k_i = |X_i| = \binom{d}{i} (q-1)^i$.

Let $\hat{\mathbb{F}}_q = \{\psi_c \mid c \in \mathbb{F}_q\}$ be the character group of $\mathbb{F}_q$. We can index these characters such that $\psi_c(x) = \psi_x(c)$. The characters of the product group X are then given by the tensor product of the component characters. For $\mathbf{u} \in X$, we define the character $\Psi_{\mathbf{u}} : X \rightarrow \mathbb{C}^\times$ by:

$$\Psi_{\mathbf{u}}(\mathbf{x}) = \prod_{\nu=1}^d \psi_{u_\nu}(x_\nu)$$

Lemma 3.73. *Let $\mathbf{u} \in X_j$ be an element of weight j . The sum of the character $\Psi_{\mathbf{u}}$ over all elements of weight k is exactly the Kravchuk polynomial $K_k(j; d, q)$:*

$$\sum_{\mathbf{x} \in X_k} \Psi_{\mathbf{u}}(\mathbf{x}) = K_k(j)$$

Proof. We partition the set X_k based on the exact support (the set of non-zero coordinates) of its elements. For any subset $M \subseteq \{1, \dots, d\}$ of size k , let $X_M = \{\mathbf{x} \in X \mid x_\nu \neq 0 \iff \nu \in M\}$. Then X_k is the disjoint union of X_M over all $\binom{d}{k}$ such subsets.

For a fixed M , we evaluate the sum over X_M . Since $x_\nu = 0$ for $\nu \notin M$, and $\psi_c(0) = 1$, the product defining the character restricts to M :

$$\sum_{\mathbf{x} \in X_M} \Psi_{\mathbf{u}}(\mathbf{x}) = \sum_{\mathbf{x} \in X_M} \prod_{\nu \in M} \psi_{u_\nu}(x_\nu) = \prod_{\nu \in M} \left(\sum_{c \in \mathbb{F}_q^\times} \psi_{u_\nu}(c) \right)$$

By the orthogonality of characters on the group $\mathbb{F}_q$, the sum of ψ_{u_ν} over all of $\mathbb{F}_q$ is q if $u_\nu = 0$, and 0 otherwise. Since $\psi_{u_\nu}(0) = 1$, subtracting this from the total sum gives the sum over the non-zero elements $\mathbb{F}_q^\times$:

$$\sum_{c \in \mathbb{F}_q^\times} \psi_{u_\nu}(c) = \begin{cases} q-1 & \text{if } u_\nu = 0 \\ -1 & \text{if } u_\nu \neq 0 \end{cases}$$

Let i be the number of coordinates in M where $\mathbf{u}$ is non-zero, i.e., $i = |\{\nu \in M \mid u_\nu \neq 0\}|$. The product over M then evaluates to $(-1)^i (q-1)^{k-i}$.

To sum over all possible supports M of size k , we group them by this value i . Since $\mathbf{u}$ has total weight j , choosing a support M that overlaps with the support of $\mathbf{u}$ in exactly i places corresponds to choosing i elements from the j non-zero coordinates of $\mathbf{u}$, and the remaining $k-i$ elements from the $d-j$ zero coordinates of $\mathbf{u}$. The number of ways to do this is $\binom{j}{i} \binom{d-j}{k-i}$.

Summing over all possible values of the overlap i yields:

$$\sum_{\mathbf{x} \in X_k} \Psi_{\mathbf{u}}(\mathbf{x}) = \sum_{i=0}^k \binom{j}{i} \binom{d-j}{k-i} (-1)^i (q-1)^{k-i}$$

which is precisely the definition of $K_k(j; d, q)$. ■

The foundational connection between classical coding theory and association schemes is established by the following theorem.

Theorem 3.74. *Let P and Q be the first and second eigenmatrices of the Hamming scheme $H(d, q)$. Then their entries are exactly given by the Kravchuk polynomials evaluated at the integers:*

$$P_k(i) = Q_k(i) = K_k(i; d, q)$$

Proof. Define a matrix H_k of size $|X| \times |X_k|$ whose rows are indexed by X and columns by X_k , with entries given by the characters: $H_k(\mathbf{x}, \mathbf{y}) = \Psi_{\mathbf{x}}(\mathbf{y})$.

We evaluate the matrix product $H_k H_k^*$, which is a $|X| \times |X|$ matrix. Its $(\mathbf{x}, \mathbf{w})$ -entry is:

$$(H_k H_k^*)(\mathbf{x}, \mathbf{w}) = \sum_{\mathbf{y} \in X_k} \Psi_{\mathbf{x}}(\mathbf{y}) \overline{\Psi_{\mathbf{w}}(\mathbf{y})} = \sum_{\mathbf{y} \in X_k} \Psi_{\mathbf{x}}(\mathbf{y}) \Psi_{-\mathbf{w}}(\mathbf{y}) = \sum_{\mathbf{y} \in X_k} \Psi_{\mathbf{x}-\mathbf{w}}(\mathbf{y})$$

Suppose $(\mathbf{x}, \mathbf{w}) \in R_i$, which means the vector $\mathbf{x} - \mathbf{w}$ has weight i . By Lemma 3.73, the sum evaluates exactly to $K_k(i)$. This implies that the matrix $H_k H_k^*$ is constant on the relations R_i . We can rewrite it using the adjacency matrices A_i :

$$H_k H_k^* = \sum_{i=0}^d K_k(i) A_i$$

Now we evaluate $H_k^* H_j$, which is an $|X_k| \times |X_j|$ matrix. For $\mathbf{y} \in X_k$ and $\mathbf{z} \in X_j$:

$$(H_k^* H_j)(\mathbf{y}, \mathbf{z}) = \sum_{\mathbf{x} \in X} \overline{\Psi_{\mathbf{x}}(\mathbf{y})} \Psi_{\mathbf{x}}(\mathbf{z}) = \sum_{\mathbf{x} \in X} \Psi_{\mathbf{x}}(\mathbf{z} - \mathbf{y})$$

By the orthogonality of characters over the full group X , this sum is $|X|$ if $\mathbf{z} - \mathbf{y} = \mathbf{0}$, and 0 otherwise. Thus, $H_k^* H_j = \delta_{k,j} |X| I$.

Consider the matrix $E_j = \frac{1}{|X|} H_j H_j^*$. Then:

$$E_k E_j = \frac{1}{|X|^2} H_k (H_k^* H_j) H_j^* = \frac{1}{|X|^2} H_k (\delta_{k,j} |X| I) H_j^* = \delta_{k,j} \frac{1}{|X|} H_k H_k^* = \delta_{k,j} E_k$$

This proves that $E_0, \dots, E_d$ form a set of mutually orthogonal idempotents. Since they are linear combinations of the adjacency matrices, they must be the primitive idempotents of the Bose-Mesner algebra. Our expansion then reads:

$$E_k = \frac{1}{|X|} \sum_{i=0}^d K_k(i) A_i$$

Comparing this directly with the definition of the dual character table $E_k = \frac{1}{|X|} \sum_{i=0}^d Q_k(i) A_i$, we immediately obtain $Q_k(i) = K_k(i)$.

To find the first eigenmatrix $P_k(i)$, we use the relation $P_k(i) = \frac{k_k}{m_i} \overline{Q_i(k)}$. Since the Kravchuk polynomials are real, $Q_i(k) = K_i(k)$. From $m_i = Q_i(0) = K_i(0) = \binom{d}{i} (q-1)^i$, we see that $m_i = k_i$. Therefore:

$$P_k(i) = \frac{k_k}{k_i} K_i(k)$$

Applying the symmetry property of Kravchuk polynomials (which can be derived directly from the generating function or Proposition 3.72(ii)), we have $k_k K_i(k) = k_i K_k(i)$. Substituting this yields $P_k(i) = K_k(i)$, completing the proof. ■

Corollary 3.75. *Since $P = Q$, the Hamming scheme $H(d, q)$ is self-dual. Furthermore, it is a P-polynomial scheme.*

Proof. Because $P = Q = \overline{Q}$ (since the entries are real Kravchuk polynomials), the condition of Theorem 3.57 is satisfied with the trivial reordering, meaning the Hamming scheme is self-dual. This provides a proof for the first part of Proposition 3.55.

Moreover, because $P_i(j) = K_i(j)$ and $K_i(x)$ is a polynomial of degree exactly i , the matrix P satisfies the third condition of Theorem 3.68. This proves that $H(d, q)$ is a P-polynomial scheme. ■

The following completes the proof of Proposition 3.55.

Proposition 3.76. *The association scheme constructed from the conjugacy classes of a finite abelian group G is self-dual.*

Proof. Let G be a finite abelian group. Thus, its conjugacy classes are simply the singleton sets $C_g = \{g\}$ for each $g \in G$. Hence, the relations of the association scheme are $R_g = \{(x, y) \in G \times G \mid xy^{-1} = g\}$. Note that the number of relations is exactly $|G|$, and the valency of each relation is $k_g = 1$.

Let $\hat{G}$ be the character group of G , consisting of all irreducible linear characters $\chi : G \rightarrow \mathbb{C}^\times$. It is a standard result in the representation theory of finite abelian groups that $\hat{G} \cong G$. We can index the characters as $\hat{G} = \{\chi_h \mid h \in G\}$ such that the isomorphism respects the group structure. Furthermore, the multiplicities of these characters are all $m_\chi = 1$.

The first eigenmatrix P evaluates these characters. With rows indexed by the group elements $g \in G$ and columns indexed by the characters $\chi \in \hat{G}$, the entries are exactly $P_g(\chi) = \chi(g)$.

To find the entries of the second eigenmatrix Q , we use the standard parameter relation derived from the character tables:

$$Q_\chi(g) = \frac{m_\chi \overline{P_g(\chi)}}{k_g}$$

Since both the valencies k_g and the multiplicities m_χ are exactly 1 for all elements and characters, this simplifies to:

$$Q_\chi(g) = \overline{P_g(\chi)} = \overline{\chi(g)}$$

This demonstrates that the matrices P and Q satisfy the equation $P = \overline{Q}$ under the natural isomorphism between G and $\hat{G}$. By Theorem 3.57, this property guarantees that the association scheme is self-dual. ■

3.10 Codes through Association Schemes

In this section, we explore one of the most beautiful applications of association schemes: Delsarte's theory of codes. By viewing a code simply as a subset of vertices in an association scheme, we can unify and generalise many classical results from algebraic coding theory.

Let $X = (X, \{R_i\}_{0 \leq i \leq d})$ be a symmetric association scheme. In this framework, a **code** is simply a non-empty subset $Y \subseteq X$. The metric properties of this code are captured by examining how pairs of codewords are distributed across the relations of the scheme.

Definition 3.77. Let $Y \subseteq X$. The **inner distribution** of Y is the $(d+1)$ -dimensional row vector $a_Y = (a_0, a_1, \dots, a_d)$ whose entries are defined as:

$$a_i = \frac{1}{|Y|} |R_i \cap (Y \times Y)| \quad \text{for } i \in \{0, 1, \dots, d\}$$

In a P-polynomial scheme (like the Hamming scheme), a_i precisely counts the average number of codewords at distance i from a given codeword. This is exactly the classical distance distribution of the code.

Definition 3.78. Let Q be the second eigenmatrix of X . We define the **dual distribution** of the code Y as the row vector $a_Y^* = (a_0^*, a_1^*, \dots, a_d^*)$ given by:

$$a_Y^* = a_Y Q$$

At first glance, the dual distribution is a purely algebraic construct. However, a foundational result of Delsarte demonstrates that it behaves much like a combinatorial distribution.

Theorem 3.79. For any code $Y \subseteq X$ in a symmetric association scheme, every entry a_k^* of the dual distribution is a non-negative real number:

$$a_Y^* \geq 0$$

Proof. We introduce the **outer distribution matrix** B_Y of size $|X| \times (d+1)$, indexed by $x \in X$ and $i \in \{0, 1, \dots, d\}$. We define its entries as $B_Y(x, i) = |Y \cap \Gamma_i(x)|$. Let ψ_Y be the characteristic column vector of Y (so $\psi_Y(z) = 1$ if $z \in Y$ and 0 otherwise). Observe that $B_Y(x, i) = (A_i \psi_Y)_x$, meaning the i -th column of B_Y is exactly $A_i \psi_Y$.

Consider the matrix $B_Y^\top B_Y$. Its (i, j) -th entry is:

$$(B_Y^\top B_Y)_{i,j} = \psi_Y^\top A_i A_j \psi_Y = \sum_{k=0}^d p_{i,j}^k \psi_Y^\top A_k \psi_Y = |Y| \sum_{k=0}^d p_{i,j}^k a_k$$

Substituting the inverse relation $a_k = \frac{1}{|X|} \sum_{\ell=0}^d a_\ell^* P_k(\ell)$ yields:

$$(B_Y^\top B_Y)_{i,j} = \frac{|Y|}{|X|} \sum_{\ell=0}^d a_\ell^* \sum_{k=0}^d p_{i,j}^k P_k(\ell) = \frac{|Y|}{|X|} \sum_{\ell=0}^d a_\ell^* P_i(\ell) P_j(\ell)$$

where the second equality follows from Proposition 3.38 (vii). As matrices, $B_Y^\top B_Y = \frac{|Y|}{|X|} P^\top \Delta_{a_Y^*} P$, where $\Delta_{a_Y^*}$ is the diagonal matrix with the entries of a_Y^* on the diagonal.

Now, we pre- and post-multiply by $Q^\top$ and Q respectively. Since $PQ = |X|I$, we have $P^\top Q^\top = |X|I$. Thus:

$$Q^\top B_Y^\top B_Y Q = \frac{|Y|}{|X|} (P^\top Q^\top)^\top \Delta_{a_Y^*} (PQ) = |X| |Y| \Delta_{a_Y^*}$$

The matrix on the left is of the form $M^\top M$ for $M = B_Y Q$, which is necessarily positive semi-definite. Therefore, its diagonal entries must be non-negative. Equating the k -th diagonal entry gives $\sum_{x \in X} M(x, k)^2 = |X| |Y| a_k^*$. Since $|X|, |Y| > 0$ and a sum of squares is non-negative, we conclude $a_k^* \geq 0$. ■

Definition 3.80. Let a_Y and a_Y^* be the inner and dual distributions of a code Y . We define the following fundamental parameters:

- (i) **Minimum Distance:** $\delta = \min \{i \mid a_i \neq 0, i \geq 1\}$
- (ii) **Degree:** $s = |\{i \mid a_i \neq 0, i \geq 1\}|$
- (iii) **Dual Degree:** $s^* = |\{i \mid a_i^* \neq 0, i \geq 1\}|$

For an arbitrary vector $\mathbf{u} = (u_0, u_1, \dots, u_d)$ with $u_0 \neq 0$, let $t(\mathbf{u})$ denote the number of leading zeros after u_0 :

$$t(\mathbf{u}) = \max \{i \mid u_1 = u_2 = \dots = u_i = 0\}$$

and let $s(\mathbf{u})$ be the number of non-zero entries for $i \geq 1$.

Theorem 3.81 (MacWilliams Inequality). *Let X be a P -polynomial scheme and Q be its second eigenmatrix. For any vector $\mathbf{u}$ with $u_0 \neq 0$, we have:*

$$s(\mathbf{u}Q) \geq \left\lfloor \frac{t(\mathbf{u})}{2} \right\rfloor$$

Proof. Let $\theta_i = P_1(i)$. Since X is P -polynomial, there exist polynomials $v_k(z)$ of degree k such that $P_k(i) = v_k(\theta_i)$. Suppose, for the sake of contradiction, that $s(\mathbf{u}Q) < \lfloor t(\mathbf{u})/2 \rfloor$. Let $s = s(\mathbf{u}Q)$, and let the strictly positive indices where $(\mathbf{u}Q)$ is non-zero be $\{\nu_1, \dots, \nu_s\}$.

We construct an annihilating polynomial. Because $s < \lfloor t(\mathbf{u})/2 \rfloor$, we have $s + 1 \leq \lfloor t(\mathbf{u})/2 \rfloor$. Define the polynomial $f(z)$ of degree $s + 1$ by:

$$f(z) = (z - \theta_0) \prod_{j=1}^s (z - \theta_{\nu_j})$$

By construction, $f(\theta_j) = 0$ precisely when $j \in \{0, \nu_1, \dots, \nu_s\}$. The square $f(z)^2$ has degree $2s + 2 \leq 2\lfloor t(\mathbf{u})/2 \rfloor \leq t(\mathbf{u})$. Since the orthogonal polynomials $v_k(z)$ form a basis for the polynomial ring, $f(z)^2$ can be uniquely expanded as $f(z)^2 = \sum_{k=0}^{t(\mathbf{u})} b_k v_k(z)$.

By orthogonal projection, the leading coefficient is given by $b_0 = \frac{1}{|X|} \sum_{j=0}^d m_j f(\theta_j)^2$. Because $f(z)$ is not identically zero and the eigenvalues θ_j are mutually distinct (Proposition 3.70), $f(\theta_j)^2 > 0$ for at least one index j . Since the multiplicities m_j are strictly positive, this guarantees that $b_0 > 0$.

Now, consider the matrix product $\mathbf{u}Q P \mathbf{b}$, where $\mathbf{b} = (b_0, \dots, b_{t(\mathbf{u})}, 0, \dots, 0)^\top$. Since $QP = |X|I$, this evaluates algebraically to:

$$|X| \mathbf{u} \mathbf{b} = |X| \sum_{k=0}^{t(\mathbf{u})} u_k b_k$$

Because $u_k = 0$ for $1 \leq k \leq t(\mathbf{u})$ (by the definition of $t(\mathbf{u})$), the sum collapses entirely to $|X| u_0 b_0$.

Alternatively, evaluating $P\mathbf{b}$ yields a vector whose j -th entry is $\sum_k b_k P_k(j) = \sum_k b_k v_k(\theta_j) = f(\theta_j)^2$. Thus, the product can also be evaluated as:

$$\sum_{j=0}^d (\mathbf{u}Q)_j f(\theta_j)^2$$

Observe the terms of this sum. If $j \notin \{0, \nu_1, \dots, \nu_s\}$, then by definition $(\mathbf{u}Q)_j = 0$. If $j \in \{0, \nu_1, \dots, \nu_s\}$, then by our construction $f(\theta_j) = 0$. Therefore, every single term in the sum vanishes, meaning the product is exactly 0.

Equating the two evaluations gives $|X| u_0 b_0 = 0$. Since $|X| \neq 0$ and $u_0 \neq 0$, this implies $b_0 = 0$. However, we independently established that $b_0 > 0$. This provides the necessary contradiction, proving the theorem. ■

Applying this inequality to the inner distribution $\mathbf{u} = a_Y$ gives $t(a_Y) = \delta - 1$ and $s(a_Y Q) = s^*$. Thus, the inequality establishes that $s^* \geq \lfloor (\delta - 1)/2 \rfloor$, cleanly linking the dual degree to the code's error-correcting capability.

Let X be a P-polynomial scheme. We adopt the shortest-path distance $\partial(x, y)$ from its associated distance-regular graph (X, R_1) .

Let $Y \subseteq X$ be a code with minimum distance δ . We define its error-correcting capability as:

$$e = \left\lfloor \frac{\delta - 1}{2} \right\rfloor$$

Lemma 3.82. *The number of vertices in a ball of radius e centred at any point in X is strictly determined by the valencies of the scheme, and is equal to:*

$$\sum_{i=0}^e k_i$$

Proof. In a P-polynomial scheme, the relations R_i correspond precisely to the shortest-path distance i in the associated distance-regular graph. For any fixed vertex $x \in X$, the ball of radius e centred at x , denoted $B_e(x)$, is the disjoint union of the spheres of radius i for $0 \leq i \leq e$.

By definition, the sphere of radius i centred at x is exactly the neighbourhood $\Gamma_i(x) = \{y \in X \mid (x, y) \in R_i\}$. Recall that the valency of the relation R_i is $k_i = |\Gamma_i(x)|$, which is constant and independent of the choice of x .

Since the subsets $\Gamma_i(x)$ are pairwise disjoint for distinct distances i , the cardinality of the ball is simply the sum of the cardinalities of these concentric spheres:

$$|B_e(x)| = \left| \bigcup_{i=0}^e \Gamma_i(x) \right| = \sum_{i=0}^e |\Gamma_i(x)| = \sum_{i=0}^e k_i$$

■

Proposition 3.83 (Sphere Packing Bound). *Because balls of radius e centred at distinct codewords of Y must be strictly disjoint, we naturally obtain the bound:*

$$|Y| \sum_{i=0}^e k_i \leq |X|$$

Proof. Let $Y \subseteq X$ be a code with minimum distance δ , and let $e = \lfloor (\delta - 1)/2 \rfloor$. By the definition of minimum distance, for any two distinct codewords $y_1, y_2 \in Y$, the distance between them is at least $\delta \geq 2e + 1$.

Suppose, for the sake of contradiction, that the balls $B_e(y_1)$ and $B_e(y_2)$ intersect. Then there exists some vertex $z \in B_e(y_1) \cap B_e(y_2)$. By the triangle inequality in the metric space of the distance-regular graph, we have:

$$\partial(y_1, y_2) \leq \partial(y_1, z) + \partial(z, y_2) \leq e + e = 2e \leq \delta - 1$$

This contradicts the fact that the minimum distance is δ . Therefore, the balls $B_e(y)$ for all $y \in Y$ must be strictly pairwise disjoint.

The total number of vertices contained in the union of all these balls is the sum of their individual sizes. Since there are $|Y|$ such balls, and each contains exactly $\sum_{i=0}^e k_i$ vertices (by the previous lemma), the union contains exactly $|Y| \sum_{i=0}^e k_i$ vertices. Because this union is a subset of the entire space X , its size cannot exceed the total number of vertices in the space, $|X|$. Thus, the bound holds. ■

Definition 3.84. A code Y is called a **perfect e -code** if it achieves equality in the bound above.

Equivalently, Y is a perfect code if the balls of radius e centred at the codewords perfectly partition the entire space X .

Before stating the main theorem, we define a special polynomial constructed from the sequence of orthogonal polynomials that generate the P-polynomial scheme.

Definition 3.85. Let $v_i(z)$ be the polynomial of degree i such that $A_i = v_i(A_1)$. For a given integer e , we define the **Lloyd polynomial** of degree e as the sum:

$$\Psi_e(z) = v_0(z) + v_1(z) + \cdots + v_e(z)$$

Proposition 3.86. For the Hamming scheme $H(d, q)$, the orthogonal polynomials $v_i(z)$ that define the P-polynomial structure are exactly the Kravchuk polynomials up to a linear transformation of the domain. Specifically:

$$v_i(z) = K_i \left(\frac{(q-1)d - z}{q}; d, q \right)$$

Proof. By Theorem 3.68, the polynomials $v_i(z)$ defining the P-polynomial scheme satisfy the evaluation identity $P_i(j) = v_i(\theta_j)$, where $\theta_j = P_1(j)$ are the distinct eigenvalues of A_1 .

For the Hamming scheme $H(d, q)$, Theorem 3.74 established that the first eigenmatrix entries are exactly the Kravchuk polynomials evaluated at the distance indices: $P_i(j) = K_i(j; d, q)$.

Equating these two expressions, we must have $v_i(\theta_j) = K_i(j; d, q)$.

We explicitly compute the eigenvalue $\theta_j = K_1(j; d, q)$. Applying Definition 3.71 for $k = 1$, we get:

$$\begin{aligned} \theta_j &= \sum_{r=0}^1 (-1)^r (q-1)^{1-r} \binom{j}{r} \binom{d-j}{1-r} \\ &= (-1)^0 (q-1)^1 \binom{j}{0} \binom{d-j}{1} + (-1)^1 (q-1)^0 \binom{j}{1} \binom{d-j}{0} \\ &= (q-1)(d-j) - j \\ &= (q-1)d - qj \end{aligned}$$

Thus, $\theta_j = (q-1)d - qj$. Treating the eigenvalue as the continuous variable $z = \theta_j$ and inverting the relation to solve for j yields:

$$j = \frac{(q-1)d - z}{q}$$

Substituting this back into our evaluation identity gives exactly:

$$v_i(z) = K_i \left(\frac{(q-1)d - z}{q}; d, q \right)$$

Therefore, the polynomial sequence generating the Bose-Mesner algebra of the Hamming scheme is precisely the Kravchuk polynomial sequence under an affine shift of the argument. ■

By applying Delsarte's elegant algebraic formulation—combining the properties of the dual distribution with the Lloyd polynomial—we can prove the sphere packing bound analytically and simultaneously classify perfect codes.

Theorem 3.87 (Lloyd Type Theorem). *Let Y be a code of minimum distance δ in a P -polynomial scheme X , and let $e = \lfloor \frac{\delta-1}{2} \rfloor$.*

If Y is a perfect e -code (i.e., $|Y| \sum_{i=0}^e k_i = |X|$), then the following hold:

- (i) *The dual degree of the code is exactly $s^* = e$.*
- (ii) *The e distinct eigenvalues θ_k for which the dual distribution entries are strictly positive ($a_k^* > 0$) are exactly the roots of the Lloyd polynomial $\Psi_e(z)$.*

Proof. Consider the polynomial $\Psi_e(z)^2$. It has degree $2e \leq \delta - 1$. Expanding it in the orthogonal polynomial basis yields $\Psi_e(z)^2 = \sum_{\ell=0}^{2e} c_\ell v_\ell(z)$.

We define an auxiliary test vector $\alpha = (\alpha_0, \dots, \alpha_d)$ by $\alpha_j = \left(\frac{\Psi_e(\theta_j)}{K_e} \right)^2$, where $K_e = \Psi_e(\theta_0) = \sum_{i=0}^e k_i$. Clearly, $\alpha_0 = 1$ and $\alpha_j \geq 0$ for all j . Using $P_\ell(j) = v_\ell(\theta_j)$, we can express α_j as $\frac{1}{K_e^2} \sum_{\ell=0}^{2e} c_\ell P_\ell(j)$.

Evaluate the sum $S = \sum_{j=0}^d \alpha_j a_j^*$ in two ways. First, separating the $j = 0$ term:

$$S = \alpha_0 a_0^* + \sum_{j=1}^d \alpha_j a_j^* \geq 1 \cdot |Y|$$

Second, substituting $a_j^* = \sum_{i=0}^d a_i Q_j(i)$:

$$S = \sum_{i=0}^d a_i \left(\sum_{j=0}^d \alpha_j Q_j(i) \right) = \sum_{i=0}^d a_i \left(\frac{1}{K_e^2} \sum_{\ell=0}^{2e} c_\ell \sum_{j=0}^d P_\ell(j) Q_j(i) \right)$$

Since $PQ = |X|I$, the innermost sum is $|X| \delta_{\ell,i}$. The expression collapses to $S = \frac{|X|}{K_e^2} \sum_{i=0}^{2e} c_i a_i$. Because $2e < \delta$ and the code has minimum distance δ , the inner distribution elements a_i are identically zero for $1 \leq i \leq 2e$. Only the $i = 0$ term survives, yielding $S = \frac{|X|}{K_e^2} c_0 a_0$.

By orthogonal projection, $c_0 = \frac{1}{|X|} \sum_{j=0}^d m_j \Psi_e(\theta_j)^2$. Expanding Ψ_e gives

$$\frac{1}{|X|} \sum_{u=0}^e \sum_{v=0}^e \sum_{j=0}^d m_j P_u(j) P_v(j)$$

By the Second Orthogonality Relation, this evaluates to $\sum_{u=0}^e k_u = K_e$. Thus, $S = \frac{|X|}{K_e^2} (K_e)(1) = \frac{|X|}{K_e}$.

Combining our two evaluations of S yields the analytic proof of the sphere packing bound: $|Y| \leq \frac{|X|}{K_e}$.

If Y is a perfect code, equality holds, which forces $\sum_{j=1}^d \alpha_j a_j^* = 0$. Because both $\alpha_j \geq 0$ and $a_j^* \geq 0$, it must be that $\alpha_j = 0$ whenever $a_j^* > 0$ for $j \geq 1$. Since $\alpha_j = 0$ is equivalent to $\Psi_e(\theta_j) = 0$, every non-zero dual distribution parameter corresponds to a root of the Lloyd polynomial. As $\Psi_e(z)$ has degree e , it has at most e roots, implying $s^* \leq e$. However, the MacWilliams inequality guarantees $s^* \geq \lfloor (\delta-1)/2 \rfloor = e$. Thus $s^* = e$, and the e distinct non-zero dual parameters occur precisely at the roots of $\Psi_e(z)$. ■

3.11 Delsarte's Linear Programming Bound and Classical Codes

While we successfully derived the sphere packing bound and classified perfect codes using the algebraic orthogonality of the dual distribution, the true power of Delsarte's framework lies in its generalisation. By

viewing the conditions on the inner distribution as a system of linear constraints, Delsarte introduced Linear Programming (LP) into combinatorics, providing a universal, computable method to bound the size of any code.

Theorem 3.88 (Delsarte's Linear Programming Bound). *Let X be a symmetric association scheme with second eigenmatrix Q . If $Y \subseteq X$ is a code with minimum distance δ , then $|Y| \leq g_{max}$, where g_{max} is the optimal value of the following linear program:*

Maximise: $g = \sum_{i=0}^d a_i$

Subject to the constraints:

$$(i) \ a_0 = 1$$

$$(ii) \ a_i = 0 \quad \text{for } 1 \leq i \leq \delta - 1$$

$$(iii) \ a_i \geq 0 \quad \text{for } \delta \leq i \leq d$$

$$(iv) \ \sum_{i=0}^d a_i Q_k(i) \geq 0 \quad \text{for } 1 \leq k \leq d$$

Proof. The proof follows directly from the existence of the inner distribution. Let $a_Y = (a_0, a_1, \dots, a_d)$ be the inner distribution of our code Y . We check that a_Y satisfies all the constraints of the linear program.

By Definition 3.77, $a_0 = \frac{1}{|Y|} |R_0 \cap (Y \times Y)|$. Since R_0 is the diagonal relation, $(y, y) \in R_0$ for every $y \in Y$, giving exactly $|Y|$ pairs. Thus $a_0 = 1$, satisfying (i). Because the code has minimum distance δ , there are strictly zero pairs of distinct codewords at distances 1 through $\delta - 1$, meaning $a_i = 0$ in this range, satisfying (ii). As a_i counts cardinalities, $a_i \geq 0$ inherently holds, satisfying (iii).

Finally, the quantity $\sum_{i=0}^d a_i Q_k(i)$ is precisely the k -th entry of the dual distribution a_Y^* . By Theorem 3.79, every entry of the dual distribution of any code in an association scheme is non-negative, which exactly provides condition (iv).

Therefore, the inner distribution a_Y is a valid, feasible solution to this linear program. The objective function evaluated at a_Y is $g = \sum_{i=0}^d a_i$. Since the relations R_i partition $X \times X$, summing the inner distribution entries counts all possible pairs in $Y \times Y$ divided by $|Y|$, which evaluates strictly to $\frac{|Y|^2}{|Y|} = |Y|$. Because a_Y is just one feasible solution, the absolute maximum g_{max} of the linear program must be an upper bound for $|Y|$. ■

We conclude by tying Delsarte's purely algebraic theory back to its roots in classical linear coding theory. Let C be a linear code of length n over $\mathbb{F}_q$. In the classical view, C is a vector subspace of $\mathbb{F}_q^n$, and its dual code $C^\perp = \{x \in \mathbb{F}_q^n \mid x \cdot y = 0 \ \forall y \in C\}$ is its orthogonal complement, satisfying $|C| |C^\perp| = q^n$.

The classical weight enumerator of C relies on the spectrum A_i , which counts the number of codewords of Hamming weight i . In the Hamming association scheme $X = H(n, q)$, since C is a linear subspace, the distance between codewords is translation invariant ($d_H(x, y) = w(x - y)$). Thus, the inner distribution a_C perfectly mirrors the classical spectrum: $a_i = A_i$.

The famous MacWilliams Identity relates the spectrum of C to the spectrum of $C^\perp$. In Delsarte's framework, this identity ceases to be a tedious polynomial manipulation and is revealed simply as a change of basis in the Bose-Mesner algebra.

Theorem 3.89 (Algebraic MacWilliams Identity). *Let $X = H(n, q)$ be the Hamming scheme. If C is a linear code in X , its algebraic dual distribution a_C^* is strictly proportional to the inner distribution of the orthogonal dual code $C^\perp$. Specifically:*

$$a_C^* = |C| a_{C^\perp}$$

Proof. Recall from our construction of the character table of $H(n, q)$ that we defined the group characters $\Psi_x(y) = \prod \psi_{x_i}(y_i)$. Because C is a linear subspace of X , the restriction of Ψ_x to C is a character of the subgroup C .

By the standard orthogonality relations of characters on finite groups, summing a character over a subgroup C yields $|C|$ if the character is trivial on C , and 0 otherwise. The character Ψ_x is trivial on C if and only if $\Psi_x(y) = 1$ for all $y \in C$, which, by the definition of the standard inner product on $\mathbb{F}_q^n$, occurs if and only if $x \in C^\perp$. Thus:

$$\sum_{y \in C} \Psi_x(y) = \begin{cases} |C| & \text{if } x \in C^\perp \\ 0 & \text{if } x \notin C^\perp \end{cases}$$

We now evaluate the k -th entry of the dual distribution a_k^* . We apply the definition of a_k^* , substitute a_i , and use the fact that $Q_k(i) = K_k(i)$:

$$a_k^* = \sum_{i=0}^n a_i Q_k(i) = \frac{1}{|C|} \sum_{i=0}^n |R_i \cap (C \times C)| K_k(i)$$

Since $K_k(i) = \sum_{z \in X_k} \Psi_{x-y}(z)$ for any $(x, y) \in R_i$ (by Lemma 3.73), we can expand the sum over the relations into a sum over all pairs in the code:

$$a_k^* = \frac{1}{|C|} \sum_{x \in C} \sum_{y \in C} \sum_{z \in X_k} \Psi_{x-y}(z) = \frac{1}{|C|} \sum_{z \in X_k} \left(\sum_{x \in C} \Psi_x(z) \right) \left(\sum_{y \in C} \overline{\Psi_y(z)} \right)$$

Because C is a subspace, it is closed under negation, so the sum over $y \in C$ of $\overline{\Psi_y(z)}$ is identical to the sum of $\Psi_y(z)$. Applying our orthogonality result, the product of the two bracketed sums is $|C|^2$ if $z \in C^\perp$, and 0 otherwise. Therefore:

$$a_k^* = \frac{1}{|C|} \sum_{z \in X_k \cap C^\perp} |C|^2 = |C| |X_k \cap C^\perp|$$

The quantity $|X_k \cap C^\perp|$ is precisely the number of codewords in $C^\perp$ of weight k , which is the k -th entry of the inner distribution $a_{C^\perp}$. Hence, $a_k^* = |C| (a_{C^\perp})_k$, proving the theorem. $\blacksquare$

Chapter 4

Analysis of some research papers

4.1 Bilinear forms and their applications to coding theory

This section analyses Delsarte's work on bilinear forms in relation to coding theory and association schemes. This is found in his paper [10].

4.1.1 Bilinear forms

Throughout, we work over a finite field $\mathbb{F}_q$ where q is some power of a prime number. All vector spaces shall be taken over $\mathbb{F}_q$.

Definition 4.1. Let V and V' be two non-zero finite-dimensional vector spaces. A **bilinear form on V and V'** is a map $f : V \times V' \rightarrow \mathbb{F}_q$ satisfying

$$f\left(\sum_i x_i \mu_i, \sum_j x'_j \mu'_j\right) = \sum_{i,j} x_i f(\mu_i, \mu'_j) x'_j = \sum_{i,j} x_i x'_j f(\mu_i, \mu'_j)$$

for all scalars $x_i, x'_j \in \mathbb{F}_q$ and all vectors $(\mu_i, \mu'_j) \in V \times V'$.

It is clear that the set of all bilinear forms on V (m -dimensional) and V' (m' -dimensional) forms an mm' -dimensional subspace of the set of all maps $V \times V' \rightarrow \mathbb{F}_q$. Note that the whole space of all maps has dimension $q^{m+m'}$. We denote this subspace by $\Omega = \Omega(m, m')$ and the full space by $\mathbb{F}_q^{\mathbf{V} \times \mathbf{V}'}$.

Definition 4.2. The **left radical**, $\mathbf{Rad}_L(f)$ of any $f \in \Omega$ is the subspace of V consisting of all $x \in V$ such that $f(x, x') = 0$ for every $x' \in V'$. A right radical $\mathbf{Rad}_R(f)$ may be defined similarly.

Proposition 4.3. Both the left and right radicals of a bilinear form $f \in \Omega$ have the same codimension, which is equal to the rank of the associated matrix.

Proof. We shall use a matrix associated with the bilinear form f . Let $\{v_1, \dots, v_m\}$ and $\{w_1, \dots, w_{m'}\}$ be bases for V and V' respectively.

Construct the $m \times m'$ matrix A with entries in $\mathbb{F}_q$ defined by

$$A_{i,j} = f(v_i, w_j) \quad \text{for all } 1 \leq i \leq m \text{ and } 1 \leq j \leq m'.$$

It is then straightforward to verify that the bilinear form is given by

$$f(v, w) = v^\top A w,$$

where $v \in V$ and $w \in V'$ are represented as coordinate column vectors in our chosen bases.

It directly follows that

$$\text{Rad}_L(f) = \{v \in V \mid v^\top A = 0\} \quad \text{and} \quad \text{Rad}_R(f) = \{w \in V' \mid Aw = 0\}.$$

Equivalently,

$$\text{Rad}_L(f) = \ker(A^\top) \quad \text{and} \quad \text{Rad}_R(f) = \ker(A).$$

Consequently, the codimensions of the radicals are identical, since

$$\begin{aligned} \text{codim}(\text{Rad}_L(f)) &= m - \dim \ker(A^\top) \\ &= m - (m - \text{rank}(A^\top)) \\ &= \text{rank}(A^\top), \end{aligned}$$

and similarly,

$$\begin{aligned} \text{codim}(\text{Rad}_R(f)) &= m' - \dim \ker(A) \\ &= m' - (m' - \text{rank}(A)) \\ &= \text{rank}(A). \end{aligned}$$

As $\text{rank}(A^\top) = \text{rank}(A)$, the result follows. ■

Definition 4.4. The above defined matrix A is called the *matrix associated to the bilinear form f* .

Thus, we may now talk of the rank of f which is simply the rank of this matrix and also the codimension of both radicals.

Definition 4.5. Let $G = \text{Aut}_{\mathbb{F}_q}(V)$ and $G' = \text{Aut}_{\mathbb{F}_q}(V')$ be the automorphism groups of V and V' . Let $H = \text{Aut}_{\mathbb{F}_q}(V) \times \text{Aut}_{\mathbb{F}_q}(V')$. We define an *action of H on Ω* in the following way. For every $(\alpha, \alpha') \in H$, the image of $f \in \Omega$ is the bilinear form h given by

$$h(x, x') = f(\alpha x, \alpha' x')$$

Proposition 4.6. *For the above defined action, the H -orbits of Ω are the sets of bilinear forms with constant rank, that is,*

$$\Omega_r = \{f \in \Omega \mid \text{rank}(f) = r\}$$

Proof. Let A be the matrix associated to the bilinear form f . Then the action of $(\alpha, \alpha') \in H$ is simply

$$A \mapsto \alpha^\top A \alpha'$$

Since α and α' are invertible square matrices, the rank is preserved under this action. Conversely, given a matrix A of rank r , by the rank factorisation we can find invertible matrices P and Q such that

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Similarly given a matrix B of rank r , we can obtain P' and Q' . Note that $P, P' \in G$ and $Q, Q' \in G'$. Thus, A and B are in the same orbit. ■

4.1.2 Schur rings

The foundational material in this section is primarily drawn from [24].

In this chapter, all groups shall be treated as finite groups unless stated otherwise. Given a finite set of elements X , let $\underline{X}$ denote the formal sum $\sum_{x \in X} x$. Given a set $X \subseteq G$ for a group G , let X^m denote $\{x^m \mid x \in X\}$.

Definition 4.7. Let $\mathbb{C}[G]$ denote the group ring of a finite group G over $\mathbb{C}$. A ring $\mathcal{A} \subseteq \mathbb{C}[G]$ is called a **Schur ring** over G if there exists a partition $\mathcal{S}$ of G satisfying:

- (i) $\{\text{id}\} \in \mathcal{S}$
- (ii) $X^{-1} \in \mathcal{S}$ for every $X \in \mathcal{S}$
- (iii) $\underline{\mathcal{S}}$ is a basis for $\mathcal{A}$

Equivalently, a linear subspace $\mathcal{A} \subseteq \mathbb{C}[G]$ is a Schur ring over G if and only if it satisfies the following conditions:

- $\mathcal{A}$ is a subalgebra under ordinary multiplication containing the identity $\{\text{id}\}$ and the all-ones vector $\underline{G}$.
- $\mathcal{A}$ is closed under the Hadamard (component-wise) product, denoted by $\circ$.
- $\mathcal{A}$ is closed under the inversion map $\sum_x a_x x \mapsto \sum_x a_x x^{-1}$.

Question. Given a Schur ring $\mathcal{A} \subseteq \mathbb{C}[G]$, how does one obtain a partition $\mathcal{S}$? Is this unique?

Ans. Observe that under component-wise multiplication, that is,

$$\left(\sum_x a_x x \right) \left(\sum_x b_x x \right) = \sum_x a_x b_x x$$

we have that $\mathcal{A}$ is a commutative algebra with identity $\underline{G}$.

Let I denote the set of primitive idempotents of this commutative algebra, that is the set of all $x \in \mathcal{A} \setminus \{0\}$ such that $x^2 = x$ and x cannot be written as a sum $e_1 + e_2$ for any two idempotents with $e_1 e_2 = 0$.

Now idempotents of $\mathcal{A}$ are of the form $\sum_x a_x x$ with $a_x^2 = a_x$ or simply, $a_x = 1$ by neglecting 0 terms.

We claim that this set I is precisely $\underline{\mathcal{S}} = \{\underline{X} \mid X \in \mathcal{S}\}$.

As noted before, every element of I is of the form $\underline{Y}$ for some $Y \subseteq G$. Two idempotents $\underline{Y}_1$ and $\underline{Y}_2$ are orthogonal, that is, $\underline{Y}_1 \underline{Y}_2 = 0$, if and only if their supports are disjoint, that is, $Y_1 \cap Y_2 = \emptyset$.

Let $\underline{\mathcal{S}}$ be the basis and A be a primitive idempotent. Then we can write

$$A = \sum_i c_i \underline{X}_i$$

for $\underline{X}_i \in \underline{\mathcal{S}}$.

But as seen, A must have all coefficient as 1 and hence all the c_i are either 0 or 1. But if two or more c_i are 1, then A is being expressed as a sum of two idempotents which have disjoint support which is not possible. Hence exactly one c_i must be 1 which shows that A is in the basis.

Next, we show that every basis element is a primitive idempotent. Let $\underline{X} \in \underline{\mathcal{S}}$. It is clear that $\underline{X}$ is a non-zero idempotent.

Suppose we are able to write $\underline{X} = A + B$ for two orthogonal idempotents A and B . The supports of A and B are disjoint since they are orthogonal.

If both A and B are primitive, then by the above argument, since primitive idempotents are in the basis, both A and B must be in the basis which contradicts the linear independence of the basis.

Without loss of generality if X is idempotent but not primitive, replacing X by A above again leads to a contradiction. Thus, the idempotent basis element X must be primitive.

Obtaining $\mathcal{S}$ from $\underline{\mathcal{S}}$ is trivial. Clearly, it is unique. □

Definition 4.8. A partition $\mathcal{S}$ of a finite group G is called a **Schur partition** if it satisfies conditions (i) and (ii) in Definition 4.7 and if additionally $\langle \underline{\mathcal{S}} \rangle$ is a subalgebra of $\mathbb{C}[G]$.

Proposition 4.9. *There is a one-one correspondence between Schur partitions of a group and Schur rings over the group.*

Proof. Given a Schur ring, we can obtain the Schur partition as illustrated in the question above. Conversely, given a Schur partition, simply generate the ring to be the span of $\underline{\mathcal{S}}$. By definition of a Schur partition, this is a subalgebra and hence is a Schur ring.

The fact that a Schur partition needs to have its span as a subalgebra is required because we want our final ring to be closed under multiplication. ■

Definition 4.10. Given a Schur ring $\mathcal{A}$, the elements of $\mathcal{S}$ are called **basic sets** of $\mathcal{A}$ and the number of basic sets is called the **rank** of $\mathcal{A}$. Any union of basic sets of $\mathcal{A}$ is called an $\mathcal{A}$ -set.

It is clear that $X \subseteq G$ is an $\mathcal{A}$ -set if and only if $\underline{X} \in \mathcal{A}$.

Definition 4.11. A Schur ring $\mathcal{A}$ is said to be **imprimitive** if there is a proper non-trivial $\mathcal{A}$ -subgroup of G , that is, a subgroup which is an $\mathcal{A}$ -set. $\mathcal{A}$ is said to be **primitive**, otherwise.

Definition 4.12. Let $\mathcal{A}$ be a Schur ring over G and $\mathcal{A}'$ be a Schur ring over G' . A group isomorphism $f : G \rightarrow G'$ is called a **Cayley isomorphism from $\mathcal{A}$ to $\mathcal{A}'$** if the induced ring isomorphism $\tilde{f} : \mathbb{C}[G] \rightarrow \mathbb{C}[G']$ maps the Schur ring $\mathcal{A}$ into $\mathcal{A}'$.

The group of all such Cayley isomorphisms is denoted by $\mathbf{Iso}_{\text{cay}}(\mathcal{A}, \mathcal{A}')$. The automorphism group is denoted $\mathbf{Aut}_{\text{cay}}(\mathcal{A})$.

Definition 4.13. An algebra isomorphism $\varphi : \mathcal{A} \rightarrow \mathcal{A}'$ is called an **algebraic isomorphism** if it maps the standard basis of $\mathcal{A}$ into that of $\mathcal{A}'$.

Let f be a Cayley isomorphism from $\mathcal{A} \rightarrow \mathcal{A}'$. One can easily verify that the map $\varphi(\underline{X}) := \underline{f(X)}$ is an algebraic isomorphism. However, not every algebraic isomorphism is induced by a Cayley isomorphism. Giving a counter-example to this is non-trivial.

We now discuss how Schur rings naturally give way to association schemes. For requisite definitions and background on association schemes, the reader is urged to look at my notes on association schemes.

Proposition 4.14. Let G be a multiplicative group and $\mathcal{A}$ be a Schur ring over the group G . Suppose $\mathcal{S} = \{X_0, X_1, \dots, X_d\}$ is the associated Schur partition of G with $X_0 = \{\text{id}\}$. Then, $\left(G, \{R_i\}_{i=0}^d\right)$ is a symmetric association scheme of class d where

$$R_i = \{(x, y) \in G \times G \mid xy^{-1} \in X_i\}$$

for all $i = 0, 1, \dots, d$.

Proof. Clearly R_0 is the diagonal $\{(g, g) \mid g \in G\}$. Further, the R_i 's are clearly disjoint and every element of $G \times G$ must necessarily be in one of the R_i .

Since $\mathcal{S}$ is a Schur partition, we have that for $0 \leq i \leq d$, $X_i^{-1} = X_j$ for some j and hence $R_i^\top = R_j$.

We only need to verify the intersection number property. To this end, fix $(x, y) \in R_k$. Let $xy^{-1} = g \in X_k$.

$$\mathcal{A} = \{z \in G \mid (x, z) \in R_i, (z, y) \in R_j\}$$

$$\mathcal{B} = \{(a, b) \in X_i \times X_j \mid ab = g\}$$

One can check that $\phi : \mathcal{A} \rightarrow \mathcal{B}$ given by $z \mapsto (xz^{-1}, zy^{-1})$ is a bijection.

We now use the fact that the partition $\mathcal{S}$ is a Schur partition. Since $\underline{\mathcal{S}}$ is a basis for the subalgebra $\langle \underline{\mathcal{S}} \rangle \subseteq \mathbb{C}[G]$, for $0 \leq i, j \leq d$ we have

$$\underline{X_i} \underline{X_j} = \sum_{t=0}^d p_{i,j}^t \underline{X_t}$$

for some uniquely determined constants $p_{i,j}^t \in \mathbb{C}$.

Now we expand the LHS to get

$$\underline{X_i} \underline{X_j} = \sum_{a \in X_i} \sum_{b \in X_j} ab$$

On the RHS, the coefficient of $g \in G$ is simply $p_{i,j}^k$ where $g \in X_k$. On the LHS the coefficient of the same g is the cardinality of the set $\mathcal{B}$.

Hence, the cardinality of $\mathcal{A}$ which is the cardinality of $\mathcal{B}$ is a constant $p_{i,j}^k$ which does not depend on the choice of $(x, y) \in R_k$. Thus, this verifies the intersection number property of association schemes.

Finally, we have shown that $(G, \{R_i\}_{i=0}^d)$ is a symmetric association scheme of class d . ■

We now briefly cover some concepts related to the duality of Schur rings.

Definition 4.15. Let G be a finite abelian group. The **dual group** $\hat{G}$ is defined to be the set of all irreducible complex characters of G under pointwise multiplication.

Proposition 4.16. *The dual group of a finite abelian group G is again a finite abelian group, which is isomorphic to G .*

Proof. Firstly, the irreducible characters of a finite abelian group are simply group homomorphisms $G \rightarrow \mathbb{C}^\times$. Under pointwise multiplication it is clear that they form a finite abelian group.

The finiteness follows from the fact that G and $\hat{G}$ are isomorphic as groups. This isomorphism is not canonical and one may proceed by using the structure theorem for finite abelian groups and analysing characters of cyclic groups. ■

Proposition 4.17. *The dual of the dual of a finite abelian group G is G itself.*

Proof. Let H denote $\hat{\hat{G}}$ for simplicity. Construct the map $\phi : G \rightarrow H$ given by

$$\phi(g) = \text{ev}_g \in H$$

where $\text{ev}_g(\chi) = \chi(g)$ for every $\chi \in \hat{G}$. One may check that ev_g is well-defined and indeed a group homomorphism from $\hat{G}$ to $\mathbb{C}^\times$ and hence a character of $\hat{G}$. It is also immediate that ϕ itself is a group homomorphism.

Using that a group and its dual are isomorphic, we have the same cardinalities for H and G . Thus ϕ is a group homomorphism between groups of the same cardinalities. We only need to show that it is injective.

Suppose $g \in \ker(\phi)$. Then ev_g is the identity map, that is $\text{ev}_g(\chi) = 1$ for every character χ of G . Thus $\chi(g) = 1$ for every χ . From basic representation theory of finite groups we know that a non-trivial character separates the points of G , that is, for $g \in G$ not the identity, there exists a character such that $\chi(g)$ is not the identity matrix (that is 1). Here, it will not be 1.

This forces g to be the identity element showing that $\ker(\phi) = \{\text{id}\}$ finishing the proof. ■

Definition 4.18. Given a Schur ring $R \subseteq \mathbb{C}[G]$ with the corresponding partition $\mathcal{S}$, set $\chi_1 \sim \chi_2$ for $\chi_1, \chi_2 \in \hat{G}$ if $\chi_1(\underline{X}) = \chi_2(\underline{X})$ for every $X \in \mathcal{S}$. This is an equivalence relation whose classes form the set, say, $\hat{\mathcal{S}}$. Then the Schur ring corresponding to $\hat{\mathcal{S}}$ is called the **dual Schur ring**.

We have two propositions that we state without proof as the proofs are quite involved.

Proposition 4.19. *$\hat{\mathcal{S}}$ is a Schur partition of $\hat{G}$.*

Proposition 4.20. *The dual of the dual Schur ring of a Schur ring $R \subseteq \mathbb{C}[G]$ is R , itself.*

4.1.3 Bilinear forms and association schemes

We now establish the connection of Schur rings and bilinear forms. Recall that Ω was the subspace of all bilinear forms on V (m -dimensional) and V' (m' -dimensional). Without loss of generality, throughout we shall assume that $m \leq m'$. Then, the rank of the matrix associated to the form can vary from 0 to m .

Recall that Ω_r was an orbit under the action of $H = \text{Aut}_{\mathbb{F}_q}(V) \times \text{Aut}_{\mathbb{F}_q}(V')$ on Ω . Each orbit has bilinear forms of the same rank $r \in \{0, 1, \dots, m\}$.

Definition 4.21. We define the *group ring* $\mathbb{Z}\Omega$ to be the additive group $(\Omega, +)$ over $\mathbb{Z}$. An element of this ring is of the form

$$A = \sum_{f \in \Omega} a_f f$$

where $a_f \in \mathbb{Z}$ and $f \in \Omega$ such that all but finitely many a_f are zero, that is, the sum is finitely supported.

Firstly note that $(\Omega, +)$ is indeed an abelian group. Further, observe that the multiplication in the ring is actually the addition of the bilinear forms. Thus there are two ‘addition’ operations in some sense.

We make the sum and product explicit below.

$$\begin{aligned} \sum_{f \in \Omega} a_f f + \sum_{f \in \Omega} b_f f &= \sum_{f \in \Omega} (a_f + b_f) f \\ \left(\sum_{f \in \Omega} a_f f \right) \left(\sum_{f \in \Omega} b_f f \right) &= \sum_{f, g} a_f b_g (f + g) \end{aligned}$$

Proposition 4.22. Consider the abelian group Ω under addition. The partition $(\Omega_r)_{r=0}^m$ is a Schur partition.

Proof. Condition (i) is clearly satisfied since the identity element of the group, which is the zero bilinear map, lies in Ω_0 and every bilinear map of rank 0 must be the zero map thus showing that $\Omega_0 = \{\text{id}\}$.

We now check condition (ii). Consider Ω_r for a fixed r and take the inverse of every bilinear map in it. In the additive group considered, the inverse is simply the negative of the map. If f has rank r , then $-f$ also has rank r and it is clear that $\Omega_r^{-1} = \Omega_r$.

The last condition to check is if $\langle \underline{\mathcal{S}} \rangle$ forms a subalgebra of $\mathbb{Z}[\Omega]$. Here we have,

$$\underline{\mathcal{S}} = \left\{ \sum_{f \in \Omega_r} f : 0 \leq r \leq m \right\}$$

We only need to check if the identity lies in the $\mathbb{Z}$ -span and if the $\mathbb{Z}$ -span is closed. The first condition is clearly true.

To check the second one, consider

$$\left(\sum_{f \in \Omega_a} f \right) \left(\sum_{g \in \Omega_b} g \right) = \sum_{f \in \Omega_a} \sum_{g \in \Omega_b} f + g = \sum_{h \in \Omega} N_{a,b}(h) h$$

where $N_{a,b}(h)$ denotes the number of pairs $(f, g) \in \Omega_a \times \Omega_b$ such that $f + g = h$.

We claim that for every $h \in \Omega_r$, this is independent of h and only depends on r . To this end, let $h_1, h_2 \in \Omega_r$. Suppose $x \in H$ is such that $xh_1 = h_2$.

It is clear that given any (f, g) such that $f + g = h_1$, we have $(xf) + (xg) = x(f + g) = xh_1 = h_2$ and conversely we may use x^{-1} to go back to (f, g) from a pair (f', g') such that $f' + g' = h_2$.

This shows that $N_{a,b}(h)$ is constant on Ω_r . Suppose this constant is $p(a,b,r) \geq 0$. Then we have

$$\left(\sum_{f \in \Omega_a} f \right) \left(\sum_{g \in \Omega_b} g \right) = \sum_{r=0}^m p(a,b,r) \sum_{h \in \Omega_r} h$$

This shows the span is closed under multiplication of the group ring and hence we get a subalgebra. Thus, the partition in question is indeed a Schur partition. $\blacksquare$

Proposition 4.23. *Let R_i be the set of all $(f,g) \in \Omega$ such that $\text{rank}(f-g) = i$. Then $(\Omega, \{R_i\}_{i=0}^m)$ forms a symmetric association scheme of class m whose intersection numbers are the constant $p(a,b,k)$ from the above Proposition 4.22.*

Proof. The fact that it is an association scheme follows from Proposition 4.14 since $fg^{-1} := f - g$ in our additive group Ω . The group partition in Proposition 4.14 is simply chosen to be $\{\Omega_r\}_{r=0}^m$. From the above Proposition 4.22 it is known that $p(a,b,r) = |\{(f,g) \in \Omega_a \times \Omega_b \mid f+g=h\}|$ for some $h \in \Omega_r$ is independent of the h chosen. Comparing with Proposition 4.14, this is the cardinality of $\mathcal{B}$ which is $p_{i,j}^r$ and we are done. $\blacksquare$

We also have an explicit formula for the sizes of the classes in the Schur partition.

Proposition 4.24. *For $0 \leq r \leq m$, we have*

$$|\Omega_r| = \prod_{i=0}^{r-1} \frac{(q^m - q^i)(q^{m'} - q^i)}{q^r - q^i}$$

Proof. Ω_r is the collection of all bilinear maps of rank r . Essentially, we are required to find the number of $m \times m'$ matrices over $\mathbb{F}_q$ of rank r .

Recall the rank canonical factorisation of matrices: If an $m \times n$ matrix M over a field $\mathbb{F}$ has rank r , then r is the smallest positive integer such that $M = XY^\top$ for some $m \times r$ matrix X over $\mathbb{F}$ and $n \times r$ matrix Y over $\mathbb{F}$. (For a proof, the reader is referred to Proposition 4 from my presentation notes on ‘Communication over Graphs’).

Note that $r = \text{rank}(M) = \text{rank}(XY^\top) \leq \text{rank}(X) \leq r$. Thus, $\text{rank}(X) = r$. Similarly, $\text{rank}(Y) = r$.

Let $S_{a,b,r} \subseteq M_{a,b}(\mathbb{F}_q)$ denote the set of all $a \times b$ matrices of rank r over $\mathbb{F}_q$. Consider the map $\varphi : S_{m,r,r} \times S_{m',r,r} \rightarrow M_{m,m'}(\mathbb{F}_q)$ given by $\varphi(A,B) = AB^\top$.

We claim that the image of φ is exactly $S_{m,m',r}$. By rank canonical factorisation, any matrix in $S_{m,m',r}$ can be written as a product of matrices in $S_{m,r,r}$ and $S_{m',r,r}$. Conversely, let $A \in S_{m,r,r}$ and $B \in S_{m',r,r}$. Viewing $B^\top$ as a linear map from $\mathbb{F}_q^r \rightarrow \mathbb{F}_q^{m'}$ and A as a linear map from $\mathbb{F}_q^r \rightarrow \mathbb{F}_q^m$, both maps have trivial kernel because they have full column rank r . Therefore, for any non-zero $x \in \mathbb{F}_q^r$, $B^\top x \neq 0$. Furthermore, the image of $B^\top$ has dimension r . Since A is injective, multiplying by A preserves the dimension of this subspace. Thus, the image of $AB^\top$ has dimension r , meaning $\text{rank}(AB^\top) = r$.

We now determine the cardinality of this image set. Given a matrix M in the image set, the decomposition is unique up to right-multiplication by invertible $r \times r$ matrices. More precisely, if $AB^\top = CD^\top = M$, then there exists a matrix $P \in GL_r(\mathbb{F}_q)$ such that $AP = C$ and $B^\top = PD^\top$.

Indeed, the columns of A and C form two ordered bases for the r -dimensional column space of M . Thus, there exists a unique change-of-basis matrix $P \in GL_r(\mathbb{F}_q)$ such that $C = AP$. Substituting this into the equality yields

$$A(B^\top - PD^\top) = 0.$$

Because A has full column rank, it represents an injective linear transformation, meaning its null space is trivial. This forces the columns of the right-hand factor to be identically zero, yielding $B^\top - PD^\top = 0$, so $B^\top = PD^\top$.

Consequently, a single matrix M in the image has exactly $|GL_r(\mathbb{F}_q)|$ pre-images. The required formula is therefore

$$|S_{m,m',r}| = \frac{|S_{m,r,r}| |S_{m',r,r}|}{|GL_r(\mathbb{F}_q)|}.$$

The proposition follows by applying standard combinatorial counting arguments to construct the full-rank matrices $S_{m,r,r}$ and $S_{m',r,r}$ column by column. $\blacksquare$

4.1.4 Detour: Representation theory and fields

This chapter covers some useful results from field theory and we aim to eventually build a character for our group $(\Omega, +)$. Certain facts from representation theory will be used without proof. The reader is referred to my notes on the ‘Representation Theory of Finite Groups’.

By a **character of an abelian group**, we will simply mean a group homomorphism $G \rightarrow \mathbb{C}^\times$. This is because every irreducible representation of an abelian group is necessarily 1-dimensional.

We now define an important map for fields and field extensions - the field trace.

Definition 4.25. Given a finite field extension L over the field K , we define the **trace of the field L over K** as $\text{tr}_{L/K} : L \rightarrow K$ with $\text{tr}(\alpha)$ being the trace of the K -linear map $m_\alpha : L \rightarrow L$ mapping $m_\alpha(x) = \alpha x$.

Lemma 4.26. Suppose $L/E/F$ is a finite field extension tower. Then the field trace has the property:

$$\text{tr}_{L/F} = \text{tr}_{E/F} \circ \text{tr}_{L/E}$$

Proof. Fix some $\alpha \in L$. Consider the map $m_\alpha : L \rightarrow L$ given by $m_\alpha(x) = \alpha x$. This is an E -linear map, as well as an F -linear map.

Choose an F -basis $\{e_1, \dots, e_n\}$ of E and an E -basis $\{\ell_1, \dots, \ell_m\}$ of L . Then one may check that the following is an F -basis for L : $\{e_i \ell_j : 1 \leq i \leq n, 1 \leq j \leq m\}$.

Now consider a standard ordering of this basis given by

$$(\ell_1 e_1, \ell_1 e_2, \dots, \ell_1 e_n; \ell_2 e_1, \ell_2 e_2, \dots, \ell_2 e_n; \dots; \ell_m e_1, \ell_m e_2, \dots, \ell_m e_n)$$

Consider the action on this basis:

$$m_\alpha(e_i \ell_j) = \alpha e_i \ell_j = (\alpha \ell_j) e_i$$

Suppose we have for some $c_{jt} \in E$ ($1 \leq j, t \leq m$),

$$\alpha \ell_j = \sum_{t=1}^m c_{jt} \ell_t$$

Thus we get

$$\alpha e_i \ell_j = \sum_{t=1}^m (c_{jt} e_i) \ell_t$$

We have to express this as an F -linear sum to write down the matrix. Suppose we have for some $a_v^{ji} \in F$ ($1 \leq j, t \leq m, 1 \leq v, i \leq n$),

$$c_{jt} e_i = \sum_{v=1}^n a_v^{ji} e_v$$

Finally we get

$$m_\alpha(e_i \ell_j) = \alpha e_i \ell_j = \sum_{t=1}^m \sum_{v=1}^n a_v^{ji} e_v \ell_t$$

Thus the $mn \times mn$ matrix of the F -linear map $m_\alpha : L \rightarrow L$ is given by

$$M = \begin{pmatrix} a_1^{111} & \dots & a_1^{11n} & a_1^{211} & \dots & a_1^{21n} & \dots & a_1^{m11} & \dots & a_1^{m1n} \\ \vdots & & \vdots & \vdots & & \vdots & \dots & \vdots & & \vdots \\ a_n^{111} & \dots & a_n^{11n} & a_n^{211} & \dots & a_n^{21n} & \dots & a_n^{m11} & \dots & a_n^{m1n} \\ \hline a_1^{121} & \dots & a_1^{12n} & a_1^{221} & \dots & a_1^{22n} & \dots & a_1^{m21} & \dots & a_1^{m2n} \\ \vdots & & \vdots & \vdots & & \vdots & \dots & \vdots & & \vdots \\ a_n^{121} & \dots & a_n^{12n} & a_n^{221} & \dots & a_n^{22n} & \dots & a_n^{m21} & \dots & a_n^{m2n} \\ \hline \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \hline a_1^{1m1} & \dots & a_1^{1mn} & a_1^{2m1} & \dots & a_1^{2mn} & \dots & a_1^{mm1} & \dots & a_1^{mmn} \\ \vdots & & \vdots & \vdots & & \vdots & \dots & \vdots & & \vdots \\ a_n^{1m1} & \dots & a_n^{1mn} & a_n^{2m1} & \dots & a_n^{2mn} & \dots & a_n^{mm1} & \dots & a_n^{mmn} \end{pmatrix}$$

Now let us write down the $m \times m$ matrix of the E -linear map m_α . We know that the action on the E -basis of L is given by

$$\alpha \ell_j = \sum_{t=1}^m c_{jt} \ell_t$$

Thus the corresponding matrix is

$$C = \begin{pmatrix} c_{11} & c_{21} & \dots & c_{m1} \\ c_{12} & c_{22} & \dots & c_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ c_{1m} & c_{2m} & \dots & c_{mm} \end{pmatrix}$$

Hence we have

$$\text{tr}_{L/E}(\alpha) = \text{tr}(C) = \sum_{t=1}^m c_{tt}$$

Now consider the F -linear map $m_{c_{xy}} : E \rightarrow E$. The action on the F -basis of E is given by

$$c_{xy} e_j = \sum_{z=1}^n a_z^{xyj} e_z$$

The corresponding matrix is hence given by

$$A^{xy} = \begin{pmatrix} a_1^{xy1} & a_1^{xy2} & \dots & a_1^{xyn} \\ a_2^{xy1} & a_2^{xy2} & \dots & a_2^{xyn} \\ \vdots & \vdots & \ddots & \vdots \\ a_n^{xy1} & a_n^{xy2} & \dots & a_n^{xyn} \end{pmatrix}$$

Hence we get

$$\text{tr}_{E/F}(c_{xy}) = \text{tr}(A^{xy})$$

Now observe that A^{yx} is the (x, y) -th block of the bigger matrix M . Finally we get the full trace to be

$$\begin{aligned}
\mathrm{tr}_{L/F}(\alpha) &= \sum_{k=1}^m \mathrm{tr}(A^{kk}) \\
&= \sum_{k=1}^m \mathrm{tr}_{E/F}(c_{kk}) \\
&= \mathrm{tr}_{E/F} \left(\sum_{k=1}^m c_{kk} \right) \\
&= \mathrm{tr}_{E/F} (\mathrm{tr}_{L/E}(\alpha)) \\
&= (\mathrm{tr}_{E/F} \circ \mathrm{tr}_{L/E}) (\alpha)
\end{aligned}$$

This completes the proof. ■

Proposition 4.27. *Suppose $\sigma_1, \dots, \sigma_n$ are the roots of the minimal polynomial of α over K , then the trace is given by*

$$\mathrm{tr}_{L/K}(\alpha) = [L : K(\alpha)] \sum_{i=1}^n \sigma_i$$

Proof. Let $p(x) = \prod_{i=1}^n (x - \sigma_i) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be the minimal polynomial of α over K . Consider the basis $\{1, \alpha, \dots, \alpha^{n-1}\}$ of the extension $K(\alpha)$ over K .

Then the matrix of the K -linear map $m_\alpha : K(\alpha) \rightarrow K(\alpha)$ is simply

$$\begin{pmatrix}
0 & 0 & \cdots & 0 & -a_0 \\
1 & 0 & \cdots & 0 & -a_1 \\
0 & 1 & \cdots & 0 & -a_2 \\
\vdots & \vdots & \ddots & \vdots & \vdots \\
0 & 0 & \cdots & 1 & -a_{n-1}
\end{pmatrix}$$

Thus we get

$$\mathrm{tr}_{K(\alpha)/K}(\alpha) = -a_{n-1} = \sum_{i=1}^n \sigma_i$$

Now we compute $\mathrm{tr}_{L/K(\alpha)}(\alpha)$. This is a special case of the trace since the element whose trace is to be computed is already in the base field. We know that the action of $\alpha \in K(\alpha)$ on the $K(\alpha)$ -basis $\{p_1, \dots, p_t\}$ of L is

$$\alpha p_i = \alpha p_i$$

Thus the matrix is simply $\alpha I_{t \times t}$ and the trace is $t\alpha = [L : K(\alpha)]\alpha$.

Using Lemma 4.26 we get

$$\mathrm{tr}_{L/K}(\alpha) = \mathrm{tr}_{K(\alpha)/K}([L : K(\alpha)]\alpha)$$

Using linearity, we get our result. ■

Corollary 4.28. *For the finite field $\mathbb{F}_q$ (q being a prime power p^n) considered as an extension of $\mathbb{F}_p$, the trace is given by*

$$\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(\alpha) = \alpha + \alpha^p + \alpha^{p^2} + \dots + \alpha^{p^{n-1}}$$

Proof. Firstly let $\alpha \in \mathbb{F}_q$. We want to find the roots of the minimal polynomial f of α taken over $\mathbb{F}_p$. We claim that if degree of f is d , then the following is the set of roots of f :

$$\alpha, \alpha^p, \alpha^{p^2}, \dots, \alpha^{p^{d-1}}$$

To see why this is true observe that since f has coefficients in $\mathbb{F}_p$, we have $f(\alpha)^p = f(\alpha^p)$.

Further by finiteness of $\mathbb{F}_q$ with $\alpha^{p^n} = \alpha$, if m is the minimum positive integer t with $\alpha^{p^t} = \alpha$, then the length of the orbit is m . We show that $m = d$.

By the minimality of m , $\mathbb{F}_p(\alpha)$ has size p^m and hence $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = m$. But this is also the degree of the minimal polynomial of α and hence $m = d$.

Now that we have obtained the roots, using Proposition 4.27, we get

$$\begin{aligned} \text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(\alpha) &= [\mathbb{F}_q : \mathbb{F}_p(\alpha)] \sum_{i=0}^{d-1} \alpha^{p^i} \\ &= \frac{n}{d} \sum_{i=0}^{d-1} \alpha^{p^i} \\ &= \sum_{i=0}^{n-1} \alpha^{p^i} \end{aligned}$$

The last equality is a well-known identity and follows directly through the period of the orbits. ■

The theorem below characterises the characters of $(\mathbb{F}_q, +)$.

Theorem 4.29. *Consider the additive group $(\mathbb{F}_q, +)$ of the finite field $\mathbb{F}_q$ having $q = p^n$ elements for some prime p and $n \geq 1$. Fix a primitive p -th root of unity $\omega \in \mathbb{C}$.*

For every $a \in \mathbb{F}_q$ define the map $\chi_a : \mathbb{F}_q \rightarrow \mathbb{C}^\times$ given by

$$\chi_a(x) = \omega^{\text{tr}(ax)} = \omega^{(ax) + (ax)^p + \dots + (ax)^{p^{n-1}}}$$

where $\text{tr}(x)$ denotes $\text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x)$.

Then every χ_a is a character of $(\mathbb{F}_q, +)$. Conversely, every character of $(\mathbb{F}_q, +)$ is of the form of χ_a for some unique a .

Proof. Firstly note that it might seem ambiguous when we say that every character will be of the form of χ_a with a fixed ω . However if ω' is another primitive p -th root of unity with $\omega' = \omega^k$ for some $k \in \{1, 2, \dots, p-1\}$ then $(\omega')^{\text{tr}(ax)} = \chi_{ka}(x)$. Thus we may fix some ω throughout without ambiguity.

Using linearity of the trace it is clear that for every $x, y \in \mathbb{F}_q$,

$$\chi_a(x+y) = \chi_a(x)\chi_a(y)$$

Thus χ_a is indeed a character, that is, a group homomorphism.

Conversely, let $\eta : (\mathbb{F}_q, +) \rightarrow \mathbb{C}^\times$ be a character. Then,

$$\eta(px) = (\eta(x))^p$$

But $px = 0$ and hence $\eta(px) = \eta(0) = 1$. Thus, $(\eta(x))^p = 1$. Hence, the image of η consists of p -th roots of unity.

Hence for some $f : \mathbb{F}_q \rightarrow \mathbb{F}_p$, the character η has the following form:

$$\eta(x) = \omega^{f(x)} \quad \forall x \in \mathbb{F}_q$$

Since η is a group homomorphism we have

$$f(x+y) = f(x) + f(y) \quad (\text{over } \mathbb{F}_p)$$

Thus f is an $\mathbb{F}_p$ -linear map. The following is a well-known fact which characterises such maps.

Lemma. *The map $\Phi : \mathbb{F}_q \rightarrow \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_q, \mathbb{F}_p)$ given by $(\Phi(a))(x) = \text{tr}(ax)$ is an $\mathbb{F}_p$ -linear isomorphism.*

Using the lemma we have that f must be of the form $f(x) = \text{tr}(ax)$ for some $a \in \mathbb{F}_q$.

We now show uniqueness. Suppose $\chi_a = \chi_b$ then we have

$$\text{tr}((a-b)x) = 0 \quad \forall x \in \mathbb{F}_q \quad (\text{over } \mathbb{F}_p)$$

Suppose for some fixed $c \in \mathbb{F}_q$ if $\text{tr}(cx) = 0$ for every $x \in \mathbb{F}_q$, then we must show that $c = 0$.

$$\text{tr}(cx) = \sum_{i=0}^{n-1} (cx)^{p^i} = \sum_{i=0}^{n-1} c^{p^i} x^{p^i}$$

Define the polynomial

$$L_c(X) = \sum_{i=0}^{n-1} c^{p^i} X^{p^i} \in \mathbb{F}_q[X]$$

This polynomial, if non-zero, cannot have more than p^{n-1} roots. However, we say that $L_c(x) = 0$ for every $x \in \mathbb{F}_q$ which is a contradiction. Thus the polynomial must be the zero polynomial and this is clearly impossible if $c \neq 0$. ■

4.1.5 MacWilliams identities and inequalities

We begin by defining an inner product on Ω .

Proposition 4.30. *The map $\varphi : \Omega \times \Omega \rightarrow \mathbb{F}_q$ given by $(f, g) \mapsto \text{tr}(fg^\top)$ is an inner product on Ω .*

Proof. There is no notion of positivity since we are working with $\mathbb{F}_q$. Thus we need to verify that this is a symmetric, bilinear and non-degenerate map. The symmetric nature and bilinearity are obvious.

Let $\varphi(f, g) = 0$ for every $g \in \Omega$. We need to show that f is the zero matrix. To this end, let g be the matrix with 1 in the (i, j) -th entry and 0 elsewhere (i.e., $g = E_{i,j}$). Then $fg^\top = fE_{j,i}$, which is a matrix whose only non-zero column is its i -th column, and this column is exactly the j -th column of f . The trace of this matrix is simply its (i, i) -th entry, which evaluates exactly to $f_{i,j}$. Since this trace is 0 for every $1 \leq i \leq m, 1 \leq j \leq m'$, f must be the zero matrix. ■

Proposition 4.31. *Fix a character χ of the group $(\mathbb{F}_q, +)$. For a fixed $g \in \Omega$, the following is a character of $(\Omega, +)$. Define the map $\psi_g : (\Omega, +) \rightarrow \mathbb{C}^\times$ to be*

$$\psi_g(f) = \chi(\text{tr}(fg^\top))$$

Conversely, every character of Ω must be of this form for some unique g .

Proof. It is clear that ψ_g is a group homomorphism. Indeed,

$$\psi_g(f + f') = \chi(\text{tr}(fg^\top + f'g^\top)) = \chi(\text{tr}(fg^\top) + \text{tr}(f'g^\top)) = \psi_g(f) + \psi_g(f')$$

Conversely, suppose $\Psi : (\Omega, +) \rightarrow \mathbb{C}^\times$ is a character. By definition,

$$\Psi(f + f') = \Psi(f)\Psi(f')$$

Now Ω is the space of all $m \times m'$ matrices over $\mathbb{F}_q$. Fix some (i, j) with $1 \leq i \leq m, 1 \leq j \leq m'$. Consider $f = xE_{i,j}$ and $f' = x'E_{i,j}$ where $E_{i,j}$ denotes the matrix having the only non-zero entry 1 in the (i, j) -th position. We have

$$\Psi((x + x')E_{i,j}) = \Psi(xE_{i,j})\Psi(x'E_{i,j})$$

Thus Ψ restricted to the (i, j) -th position of the matrix is a character of $(\mathbb{F}_q, +)$ and by Theorem 4.29 we have

$$\Psi(xE_{i,j}) = \chi_{a_{i,j}}(x)$$

for some unique a_{ij} (with respect to a fixed p -th root of unity used for defining χ). Thus we have

$$\begin{aligned}\Psi(f) &= \Psi\left(\sum_{i,j} f_{ij} E_{ij}\right) \\ &= \prod_{i,j} \Psi(f_{ij} E_{ij}) \\ &= \prod_{i,j} \chi_{a_{ij}}(f_{ij}) \\ &= \chi\left(\sum_{i,j} a_{ij} f_{ij}\right)\end{aligned}$$

Choosing g to be the matrix with entries a_{ij} , it is clear that

$$\Psi(f) = \chi(\text{tr}(fg^\top))$$

The uniqueness of g follows from the uniqueness of the a_{ij} . A direct argument is not too difficult, either. $\blacksquare$

We usually write this as $\psi_g(f) = \langle f, g \rangle = \psi_f(g) = \langle g, f \rangle$. Note that despite the notation this is not an inner product on Ω .

Fix a $g \in \Omega$. We extend this character to a ring homomorphism from the group ring to the scalars by linearity as

$$A \mapsto \langle g, A \rangle := \sum_{f \in \Omega} a_f \langle g, f \rangle \in \mathbb{Z}[\omega]$$

where $A = \sum_{f \in \Omega} a_f f \in \mathbb{Z}\Omega$ and ω is a p -th primitive root of unity.

Definition 4.32. We define the *rank k -element* $\Delta_k \in \mathbb{Z}[\omega]\Omega$ as

$$\Delta_k = \sum_{f \in \Omega} \langle f, \Omega_k \rangle f = \sum_{f \in \Omega} \sum_{g \in \Omega_k} \langle f, g \rangle f$$

The coefficients are denoted as $\Delta_k(f) = \langle f, \Omega_k \rangle$.

The following proposition gives that $\Delta_k \in \mathbb{Z}\Omega$ and $\Delta_k(f)$ only depends on the rank of f .

Proposition 4.33. Δ_k lies in the Schur ring generated by the orbits Ω_r .

Proof. We need to show that $\Delta_k(f)$ is an integer and is only dependent on the rank of f .

For any $u \in \mathbb{Z}[\omega]$ and $a \in \{0, 1, \dots, p-1\}$, let $u^{(a)}$ denote the a -th conjugate of u , that is, the image of u under the map $\omega \mapsto \omega^a$.

We then have,

$$\begin{aligned}(\Delta_k(f))^{(a)} &= \left(\sum_{g \in \Omega_k} \chi(\text{tr}(fg^\top)) \right)^{(a)} \\ &= \left(\sum_{g \in \Omega_k} \omega^{\text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(\text{tr}(fg^\top))} \right)^{(a)} \\ &= \sum_{g \in \Omega_k} \omega^{a \text{tr}_{\mathbb{F}_q/\mathbb{F}_p}(\text{tr}(fg^\top))} \\ &= \sum_{g \in \Omega_k} \chi(\text{tr}(f(ag)^\top)) \\ &= \sum_{g \in \Omega_k} \langle f, ag \rangle\end{aligned}$$

But $\text{rank}(ag) = \text{rank}(g)$ and hence changing the summation index we get that the sum is simply $\Delta_k(f)$. Thus all conjugates of the number $\Delta_k(f) \in \mathbb{Z}[\omega]$ are equal implying that it has no complex part. Hence, $\Delta_k(f) \in \mathbb{Z} \subseteq \mathbb{Z}[\omega]$ is an integer.

Now suppose f and h have the same rank. Since they are in the same orbit, we have some $\alpha \in G$ and $\alpha' \in G'$ such that $h = \alpha^\top f \alpha'$. Thus,

$$\Delta_k(h) = \sum_{g \in \Omega_k} \langle h, g \rangle = \sum_{g \in \Omega_k} \langle \alpha^\top f \alpha', g \rangle = \sum_{g \in \Omega_k} \langle f, \alpha g (\alpha')^\top \rangle = \sum_{g \in \Omega_k} \langle f, g \rangle = \Delta_k(f)$$

This shows that the coefficients $\Delta_k(f)$ only depend on the rank of f . This completes the proof. $\blacksquare$

We have some orthogonality relations for these rank k -elements.

Proposition 4.34. *For $k, l \in \{0, 1, \dots, m\}$ we have*

$$\Delta_k \Delta_l = \delta_{k,l} |\Omega| \Delta_k$$

Proof. We use the product of $(\Omega, +)$ to get

$$\begin{aligned} \Delta_k \Delta_l &= \left(\sum_{f \in \Omega} \langle f, \Omega_k \rangle f \right) \left(\sum_{f \in \Omega} \langle f, \Omega_l \rangle f \right) \\ &= \sum_{f, g \in \Omega} \langle f, \Omega_k \rangle \langle g, \Omega_l \rangle (f + g) \\ &= \sum_{h \in \Omega} \langle f - h, \Omega_k \rangle \langle h, \Omega_l \rangle f \end{aligned}$$

We show that the coefficients are the same on both sides.

$$\begin{aligned} (\Delta_k \Delta_l)(f) &= \sum_{h \in \Omega} \langle f - h, \Omega_k \rangle \langle h, \Omega_l \rangle \\ &= \sum_{h \in \Omega} \sum_{g \in \Omega_k} \langle f - h, g \rangle \langle h, \Omega_l \rangle \end{aligned}$$

Now note that

$$\langle f - h, g \rangle = \chi(\text{tr}((f - h)g^\top)) = \chi(\text{tr}(fg^\top) - \text{tr}((-h)g^\top)) = \langle f, g \rangle \langle -h, g \rangle$$

Thus we reach

$$\begin{aligned} (\Delta_k \Delta_l)(f) &= \sum_{h \in \Omega} \sum_{g \in \Omega_k} \langle f, g \rangle \langle -h, g \rangle \langle h, \Omega_l \rangle \\ &= \sum_{g \in \Omega_k} \langle f, g \rangle \sum_{h \in \Omega} \langle h, -g \rangle \sum_{u \in \Omega_l} \langle h, u \rangle \\ &= \sum_{g \in \Omega_k} \langle f, g \rangle \sum_{u \in \Omega_l} \sum_{h \in \Omega} \langle h, u - g \rangle \end{aligned}$$

Recall that for a character χ of a finite group G , we have the orthogonality relation:

$$\sum_{g \in G} \chi(g) = \begin{cases} |G| & \text{if } \chi \text{ is the trivial character} \\ 0 & \text{otherwise} \end{cases}$$

Applying this to the innermost sum over $h \in \Omega$, the sum evaluates to $|\Omega|$ if $u - g = 0$ (i.e., $u = g$) and 0 otherwise. Thus:

$$\sum_{u \in \Omega_l} \sum_{h \in \Omega} \langle h, u - g \rangle = |\Omega| |\{u \in \Omega_l \mid u = g\}| = |\Omega| \delta_{k,l}$$

Substituting this back into our coefficient expression, we obtain:

$$(\Delta_k \Delta_l)(f) = \sum_{g \in \Omega_k} \langle f, g \rangle |\Omega| \delta_{k,l} = \delta_{k,l} |\Omega| \Delta_k(f)$$

This finishes the proof. ■

Corollary 4.35. *The elements $E_k = |\Omega|^{-1} \Delta_k$ form the full set of primitive idempotents of the Schur ring.*

Proof. We have the orthogonality relations

$$E_k E_l = \delta_{k,l} E_k \quad \forall k, l = 0, 1, \dots, m$$

One may note that the E_k 's are linearly independent and sum to the identity of the Schur ring (and hence form a basis of the ring). Further, each of them generates a one-dimensional ideal and hence must be primitive. ■

We have explicit formulas for the coefficients $\Delta_k(f)$ which we cover in a separate chapter following this one.

Definition 4.36. We set $\Delta_k(f) = P_k(i)$ for every $f \in \Omega_i$ in accordance with Proposition 4.33. The $(m+1) \times (m+1)$ matrix P whose (i, k) -th entry is $P_k(i)$ is defined to be the **character matrix of the Schur ring**.

We have the following lemmas which prove Theorem 4.39. This theorem is not related to any of the others and is stated and proved as a bonus.

Lemma 4.37. *For any $k < i$ integers in $\{0, 1, \dots, m\}$ we have*

$$\frac{P_i(k)}{P_k(i)} = \frac{|\Omega_k|}{|\Omega_i|}$$

Proof. Consider the double sum

$$S = \sum_{f \in \Omega_k} \sum_{g \in \Omega_i} \langle f, g \rangle$$

We can write it in two ways as

$$\sum_{f \in \Omega_k} \langle f, \Omega_i \rangle = S = \sum_{g \in \Omega_i} \langle g, \Omega_k \rangle$$

From Proposition 4.33 we know that $\langle f, \Omega_r \rangle$ is constant for every $f \in \Omega_t$ and this constant is precisely $P_r(t)$. Thus we get

$$|\Omega_k| P_i(k) = S = |\Omega_i| P_k(i)$$
■

Lemma 4.38. *For all $k, l \in \{0, 1, \dots, m\}$ we have*

$$\sum_{i=0}^m |\Omega_i| P_k(i) P_l(i) = |\Omega| |\Omega_k| \delta_{k,l}$$

Equivalently, in matrix form we have

$$P^\top D P = |\Omega| D$$

where D is the diagonal $(m+1) \times (m+1)$ matrix with entries $(|\Omega_0|, \dots, |\Omega_m|)$

Proof. We have

$$\sum_{i=0}^m |\Omega_i| P_k(i) P_l(i) = \sum_{i=0}^m \sum_{f \in \Omega_i} \left(\sum_{g \in \Omega_k} \langle f, g \rangle \right) \left(\sum_{h \in \Omega_l} \langle f, h \rangle \right)$$

Using that the character is a group homomorphism we get

$$\begin{aligned} \sum_{i=0}^m |\Omega_i| P_k(i) P_l(i) &= \sum_{f \in \Omega} \sum_{g \in \Omega_k} \sum_{h \in \Omega_l} \langle f, g+h \rangle \\ &= \sum_{g \in \Omega_k} \sum_{h \in \Omega_l} \sum_{f \in \Omega} \langle f, g+h \rangle \end{aligned}$$

Now recall the following property of group characters:

$$\sum_{f \in \Omega} \langle f, x \rangle = \begin{cases} |\Omega| & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

Thus we count $|\Omega|$ for every pair $(g, h) \in \Omega_k \times \Omega_l$ with $g + h = 0$. If $k \neq l$, the number of such pairs is zero. If $k = l$, the number of such pairs is exactly the number of choices for, say, g ; which is $|\Omega_k|$. ■

Theorem 4.39. *The matrix P satisfies $P^2 = |\Omega| I = q^{mm'} I$.*

Proof. Define the $(m+1) \times (m+1)$ matrix M whose (i, k) -th entry is $P_k(i)/|\Omega_k|$, that is, it is a column scaled version of P .

We first have that M is symmetric. The symmetric nature follows from Lemma 4.37.

Consider the matrix $D = \text{diag}(|\Omega_0|, \dots, |\Omega_m|)$ as before. Set

$$S = D^{1/2} M D^{1/2}$$

where $D^{1/2}$ is the diagonal matrix with entries being positive square roots of the entries of D , which are conveniently non-negative; in fact, positive. Thus we may talk about D^{-1} and $D^{1/2}$, too.

Note that the relation between P and M is given by $M = P D^{-1}$.

Thus $S = D^{1/2} P D^{-1/2}$ and $P = D^{-1/2} S D^{1/2}$.

Clearly S is a symmetric matrix since M is, too. We compute $S^2 = S^\top S$ below.

$$\begin{aligned} S^2 &= S^\top S = D^{1/2} M^\top D^{1/2} D^{1/2} M D^{1/2} \\ &= D^{1/2} D^{-1} P^\top D P D^{-1} D^{1/2} \\ &= D^{-1/2} (P^\top D P) D^{-1/2} \end{aligned}$$

Using Lemma 4.38 we get $S^2 = |\Omega| I$. Thus,

$$P^2 = D^{-1/2} S^2 D^{1/2} = |\Omega| I$$

This finishes the proof. ■

We now build definitions to state and prove the Mac-Williams inequality and identity.

Definition 4.40. Given a non-empty subset $X \subseteq \Omega$ we define the **q-distance enumerator** of X to be the formal polynomial

$$a_X(z) = \frac{1}{|X|} \sum_{f, g \in X} z^{\text{rank}(f-g)}$$

Observe that the ‘polynomial’ is indeed a polynomial, that is, it is a finite formal sum. Let us write it as

$$a_X(z) = a_0 + a_1 z + \cdots + a_m z^m$$

Note that the coefficient of z^r in $a_X(z)$ is the average number of elements $g \in X$ at a distance of r from a fixed element $f \in X$.

In particular, also note that if X is a subgroup of $(\Omega, +)$, then we may as well sum over a single variable f and we get that the coefficient of z^r in $a_X(z)$ is $|X \cap \Omega_r|$.

Theorem 4.41 (Mac-Williams Inequalities). *The character matrix P of the Schur ring R generated by the Ω_r satisfies*

$$\sum_{i=0}^m a_i P_k(i) \geq 0 \quad \forall k$$

for any choice of subset X and its q -distance enumerator polynomial $a_X(z)$.

Proof. Consider $\underline{X}$ and $\underline{X}^{-1}$. Taking their product we get

$$\begin{aligned} \underline{X} \underline{X}^{-1} &= \left(\sum_{f \in X} f \right) \left(\sum_{g \in X} (-g) \right) \\ &= \sum_{h \in \Omega} N_h h \end{aligned}$$

where N_h is the number of ordered pairs $(f, g) \in X \times X$ such that $f - g = h$. Suppose h has rank t . Then we have

$$\begin{aligned} \sum_{h \in \Omega_t} N_h &= \sum_{h \in \Omega_t} |\{(f, g) \in X \times X : f - g = h\}| \\ &= |X| a_t \end{aligned}$$

The above follows by observing that the sets whose cardinalities are being summed are all disjoint. The rest follows from the definition of a_t as

$$a_t = \frac{1}{|X|} |\{(f, g) \in X \times X : \text{rank}(f - g) = t\}|$$

Now we have the following conclusions

$$\begin{aligned} |X| \sum_{i=0}^m a_i P_k(i) &= \sum_{i=0}^m \left(\sum_{h \in \Omega_i} N_h \right) P_k(i) \\ &= \sum_{i=0}^m \sum_{h \in \Omega_i} N_h \langle h, \Omega_k \rangle \\ &= \sum_{h \in \Omega} N_h \langle h, \Omega_k \rangle \\ &= \sum_{g \in \Omega_k} \sum_{h \in \Omega} \langle N_h h, g \rangle \\ &= \sum_{g \in \Omega_k} \langle \underline{X} \underline{X}^{-1}, g \rangle \end{aligned}$$

We now analyse the terms inside the summation. We have for each $g \in \Omega_k$,

$$\begin{aligned} \langle g, \underline{X} \underline{X}^{-1} \rangle &= \sum_{h_1 \in X} \sum_{h_2 \in X} \langle g, h_1 - h_2 \rangle \\ &= \langle g, X \rangle \sum_{h \in X} \langle g, -h \rangle \end{aligned}$$

Recall that $\langle f, g \rangle = \chi(\text{tr}(fg^\top))$. Hence, $\langle g, -h \rangle = \langle g, h \rangle^{-1}$. However, the character χ of the finite group $(\mathbb{F}_q, +)$ only takes values on the complex unit circle and hence $\chi(a)^{-1} = \overline{\chi(a)}$ for every $a \in \mathbb{F}_q$.

Thus we get

$$\langle g, \underline{X} \underline{X}^{-1} \rangle = \langle g, X \rangle \overline{\langle g, X \rangle} = |\langle g, X \rangle|^2 \geq 0$$

This finishes the proof of the inequality since we are summing over all non-negative terms and the $|X|$ term on the LHS is strictly positive. $\blacksquare$

Definition 4.42. When $X \subseteq \Omega$ is a subgroup of Ω , then we may define the **dual subgroup** $X^\perp$ to be

$$X^\perp = \{g \in \Omega : \langle f, g \rangle = 1 \ \forall f \in X\}$$

One may verify that this is indeed a subgroup and that we need X to be a subgroup for this to be a subgroup. Further, it is also clear that $(X^\perp)^\perp = X$.

Theorem 4.43 (Mac-Williams Identities). *The q -distance enumerators $a_X(z)$ and $a_{X^\perp}(z)$ of a subgroup $X \subseteq \Omega$ and its dual $X^\perp$ are related by*

$$a_k^\perp = \frac{1}{|X|} \sum_{i=0}^m a_i P_k(i) \quad \forall k$$

Proof. Since X is a subgroup we have $a_i = |X \cap \Omega_i|$ as noted earlier. Now consider the following conclusions.

$$\begin{aligned} \sum_{i=0}^m a_i P_k(i) &= \sum_{i=0}^m |X \cap \Omega_i| P_k(i) \\ &= \sum_{i=0}^m \sum_{f \in X \cap \Omega_i} \langle f, \Omega_k \rangle \\ &= \sum_{i=0}^m \sum_{g \in \Omega_k} \sum_{f \in X \cap \Omega_i} \langle f, g \rangle \\ &= \sum_{g \in \Omega_k} \sum_{f \in X} \langle g, f \rangle \\ &= \sum_{g \in \Omega_k} \langle g, X \rangle \end{aligned}$$

Now we analyse $\langle g, X \rangle$. If $g \in X^\perp$, then clearly this is $|X|$, by definition of $X^\perp$.

Suppose $g \notin X^\perp$. Then there exists some $h_0 \in X$ with $\langle g, h_0 \rangle \neq 1$. Set $S = \langle g, X \rangle$. We have,

$$\begin{aligned} \langle g, h_0 \rangle S &= \sum_{f \in X} \langle g, h_0 \rangle \langle g, f \rangle \\ &= \sum_{f \in X} \langle g, f + h_0 \rangle \\ &= \sum_{f' \in X} \langle g, f' \rangle \\ &= S \end{aligned}$$

Note that the penultimate step uses the fact that the summation can be reordered since X is a subgroup and hence $X + h_0 = X$. Since $\langle g, h_0 \rangle \neq 1$, we must have $S = 0$. Thus,

$$\langle g, X \rangle = \begin{cases} |X| & \text{if } g \in X^\perp \\ 0 & \text{otherwise} \end{cases}$$

Using this we have

$$\sum_{i=0}^m a_i P_k(i) = |X| |\Omega_k \cap X^\perp|$$

But since $X^\perp$ is a subgroup, too, we have $a_i^\perp = |X^\perp \cap \Omega_i|$. This finishes the proof. $\blacksquare$

4.1.6 Applications to coding theory

We think of codes in the following way. Consider a code of length $n = q^{m+m'}$ over the alphabet $\mathbb{F}_q$. It can be thought of as lying in the space $\mathbb{F}_q^{V \times V'}$. That is, the vector of the codeword x is indexed by elements of $V \times V'$. Let us denote $\mathbb{F}_q^{V \times V'}$ by W .

We briefly recall the concepts of weight enumerator and distance enumerator polynomials. The definition of weight enumerator polynomials is given by Definition 1.35.

Definition 4.44. Given a code $C \subseteq W$, we define the *distance enumerator polynomial* to be

$$B(z) = \frac{1}{|C|} \sum_{f,g \in C} z^{\text{wt}(f-g)} = \sum_{i=0}^n B_i z^i$$

where B_i is the average number of codewords $f \in C$ at a distance of i from a fixed codeword $g \in C$.

If C happens to be a linear code, that is, a subspace of W , then it is clear that $A(z) = B(z)$. Here, $A(z)$ is the weight enumerator polynomial.

In this exposition, we discuss Reed-Muller codes. In our setting we define these codes as follows.

Definition 4.45. Fix a non-negative integer r . Set $n = q^{m+m'}$. We define the *r -th order generalised Reed-Muller code*, denoted by C_r , as the following subspace of W :

$$C_r = \{f : V \times V' \rightarrow \mathbb{F}_q \mid f \text{ is a polynomial on } V \times V' \text{ with total degree } \leq r\}$$

We use the notation of C_r in place of $GRM_q(r, n)$. One should contrast this with the classical Definition 1.32.

Note that by polynomials over $V \times V'$ we simply mean that we choose a basis of $V \times V' \cong \mathbb{F}_q^{m+m'}$ then every vector of $V \times V'$ may be expressed as an $(m+m')$ -tuple over $\mathbb{F}_q$. Then the polynomial f over $V \times V'$ is defined to be

$$f(X_1, \dots, X_{m+m'}) = \sum c_{i_1, \dots, i_{m+m'}} X_1^{i_1} \cdots X_{m+m'}^{i_{m+m'}}$$

with $c_{i_1, \dots, i_{m+m'}} \in \mathbb{F}_q$.

One may check that C_r is a linear code. Below, we classify the code for $r = 1$.

Proposition 4.46. C_1 is the set of all maps of the form $\psi + \psi' + c$ where $\psi : V \rightarrow \mathbb{F}_q$ and $\psi' : V' \rightarrow \mathbb{F}_q$ are linear forms on V and V' and c is a constant function.

Proof. We want the total degree of $f : V \times V' \rightarrow \mathbb{F}_q$ to be less than or equal to 1. Fixing the standard basis of $\mathbb{F}_q$ we write

$$f(X_1, \dots, X_{m+m'}) = \sum c_{i_1, \dots, i_{m+m'}} X_1^{i_1} \cdots X_{m+m'}^{i_{m+m'}}$$

Since we require total degree at most 1, it must be the case that either all i_t are 0 or one of the i_t is 1 while the others are 0.

In the first case we get a constant $c_{0, \dots, 0}$. In the latter, we get for every $t \in [m+m']$,

$$f(X_1, \dots, X_{m+m'}) = c_{0, \dots, 0, 1, 0, \dots, 0} X_t$$

Depending on whether $1 \leq t \leq m$ or $m+1 \leq t \leq m+m'$ we have that the RHS is clearly a linear form on V or V' .

Thus every monomial of degree at most one is either a constant or a linear form on V or V' . Thus adding over the monomials we get that every map in C_1 must have the desired form.

Conversely, it is clear that every map of the desired form is in C_1 . This finishes the proof. $\blacksquare$

For future use we define the trinomial $A_r(z)$ below.

Definition 4.47. Let $r \in \{0, \dots, m\}$. We define

$$A_r(z) = A_{r,0}z^{w_{r,0}} + A_{r,1}z^{w_{r,1}} + A_{r,2}z^{w_{r,2}}$$

where for $n = q^{m+m'}$ we define

$$\begin{aligned} w_{r,0} &= (1 - q^{-1})n & A_{r,0} &= q(n - q^{2r}), \\ w_{r,1} &= (1 - q^{-1})(1 - q^{-r})n & A_{r,1} &= q^{2r}, \\ w_{r,2} &= (1 - q^{-1} + q^{-1-r})n & A_{r,2} &= (q - 1)q^{2r} \end{aligned}$$

Lemma 4.48. *The weight enumerator polynomial of $C_1 + f$ for some $f \in \Omega_r$ is precisely $A_r(z)$.*

Proof. Choose bases $\{\mu\}$ and $\{\mu'\}$ of V and V' , respectively, so that the matrix of the bilinear form is in the canonical form

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Then any $g \in C$ will necessarily have the form

$$g(x, x') = \sum_{i=1}^m b_i x_i + \sum_{j=1}^{m'} b'_j x'_j + c + \sum_{i=1}^r x_i x'_i$$

for some $b_i, b'_j, c \in \mathbb{F}_q$ arbitrary. Conversely, every such g will lie in C .

We now proceed to compute the possible weights of such a $g \in C$. For this, we must determine for how many pairs (x, x') the function g evaluates to zero.

Suppose $r < m$ and some index $m \geq t > r$ so that $b_t \neq 0$, we can separate out the linear x_t and write

$$x_t = \psi(x_1, \dots, x_{t-1}, x_{t+1}, \dots, x_m, x'_1, \dots, x'_{m'})$$

A similar statement can be made for an index $m' \geq t' > r$ with $b'_{t'} \neq 0$. Thus choosing the remaining $m + m' - 1$ variables fixes this isolated variable and we get $g(x, x') = 0$.

Thus, the total number of zeroes in this case is $q^{m+m'-1} = nq^{-1}$ and hence we have a possible weight of g as $(1 - q^{-1})n$.

Now suppose we are not in this case. Then $r = m$ or for every index $m \geq t > r$ and index $m' \geq t' > r$ we have $b_t = 0 = b'_{t'}$. In either case we may write

$$\begin{aligned} g(x, x') &= \sum_{i=1}^r (x'_i + b_i)x_i + \sum_{j=1}^r b'_j x'_j + c \\ &= \sum_{i=1}^r (x'_i + b_i)(x_i + b'_i) - \sum_{i=1}^r b_i b'_i + c \end{aligned}$$

Letting $y'_i = x'_i + b_i$, $y_i = x_i + b'_i$ and $K = \sum_{i=1}^r b_i b'_i - c$ we are simply trying to obtain number of solutions to

$$\sum_{i=1}^r y_i y'_i = K$$

We prove a lemma for this below.

Lemma. The number of solutions to $\sum_{i=1}^r y_i y'_i = K$ over $\mathbb{F}_q$ is given by

$$\begin{cases} q^{2r-1} - q^{r-1} & \text{if } K \neq 0 \\ q^r + q^{2r-1} - q^{r-1} & \text{if } K = 0 \end{cases}$$

Proof of Lemma. Suppose $K \neq 0$. Then clearly we cannot choose $y = (y_1, \dots, y_r)$ to be the zero vector. However, fix any non-zero y in $q^r - 1$ choices and then we get a hyperplane, that is, a linear type equation in the y' vector which has q^{r-1} number of solutions. This gives the desired result in this case.

Now suppose $K = 0$. Suppose y is the zero vector, then any of the q^r choices of y' are valid. Suppose y' is not the zero vector. Then by the same logic as the previous case we get $(q^r - 1)q^{r-1}$ solutions. Adding these gives us our result for the second case, too. $\backslash\backslash\backslash$

Now observe that g is only in $2r$ variables and hence the remaining $m + m' - 2r$ variables can be chosen in $q^{m+m'-2r}$ number of ways freely.

In short we have the following possible weights of g .

$$\begin{cases} w_{r,0} = (1 - q^{-1})n & \text{if there exists at least one } r < t \leq m \text{ with } b_t \neq 0 \\ w_{r,0} = (1 - q^{-1})n & \text{if there exists at least one } r < t' \leq m' \text{ with } b'_{t'} \neq 0 \\ w_{r,1} = (1 - q^{-1})(1 - q^{-r})n & \text{if } b_t = 0 = b'_{t'} \quad \forall \quad r < t \leq m, r < t' \leq m' \text{ AND } c = \sum_{i=1}^r b_i b'_i \\ w_{r,2} = (1 - q^{-1} + q^{-1-r})n & \text{if } b_t = 0 = b'_{t'} \quad \forall \quad r < t \leq m, r < t' \leq m' \text{ AND } c \neq \sum_{i=1}^r b_i b'_i \end{cases}$$

This establishes that the weight polynomial is in fact a trinomial. Now all we have to do is compute the coefficients $A_{r,i}$.

We compute $A_{r,1}$ first. We make any choice of $b_1, \dots, b_r, b'_1, \dots, b'_r$ and the constant c gets fixed so that $c = \sum_{i=1}^r b_i b'_i$. Thus $A_{r,1}$ is indeed q^{2r} .

Now we compute $A_{r,2}$. For this case, after making one of the q^{2r} choices as above, the c is chosen so that it is not $\sum_{i=1}^r b_i b'_i$. This has $q - 1$ choices in $\mathbb{F}_q$. Thus we see that $A_{r,2}$ is indeed $(q - 1)q^{2r}$.

$A_{r,0}$ is simply $q^{m+m'+1} - A_{r,1} - A_{r,2}$ which again matches with the definition. Note that we obtained $m + m' + 1$ since b_i, b'_i and c are arbitrary in $\mathbb{F}_q$. This proves that the weight polynomial is indeed the above defined trinomial and we are done. $\blacksquare$

Let $X \subseteq \Omega$ be non-empty. Consider the code $C = C_1 + X$. Since every bilinear map is homogeneous of degree 2, we have that every $g \in C$ is a polynomial of degree at most 2 in $m + m'$ variables. Thus C may be regarded as a subset of C_2 .

It is however not always a subcode of C_2 . This code C may not be linear due to the X . One can verify that C is linear if and only if X is a subgroup of Ω .

We now have the theorem we would like to prove in this chapter.

Theorem 4.49. The distance enumerator polynomial of $C = C_1 + X$ is given by

$$B(z) = \sum_{r=0}^m a_r A_r(z)$$

where $\sum_{r=0}^m a_r z^r$ is the q -distance enumerator polynomial of $X \subseteq \Omega$.

Proof. First we compute the cardinality of our code C . For this, note that X has bilinear forms which must be homogeneous of degree exactly 2 while C_1 has polynomials of degree at most 1. Thus $C_1 \cap X = \emptyset$ and consequently $|C| = |C_1| |X|$.

By definition of the distance enumerator polynomial, if $B(z)$ is the distance enumerator of C we have

$$\begin{aligned}
B(z) &= \frac{1}{|X||C_1|} \sum_{f,g \in C} z^{\text{wt}(f-g)} \\
&= \frac{1}{|X|} \frac{1}{|C_1|} \sum_{h_1, h_2 \in C_1} \sum_{f,g \in X} z^{\text{wt}(h_1-h_2+f-g)} \\
&= \frac{1}{|X|} \frac{1}{|C_1|} |C_1| \sum_{h' \in C_1} \sum_{f,g \in X} z^{\text{wt}(h'+f-g)} \\
&= \frac{1}{|X|} \sum_{h \in C_1} \sum_{f,g \in X} z^{\text{wt}(h+f-g)}
\end{aligned}$$

From Lemma 4.48 we have that if $\text{rank}(f-g) = t$, then

$$\sum_{h \in C_1} z^{\text{wt}(h+f-g)} = A_t(z)$$

Thus if $N(r)$ is the number of pairs $(f, g) \in X^2$ with $\text{rank}(f-g) = r$, we have

$$B(z) = \frac{1}{|X|} \sum_{r=0}^m N(r) A_r(z)$$

Recall that the q -distance enumerator of X is

$$\sum_{r=0}^m a_r z^r = \frac{1}{|X|} \sum_{f,g \in X} z^{\text{rank}(f-g)}$$

Then clearly $|X|a_r = N(r)$. Thus we have

$$B(z) = \sum_{r=0}^m a_r A_r(z)$$

This completes the proof. ■

As another application we highlight another method to construct codes from bilinear forms.

Definition 4.50. Let $f \in \Omega$. We define the function $\mathbf{f}^{(k)} : \Omega_k \rightarrow \mathbb{F}_q$ given by

$$f^{(k)}(g) = \text{tr}(fg^\top)$$

Proposition 4.51. For a fixed $k \in \{1, \dots, m\}$, the mapping $f \mapsto f^{(k)}$ is injective.

Proof. Firstly note that for $k = 0$, it is clearly not injective since it is simply the zero map.

Let $f^{(k)} = g^{(k)}$. Then for every $h \in \Omega_k$ we have

$$\text{tr}(fh^\top) = \text{tr}(gh^\top)$$

$$\therefore \text{tr}((f-g)h^\top) = 0$$

Lemma. All matrices here are $m \times m'$ over $\mathbb{F}_q$. Fix a $k \in \{1, \dots, m\}$. Suppose for a matrix f we have

$$\text{tr}(fg^\top) = 0$$

for every g of rank k , then f must be the 0 matrix.

Proof of Lemma. We have already done this for $k = 1$. Simply choose g to be the matrix with the only non-zero entry being 1 in the (i, j) -th entry. The trace condition gave $f_{ij} = 0$.

Here, we prove it more generally. Let f have rank r . Decompose f into canonical form

$$AfB = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = J \quad (\text{say})$$

for some square matrices A and B of order m and m' respectively.

Using the commutativity of the trace, we have

$$\text{tr}(fg^\top) = \text{tr}(A^{-1}JB^{-1}g^\top) = \text{tr}(JB^{-1}g^\top A^{-1})$$

As g varies over all matrices of rank k exactly once, so does $(A^{-1})^\top g(B^{-1})^\top$. Hence, the condition in question is equivalent to

$$\text{tr}(Jg^\top) = 0$$

for every rank k matrix g . This condition simply implies that

$$\sum_{t=1}^r g_{tt} = 0$$

for every g of rank k .

Suppose, for the sake of contradiction, that $r \geq 1$. We construct a rank k matrix g such that $\text{tr}(Jg^\top) \neq 0$. Set $g_{11} = 1$ and $g_{tt} = 0$ for all $2 \leq t \leq r$. This guarantees the sum is $1 \neq 0$ over $\mathbb{F}_q$. To ensure the overall matrix g has rank k , we can distribute $k - 1$ additional 1s either on the main diagonal strictly outside the top-left $r \times r$ block, or on the off-diagonal entries (e.g., $g_{t,t+1}$) such that they do not intersect or alter the main diagonal of the top-left block. This construction yields a valid rank k matrix where $\text{tr}(Jg^\top) = 1$, which contradicts our hypothesis.

Thus, r must be 0, which forces $f = 0$. \\

Applying the lemma to our case we get $f - g = 0$ giving $f = g$ which shows injectivity of the given map. ■

Proposition 4.52. Fix k from $\{1, 2, \dots, m\}$. Consider a set $X \subseteq \Omega$ and consider the image $X^{(k)} = \{f^{(k)} \mid f \in X\}$ taken with respect to the map from the above proposition. This is a q -ary code of length $|\Omega_k|$ and size $|X|$.

Proof. The size is $|X|$ because of the injectivity proved above. The codewords have been described as $f^{(k)}$. Now each of these is a map $\Omega_k \rightarrow \mathbb{F}_q$ and hence may be represented as a vector of length $|\Omega_k|$ with values in the finite field $\mathbb{F}_q$. One may check that it is a linear code if and only if X is a subgroup of Ω . ■

We have a seemingly unrelated combinatorial result which plays a central role in obtaining the distance enumerator polynomial of $X^{(k)}$.

Proposition 4.53. Fix $i, k \in \{0, 1, \dots, m\}$. Let $N_k(i)$ denote the number of $m \times m'$ matrices g with entries in $\mathbb{F}_q$ of rank k such that the i -trace, $g_{11} + g_{22} + \dots + g_{ii}$ is non-zero. Then

$$N_k(i) = q^{-1}(q - 1) \left(|\Omega_k| - P_k(i) \right)$$

Proof. Recall the trick to find the sum of characters used in Theorem 4.43: For $x \in \mathbb{F}_q$,

$$\sum_{a \in \mathbb{F}_q} \chi(ax) = \begin{cases} 0 & \text{if } x \neq 0 \\ q & \text{if } x = 0 \end{cases}$$

Let $Z_k(i)$ denote the number of $g \in \Omega_k$ with the i -trace, $\text{tr}_i(g)$, being 0. Thus we may write

$$\begin{aligned} Z_k(i) &= \sum_{g \in \Omega_k} \left(\frac{1}{q} \sum_{a \in \mathbb{F}_q} \chi(a \text{tr}_i(g)) \right) \\ &= \frac{1}{q} \sum_{a \in \mathbb{F}_q} \sum_{g \in \Omega_k} \chi(a \text{tr}_i(g)) \\ &= \frac{1}{q} \left(|\Omega_k| + \sum_{a \in \mathbb{F}_q^*} \sum_{g \in \Omega_k} \chi(a \text{tr}_i(g)) \right) \end{aligned}$$

Choose $F_i(\lambda)$ to be the matrix

$$\begin{bmatrix} \lambda I_i & 0 \\ 0 & 0 \end{bmatrix}$$

Firstly, for $a \neq 0$, $F_i(a)$ has rank i . We also have $\text{tr}(F_i(\lambda)g^\top) = \lambda \text{tr}_i(g)$. Thus,

$$\begin{aligned} \sum_{g \in \Omega_k} \chi(a \text{tr}_i(g)) &= \sum_{g \in \Omega_k} \chi(F_i(a)g^\top) \\ &= \langle F_i(a), g \rangle \\ &= P_k(i) \end{aligned}$$

Thus we reach:

$$Z_k(i) = \frac{1}{q} (|\Omega_k| + (q-1)P_k(i))$$

Since $N_k(i) = |\Omega_k| - Z_k(i)$, we have:

$$N_k(i) = |\Omega_k| - \frac{1}{q} |\Omega_k| - \frac{q-1}{q} P_k(i) = \frac{q-1}{q} |\Omega_k| - \frac{q-1}{q} P_k(i)$$

Factoring out $q^{-1}(q-1)$ yields $N_k(i) = q^{-1}(q-1) (|\Omega_k| - P_k(i))$, which completes the proof. ■

We now work towards proving the main theorem on distance enumerator of $X^{(k)}$. We need a few results before we prove this theorem.

Lemma 4.54. *Given a bilinear form f of rank i , the weight of the codeword $f^{(k)}$ is exactly $N_k(i)$.*

Proof. We have

$$\begin{aligned} \text{wt}(f^{(k)}) &= |\{g \in \Omega_k \mid \text{tr}(fg^\top) \neq 0\}| \\ N_k(i) &= |\{g \in \Omega_k \mid \text{tr}_i(g) \neq 0\}| \end{aligned}$$

We convert f to its canonical form J . We can clearly see that the weight will remain preserved since the trace is commutative. Thus,

$$\begin{aligned} \text{wt}(f^{(k)}) &= \text{wt}(J^{(k)}) \\ &= |\{g \in \Omega_k \mid \text{tr}(Jg^\top) \neq 0\}| \\ &= |\{g \in \Omega_k \mid g_{11} + \cdots + g_{ii} \neq 0\}| \\ &= |\{g \in \Omega_k \mid \text{tr}_i(g) \neq 0\}| \\ &= N_k(i) \end{aligned}$$
■

Theorem 4.55. *The distance enumerator polynomial of $X^{(k)}$ is given by*

$$B^{(k)}(z) = \sum_{i=0}^m a_i z^{N_k(i)}$$

Proof. The proof is analogous to Theorem 4.49. We begin with the definition

$$\begin{aligned} B^{(k)}(z) &= \frac{1}{|X^{(k)}|} \sum_{f,g \in X^{(k)}} z^{\text{wt}(f-g)} \\ &= \frac{1}{|X|} \sum_{f,g \in X} z^{\text{wt}((f-g)^{(k)})} \end{aligned}$$

Sorting by rank of $f - g$ we get

$$\begin{aligned} B^{(k)}(z) &= \frac{1}{|X|} \sum_{i=0}^m \left(\sum_{\substack{f,g \in X \\ \text{rank}(f-g)=i}} z^{N_k(i)} \right) \\ &= \sum_{i=0}^m a_i z^{N_k(i)} \end{aligned}$$

■

4.1.7 The character matrix of the Schur ring

The only goal of this section is to make the characters $P_k(i)$ explicit. In doing so, we also make $N_k(i)$ explicit since these directly depend on $P_k(i)$.

Definition 4.56. Let $c \in \mathbb{F}_q$ be a scalar and f be an $m \times m'$ matrix over $\mathbb{F}_q$. We define the **direct sum of a scalar and a matrix** to be

$$c \dot{+} f = \begin{bmatrix} c & 0 \\ 0^\top & f \end{bmatrix}$$

Now, recall the rank k -elements $\Delta_k \in \mathbb{Z}[\omega]\Omega$ from Definition 4.32. In fact, from Proposition 4.33 we saw that $\Delta_k \in \mathbb{Z}\Omega$ and these elements lie in the Schur ring.

Lemma 4.57. Given $i, k \in \{0, \dots, m\}$ and $f \in \Omega_i(m, m')$, one has

$$\Delta_{k+1}(0 \dot{+} f) - \Delta_{k+1}(1 \dot{+} f) = q^{m+m'-i+1} \Delta_k(f)$$

Proof. Let the LHS be denoted by $\mu_k(f)$. We denote $\Omega_{k+1}(m+1, m'+1)$ by Ω_{k+1} and $\Omega_k(m, m')$ by Ω , for simplicity. The set of all $a \times b$ matrices over $\mathbb{F}_q$ shall be denoted by $M_{a,b}$. We simplify the LHS as follows.

$$\begin{aligned} \mu_k(f) &= \sum_{\hat{g} \in \Omega_{k+1}} \langle 0 \dot{+} f, \hat{g} \rangle - \langle 1 \dot{+} f, \hat{g} \rangle \\ &= \sum_{c \in \mathbb{F}_q} \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \sum_{\substack{g \in M_{m,m'} \\ \text{rank}(\hat{g})=k+1}} \langle f, g \rangle - \chi(c) \langle f, g \rangle \\ &= \sum_{c \in \mathbb{F}_q^*} (1 - \chi(c)) \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \sum_{\substack{g \in M_{m,m'} \\ \text{rank}(g - c^{-1}yx^\top)=k}} \langle f, g \rangle \\ &= \sum_{c \in \mathbb{F}_q^*} (1 - \chi(c)) \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \sum_{\substack{t \in M_{m,m'} \\ \text{rank}(t)=k}} \langle f, t + c^{-1}yx^\top \rangle \\ &= \sum_{c \in \mathbb{F}_q^*} (1 - \chi(c)) \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \sum_{\substack{t \in M_{m,m'} \\ \text{rank}(t)=k}} \langle f, t \rangle \langle f, c^{-1}yx^\top \rangle \\ &= \Delta_k(f) \sum_{c \in \mathbb{F}_q^*} (1 - \chi(c)) \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \langle f, c^{-1}yx^\top \rangle \end{aligned}$$

The equalities are explained as follows. The first one is simply the definition. For the second one, we write the matrix $\hat{g} \in M_{m+1, m'+1}$ as follows

$$\hat{g} = \begin{bmatrix} c & x^\top \\ y & g \end{bmatrix}$$

then use simple matrix multiplication to find the trace and then make use of the property that $\langle f_1 + f_2, g \rangle = \langle f_1, g \rangle \langle f_2, g \rangle$. For the third step, we have made use of the fact that if $c = 0$, then the summand is zero since $\chi(0) = 1$. Further, since $c \neq 0$, it is invertible and we may use the Schur complement formula for an invertible matrix A :

$$\text{rank} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \text{rank}(A) + \text{rank}(D - CA^{-1}B)$$

The fourth step is simply substitution of a new variable $t = g - c^{-1}yx^\top$. The fifth step is again the use of the property $\langle f_1 + f_2, g \rangle = \langle f_1, g \rangle \langle f_2, g \rangle$. The last and sixth step is simply pulling out the common and independent $\Delta_k(f) = \sum_{t \in \Omega_k} \langle f, t \rangle$.

Now we analyse the innermost term as follows.

$$\begin{aligned} \sum_{x \in \mathbb{F}_q^{m'}} \sum_{y \in \mathbb{F}_q^m} \langle f, c^{-1}yx^\top \rangle &= \sum_{\substack{u \in M_{m, m'} \\ \exists x \in \mathbb{F}_q^{m'}, \exists y \in \mathbb{F}_q^m \\ yx^\top = u}} \langle c^{-1}f, u \rangle \\ &= N_0 \Delta_0(f) + N_1 \Delta_1(f) \end{aligned}$$

where N_i is the number of pairs $(y, x) \in \mathbb{F}_q^m \times \mathbb{F}_q^{m'}$ such that $yx^\top$ has rank i . Note that given u and u' of rank i , the number of pairs (y, x) such that $yx^\top = u$ and the number of pairs (w, v) such that $wv = u'$ is the same and hence independent of the product matrix u .

The second equality is explained as follows. Now firstly, any $u = yx^\top$ will necessarily have rank at most 1 since rank of x and y is at most 1. If u has rank 0, it is the 0 matrix. If it has rank 1, then we simply get

$$\sum_{\substack{u \in M_{m, m'} \\ \text{rank}(u)=1}} \langle c^{-1}f, u \rangle = \Delta_1(c^{-1}f) = \Delta_1(f)$$

The last part follows since $c \neq 0$.

We now compute values of N_0 and N_1 . For N_0 we want pairs (y, x) such that $yx^\top = 0$. This can only happen when one of the vectors is entirely zero. Thus the total number of such pairs is $q^m + q^{m'} - 1$. We subtracted 1 corresponding to both x and y being 0. For N_1 , let us fix the rank 1 matrix $u = yx^\top$. From rank canonical decomposition we know that $u = y_0 x_0^\top$ for some $y_0 \in \mathbb{F}_q^m$ and $x_0 \in \mathbb{F}_q^{m'}$. Thus we want the number of solutions of $yx^\top = y_0 x_0^\top$. Simple linear algebraic techniques show that $y = \lambda y_0$ and $x = \lambda^{-1} x_0$ are all the solutions for varying $\lambda \in \mathbb{F}_q$. However, the rank 1 condition gives that $\lambda \neq 0$. Thus, we have $N_1 = q - 1$.

We also have $\sum_{c \in \mathbb{F}_q^*} (1 - \chi(c)) = (q - 1) + \sum_{c \in \mathbb{F}_q^*} \chi(c) = q - 1 - (-1) = q$ as can be verified easily. Putting everything together we get

$$\mu_k(f) = \Delta_k(f) \left(q(q^m + q^{m'} - 1) \Delta_0(f) + q(q - 1) \Delta_1(f) \right)$$

To find the formula for $\Delta_1(f)$ we have the following lemma.

Lemma. $P_1(i) = \Delta_1(f)$ is given by the formula

$$\frac{1}{q-1} \left(q^{m+m'-i} - q^m - q^{m'} + 1 \right)$$

Proof of Lemma. We have

$$\begin{aligned}
\Delta_1(f) &= \sum_{u \in \Omega_1} \chi(\text{tr}(fu^\top)) \\
&= \frac{1}{q-1} \sum_{\substack{x \in \mathbb{F}_q^{m'} \\ x \neq 0}} \sum_{\substack{y \in \mathbb{F}_q^m \\ y \neq 0}} \chi(\text{tr}(f(yx^\top)^\top)) \\
&= \frac{1}{q-1} \sum_{x \neq 0} \sum_{y \neq 0} \chi(\text{tr}(fxy^\top)) \\
&= \frac{1}{q-1} \sum_{x \neq 0} \sum_{y \neq 0} \chi(\text{tr}(y^\top fx)) \\
&= \frac{1}{q-1} \sum_{x \neq 0} \sum_{y \neq 0} \chi(y^\top fx)
\end{aligned}$$

From standard character sum properties, we have

$$\sum_{y \in \mathbb{F}_q^m} \chi(y^\top fx) = \begin{cases} q^m & \text{if } fx = 0 \\ 0 & \text{if } fx \neq 0 \end{cases}$$

Furthermore, the number of vectors x such that $fx = 0$ is the cardinality of the null space of f , which is $q^{m'-i}$ since $\text{rank}(f) = i$. Consequently, the number of strictly non-zero vectors x satisfying $fx = 0$ is $q^{m'-i} - 1$. The number of non-zero vectors outside the null space is therefore $(q^{m'} - 1) - (q^{m'-i} - 1) = q^{m'} - q^{m'-i}$.

Using these counts, the summation evaluates to:

$$\begin{aligned}
\Delta_1(f) &= \frac{1}{q-1} \left((q^m - 1)(q^{m'-i} - 1) + (-1)(q^{m'} - q^{m'-i}) \right) \\
&= \frac{1}{q-1} \left(q^{m+m'-i} - q^m - q^{m'} + 1 \right)
\end{aligned}$$

\\

Substituting the result from this lemma we get

$$\mu_k(f) = q^{m+m'-i+1} \Delta_k(f)$$

This finishes the proof of the lemma. ■

It is interesting to note that instead of directly finding the value of $\Delta_1(f)$ as in the lemma inside the lemma above, we can instead use the lemma itself on the base case of $k = 0$ and obtain a recurrence for $P_1(i)$ which we need to solve using generating functions.

Theorem 4.58. For $i, k \in \{0, 1, \dots, m\}$, we have the explicit formula in terms of q -binomial coefficients:

$$P_k(i) = \sum_{j=0}^m (-1)^{k-j} q^{jm' + \binom{k-j}{2}} \binom{m-j}{m-k}_q \binom{m-i}{j}_q$$

Proof. Let $P_k(i)$ be made more explicit in terms of the scheme they are part of as $P_k(i; m, m')$. From Lemma 4.57 we know that these satisfy

$$P_{k+1}(i; m+1, m'+1) - P_{k+1}(i+1; m+1, m'+1) = q^{m+m'-i+1} P_k(i; m, m')$$

with the base case $P_k(0; m, m') = |\Omega_k|$.

We verify the prescribed formula for the base case of $i = 0$.

$$S(k) = \sum_{j=0}^m (-1)^{k-j} q^{jm' + \binom{k-j}{2}} \binom{m-j}{m-k}_q \binom{m}{j}_q$$

We want to show that $S(k)$ is the number of rank k matrices, that is, $|\Omega_k|$. To this end, fix a k -dimensional subspace W of $\mathbb{F}_q^m$ and count the number of $m \times m'$ matrices whose column space is W . Let it be $f_W(k)$.

Firstly note that $f_W(k)$ is independent of W through base change matrix and we may simply call it $f(k)$. The number of choices for W is the q -binomial coefficient. Thus the expression for $|\Omega_k|$ is

$$|\Omega_k| = f(k) \binom{m}{k}_q$$

Now we focus on computing this $f(k)$ using the principle of inclusion and exclusion.

Fix a vector space W of dimension k . Let $g(W)$ denote the number of matrices whose column space is contained in W and $h(W)$ denote the number of matrices whose column space is exactly W . We clearly have the formulae

$$\begin{aligned} g(W) &= \sum_{S \subseteq W} h(S) \\ g(W) &= q^{\dim(W)m'} \end{aligned}$$

Using the well-known Möbius inversion formula we have

$$h(W) = \sum_{S \subseteq W} \mu(S, W) g(S)$$

From the theory of Möbius functions one may obtain that $\mu(S, W)$ is given by

$$\mu(S, W) = (-1)^{\dim W - \dim S} q^{\binom{\dim W - \dim S}{2}}$$

Substituting this back we get our formula for $h(W)$ to be

$$f(k) = h(W) = \sum_{j=0}^k \binom{k}{j}_q (-1)^{k-j} q^{\binom{k-j}{2}} q^{jm'}$$

We now make use of the identity

$$\binom{m}{k}_q \binom{k}{j}_q = \binom{m}{j}_q \binom{m-j}{k-j}_q$$

This finishes the proof of the base case, verifying the prescribed formula. If one wants to avoid the theory of Möbius functions entirely, one may simply fix a basis of W and directly compute $f(W) = f(k) = \prod_{j=0}^{k-1} (q^{m'} - q^j)$. After this, through induction we may prove the above inclusion and exclusion formula.

Now all we need to do is show that the formula given in the theorem statement indeed satisfies the recursion we had obtained. To this end, for brevity let us denote

$$A = P_{k+1}(i; m+1, m'+1); \quad B = P_{k+1}(i+1; m+1, m'+1)$$

Then we have,

$$\begin{aligned} A - B &= \sum_{j=0}^{m+1} (-1)^{k+1-j} q^{j(m'+1) + \binom{k+1-j}{2}} \binom{m+1-j}{m+1-k-1}_q \left(\binom{m+1-i}{j}_q - \binom{m+1-i-1}{j}_q \right) \\ &= \sum_{j=0}^{m+1} (-1)^{k+1-j} q^{j(m'+1) + \binom{k+1-j}{2}} \binom{m+1-j}{m-k}_q \left(q^{m-i-j+1} \binom{m-i}{j-1}_q \right) \\ &= \sum_{j=0}^m (-1)^{k-j} q^{jm' + m' + j + 1 + \binom{k-j}{2}} q^{m-i-j} \binom{m-j}{m-k}_q \binom{m-i}{j}_q \end{aligned}$$

Taking out $q^{m+m'-i+1}$ we will be left with $P_k(i; m, m')$ and this completes the proof. ■

4.1.8 Applications to design theory

Recall the classical definition of a design from Definition 2.1. We however apply this to our theory and alter it to the following definition.

Definition 4.59. A *bilinear forms t -design* is a subset $X \subseteq \Omega$ such that for every t -dimensional subspace $V_0 \subseteq V$ and for every bilinear form f_0 on $V_0 \times V'$, the number of bilinear forms $f \in X$ such that the restriction of f to $V_0 \times V'$ is f_0 is a constant depending only on X and t . This constant is called the *index* of the design.

One must note that there is no concept of a block size k or number of points v in this case. However, one may take points to be (V_0, f_0) in some sense.

Henceforth, by a design, we shall mean the bilinear one.

Proposition 4.60. For a t -design X , the index λ is given by

$$\lambda = q^{-tm'} |X|$$

Proof. This is a standard double counting trick as done with classical designs. We wish to count the cardinality of the set

$$\mathcal{I} = \{(f, (V_0, f_0)) \mid f \in X, V_0 \leq V, \dim(V_0) = t, f_0 \in \Omega(V_0, V'), f|_{V_0 \times V'} = f_0\}$$

Given an $f \in X$, the number of possible pairs (V_0, f_0) satisfying the other four conditions is $\binom{m}{t}_q$.

This is because the choice for f_0 is forced to be the restriction. We simply need to choose the t -dimensional subspace V_0 of V . The number of choices for f is $|X|$ and hence the count is

$$|X| \binom{m}{t}_q$$

We now fix a pair (V_0, f_0) such that V_0 is a t -dimensional subspace of V and $f_0 \in \Omega(V_0, V')$. By definition of a design the number of extensions f of f_0 is λ . Now the number of choices for the pairs (V_0, f_0) is $\binom{m}{t}_q q^{tm'}$. This gives the count of

$$\binom{m}{t}_q q^{tm'} \lambda$$

Equating the two counts, the result follows. ■

We would like to prove a classification result for bilinear form designs. This is Theorem 4.63.

Lemma 4.61. Let $V_0 \subseteq V$ be a vector subspace. Define $U(V_0) = \{f \in \Omega \mid f|_{V_0 \times V'} \equiv 0\}$. Then $U = U(V_0)$ is a subgroup of $(\Omega, +)$ and

$$\sum_{g \in U^\perp} \langle g, f - f_0 \rangle = \begin{cases} |U^\perp| & \text{if } f \in f_0 + U \\ 0 & \text{otherwise} \end{cases}$$

Proof. It is clear that U is a subgroup. The character property has already been proved in the last part of Theorem 4.43. ■

Lemma 4.62. Let V_0 be a t -dimensional subspace of V . Let $U = U(V_0)$ be as in the above lemma. Then the orthogonal complement of U is described as

$$U^\perp \subseteq \{g \in \Omega \mid \text{rank}(g) \leq t\}$$

Proof. Fix a basis $\{e_1, \dots, e_m\}$ for V such that $\{e_1, \dots, e_t\}$ is a basis for V_0 . Then, if $g \in U$, the first t rows of the matrix associated with g must be 0.

By definition, $h \in U^\perp$ if and only if $\chi(\text{tr}(hg^\top)) = 1$ for every $g \in U$. Because χ is a non-trivial character of $(\mathbb{F}_q, +)$, this equality holds for all $g \in U$ if and only if $\text{tr}(hg^\top) = 0$ over $\mathbb{F}_q$. Through elementary linear algebra, considering the arbitrary nature of the rows of g below the t -th row, one concludes that $h_{i,j} = 0$ for every $i > t$.

Thus, every matrix in $U^\perp$ can have non-zero entries only within its first t rows, which immediately implies that the rank of h is at most t . This finishes the proof. $\blacksquare$

Theorem 4.63. *A given $X \subseteq \Omega$ is a t -design if and only if the q -distance enumerator $a_X(z)$ satisfies*

$$\sum_{i=0}^m a_i P_k(i) = |X| \delta_{0,k} \quad \forall k = 0, 1, \dots, t$$

Proof. Firstly the $k = 0$ condition $\sum_{i=0}^m a_i P_0(i) = \sum_{i=0}^m a_i = |X|$ always holds. Thus, we may divert our attention to only $1 \leq k \leq t$.

Now fix some $1 \leq k \leq t$. We have

$$\begin{aligned} \sum_{i=0}^m a_i P_k(i) &= \frac{1}{|X|} \sum_{f,g \in X} P_k(\text{rank}(f-g)) \\ &= \frac{1}{|X|} \sum_{h \in \Omega_k} \sum_{f,g \in X} \langle h, f-g \rangle \\ &= \frac{1}{|X|} \sum_{h \in \Omega_k} \sum_{f \in X} \langle h, f \rangle \sum_{g \in X} \overline{\langle h, g \rangle} \\ &= \frac{1}{|X|} \sum_{h \in \Omega_k} \left| \sum_{f \in X} \langle f, h \rangle \right|^2 \end{aligned}$$

Thus, $\sum_{i=0}^m a_i P_k(i) = 0$ if and only if $\sum_{f \in X} \langle f, h \rangle = 0$ for every $h \in \bigcup_{i=0}^t \Omega_i$.

Now for the forward direction, assume that X is a t -design. Fix $1 \leq k \leq t$ and $h \in \Omega_k$. Let the row space of h be W . Since $\text{rank}(h) \leq t$, fix some t -dimensional $V_0 \subseteq V$ such that $W \subseteq V_0$.

Let $S(V_0, f_0)$ denote the set of all bilinear forms whose restriction to $V_0 \times V'$ is f_0 . By definition of a t -design, we have $|X \cap S(V_0, f_0)| = \lambda = \text{constant}$

Now consider the sum

$$\begin{aligned} \sum_{f \in X} \langle f, h \rangle &= \sum_{f_0 \in \Omega(V_0, V')} \sum_{f \in X \cap S(V_0, f_0)} \langle f, h \rangle \\ &= \sum_{f_0 \in \Omega(V_0, V')} |X \cap S(V_0, f_0)| \langle f_0, h \rangle \\ &= \lambda \sum_{f_0 \in \Omega(V_0, V')} \langle f_0, h \rangle \end{aligned}$$

The second inequality follows by observing that for every $f \in S(V_0, f_0)$, the character $\langle f, h \rangle$ is constant. Indeed, the difference of two functions in $S(V_0, f_0)$ will necessarily lie in $U(V_0)$ allowing us to conclude that $S(V_0, f_0) = f_0 + U(V_0)$. Further, since the row space of h lies inside V_0 , we have $h \in U(V_0)^\perp$ giving that $\langle u, h \rangle = 1$ for every $u \in U(V_0)$.

Now observe that the map $f_0 \mapsto \langle f_0, h \rangle$ is a character on the group $(\Omega(V_0, V'), +)$ and hence

$$\sum_{f_0 \in \Omega(V_0, V')} \langle f_0, h \rangle = \begin{cases} |\Omega(V_0, V')| & \text{if the character is trivial} \\ 0 & \text{otherwise} \end{cases}$$

In our case, since $h \neq 0$, the character is non-trivial. This follows from the non-degeneracy proved in Proposition 4.30. Thus in our case the sum is zero and hence we have

$$\sum_{f \in X} \langle f, h \rangle = 0$$

Since $h \in \bigcup_{i=0}^t \Omega_i$ is arbitrary we are done with the forward implication.

For the reverse implication we assume that for every $h \in \bigcup_{i=0}^t \Omega_i$, we have

$$\sum_{f \in X} \langle f, h \rangle = 0$$

Consider any arbitrary t -dimensional subspace $V_0 \subseteq V$ and an arbitrary form f_0 on $V_0 \times V'$. Let the number of extensions of f_0 to $f \in X$ be $\lambda(V_0, f_0)$. We want to show that this is a constant independent of V_0 and f_0 . We go one step further and show that it is directly

$$\lambda(V_0, f_0) = q^{-tm'} |X|$$

By definition we start with the following equality and manipulate it. Here, $1_A(f)$ is the indicator function of whether $f \in A$ or not.

$$\begin{aligned} \lambda(V_0, f_0) &= \sum_{f \in X} 1_{S(V_0, f_0)}(f) \\ &= \sum_{f \in X} \frac{1}{|U(V_0)^\perp|} \sum_{g \in U(V_0)^\perp} \langle f - f_0, g \rangle \\ &= \frac{1}{|U(V_0)^\perp|} \sum_{g \in U(V_0)^\perp} \langle -f_0, g \rangle \sum_{f \in X} \langle f, g \rangle \\ &= \frac{1}{|U(V_0)^\perp|} (|X| + 0 + \dots + 0) \\ &= q^{-tm'} |X| \end{aligned}$$

There are several equalities which need justification in words. The first one is simply the definition of $\lambda(V_0, f_0)$. The second one follows from Lemma 4.62. The third one is a rearrangement. The fourth one uses the hypothesis we began with except that when $g = 0$ we get $|X|$.

The final equality just computes the size of $U(V_0)^\perp$. Since $U(V_0)$ has matrices (after a change of basis) with the first t rows zero as seen before, the dimension of this subspace of Ω is $(m - t)m'$. Since the dimension of Ω is mm' , we have our result. Indeed, $|U(V_0)^\perp| = |\Omega| - |U(V_0)|$. ■

We now have the dual concept of a codesign.

Definition 4.64. Let $s \in \{0, 1, \dots, m\}$. A non-empty subset $X \subseteq \Omega$ is called a **s-codesign** if for all distinct $f, g \in X$, the rank of $f - g$ is greater than s . Conventionally, if X is Singleton, it is an m -codesign.

Below, we have an equivalent definition through the q -distance enumerator.

Proposition 4.65. $X \subseteq \Omega$ is an s -codesign if and only if the q -distance enumerator satisfies $a_i = \delta_{i,0}$ for every $i \leq s$.

Proof. By definition, the minimum rank of $f - g$ for distinct elements $f, g \in X$ is at least $s + 1$. Therefore, there are no pairs in X at a distance i for $1 \leq i \leq s$, implying $a_1 = \dots = a_s = 0$. The coefficient a_0 counts pairs where $f = g$, so $a_0 = 1$ is always true for any non-empty set.

Conversely, if $a_1, \dots, a_s$ are all zero, there are no distinct elements in X whose difference has a rank of s or less. Thus, the minimum distance must necessarily be strictly greater than s , satisfying the definition of an s -codesign. ■

The proposition below highlights the duality of codesigns to designs.

Proposition 4.66. *A subgroup $(\Omega, +)$ is a t -design if and only if the dual $X^\perp$ is a t -codesign.*

Proof. From Theorem 4.63 we know that X is a t -design if and only if

$$\sum_{i=0}^m a_i P_k(i) = |X| \delta_{0,k} \quad \forall k \in \{0, 1, \dots, t\}$$

From Theorem 4.43, since X is a subgroup, $X^\perp$ exists and we know that

$$a_k^\perp = \frac{1}{|X|} \sum_{i=0}^m a_i P_k(i) \quad \forall k \in \{0, 1, \dots, m\}$$

Thus, X is a t -design if and only if

$$a_k^\perp = \delta_{0,k} \quad \forall k \in \{0, 1, \dots, t\}$$

Applying Proposition 4.65 we are done. ■

We now have some bounds on the size of X is a design and a codesign. This is an important structural question in design theory.

Theorem 4.67. *Let $X \subseteq \Omega$ be a t -design and an s -codesign for the largest possible t and s . Then*

$$q^{tm'} \leq |X| \leq q^{(m-s)m'}$$

Moreover, if one of the bounds is achieved, so is the other and consequently we must have $s + t = m$.

Proof. The lower bound follows trivially from Proposition 4.60 by noting that since X is non-empty, we must have $\lambda \geq 1$.

We now prove the upper bound. If X is a Singleton, all terms in the inequalities evaluate to 1 and the result is trivial. Thus, suppose $|X| \geq 2$. Let f and g be any distinct elements of X .

By the definition of a codesign, we must have $\text{rank}(f - g) > s$. Consequently, the dimension of the left radical, $\dim(\text{Rad}_L(f - g))$, must be strictly less than $m - s$. Let V_0 be any $(m - s)$ -dimensional subspace of V .

We claim that f and g must differ when restricted to $V_0 \times V'$. Suppose for the sake of contradiction that they do not differ. Then V_0 would be entirely contained within $\text{Rad}_L(f - g)$. However, this is impossible since $\dim(V_0) = m - s$, but $\dim(\text{Rad}_L(f - g)) < m - s$. Since every element of X restricts to a unique, distinct map on $V_0 \times V'$, the size of X cannot exceed the total number of possible bilinear forms on $V_0 \times V'$. Thus, $|X| \leq |\Omega(V_0, V')| = q^{(m-s)m'}$.

We now show, with the help of the lemma below, that if one of the bounds is achieved, so is the other. It is clear that if both bounds are achieved, then $s + t = m$.

Lemma. *The upper bound is tight if and only if X is an $(m - s)$ -design with $\lambda = 1$.*

Proof of Lemma. If X is an $(m - s)$ -design with $\lambda = 1$, the upper bound is clearly tight by Proposition 4.60.

Conversely, if the upper bound is achieved, then for any $(m - s)$ -dimensional space $V_0 \subseteq V$, we know that the restriction map $X \rightarrow \Omega(V_0, V')$ is injective. Since both the domain and the codomain have the exact same cardinality $q^{(m-s)m'}$, the map is a bijection. Hence, every $f_0 \in \Omega(V_0, V')$ can be extended to a unique $f \in X$ given by the pre-image of the restriction. Thus $\lambda = 1$, and X is an $(m - s)$ -design. \\

Now suppose the upper bound is achieved. By the lemma, $\lambda = 1$, and hence the lower bound must also be achieved. Alternatively, suppose the lower bound is achieved. Then $\lambda = 1$, and we know generally that $m - s \geq t$.

We would like to prove the reverse inequality, $t \geq m - s$. For any t -dimensional subspace $U \subseteq V$, the restriction map $X \rightarrow \Omega(U, V')$ is a bijection, as observed above. Thus, there is no t -dimensional subspace on which $f - g$ is identically zero for distinct $f, g \in X$. Hence, $\dim(\text{Rad}_L(f - g)) < t$. Consequently, $\text{rank}(f - g) = m - \dim(\text{Rad}_L(f - g)) \geq m - (t - 1)$. Thus, the rank of $f - g$ is strictly greater than $m - t$. However, s was defined as the largest possible integer such that $\text{rank}(f - g) > s$ for all distinct pairs. Thus, we must have $s \geq m - t$, which implies $s + t \geq m$. Combined with $m - s \geq t$, this forces $s + t = m$. ■

We can now define Singleton systems. This is analogous to how MDS codes are defined through the Singleton bound.

Definition 4.68. A t -design $X \subseteq \Omega(m, m')$ with index 1 is called an $(\mathbf{m}, \mathbf{m}', \mathbf{t}, \mathbf{q})$ -*Singleton system*

In accordance with the above theorem, it may also be viewed as an $(m - t)$ -codesign of cardinality $q^{tm'}$. It is also clear that the dual, if X is a subgroup, is an $(m, m', m - t, q)$ -Singleton system.

Theorem 4.69. The coefficients a_{m-i} ($0 \leq i \leq t-1$) of the q -distance enumerator $a(z)$ of any (m, m', t, q) -Singleton system X are given by

$$a_{m-i} = \binom{m}{i}_q \sum_{j=0}^{t-i-1} (-1)^j q^{\binom{j}{2}} \binom{m-i}{j}_q (q^{(t-i-j)m'} - 1)$$

Proof. We would like to show that the solution to the equation in Theorem 4.63 for $|X| = q^{tm'}$ and $a_j = \delta_{j,0}$ for $j \leq s = m - t$ is uniquely given by the above mentioned formula. For $k = 0$, the equation is trivial and provides no information. For $1 \leq k \leq t$, we have the system of equations

$$|\Omega_k| + \sum_{j=m-t+1}^m a_j P_k(j) = 0 \quad \forall k \in \{1, 2, \dots, t\}$$

This is a system of t linear equations in t variables. It can be solved uniquely with the desired unique solution. The calculations are not so direct and one may refer to Theorem 4 from [11]. ■

4.1.9 Construction of Singleton systems

We construct linear (m, m', t, q) -Singleton systems explicitly in this section.

For the remainder of this section, we consider $V' = \mathbb{F}_{q^{m'}}$ and $V \subseteq V'$. Both are treated as vector spaces over $\mathbb{F}_q$ and V has dimension m . This also aligns with our global assumption of $m \leq m'$ for the paper.

Definition 4.70. Given any m -tuple $\omega = (\omega_0, \dots, \omega_{m-1})$ of elements $\omega_j \in V'$, we define the function $f_\omega : V \times V' \rightarrow \mathbb{F}_q$ given by

$$f_\omega(x, x') = \sum_{i=0}^{m-1} \text{tr}_{\mathbb{F}_{q^{m'}}/\mathbb{F}_q}(\omega_i x^{q^i} x')$$

Lemma 4.71. The mapping $\omega \mapsto f_\omega$ is an $\mathbb{F}_q$ -vector space isomorphism from $(V')^m$ to $\Omega(m, m')$.

Proof. It is clear that f_ω is a bilinear map on $V \times V'$ for every choice of ω . It is also clear from the linearity of the trace that $f_{\omega+\omega'} = f_\omega + f_{\omega'}$. Thus, to show that the linear map is an isomorphism, it suffices to show injectivity, because both spaces have the same finite dimension of mm' over $\mathbb{F}_q$.

Suppose $f_\omega = 0$. Consider the polynomial $p_\omega \in \mathbb{F}_{q^{m'}}[x]$ given by

$$p_\omega(x) = \sum_{i=0}^{m-1} \omega_i x^{q^i}$$

We can rewrite our function as $f_\omega(x, x') = \text{tr}(p_\omega(x)x')$ for every $x \in V, x' \in V'$. If f_ω is the zero bilinear map, then the trace evaluates to zero for all $x' \in V'$. Because the field trace $\text{tr}_{\mathbb{F}_{q^{m'}}/\mathbb{F}_q}$ is a non-degenerate bilinear form, this implies that $p_\omega(x) = 0$ for all $x \in V$.

The degree of the polynomial $p_\omega(x)$ is at most q^{m-1} (assuming $\omega_{m-1} \neq 0$). However, the polynomial vanishes on the entirety of V , meaning it has at least $|V| = q^m$ roots. Since a non-zero polynomial over a field cannot have more roots than its degree, $p_\omega(x)$ must be the identically zero polynomial. Therefore, $\omega_i = 0$ for all i , meaning $\omega = 0$. This proves that the kernel is trivial. ■

Lemma 4.72. *Given $r \in \{0, 1, \dots, m\}$, let $\omega = (\omega_0, \dots, \omega_{m-1})$ be a non-zero m -tuple over V' satisfying $\omega_j = 0$ for all $j > r$. Then the corresponding bilinear form f_ω has rank at least $m - r$.*

Proof. From Lemma 4.71, we established that $f_\omega(x, x') = \text{tr}(p_\omega(x)x')$. Thus, the left radical of f_ω , $\text{Rad}_L(f_\omega)$, is the set of $x \in V$ such that $p_\omega(x)x' = 0$ for all $x' \in V'$, which is precisely the set of roots of $p_\omega(x)$ in V .

Since $\omega_j = 0$ for all $j > r$, the highest possible degree of the polynomial $p_\omega(x)$ is q^r . Consequently, $p_\omega(x)$ can have at most q^r roots. Therefore, $|\text{Rad}_L(f_\omega)| \leq q^r$, which implies that the dimension of the left radical over $\mathbb{F}_q$ is at most r .

By the rank-nullity theorem, the rank of the bilinear form is $m - \dim(\text{Rad}_L(f_\omega)) \geq m - r$. The result follows. ■

Theorem 4.73. *Given $t \in \{0, 1, \dots, m\}$ we define X to be the set of bilinear forms f_ω subject to $\omega_j = 0$ whenever $j \geq t$. Then X is an (m, m', t, q) -Singleton system.*

Proof. It is easily verified that X forms a vector space over $\mathbb{F}_q$. For any distinct $f, g \in X$, their difference $f - g$ is also a non-zero element of X . Since $\omega_j = 0$ for all $j \geq t$ (which means $j > t - 1$), we can apply Lemma 4.72 with $r = t - 1$ to obtain

$$\text{rank}(f - g) \geq m - (t - 1) > m - t$$

This holds for all distinct pairs $f, g \in X$. Thus, X is an $(m - t)$ -codesign.

Furthermore, the cardinality of X is exactly $q^{tm'}$ because the isomorphism established in Lemma 4.71 maps the t freely chosen components of ω (each from $\mathbb{F}_{q^{m'}}$) uniquely to X .

By the observation following Definition 4.68, an $(m - t)$ -codesign of cardinality $q^{tm'}$ is equivalently an (m, m', t, q) -Singleton system. This finishes the proof. ■

4.2 Rank-Metric codes

This is a part of a chapter on Rank-Metric codes authored by E. Gorla [14]. This section is closely related to the previous section on bilinear forms.

4.2.1 Matrix and Vector Rank-Metric Codes

Throughout, q shall be some prime power and we shall work over $\mathbb{F}_q$. Let m and n be positive integers. We denote the $\mathbb{F}_q$ -vector space of matrices of size $m \times n$ over $\mathbb{F}_q$ by $\text{Mat}_{n \times m}(\mathbb{F}_q)$.

We restrict our attention to linear codes and all dimensions are considered over $\mathbb{F}_q$ unless stated otherwise.

Definition 4.74. We define the *weight of a matrix* $A \in \text{Mat}_{n \times m}(\mathbb{F}_q)$ to be

$$\text{wt}(A) = \text{rank}(A)$$

The *matrix rank distance* between two matrices is then naturally defined to be the weight of their difference, that is, $d(A, B) = \text{rank}(A - B)$ for $A, B \in \text{Mat}_{n \times m}(\mathbb{F}_q)$.

Definition 4.75. A *matrix rank-metric code* (MRM code) is an $\mathbb{F}_q$ -linear subspace of $\text{Mat}_{n \times m}(\mathbb{F}_q)$ equipped with the above defined weight and distance functions.

Definition 4.76. We define the *weight of a vector* $x = (x_1, \dots, x_n) \in \mathbb{F}_{q^m}^n$ to be

$$\text{wt}(x) = \text{rank}(x) = \dim_{\mathbb{F}_q} \left(\text{span}_{\mathbb{F}_q} \{x_1, \dots, x_n\} \right)$$

The *vector rank distance* between two vectors is then naturally defined to be the weight of their difference, that is, $d(u, v) = \text{rank}(u - v)$ for $u, v \in \mathbb{F}_{q^m}^n$.

Definition 4.77. A *vector rank-metric code* (VRM code) is an $\mathbb{F}_{q^m}$ -linear subspace of $\mathbb{F}_{q^m}^n$ equipped with the rank distance function where the weight of a vector is simply its rank.

To better explain the concept of the weight of a vector, we discuss a few examples. Suppose we write

$$\mathbb{F}_8 = \mathbb{F}_2[\alpha] / \langle \alpha^3 + \alpha + 1 \rangle = \{0, 1, \alpha, \alpha + 1, \alpha^2, \alpha^2 + 1, \alpha^2 + \alpha, \alpha^2 + \alpha + 1\}$$

Then the weight of the vector $(1, \alpha) \in \mathbb{F}_8^2$ is 2 because both components are linearly independent over $\mathbb{F}_2$, while the weight of $(1, \alpha, \alpha^2 + \alpha, \alpha + 1, \alpha^2) \in \mathbb{F}_8^5$ is 3 because the third and fourth components can be expressed in terms of the other three, which are linearly independent.

Now we highlight the connection between the two codes defined above.

Definition 4.78. Let $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ be a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$ and let $v = (v_1, \dots, v_n) \in \mathbb{F}_{q^m}^n$ be some vector. Express each of its components in terms of the basis vectors as

$$v_i = \sum_{j=1}^m \Gamma_{i,j}(v) \gamma_j \quad \forall i = 1, 2, \dots, n$$

We associate the matrix $\Gamma(v) \in \text{Mat}_{n \times m}(\mathbb{F}_q)$ whose (i, j) -th entry is $\Gamma_{i,j}(v)$ to the vector v . Given a VRM code C in $\mathbb{F}_{q^m}^n$, we define the set

$$\Gamma(C) = \{\Gamma(x) \mid x \in C\}$$

to be the *matrix rank-metric code associated to C with respect to the basis Γ* .

One can verify that this is indeed an MRM code since it is an $\mathbb{F}_q$ -linear subspace of $\text{Mat}_{n \times m}(\mathbb{F}_q)$.

We return to the previous example of $\mathbb{F}_8$ and consider the VRM $\text{span}\{(1, \alpha)\}$ where $(1, \alpha) \in \mathbb{F}_8^2$. We choose a basis of $\mathbb{F}_8$ over $\mathbb{F}_2$ to be $\Gamma = \{1, \alpha, \alpha^2\}$. The full VRM code $C \subseteq \mathbb{F}_8^2$ is given by

$$C = \{(0, 0), (1, \alpha), (\alpha, \alpha^2), (\alpha + 1, \alpha^2 + \alpha), (\alpha^2, \alpha + 1), (\alpha^2 + 1, 1), (\alpha^2 + \alpha, \alpha^2 + \alpha + 1), (\alpha^2 + \alpha + 1, \alpha^2 + 1)\}$$

This has 8 elements and dimensions 1 and 3 when considered over $\mathbb{F}_8$ and $\mathbb{F}_2$, respectively. Over $\mathbb{F}_2$, a basis is $\{(1, \alpha), (\alpha, \alpha^2), (\alpha^2, \alpha + 1)\}$. Thus we compute,

$$\Gamma((1, \alpha)) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \Gamma((\alpha, \alpha^2)) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \Gamma((\alpha^2, \alpha + 1)) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Hence the associated MRM code will be the linear span of these three matrices over $\mathbb{F}_2$.

Proposition 4.79. *The map $v \mapsto \Gamma(v)$ as above, is an $\mathbb{F}_q$ -linear isometry, that is, a weight-preserving $\mathbb{F}_q$ -linear map. In particular, if $C \subseteq \mathbb{F}_{q^m}^n$ is a VRM of dimension k over $\mathbb{F}_{q^m}$ then $\Gamma(C)$ is an $\mathbb{F}_q$ -linear MRM code of dimension mk over $\mathbb{F}_q$.*

Proof. The map is clearly $\mathbb{F}_q$ -linear and this can be verified quite easily. To show that the weight, that is, rank is preserved, we need to show the following for any $v \in \mathbb{F}_{q^m}^n$ mapping to a matrix with rows $R_1, \dots, R_n$:

$$\dim_{\mathbb{F}_q}(\text{span}_{\mathbb{F}_q}\{v_1, \dots, v_n\}) = \dim_{\mathbb{F}_q}(\text{span}_{\mathbb{F}_q}\{R_1, \dots, R_n\})$$

Note that the map $\Phi : \mathbb{F}_q^m \rightarrow \mathbb{F}_{q^m}$ mapping the rows of the matrix to vector components defined as

$$\Phi(R_i) = \sum_{j=1}^m \Gamma_{i,j}(v) \gamma_j = v_i$$

is an isomorphism and hence the dimension of subspaces is preserved showing that the ranks are preserved. Thus since the map is injective and C has dimension k over $\mathbb{F}_{q^m}$ we get,

$$\dim_{\mathbb{F}_q}(\Gamma(C)) = \dim_{\mathbb{F}_q}(C) = \dim_{\mathbb{F}_q}(\mathbb{F}_{q^m}) \dim_{\mathbb{F}_{q^m}}(C) = mk$$

■

Observe that a necessary condition for a given MRM code to have been obtained from some VRM code is that the dimension of the MRM code must be divisible by m . This is not however sufficient. Given any MRM code we can choose a basis and apply Γ^{-1} to get back some subspace of $\mathbb{F}_{q^m}^n$ but this will be an $\mathbb{F}_q$ -subspace of $\mathbb{F}_{q^m}^n$ whereas for the inverse to be a VRM code, we need it to be an $\mathbb{F}_{q^m}$ -subspace.

We now define the condition for two codes to be equivalent.

Definition 4.80. By an $\mathbb{F}_q$ *linear isometry of $\text{Mat}_{n \times m}(\mathbb{F}_q)$* we mean an $\mathbb{F}_q$ -linear map $\text{Mat}_{n \times m}(\mathbb{F}_q) \rightarrow \text{Mat}_{n \times m}(\mathbb{F}_q)$ which preserves ranks of matrices.

Definition 4.81. By an $\mathbb{F}_q$ *linear isometry of $\mathbb{F}_{q^m}^n$* we mean an $\mathbb{F}_{q^m}$ -linear map $\mathbb{F}_{q^m}^n \rightarrow \mathbb{F}_{q^m}^n$ which preserves ranks of vectors.

Definition 4.82. Two MRM (resp. VRM) codes C and D are said to be *equivalent* if there exists some $\mathbb{F}_q$ -linear isometry of $\text{Mat}_{n \times m}(\mathbb{F}_q)$ (resp. $\mathbb{F}_{q^m}^n$) between them.

A natural question to ask at this point is the following: Suppose we start with two equivalent VRM codes and obtain MRM codes using Proposition 4.78 then are the resultant codes equivalent. The proposition below answers this for us.

Proposition 4.83. *Let $C, D \subseteq \mathbb{F}_{q^m}^n$ be two equivalent VRM codes. Let Γ and Γ' be two bases of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. We have $\Gamma(C) \sim \Gamma'(D)$.*

Proof. Since C and D are equivalent, there is some $\mathbb{F}_{q^m}$ -linear map $\phi : \mathbb{F}_{q^m}^n \rightarrow \mathbb{F}_{q^m}^n$ which restricts to a bijection between C and $D = \phi(C)$ and also preserves ranks. It will also be $\mathbb{F}_q$ -linear since $\mathbb{F}_q$ is a subfield of $\mathbb{F}_{q^m}$.

Now we consider the composition $\psi = \Gamma' \circ \phi \circ \Gamma^{-1} : \text{Mat}_{n \times m}(\mathbb{F}_q) \rightarrow \text{Mat}_{n \times m}(\mathbb{F}_q)$. Using Proposition 4.79, this is a composition of weight-preserving $\mathbb{F}_q$ -linear bijections, and hence ψ is a weight-preserving $\mathbb{F}_q$ -linear bijection. Furthermore, $\psi(\Gamma(C)) = \Gamma'(\phi(\Gamma^{-1}(\Gamma(C)))) = \Gamma'(D)$, showing that $\Gamma(C) \sim \Gamma'(D)$. ■

We characterise the linear isometries of $\text{Mat}_{n \times m}(\mathbb{F}_q)$ and $\mathbb{F}_{q^m}^n$ in the next section.

4.2.2 Characterisation of linear isometries

We begin by classifying linear isometries of $\mathbb{F}_{q^m}^n$. This was developed in [5]. We adopt the convention of vectors being treated as row vectors to maintain consistency with coding theory.

Theorem 4.84. *Let $\psi : \mathbb{F}_{q^m}^n \rightarrow \mathbb{F}_{q^m}^n$ be an $\mathbb{F}_{q^m}$ -linear isometry with respect to the rank metric. Then there exists a scalar $\alpha \in \mathbb{F}_{q^m}^*$ and $B \in GL_n(\mathbb{F}_q)$ such that for every $v \in \mathbb{F}_{q^m}^n$:*

$$\psi(v) = \alpha v B$$

Proof. Since the map preserves ranks, in particular we have that it maps every rank 1 vector another rank 1 vector. Suppose $x = (x_1, \dots, x_n) \in \mathbb{F}_{q^m}^n$ is a rank 1 vector. Then the $\mathbb{F}_q$ -space spanned $x_1, \dots, x_n \in \mathbb{F}_{q^m}$ is one-dimensional which gives that each x_i must be some $\mathbb{F}_q$ -multiple of a fixed non-zero vector $y \in \mathbb{F}_{q^m}$ giving

$$x = (x_1, \dots, x_n) = (\lambda_1 y, \dots, \lambda_n y) = \lambda y$$

where $\lambda \in \mathbb{F}_q^n$ is non-zero and $y \in \mathbb{F}_{q^m}^*$. Note that we demand λ to be non-zero because we cannot have all the multiples be 0 else the space is $\{0\}$ which is 0-dimensional.

Fix the standard $\mathbb{F}_{q^m}$ -basis $\{e_1, \dots, e_n\}$ of $\mathbb{F}_{q^m}^n$. Since the rank is preserved and each e_j clearly has rank 1 we get

$$\psi(e_j) = \lambda_j y_j \quad \forall j = 1, 2, \dots, n$$

for some non-zero $\lambda_j \in \mathbb{F}_q^n$ and non-zero $y_j \in \mathbb{F}_{q^m}$. Now we normalise to the following map. We define $\psi' : \mathbb{F}_{q^m}^n \rightarrow \mathbb{F}_{q^m}^n$ given by

$$\psi'(v) = y_1^{-1} \psi(v)$$

Note that this is still an $\mathbb{F}_{q^m}$ -linear isometry. Under this map we get

$$\psi'(e_1) = \lambda_1 \quad \text{and} \quad \psi'(e_j) = \mu_j \lambda_j \quad \forall j = 2, 3, \dots, n$$

where $\mu_j = y_1^{-1} y_j \in \mathbb{F}_{q^m}$. We now show that each μ_j in fact lies in $\mathbb{F}_q$.

Note that for $n = 1$, the statement holds trivially with B always being chosen as the identity matrix (or possibly a scalar multiple which we can adjust with the scalar α). Thus, assume that $n \geq 2$.

For some $j \neq 1$ consider $e_1 + e_j$. The coordinates of this vector are just 0 and 1 and hence the rank is still 1. Thus by linearity of ψ' we have that the rank of $\lambda_1 + \mu_j \lambda_j$ is also 1 and hence it must be of the form γz for some non-zero $\gamma \in \mathbb{F}_q^n$ and non-zero $z \in \mathbb{F}_{q^m}$. Coordinate-wise we see at the k -th ($1 \leq k \leq n$) coordinate:

$$\lambda_1^{(k)} + \mu_j \lambda_j^{(k)} = \gamma^{(k)} z$$

In particular, considering the equality at some k and $\ell \neq k$ we get

$$\mu_j \left(\lambda_j^{(k)} \gamma^{(\ell)} - \lambda_j^{(\ell)} \gamma^{(k)} \right) = \left(\lambda_1^{(\ell)} \gamma^{(k)} - \lambda_1^{(k)} \gamma^{(\ell)} \right)$$

Observe that both the terms in the bracket lie in $\mathbb{F}_q$. Suppose for the sake of contradiction we have $\mu_j \notin \mathbb{F}_q$. Then for the above equation to hold for all pairs k and $\ell \neq k$, we must have $\lambda_j^{(k)}\gamma^{(\ell)} - \lambda_j^{(\ell)}\gamma^{(k)} = 0$ (and hence the RHS also 0). Thus for every pair $k, \ell \neq k$ of indices in $[n]$ we get

$$\frac{\lambda_j^{(k)}}{\lambda_j^{(\ell)}} = \frac{\gamma^{(k)}}{\gamma^{(\ell)}}$$

Thus λ_j and γ are $\mathbb{F}_q$ -scalar multiples of each other. From $\lambda_1^{(\ell)}\gamma^{(k)} - \lambda_1^{(k)}\gamma^{(\ell)} = 0$ we obtain that λ_1 is also an $\mathbb{F}_q$ -scalar multiple of γ .

Thus, λ_1 and λ_j are $\mathbb{F}_q$ -scalar multiples of each other. This contradicts the injectivity of the map ψ' since we get that $\psi'(e_1)$ and $\psi'(e_j)$ are $\mathbb{F}_{q^m}$ -scalar multiples while e_1 and e_j are not.

Thus we have shown that every μ_j lies in the base field $\mathbb{F}_q$. Suppose we write $\psi'(e_i) = b_i$ for every $i \in [n]$. Then we have

$$\psi'(v) = vB$$

where the rows of B are $b_1, \dots, b_n$. However since we have shown that $b_1 = \psi'(e_1) = \lambda_1 \in \mathbb{F}_q^n$ and for $j \geq 2$, $b_j = \psi'(e_j) = \mu_j \lambda_j \in \mathbb{F}_q^n$, we have that every entry of B is in $\mathbb{F}_q^n$. Since ψ' is an $\mathbb{F}_{q^m}$ -linear bijection, the matrix B is invertible over $\mathbb{F}_{q^m}$. However, all the b_i are linearly independent vectors of $\mathbb{F}_q^n$ over $\mathbb{F}_q$ and hence the matrix is in fact invertible over $\mathbb{F}_q$.

Thus putting it together we get

$$\psi(v) = y_1 \psi'(v) = y_1 vB$$

Choosing $\alpha = y_1$ finishes the proof. ■

We now turn our attention to linear isometries of rectangular matrices. This can be found in [33].

We need to set up some definitions and prove some results before the actual characterisation of these isometries. Throughout we shall assume that $\phi : \text{Mat}_{n \times m}(\mathbb{F}_q) \rightarrow \text{Mat}_{n \times m}(\mathbb{F}_q)$ is an $\mathbb{F}_q$ -linear isometry which preserves rank.

Definition 4.85. Two matrices $A, B \in \text{Mat}_{n \times m}(\mathbb{F}_q)$ are said to be **adjacent** if their difference has rank 1.

Observe that the images of adjacent matrices under ϕ will also be adjacent.

Definition 4.86. A set $M \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ is said to be **maximal set of rank 1** if every distinct pair in M is adjacent and M is the largest possible collection with this property.

We now classify these maximal sets.

Proposition 4.87. *There are only two types of maximal sets of rank 1 containing the 0 matrix given by*

(i) Type I: *Matrices with non-zero entries only in a fixed row i , that is,*

$$R_i = \{X \in \text{Mat}_{n \times m}(\mathbb{F}_q) \mid \text{every row other than row } i \text{ is zero}\}$$

(ii) Type II: *Matrices with non-zero entries only in a fixed column j , that is,*

$$C_j = \{X \in \text{Mat}_{n \times m}(\mathbb{F}_q) \mid \text{every column other than column } j \text{ is zero}\}$$

Proof. Since every matrix of M is adjacent to 0, all matrices in M must have rank 1. Suppose $A \in M$ has rank 1. Then it can be expressed as $A = u^\top v$ for some $u \in \mathbb{F}_q^n, v \in \mathbb{F}_q^m$.

Up to a change of basis we can assume that $A = E_{11}$, that is 1 in the $(1, 1)$ -th entry and 0 elsewhere. Now let $B \in M$ be some other matrix. Suppose it has the form

$$B = a^\top b = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_m \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_m \\ \vdots & \vdots & \ddots & \vdots \\ a_n b_1 & a_n b_2 & \cdots & a_n b_m \end{bmatrix}$$

Since $B - A$ also has rank 1, every size 2 minor must vanish and hence for any $2 \leq i \leq n, 2 \leq j \leq m$, we must have

$$\det \begin{bmatrix} a_1 b_1 - 1 & a_1 b_j \\ a_i b_1 & a_i b_j \end{bmatrix} = 0$$

This gives $a_i b_j = 0$ showing that only the first row and column of B could possibly be non-zero. Thus, either all of $a_2, \dots, a_n$ are all zero or $b_2, \dots, b_m$ are all zero. Thus $B \in R_1$ or $B \in C_1$.

Now we show that M must entirely be contained in one of these sets. Suppose not. Then there are matrices $X \in R_1 \setminus C_1$ and $Y \in C_1 \setminus R_1$. Then their difference clearly has rank at least 2 which is a contradiction. Since R_1 and C_1 are themselves maximal sets of rank 1, M must be exactly equal to one of these since M is also maximal. ■

Corollary 4.88. *Given any two adjacent points in $\text{Mat}_{n \times m}(\mathbb{F}_q)$, there exist exactly two unique maximal sets of rank 1 which contain both of them.*

Proof. Simply translate the two points A and B to 0 and $B - A$ and then multiply by appropriate invertible matrices P and Q on the left and right to transform to 0 and E_{11} . These points are contained in exactly two maximal sets, namely, R_1 and C_1 . We can then backtrack and invert the transformations to get two maximal sets M and N containing A and B .

One may observe that both the translation and row and column operations are bijections which preserve maximal sets of rank 1 and hence M and N are indeed maximal sets of rank 1. ■

Definition 4.89. If the intersection of two distinct maximal sets of rank 1 contains more than one point, then this intersection is called a *line*.

Corollary 4.90. *Given any two adjacent points, there is a unique line passing through them.*

Proof. Consider two adjacent points X and Y . Obtain the two unique maximal sets M and N such that they contain both the points using Corollary 4.88. Thus, X and Y lie on the line $L = M \cap N$. Suppose L' is another line containing the points. By definition, $L' = M' \cap N'$. But M and N are unique and hence M' and N' must be equal to M and N in some order. Thus we must have $M \cap N = M' \cap N' \implies L = L'$. ■

We now describe the form of these lines in the proposition below.

Proposition 4.91. *The parametric equation of a line in $\text{Mat}_{n \times m}(\mathbb{F}_q)$ is*

$$\{tp^\top q + R \mid t \in \mathbb{F}_q\}$$

where $p \in \mathbb{F}_q^n$ and $q \in \mathbb{F}_q^m$ are non-zero and $R \in \text{Mat}_{n \times m}(\mathbb{F}_q)$.

Proof. Suppose we want to find the equation of the line through the points A and B .

Let $L = R_1 \cap C_1 = \{xE_{11} \mid x \in \mathbb{F}_q\}$ be the line containing the points 0 and E_{11} . Let ψ be the unique maximal set preserving bijection mapping 0 to A and E_{11} to B . We know that this has the form

$$\psi(X) = PXQ + R$$

where $P \in GL_n(\mathbb{F}_q)$, $Q \in GL_m(\mathbb{F}_q)$ and $R = A \in \text{Mat}_{n \times m}(\mathbb{F}_q)$. Under ϕ the line through A and B is image of the line L . This can be calculated as follows:

$$\phi(xE_{11}) = xPE_{11}Q + R = xP(e_1^\top f_1)Q + R = xu^\top v + R$$

where $e_1 = (1, 0, \dots, 0) \in \mathbb{F}_q^n$ and $f_1 = (1, 0, \dots, 0) \in \mathbb{F}_q^m$; and $u = e_1 P^\top$ and $v = f_1 Q$. This proves the proposition with the choice of $p = u$ and $q = v$. ■

We can specialise to R_1 in particular. Specialisations to other R_i and also C_i are analogous.

Corollary 4.92. *The lines in R_1 are of the form*

$$\left\{ x \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \mid x \in \mathbb{F}_q \right\}$$

Proof. Using Proposition 4.91 and setting $x = 0$ we get that R must be of the form of the second matrix involving the b_i since we are in R_1 .

For $p^\top q$ to be in R_1 , we must have $p_i q_j = 0$ for every $i \geq 2$ and every j . Since q is non-zero, it means that $p_2 = \cdots = p_n = 0$. Thus $p^\top q$ has the form of the matrix involving the a_i where the factor p_1 is absorbed into t to give x . ■

Now since ϕ is an $\mathbb{F}_q$ -linear bijection which preserves adjacency and hence maps maximal sets to maximal sets, we must have that ϕ maps R_i to R_i itself or C_j and C_j to C_j itself or R_i .

Lemma 4.93. *If $n \neq m$, then ϕ maps R_i to some $R_{i'}$ and C_j to some $C_{j'}$, but if $n = m$, the types may or may not be swapped. In the event that it is swapped, we may apply a transpose map to undo the swap.*

Proof. Since ϕ is a bijection it preserves cardinality and hence R_i cannot be mapped to C_j and vice-versa as their sizes are q^m and q^n respectively. Suppose $m = n$, then the types may be swapped but if this is indeed the case, then under the map $A \mapsto A^\top$, the types may be said to map to themselves. ■

Without loss of generality, in the case of $m = n$, we assume that the types do not swap and map to themselves, keeping open the possibility of swap which we factor in at the end by using $A^\top$.

We have a result from affine geometry below. It is commonly known as the fundamental theorem of affine geometry.

Theorem 4.94. *Let $f : V \rightarrow W$ be a bijective map between vector spaces over a field K of dimension $n \geq 2$ which maps lines to lines. Then f is a semi-affine map, that is, of the form*

$$f(v) = A\sigma(v) + c$$

where $A \in GL_n(K)$, $c \in K^n$ is a translation vector and $\sigma \in \text{Aut}(K)$. By a line of the vector space V we mean sets of the type $\{u + \lambda v \mid \lambda \in K\}$ where $u, v \in V$ are fixed. Similarly, one defines lines in W .

We do not prove this here. A proof may be found in [4].

Theorem 4.95. *Let $\phi : \text{Mat}_{n \times m}(\mathbb{F}_q) \rightarrow \text{Mat}_{n \times m}(\mathbb{F}_q)$ be an $\mathbb{F}_q$ -linear bijection which preserves ranks. Then*

- (i) *If $m \neq n$, then ϕ must have the form $\phi(M) = AMB$ for some $A \in GL_n(\mathbb{F}_q)$ and $B \in GL_m(\mathbb{F}_q)$.*
- (ii) *If $m = n$, then ϕ must have the form $\phi(M) = AMB$ or $AM^\top B$ for some $A, B \in GL_n(\mathbb{F}_q)$.*

Proof. From Lemma 4.93 we have that each R_i is mapped to some $R_{i'}$ and each C_j is mapped to some $C_{j'}$ for every $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

Consider the map ϕ restricted to R_1 . It reduces to a linear map $f_1 : \mathbb{F}_q^m \rightarrow \mathbb{F}_q^m$. We show that ϕ preserves lines of the matrix space so that the restriction f_1 preserves affine lines of the vector space $\mathbb{F}_q^m$ and we may apply Theorem 4.94.

Let $L = M_1 \cap M_2$ be a matrix line. Applying ϕ gives $\phi(L) = \phi(M_1) \cap \phi(M_2)$ since ϕ is a bijection. Further, since it is a bijection, we also have that $\phi(M_1)$ and $\phi(M_2)$ are distinct because M_1 and M_2 are distinct. By definition, $\phi(L)$ is also a line and hence we have shown that ϕ preserves matrix lines.

Thus the restriction f_1 also preserves matrix lines. However, a line in R_1 is of the form described in Corollary 4.92. Thus, f_1 preserves affine lines of $\mathbb{F}_q^m$ and hence by Theorem 4.94 we have

$$f_1(v) = \sigma(v)B_1 \quad \forall v \in \mathbb{F}_q^m$$

for some $\sigma \in \text{Aut}(\mathbb{F}_q)$ and $B_1 \in GL_m(\mathbb{F}_q)$. By the assumption that ϕ (and thus f_1) is an $\mathbb{F}_q$ -linear map, we have $f_1(\lambda v) = \lambda f_1(v)$ for all $\lambda \in \mathbb{F}_q$. Applying the semi-affine form, we obtain

$$\sigma(\lambda)\sigma(v)B_1 = \lambda\sigma(v)B_1$$

This holds for every $\lambda \in \mathbb{F}_q$ and $v \in \mathbb{F}_q^m$. Thus we have $\sigma(\lambda) = \lambda$ for every $\lambda \in \mathbb{F}_q$, giving $\sigma = \text{id}$.

This gives $f_1(v) = vB_1$. Similarly $f_i(v) = vB_i$ for every row index $1 \leq i \leq n$. But we have noted that as a whole, ϕ maps R_1 to some space $R_{i'}$. Looking at this action for every i , since ϕ is a bijection, we obtain some permutation π associated with the given ϕ mapping R_i to $R_{\pi(i)}$. Thus,

$$\phi(e_i^\top v) = e_{\pi(i)}^\top f_i(v) = e_{\pi(i)}^\top vB_i$$

We show that all of these B_i are the same matrix, say, B . Consider $X = (e_1 + e_2)^\top v$ for some non-zero vector v . The image is

$$\phi(X) = e_{\pi(1)}^\top vB_1 + e_{\pi(2)}^\top vB_2$$

Since X has two rows as v and rest of the rows zero, it has rank 1 and hence the RHS has rank 1. Thus vB_1 and vB_2 must be scalar multiples. Thus B_1 and B_2 are scalar multiples of each other. Doing this for every pair $(1, i)$ we get $B_i = \alpha_i B$ for some scalars α_i and $B \in GL_m(\mathbb{F}_q)$. Now the action on each row is

$$\phi(e_i^\top v) = \alpha_i e_{\pi(i)}^\top vB$$

Now we construct A to be a monomial matrix whose entries are given by

$$A_{(\pi(i), i)} = \alpha_i$$

Then for a general matrix $M \in \text{Mat}_{n \times m}(\mathbb{F}_q)$ with rows u_i we have

$$MB = \begin{bmatrix} u_1 B \\ u_2 B \\ \vdots \\ u_n B \end{bmatrix}$$

and by definition of A we have

$$AMB = \begin{bmatrix} \alpha_{\pi^{-1}(1)} u_{\pi^{-1}(1)} B \\ \alpha_{\pi^{-1}(2)} u_{\pi^{-1}(2)} B \\ \vdots \\ \alpha_{\pi^{-1}(n)} u_{\pi^{-1}(n)} B \end{bmatrix}$$

which is what is desired since this is exactly the definition of $\phi(M)$. This proves the theorem. Note that in the case of $m = n$, rows may map to columns and this can be handled by transposing the matrix M . ■

4.2.3 The Mac-Williams extension theorem

We first state and prove the original version for classical codes.

Theorem 4.96. *Let C and D be two linear codes in $\mathbb{F}_q^n$ and let $\phi : C \rightarrow D$ be a Hamming weight preserving linear bijection. Then it extends to a weight preserving linear bijection $f : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ whose restriction to C gives exactly ϕ .*

Proof. Firstly note that the extension map f must take the form classified in Theorem 4.84.

Suppose C has dimension k . Because ϕ is a linear bijection, D also has dimension k . Let G be a $k \times n$ generator matrix of C with rows $g_1, \dots, g_k$. Consider the images $\phi(g_i)$ for every row. Since ϕ is a linear

bijection, the images $\phi(g_i)$ must be linearly independent. Thus the matrix consisting of the rows given by $\phi(g_i)$ forms a generator matrix G' of D . For any codeword $uG \in C$ (where $u \in \mathbb{F}_q^k$), we have

$$\phi(uG) = \phi(u_1g_1 + \cdots + u_kg_k) = u_1\phi(g_1) + \cdots + u_k\phi(g_k) = uG'$$

Let the columns of these matrices be denoted by $h_1, \dots, h_n$ and $h'_1, \dots, h'_n$ respectively.

Let χ be a non-trivial character of the additive group $(\mathbb{F}_q, +)$. We can express the Hamming weight of any vector $x = (x_1, \dots, x_n) \in \mathbb{F}_q^n$ using character sums:

$$\text{wt}(x) = n - \frac{1}{q} \sum_{i=1}^n \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha x_i)$$

To justify this, consider the inner sum $\sum_{\alpha \in \mathbb{F}_q} \chi(\alpha x_i)$. If $x_i = 0$, then $\chi(0) = 1$ for all α , so the sum evaluates to q . If $x_i \neq 0$, the map $\alpha \mapsto \alpha x_i$ is a bijection on $\mathbb{F}_q$, so the sum is equivalent to $\sum_{\beta \in \mathbb{F}_q} \chi(\beta)$. Because χ is non-trivial, there exists some $a \in \mathbb{F}_q$ such that $\chi(a) \neq 1$. Therefore, $\sum_{\beta \in \mathbb{F}_q} \chi(\beta) = \sum_{\beta \in \mathbb{F}_q} \chi(\beta + a) = \chi(a) \sum_{\beta \in \mathbb{F}_q} \chi(\beta)$, which forces the sum to be exactly 0. Thus, the inner sum detects the zero coordinates, and the formula accurately counts the non-zero coordinates.

Since ϕ preserves weight, the weights of uG and uG' must be identical for all $u \in \mathbb{F}_q^k$. Evaluating the coordinates via the column vectors, we get

$$n - \frac{1}{q} \sum_{i=1}^n \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha u \cdot h_i) = n - \frac{1}{q} \sum_{i=1}^n \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha u \cdot h'_i)$$

Now consider the space of all functions from $\mathbb{F}_q^k \rightarrow \mathbb{C}$. For a given $v \in \mathbb{F}_q^k$, define $\psi_v(u) = \chi(u \cdot v) = \chi(uv^\top)$. Since there are q^k many of these functions and the dimension of the space is also q^k , it suffices to show that these functions ψ_v are orthogonal under the standard inner product to claim they form a basis. The inner product is defined as

$$\langle f, g \rangle = \sum_{u \in \mathbb{F}_q^k} f(u) \overline{g(u)}$$

Applying this to our functions, we have

$$\langle \psi_a, \psi_b \rangle = \sum_{u \in \mathbb{F}_q^k} \chi(u \cdot a) \overline{\chi(u \cdot b)} = \sum_{u \in \mathbb{F}_q^k} \chi(u \cdot (a - b))$$

Let $c = a - b$. If $a \neq b$, then c is a non-zero vector in $\mathbb{F}_q^k$. We evaluate the sum:

$$\sum_{u \in \mathbb{F}_q^k} \chi(u \cdot c) = \sum_{x \in \mathbb{F}_q} |\{u \in \mathbb{F}_q^k \mid u \cdot c = x\}| \chi(x) = q^{k-1} \sum_{x \in \mathbb{F}_q} \chi(x) = 0$$

Thus for $a \neq b$ we have $\langle \psi_a, \psi_b \rangle = 0$, which proves the orthogonality. From our established weight-preserving condition, we can rewrite the equality as a linear combination of these basis functions:

$$\sum_{i=1}^n \sum_{\alpha \in \mathbb{F}_q} \psi_{\alpha h_i}(u) = \sum_{i=1}^n \sum_{\alpha \in \mathbb{F}_q} \psi_{\alpha h'_i}(u)$$

Because the ψ_v form a basis, the coefficients of the basis vectors on both sides must match exactly. This implies that the multisets of indices must be identical: $\{\alpha h_i \mid \alpha \in \mathbb{F}_q, 1 \leq i \leq n\} = \{\beta h'_i \mid \beta \in \mathbb{F}_q, 1 \leq i \leq n\}$. Consequently, the columns of G' are scalar multiples of some permutation of the columns of G . Thus there exists a monomial matrix $M \in GL_n(\mathbb{F}_q)$ such that $GM = G'$.

The required extension map f is therefore $f(v) = vM$. Clearly, $f(C) = CM = D$, so f restricts exactly to ϕ . Because M is a monomial matrix, f merely scales and permutes coordinates, making it a weight-preserving bijection on the entire space $\mathbb{F}_q^n$. $\blacksquare$

We now investigate whether an analogous theorem holds for our rank-metric codes of both types. It suffices to answer for any one type. The answer is negative. The example below highlights this fact.

Question. Show that the Mac-Williams extension property does not hold for MRM and VRM codes.

Ans. Consider the MRM code given by

$$C = \left\{ \begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix} \mid a, b, c, d \in \mathbb{F}_2 \right\} \subseteq \text{Mat}_{2 \times 3}(\mathbb{F}_2)$$

and consider the map $\varphi : C \rightarrow C$ given by

$$\varphi \left(\begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix} \right) = \begin{bmatrix} a & c & 0 \\ b & d & 0 \end{bmatrix}$$

It is clear that this is an $\mathbb{F}_2$ -linear bijection which preserves the rank of matrices. Suppose it extends to an $\mathbb{F}_2$ -linear bijection $F : \text{Mat}_{2 \times 3}(\mathbb{F}_2) \rightarrow \text{Mat}_{2 \times 3}(\mathbb{F}_2)$ such that the restriction of F to C is φ .

From Theorem 4.95 we know that such an isometry F must have the form $F(M) = AMB$ for invertible matrices A and B of sizes 2 and 3 respectively. When this equation is restricted to the first two columns (the 2×2 block), we get

$$F(M)_{\text{res}} = A' M_{\text{res}} B'$$

Since F restricts to φ on C , and the restriction of C to its first two columns spans the entire space $\text{Mat}_{2 \times 2}(\mathbb{F}_2)$, we have that for all $X \in \text{Mat}_{2 \times 2}(\mathbb{F}_2)$,

$$X^\top = A' X B'$$

This forces a contradiction: the right-hand side represents a linear automorphism of the matrix ring $\text{Mat}_{2 \times 2}(\mathbb{F}_2)$, which preserves matrix multiplication order, while the left-hand side is the transpose operator, which is an anti-automorphism that reverses multiplication order. These cannot be identically equal on the entire space.

This particular MRM code chosen above in fact arises from a VRM code and hence that could serve as a counterexample. Suppose $\{1, \alpha\}$ is an $\mathbb{F}_2$ -basis of $\mathbb{F}_4$. We choose the following code

$$D = \{(v_1, v_2, 0) \mid v_1, v_2 \in \mathbb{F}_4\} = \{(a + c\alpha, b + d\alpha, 0) \mid a, b, c, d \in \mathbb{F}_2\} \subseteq \mathbb{F}_{2^2}^3 = \mathbb{F}_4^3$$

and consider the map $\theta : D \rightarrow D$ given by

$$\theta(a + c\alpha, b + d\alpha, 0) = (a + b\alpha, c + d\alpha, 0)$$

Note that the MRM code arising from this VRM code is the same as C and we can apply the same logic as before to show that θ does not extend to a linear weight preserving bijection $\mathbb{F}_4^3 \rightarrow \mathbb{F}_4^3$. $\square$

4.2.4 Supports, MRD codes and optimal anti-codes

Just like in classical codes, we may define the support of a codeword. We can also define a generalised notion in a similar manner as we did while dealing with generalised Hamming weights.

Definition 4.97. Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM and let $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ be a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. We define the **support of a vector codeword** $v \in C$ to be

$$\text{supp}(v) = \text{colspan}(\Gamma(v)) \subseteq \mathbb{F}_q^n$$

The **support of a vector code subset** $D \subseteq C$ is defined to be

$$\text{supp}(D) = \sum_{v \in D} \text{supp}(v) \subseteq \mathbb{F}_q^n$$

where $\sum$ is the usual sum of vector subspaces.

Note that the definitions of support above are independent of the choice of basis Γ .

Definition 4.98. Let $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ be an MRM. We define the *support of a matrix codeword* $A \in C$ to be

$$\text{supp}(A) = \begin{cases} \text{colspan}(A) \subseteq \mathbb{F}_q^n & \text{if } n \leq m \\ \text{rowspan}(A) \subseteq \mathbb{F}_q^m & \text{if } n > m \end{cases}$$

The *support of a matrix code subset* $D \subseteq C$ is defined to be

$$\text{supp}(D) = \sum_{A \in D} \text{supp}(A) \subseteq \mathbb{F}_q^{\min\{n, m\}}$$

where $\sum$ is the usual sum of vector subspaces.

Note that if we begin with a VRM and convert it to an MRM, then the supports of codewords are preserved and are exactly the same provided $n \leq m$. This is however not true for $n > m$. In fact, for this case, the support of the vector codeword and the obtained matrix codeword lie in different spaces altogether and are of different lengths.

We choose to split out the $n > m$ case for several reasons. One may choose not to do so and can use their own convention. However, for the results proved in this paper, we choose to adopt the above definitions.

We define the notation for a supported codeword below.

Definition 4.99. Let $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ be an MRM code and let $V \subseteq \mathbb{F}_q^{\min\{n, m\}}$ be some vector subspace. The *subcode of C supported on V* is defined to be

$$C(V) = \{M \in C \mid \text{supp}(M) \subseteq V\}$$

Definition 4.100. Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM code and let $V \subseteq \mathbb{F}_q^n$ be some $\mathbb{F}_q$ -vector subspace. Then the *subcode of C supported on V* is defined to be

$$C(V) = \{v \in C \mid \text{supp}(v) \subseteq V\}$$

We can now discuss the minimum distance and bounds like the Singleton bound.

Definition 4.101. We define the *minimum distance of an MRM code* $0 \neq C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ to be the integer

$$d_{\min}(C) = \min \{\text{rank}(M) \mid M \in C, M \neq 0\}$$

Additionally we define $\text{rank}(0) = \min\{n, m\} + 1$ by convention.

Definition 4.102. By the *maximum rank of an MRM code* $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ we mean the maximum rank of all possible $M \in C$.

Definition 4.103. We define the *minimum distance of a VRM code* $0 \neq C \subseteq \mathbb{F}_{q^m}^n$ to be the integer

$$d_{\min}(C) = \min \{\text{rank}(v) \mid v \in C, v \neq 0\}$$

Additionally we define $\text{rank}(0) = n + 1$ by convention.

Definition 4.104. By the *maximum rank of a VRM code* $C \subseteq \mathbb{F}_{q^m}^n$ we mean the maximum rank of all possible $x \in C$.

Note that by Proposition 4.79 the minimum distance and maximum rank of a VRM code and the associated MRM code are the same since the weights, that is, the ranks, are preserved in the transformation.

Below we have the *Singleton bounds* and *anti-code* bounds for both MRM codes and VRM codes.

Theorem 4.105. *Let $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ be an MRM code. Then we have the following two bounds*

$$\dim(C) \leq \max\{n, m\} (\min\{n, m\} - d_{\min}(C) + 1)$$

$$\dim(C) \leq \max\{n, m\} \max\text{rank}(C)$$

Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM code. Then we have the following two bounds

$$\dim_{\mathbb{F}_{q^m}}(C) \leq n - d_{\min}(C) + 1$$

$$\dim_{\mathbb{F}_{q^m}}(C) \leq \max\text{rank}(C)$$

with the last bound holding if and only if $\dim_{\mathbb{F}_{q^m}}(C) \leq m$.

Proof. We first prove the Singleton bound for matrices. The proof is analogous to the proof of Proposition 1.9. Consider an MRM code of minimum distance d . Without loss of generality assume $m \leq n$. The proof of the other case is exactly the same.

Consider the function $\pi : C \rightarrow \text{Mat}_{(n-d+1) \times m}(\mathbb{F}_q)$ which deletes the bottom $d-1$ rows of a matrix $A \in C$. We claim that this is injective. Suppose not.

Then there are two matrices A and B mapping to the same deleted matrix. Thus A and B have the same first $n-d+1$ rows and hence $A-B \in C$ has its first $n-d+1$ rows zero. Thus the rank of $A-B$ can be at most $d-1$, being contributed by the $d-1$ bottom rows.

This contradicts the minimum distance being d and hence the map is injective. Thus the dimension of the domain is at most the dimension of the target space which is $m(n-d+1)$ which is what we wanted to show.

We now prove the anti-code bound for matrix codes. This uses the existence of MRM codes satisfying the Singleton bound proved just after. Thus, let C be some MRM code of minimum distance d . Without loss of generality assume that $n \leq m$. Let the maximum rank of C be r . Note that $0 \leq r \leq n$. Suppose $r = n$, then the bound becomes $\dim(C) \leq mn$ which is trivially true. Thus suppose that $r \leq n-1$.

Obtain, using Proposition 4.108, an MRM code D of minimum distance $r+1$ (noting that $1 \leq r+1 \leq n$) which satisfies the MRM Singleton bound. Since the minimum rank in D is $r+1$ while the maximum rank in C is r (both achieved, by the way), we have $C \cap D = \emptyset$.

Now consider the subspace $C \oplus D$ of $\text{Mat}_{n \times m}(\mathbb{F}_q)$ of dimension $\dim(C) + \dim(D) = \dim(C) + m(n - (r+1) + 1) = \dim(C) + m(n-r)$. This has to be bounded above by the dimension of the whole space which is mn and hence we get

$$\dim(C) + m(n-r) \leq mn \implies \dim(C) \leq mr$$

which proves the anti-code bound for MRM codes.

The Singleton bound for VRM codes is simply Proposition 1.9 with the base field now to be taken as $\mathbb{F}_{q^m}$.

We now prove the last bound, that is, the anti-code bound for VRM codes. Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM code with maximum rank r . Suppose the inequality holds. Then we have

$$\dim_{\mathbb{F}_{q^m}}(C) \leq \max\text{rank}(C) = r \leq m$$

Indeed, the dimension of any set of vectors of $\mathbb{F}_{q^m}$ considered over $\mathbb{F}_q$ cannot exceed m , giving $r \leq m$. Note additionally that the number of vectors is n and we must also have $r \leq n$.

Now suppose that $\dim_{\mathbb{F}_{q^m}}(C) \leq m$. Let $\dim_{\mathbb{F}_{q^m}}(C) = k$. Fix any basis Γ of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$ and consider the associated MRM code $\Gamma(C)$. From Proposition 4.79 we know that $\dim_{\mathbb{F}_q}(\Gamma(C)) = mk$. Using the anti-code bound on this MRM code we get

$$mk \leq \max\{n, m\} r$$

Note that the maximum rank remains preserved. Thus if $n \leq m$, we are done.

Now suppose that $n > m$. The assumption of $k \leq m$ continues to hold. We want to show that $k \leq r$, that is, there exists some vector in C of rank at least k .

Since C has dimension k , there exist some k coordinates so that the projection onto these is an injective map. The projection here is the same as puncturing the code outside these k positions, which are in fact given by the indices of the k linearly independent columns of a generator matrix of C .

Pick a basis Γ of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$ and since $k \leq m$, pick some k elements $\gamma_1, \dots, \gamma_k$ from here and consider the vector $u = (\gamma_1, \dots, \gamma_k)$. Then the rank of this vector u is exactly k .

Now since the projection map is injective and both the domain C and the codomain $\mathbb{F}_{q^m}^k$ have q^{mk} elements, it is a bijection and hence there exists a unique codeword $x \in C$ which projects to the codeword u . Since u has rank k , adding coordinates will only possibly increase the rank, giving that $\text{rank}(x) \geq k$.

Thus we have found a codeword with rank at least k and we are done. $\blacksquare$

We now analyse the equality cases of the four bounds from above. These are given special names as below.

Definition 4.106. An MRM code which achieves the MRM Singleton bound is called a **maximum rank distance** (MRD) code while it is called an **optimal matrix anti-code** if it satisfies the MRM anti-code bound.

Definition 4.107. A VRM code which achieves the VRM Singleton bound is called a **maximum rank distance** (MRD) code while it is called an **optimal vector anti-code** if it satisfies the VRM anti-code bound.

We now analyse their existence properties.

Proposition 4.108. *For any integer $1 \leq d \leq \min\{n, m\} + 1$ there exists an MRD MRM code of minimum distance d .*

Proof. This uses some theory from the previously discussed section on bilinear forms. Assume without loss of generality that $n \leq m$. Aligning with the convention followed in the previous bilinear form analysis, we have $m \leq m'$. Recall Theorem 4.73. For any $0 \leq t \leq n$, we can obtain an (n, m, t, q) -Singleton system X .

By definition, this is a t -design $X \subseteq \Omega(n, m)$ with index 1. Alternatively and equivalently, it is also an $(n - t)$ -codesign of cardinality $q^{tm'}$.

Invoking Theorem 4.67, since the lower bound is achieved by X , so is the upper bound and hence the dimension of X is $(n - s)m = mt$ since $s + t = n$. The MRM code Singleton bound is $m(n - d + 1)$.

Having an $(n - t)$ -codesign, that is, an s -codesign, by definition means that the rank of all possible differences of matrices in X is strictly greater than s . Further s is the largest such integer, giving that the minimum distance of X , viewed as an MRM code, is $s + 1$ and hence $d = s + 1$. Note that from Theorem 4.73, the constructed X is an $\mathbb{F}_q$ -subspace and is hence a code. Not all Singleton systems are codes.

Thus the bound is $m(n - d + 1) = m(n - (s + 1) + 1) = m(n - s) = mt$. This is exactly the dimension of the code and hence the bound is achieved. Also note that as t varies between 0 and n , the corresponding $d = s + 1$ varies between 1 and $n + 1$. By Definition 4.101, the dimension of the zero code is $n + 1$ and here, too, the code X corresponding to $d = n + 1$, that is, $t = 0$ and $s = n$, is the zero code. $\blacksquare$

Proposition 4.109. *If $n > m$, then the trivial codes are the only MRD VRM codes. If $n \leq m$, then for any $1 \leq t \leq n$, there exists an MRD VRM code of minimum distance t .*

Proof. Suppose $n > m$. The zero VRM code has dimension 0 and minimum distance $n + 1$. Thus this satisfies the Singleton bound. Similarly the full space has dimension n and minimum distance 1 and is also MRD.

Suppose there is some non-trivial MRD VRM code. Then the minimum distance must satisfy $2 \leq d \leq n$ with dimension satisfying $1 \leq k \leq n - 1$. Also since it is MRD we have $k = n - d + 1$.

Obtain the MRM code from this VRM code. The dimension will be mk in accordance with Proposition 4.79. From the Singleton bound for matrices we have

$$mk = m(n - d + 1) \leq n(m - d + 1) \implies (d - 1)(n - m) \leq 0$$

This is a contradiction since $2 \leq d$ and $n > m$.

Now suppose $n \leq m$. Fix some integer $t \in [1, n]$. The construction of the subsequent code, called the **Gabidulin code**, is very similar to that of the Reed-Solomon code.

Fix a basis $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. Since $n \leq m$, fix some distinct n elements from this set and call them $a_1, \dots, a_n$. Set $k = n - t + 1$. We want to construct a code of dimension k and minimum distance t . Consider the Gabidulin code given by

$$G_t(a_1, \dots, a_n) = \left\{ (f(a_1), \dots, f(a_n)) \mid f(x) = \sum_{j=0}^{k-1} a_j x^{q^j}, a_j \in \mathbb{F}_{q^m} \right\}$$

This is clearly a linear subspace of $\mathbb{F}_{q^m}^n$. The dimension of this code corresponds to the number of independent coefficients a_j in the linearised polynomials of q -degree strictly less than k , which is exactly k .

We want to show that the minimum distance of this code is t so that the VRM Singleton bound is satisfied. Pick any non-zero codeword c corresponding to a unique polynomial f so that

$$c = (c_1, \dots, c_n) = (f(a_1), \dots, f(a_n))$$

By definition, the rank of c is given by

$$\text{rank}(c) = \dim_{\mathbb{F}_q}(\text{span}\{f(a_1), \dots, f(a_n)\}) = \dim_{\mathbb{F}_q}(f(\text{span}\{a_1, \dots, a_n\}))$$

Let $V = \text{span}\{a_1, \dots, a_n\}$ so that $\text{rank}(c) = \dim_{\mathbb{F}_q}(f(V))$. But from the rank-nullity theorem we have

$$\begin{aligned} \text{rank}(c) &= \dim_{\mathbb{F}_q}(f(V)) = \dim_{\mathbb{F}_q}(V) - \dim_{\mathbb{F}_q}(V \cap \ker(f)) \\ &\geq \dim_{\mathbb{F}_q}(V) - \dim_{\mathbb{F}_q}(\ker(f)) \\ &= n - \dim_{\mathbb{F}_q}(\ker(f)) \\ &\geq n - (k - 1) \\ &= n - k + 1 = t \end{aligned}$$

The inequality $\dim_{\mathbb{F}_q}(\ker(f)) \leq k - 1$ holds because f has degree at most q^{k-1} and thus can have at most q^{k-1} roots over $\mathbb{F}_{q^m}$. Thus the minimum distance of the code is at least t . But from the VRM Singleton bound, it can be at most t and hence it is exactly equal to t . Alternatively, finding an explicit polynomial with $\text{rank}(c_f)$ being exactly t is also not too difficult. ■

We now classify optimal anti-codes for both the matrix and vector cases. The proofs of the following theorems are very involved and we refer the reader to their respective sources.

The following is a classification theorem for MRM optimal anti-codes. This was proved by de Seguins Pazzis in [29] as theorems 4 and 6.

Proposition 4.110. *The optimal anti-codes of $\text{Mat}_{n \times m}(\mathbb{F}_q)$ are exactly the following*

- (i) $\text{Mat}_{n \times m}(\mathbb{F}_q)(V)$ for some $V \subseteq \mathbb{F}_q^{\min\{n, m\}}$, if $m \neq n$
- (ii) $\text{Mat}_{n \times n}(\mathbb{F}_q)(V)$ and $\text{Mat}_{n \times n}(\mathbb{F}_q)(V)^\top$ for some $V \subseteq \mathbb{F}_q^n$, if $m = n$

with notation as in Definition 4.99.

The proposition below classifies vector optimal anti-codes. This appeared in [27] as theorem 18.

Proposition 4.111. *Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM with $k = \dim_{\mathbb{F}_{q^m}}(C) \leq m$. Then the following are equivalent:*

- (i) C is an optimal anti-code
- (ii) C has a basis consisting of vectors with entries in $\mathbb{F}_q$
- (iii) $C \sim \text{span}_{\mathbb{F}_{q^m}}\{e_1, e_2, \dots, e_k\}$ with equivalence as in Definition 4.82

It is now natural to wonder whether these classes of codes are preserved when we pass from VRM codes to MRM codes. The following proposition handles this.

Proposition 4.112. *Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM code. Let Γ be a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. Consider the associated MRM code $\Gamma(C) \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$.*

If $n \leq m$, then C is MRD if and only if $\Gamma(C)$ is MRD and C is an optimal anti-code if and only if $\Gamma(C)$ is an optimal anti-code.

If $n > m$, then C and $\Gamma(C)$ are both MRD if and only if C is one of the two trivial codes and C and $\Gamma(C)$ are optimal anti-codes if and only if $C = \{0\}$.

Proof. In the case of $n \leq m$, the results follow immediately from Theorem 4.105 since the MRM bounds are simply scaled by m while the dimension is also scaled by m from Proposition 4.79.

Now suppose $n > m$. For C to be an MRD code, from Proposition 4.109 C can only be one of the trivial codes and $\Gamma(C)$ will also be the corresponding trivial code (either zero or full space) in this case. Conversely, these trivial codes are MRD in both the matrix and the vector cases.

The 0 code is also an optimal anti-code in both the matrix and vector cases. Now suppose C and $\Gamma(C)$ are both optimal anti-codes. Since C is an optimal anti-code we have

$$\dim(C) = \text{maxrank}(C)$$

Since $\Gamma(C)$ is also an optimal anti-code we have

$$\dim(\Gamma(C)) = m \dim(C) = n \cdot \text{maxrank}(\Gamma(C)) = n \cdot \text{maxrank}(C)$$

Thus we have $m \cdot \text{maxrank}(C) = n \cdot \text{maxrank}(C)$ which gives that the maximum rank of C must be 0 because $n > m$. This shows that C must be the zero code. ■

4.2.5 Duality and MacWilliams Identities

We define the dual codes with respect to a certain standard inner product.

Definition 4.113. We define the **dual code of an MRM code** $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$ to be

$$C^\perp = \{M \in \text{Mat}_{n \times m}(\mathbb{F}_q) \mid \langle M, N \rangle := \text{tr}(MN^\top) = 0 \ \forall N \in C\}$$

We define the **dual code of a VRM code** $C \subseteq \mathbb{F}_{q^m}^n$ to be

$$C^\perp = \{w \in \mathbb{F}_{q^m}^n \mid \langle v, w \rangle = 0 \ \forall v \in C\}$$

where the inner product of two vectors is the standard Euclidean inner product.

It is not hard to check that the dual is also a code.

These notions are compatible with the definition of an MRM arising from a VRM. We need another notion of bases being orthogonal to prove this.

Definition 4.114. Two bases $\Gamma = \{\gamma_1, \dots, \gamma_m\}$ and $\Gamma' = \{\gamma'_1, \dots, \gamma'_m\}$ of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$ are said to be orthogonal if

$$\text{tr}_{\mathbb{F}_{q^m}/\mathbb{F}_q}(\gamma_i \gamma'_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

We now have the compatibility proposition.

Proposition 4.115. *Let $C \subseteq \mathbb{F}_{q^m}^n$ be a VRM and let Γ and Γ' be orthogonal bases of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. Then,*

$$\Gamma(C)^\perp = \Gamma'(C^\perp)$$

Proof. From the subsequent Proposition 4.116 it suffices to show one containment. We show that $\Gamma'(C^\perp) \subseteq \Gamma(C)^\perp$. To this end fix an arbitrary $A \in \Gamma'(C^\perp)$. Write out

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \in \text{Mat}_{n \times m}(\mathbb{F}_q)$$

Then the corresponding codeword $x_A \in C^\perp$ will be given by

$$x_A = \left(\sum_{i=1}^m a_{1i} \gamma'_i, \sum_{i=1}^m a_{2i} \gamma'_i, \dots, \sum_{i=1}^m a_{ni} \gamma'_i \right) \in C^\perp \subseteq \mathbb{F}_{q^m}^n$$

Pick any arbitrary matrix $B \in \Gamma(C)$. Let it be written as

$$B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix} \in \text{Mat}_{n \times m}(\mathbb{F}_q)$$

Obtain the corresponding codeword

$$y_B = \left(\sum_{j=1}^m b_{1j} \gamma_j, \sum_{j=1}^m b_{2j} \gamma_j, \dots, \sum_{j=1}^m b_{nj} \gamma_j \right) \in C \subseteq \mathbb{F}_{q^m}^n$$

By definition we must have $\langle x_A, y_B \rangle = 0$ over $\mathbb{F}_{q^m}$, and hence

$$\sum_{t=1}^n \left(\sum_{i=1}^m a_{ti} \gamma'_i \right) \left(\sum_{j=1}^m b_{tj} \gamma_j \right) = 0$$

Because this equality holds in $\mathbb{F}_{q^m}$, applying the $\mathbb{F}_q$ -linear field trace $\text{tr}_{\mathbb{F}_{q^m}/\mathbb{F}_q}$ to both sides must also yield $0 \in \mathbb{F}_q$. Distributing the trace over the sums and the $\mathbb{F}_q$ -scalars a_{ti} and b_{tj} , we obtain

$$\sum_{t=1}^n \sum_{i=1}^m \sum_{j=1}^m a_{ti} b_{tj} \text{tr}_{\mathbb{F}_{q^m}/\mathbb{F}_q}(\gamma'_i \gamma_j) = 0$$

Using the definition of orthogonal bases, $\text{tr}_{\mathbb{F}_{q^m}/\mathbb{F}_q}(\gamma'_i \gamma_j) = \delta_{i,j}$. The sum collapses to exactly the terms where $i = j$, yielding

$$\sum_{t=1}^n \sum_{i=1}^m a_{ti} b_{ti} = 0$$

This double sum is precisely the standard definition of $\text{tr}(AB^\top)$. Since $B \in \Gamma(C)$ is arbitrary, $\text{tr}(AB^\top) = 0$ holds for all $B \in \Gamma(C)$. Thus by the definition of the dual of an MRM code, we have shown that $A \in \Gamma(C)^\perp$. $\blacksquare$

We now analyse the dimension of the dual code.

Proposition 4.116. *For an MRM code $C \subseteq \text{Mat}_{n \times m}(\mathbb{F}_q)$, we have*

$$\dim(C) + \dim(C^\perp) = mn$$

For a VRM code $C \subseteq \mathbb{F}_{q^m}^n$, we have

$$\dim(C) + \dim(C^\perp) = n$$

Proof. Recall the following theorem from linear algebra:

Lemma. *Let W be a subspace of a finite-dimensional vector space V equipped with a non-degenerate bilinear form. Then we have*

$$\dim(W) + \dim(W^\perp) = \dim(V)$$

where $W^\perp$ is defined using the bilinear form.

Applying this to our scenario with the bilinear forms $\text{tr}(AB^\top)$ and $\langle v, w \rangle$ in the matrix and vector cases, respectively, we get the desired results. Recall that the dimension of a VRM code is taken over $\mathbb{F}_{q^m}$. ■

4.3 Bounds on binary code sizes

We discuss an important bound on binary code sizes given by Delsarte. This appears in his thesis [12].

4.3.1 Introduction and some Graph Theory

Notation 4.117. Given any length n string or sequence x consisting of data at each of its n coordinates, by x_i we mean the value at the i^{th} coordinate for $i = 1, \dots, n$ so that $x = (x_1, \dots, x_n)$.

Notation 4.118. The full code of size 2^n given by the set of all possible binary strings of length n is denoted $\mathcal{B}(n)$. One may also think of this as the power set of $[n]$.

The aim of this article is to prove Delsarte's bound on the maximum size of the code under certain restrictions. In particular, we would like to find the maximum possible size of a binary code C of length n whose minimum distance is at least d . That is, we want to determine $\mathcal{A}_2(n, d)$.

Consider a simple undirected graph $G = (V, E)$ with vertex set V and edge set E .

Definition 4.119. By an *independent set* of vertices of G we mean a subset $\emptyset \neq S \subseteq V$ such that there are no edges between any two vertices of S .

Definition 4.120. Given any $A \subseteq V$ we can define the binary vector $\chi(A) \in \mathbb{F}_2^V$ of length $|V|$, which we refer to as the *characteristic vector* of A , to be

$$\chi(A)_v = \begin{cases} 1 & \text{if } v \in A \\ 0 & \text{otherwise} \end{cases}$$

Proposition 4.121. For any independent set S the $|V| \times |V|$ matrix M given by

$$M = \frac{1}{|S|} \chi(S) \chi(S)^\top$$

satisfies the following properties.

- (i) M is positive semi-definite (PSD).
- (ii) $M_{v,w} = 0$ whenever $\{v, w\}$ is an edge.
- (iii) The trace of M is $\text{tr } M = 1$.
- (iv) The sum of all entries of M is $\sum_{v,w \in V} M_{v,w} = |S|$.

Proof. The fact that M is PSD follows since M is symmetric and in particular of the form $AA^\top$ for some matrix A . Such matrices are always PSD. For the other three facts we explicitly write down every entry of this matrix M :

$$M_{v,w} = \frac{1}{|S|} \chi(S)_v \chi(S)_w = \begin{cases} \frac{1}{|S|} & \text{if } v, w \in S \\ 0 & \text{otherwise} \end{cases}$$

One can now easily check that the other three items follow. ■

Definition 4.122. By the *independence number* $\alpha(G)$ of an undirected simple graph G we mean the size of the largest independent set of vertices in the graph.

Proposition 4.123. The following is a bound on the independence number:

$$\alpha(G) \leq \max_M \sum_{v,w} M_{v,w}$$

where M varies over all $|V| \times |V|$ PSD matrices such that $\text{tr } M = 1$ and $M_{v,w} = 0$ whenever $\{v, w\} \in E$.

Proof. From Proposition 4.121 we know that for any independent set S there will exist a matrix M (given as in the proposition) which is PSD, has unit trace and its (v, w) -th entry is 0 whenever $\{v, w\} \in E$ such that the sum of all its entries is $|S|$.

In particular we have

$$|S| \leq \max_M \sum_{v,w} M_{v,w}$$

and hence the result follows. ■

4.3.2 Connecting graphs and codes

Notation 4.124. We denote the set of all real $n \times n$ symmetric matrices by $\mathbb{S}^n$.

Notation 4.125. If a matrix A is PSD, we denote it as $A \succeq 0$.

Definition 4.126. A *Semi-definite program* (SDP) is an optimisation problem of the form

$$\begin{aligned} & \text{maximise} && \langle C, X \rangle \\ & \text{subject to} && \langle A_i, X \rangle = b_i, \quad i = 1, \dots, m, \\ & && X \succeq 0, \end{aligned}$$

where $C, A_1, \dots, A_m \in \mathbb{S}^n$ are given symmetric matrices, $b \in \mathbb{R}^m$ is a given vector, and $X \in \mathbb{S}^n$ is the symmetric matrix variable constrained to be positive semidefinite.

The upper bound from the previous proposition is known as the **Lovász number**, $\vartheta(G)$. It was introduced by Lovász to get bounds on the Shannon capacity of a graph [21].

Definition 4.127. The **Lovász number** of a simple undirected graph is defined to be

$$\vartheta(G) = \max \left\{ \sum_{v,w} M_{v,w} \mid M \succeq 0, \text{tr}(M) = 1, M_{v,w} = 0 \ \forall \{v, w\} \in E \right\}$$

This is an SDP and all SDPs are solvable in polynomial time up to any desired error $\varepsilon > 0$, most commonly using interior-point methods (though the ellipsoid method was a key theoretical breakthrough).

We construct the graph G_d as follows. The set of vertices is given by $V = \mathcal{B}(n)$. We join $x, y \in \mathcal{B}(n)$ if and only if $d(x, y) \leq d - 1$.

Now any independent set will have vertices with the distance between any two being at least d . Thus the equivalent aim of this article is to bound the independence number of this graph.

We can apply the bound obtained earlier which is computed by solving an SDP. However, the complexity of this bound in our case is not polynomial time because our graph G_d has 2^n vertices. Consequently, we have a polynomial in 2^n . Thus this method is not viable.

A ray of hope: We can however exploit the symmetry and in fact obtain a linear programming model.

4.3.3 Symmetry of the graph

Notation 4.128. Let H_n denote the set of all distance (Hamming) preserving bijections $\mathcal{B}(n) \rightarrow \mathcal{B}(n)$.

Proposition 4.129. The size of H_n is $|H_n| = 2^n n!$.

Proof. We claim that every such bijection is a permutation of the n entries followed by individually flipping the bits $0 \leftrightarrow 1$. It is clear that every map of the latter form is a distance preserving bijection and that there are $2^n n!$ of these.

Now let σ be such a bijection. Consider $x = \sigma(0)$. This will give us the information about the flipping operation. Let $A \subseteq [n]$ be the set of indices of x where we have a 1. Let f_A denote the flipping operation performed on a given set $A \subseteq [n]$ of indices. Observe that $f_A \circ f_A = \text{id}$.

Consider $\tau = \sigma \circ f_A$. We want to show that τ is a permutation so that we are done. Firstly, if the Hamming weight $\text{wt}(x)$ of a codeword $x \in \mathcal{B}(n)$ is defined to be its distance from the 0 codeword, then

$$\text{wt}(\tau(x)) = d(\tau(x), 0) = d(\tau(x), \tau(0)) = d(x, 0) = \text{wt}(x)$$

Thus τ preserves weight. Now let $e_1, \dots, e_n$ be the vectors with 1 in the i -th place of e_i and 0 elsewhere. Then it is clear that τ permutes these e_i . This follows since the weight of each e_i is 1, every weight 1 codeword must be some e_i and τ is a bijection being a composition of two bijections.

Suppose this permutation is given by $\pi : [n] \rightarrow [n]$ so that

$$\tau(e_i) = e_{\pi(i)} = \pi(e_i) \quad \forall i \in [n]$$

We need to show that $\tau(x) = \pi(x)$ for every $x \in \mathcal{B}(n)$; not just the weight 1 codewords.

To this end let $x \in \mathcal{B}(n)$ have weight k . Then $y = \tau(x)$ also has weight k . Now we have

$$d(x, e_i) = \begin{cases} k-1 & \text{if } x_i = 1 \\ k+1 & \text{otherwise} \end{cases}$$

Since τ is distance preserving, we must have

$$d(y, \pi(e_i)) = d(y, e_{\pi(i)}) = \begin{cases} k-1 & \text{if } x_i = 1 \\ k+1 & \text{otherwise} \end{cases}$$

But we know that

$$d(y, e_j) = \begin{cases} k-1 & \text{if } y_j = 1 \\ k+1 & \text{otherwise} \end{cases}$$

Thus we have $x_i = y_{\pi(i)}$ showing that $y = \pi(x)$. This shows that τ is a permutation and hence σ is a permutation followed by flipping. ■

As a remark, one may note that H_n is also the group of automorphisms of the hypercube graph $\mathcal{H}_n$ on 2^n vertices. The inclusion $H_n \subseteq \text{Aut}(\mathcal{H}_n)$ is trivial. The other one needs work. Alternatively one may show that $|\text{Aut}(\mathcal{H}_n)|$ is also the same $2^n n!$ to avoid proving the other inclusion.

Definition 4.130. Given a binary code C of length n , we define its *characteristic vector* $\chi(C) \subseteq \mathbb{F}_2^{2^n}$ to be

$$\chi(C)_x = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{otherwise} \end{cases}$$

Here x varies over all codewords of $\mathcal{B}(n)$ with the ordering given by the dictionary:

$$(0, \dots, 0, 0) < (0, \dots, 0, 1) < (0, \dots, 1, 0) < \dots < (1, \dots, 1, 0) < (1, \dots, 1, 1)$$

The ordering could also be interpreted as binary representations of numbers in $\{0, 1, \dots, 2^n - 1\}$ with the standard total ordering on this set.

Lemma 4.131. For a binary code C of length n and any distance preserving bijection $\tau \in H_n$,

$$\chi(\tau(C)) = P_\tau \chi(C)$$

where P_τ is the $2^n \times 2^n$ permutation matrix corresponding to the bijection $\tau : \mathcal{B}(n) \rightarrow \mathcal{B}(n)$.

Proof. Let us compare the x -th entries on both sides. On the left hand side we have

$$\chi(\tau(C))_x = \begin{cases} 1 & \text{if } x \in \tau(C) \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 1 & \text{if } \tau^{-1}(x) \in C \\ 0 & \text{otherwise} \end{cases} = \chi(C)_{\tau^{-1}(x)}$$

On the right hand side we have

$$\sum_{y \in \mathcal{B}(n)} (P_\tau)_{x,y} \chi(C)_y = \chi(C)_{\tau^{-1}(x)}$$

■

Definition 4.132. Given a binary code C of length n , we associate it to the following matrix

$$M_C = \frac{1}{|C|} \sum_{\tau \in H_n} \chi(\tau(C)) \chi(\tau(C))^\top$$

We call it the *symmetrised gram matrix* (SG matrix) associated with the code C .

One can note immediately that this is PSD.

4.3.4 Commutants and orbital basis

Notation 4.133. By $M_{m \times n}(\mathbb{F})$ we mean the set of all $m \times n$ matrices over the field $\mathbb{F}$. When $m = n$ we may use only one subscript.

Notation 4.134. Let a group G act on a set X . We denote the $|X| \times |X|$ permutation matrix of the bijection corresponding to $g \in G$ by $\rho(g) \in M_{|X|}(\mathbb{C})$.

Definition 4.135. We define the *commutant* $\text{Hom}_G(S, T)$ to be the collection

$$\text{Hom}_G(S, T) = \{A \in M_{|T| \times |S|}(\mathbb{C}) \mid A\rho(g) = \tau(g)A\}$$

where ρ corresponds to permutation matrices of the action on S while τ is reserved for the action on T .

Note that this is a vector space over $\mathbb{C}$. If $S = T$, then it becomes a $\mathbb{C}$ -algebra.

Proposition 4.136. *The SG matrix of a code is in the commutant:*

$$M_C \in \text{Hom}_{H_n}(\mathcal{B}(n), \mathcal{B}(n))$$

Proof. Let $\sigma \in H_n$. We want to show that $P_\sigma M_C = M_C P_\sigma$. Equivalently we show that $P_\sigma M_C P_\sigma^\top = M_C$.

$$\begin{aligned} P_\sigma M_C P_\sigma^\top &= \frac{1}{|C|} \sum_{\tau \in H_n} P_\sigma P_\tau \chi(C) \chi(C)^\top P_\tau^\top P_\sigma^\top \\ &= \frac{1}{|C|} \sum_{\tau \in H_n} (P_\sigma P_\tau) \chi(C) \chi(C)^\top (P_\sigma P_\tau)^\top \\ &= \frac{1}{|C|} \sum_{\tau \in H_n} P_{\sigma \circ \tau} \chi(C) \chi(C)^\top P_{\sigma \circ \tau}^\top \\ &= \frac{1}{|C|} \sum_{\tau \in H_n} \chi((\sigma \circ \tau)(C)) \chi((\sigma \circ \tau)(C))^\top \\ &= M_C \end{aligned}$$

Note that we have used Lemma 4.131 in the first step and the penultimate step. The last step is simply a reordering of summation variables. ■

Lemma 4.137. *Let $M \in M_{|T| \times |S|}$. Then $M \in \text{Hom}_G(S, T)$ if and only if*

$$M(t, s) = M(g * t, g * s) \quad \forall t \in T, s \in S, g \in G$$

Proof. We have by definition that $M \in \text{Hom}_G(S, T)$ if and only if $M\rho(g) = \tau(g)M$. Comparing the (t, s) -th entries on both sides we get $M \in \text{Hom}_G(S, T)$ if and only if

$$M(t, g * s) = M(g^{-1} * t, s)$$

This is what we wanted. ■

Definition 4.138. Let G act on sets T and S . Extend the action naturally to $T \times S$. Let $\theta_1, \dots, \theta_m$ denote the orbits of the action on $T \times S$. Consider the matrices $M_j \in M_{|T| \times |S|}(\mathbb{C})$ for $j = 1, 2, \dots, m$ defined as

$$M_j(t, s) = \begin{cases} 1 & \text{if } (t, s) \in \theta_j \\ 0 & \text{otherwise} \end{cases}$$

The set of these m matrices is called the **orbital basis** of $\text{Hom}_G(S, T)$.

Proposition 4.139. *The orbital basis of $\text{Hom}_G(S, T)$ forms a $\mathbb{C}$ -basis of this vector space.*

Proof. Firstly observe that by Lemma 4.137 it follows that since (t, s) and $(g * t, g * s)$ lie in the same orbit, each $M_j \in \mathcal{A} = \text{Hom}_G(S, T)$.

The linear independence follows from the fact that every (t, s) lies in some orbit and all the orbits form a partition of $T \times S$. Thus all the non-zero 1's of each M_j are at disjoint positions.

Let $A \in \mathcal{A}$. By Lemma 4.137 we know that the entries of A remain constant on each orbit θ_i . Let this constant be c_i . Then it is clear that

$$A = \sum_{i=1}^m c_i M_i$$

showing that the M_j span the vector space $\mathcal{A}$. ■

Corollary 4.140. *The dimension of $\text{Hom}_G(S, T)$ is the number of orbits of the action of G on $T \times S$.*

Lemma 4.141 (Gelfand's lemma). *Let a group G act on a set S and let $M_1, \dots, M_m$ be the orbital basis of $\mathcal{A} = \text{Hom}_G(S, S)$. If each M_j is symmetric, then the $\mathbb{C}$ -algebra $\mathcal{A}$ is commutative.*

Proof. The basis M_{ij} is clearly symmetric and we show that the basis commutes pairwise. Firstly observe for that for distinct i and j in $\{1, \dots, m\}$, the matrix $M_i M_j$ is symmetric.

$$\therefore M_i M_j = (M_i M_j)^\top = M_j^\top M_i^\top = M_j M_i$$

Thus the algebra is commutative since the basis commutes pairwise. ■

Theorem 4.142. $\mathcal{A} = \text{Hom}_{H_n}(\mathcal{B}(n), \mathcal{B}(n))$ is a commutative algebra of dimension $n + 1$.

Proof. We analyse the action of H_n on $\mathcal{B}(n) \times \mathcal{B}(n)$ to get information about the orbits. We make the following claim.

Lemma. (α, β) and (α', β') lie in the same orbit if and only if $d(\alpha, \beta) = d(\alpha', \beta')$.

Proof of Lemma. If they are in the same orbit, then the new pair is obtained by applying a permutation and then flipping some bits of the old pair. Both these operations do not affect the Hamming distance.

Conversely, let the distance be preserved. Suppose $d(\alpha, \beta) = d(\alpha', \beta') = k$. Apply some $\sigma \in H_n$ so that we reduce α to $(0, \dots, 0)$. Simultaneously, β will be mapped to some β'' . Now apply $\tau \in H_n$, which is a

permutation that reduces β'' to $(0, \dots, 0, 1, \dots, 1)$ so that all the k 1's are pushed to the end. Note that $\tau(0) = 0$. This can be summarised as below:

$$(\alpha, \beta) \xrightarrow{\sigma} ((0, \dots, 0), \beta'') \xrightarrow{\tau} ((0, \dots, 0), (0, \dots, 0, 1, \dots, 1))$$

Similarly (α', β') can be brought down to the exact same pair $((0, \dots, 0), (0, \dots, 0, 1, \dots, 1))$ with k 1's. Thus, by inverting and composing we can pass between the old and new pairs. $\backslash\backslash\backslash$

Now note that $d(\alpha, \beta) = d(\beta, \alpha)$ and hence the reverse of every pair lies in the same orbit as that pair. Thus, every M_j is symmetric. By Lemma 4.141 we get that the algebra is commutative.

The number of orbits will be determined by the number of possible Hamming distances which is simply $n+1$ since the distances could be any integer between (and including) 0 and n . This gets us the dimension. ■

As a remark, one may note that this is in fact the Bose-Mesner algebra of the Hamming association scheme. We return to our case of H_n acting on $\mathcal{B}(n)$ and find the orbital basis explicitly:

$$A_i(x, y) = \begin{cases} 1 & \text{if } d(x, y) = i \\ 0 & \text{otherwise} \end{cases}$$

for every $i = 0, 1, \dots, n$.

Note that each A_i is an $2^n \times 2^n$ matrix.

Notation 4.143. We denote $\lambda_i = |\{(x, y) \in C \times C \mid d(x, y) = i\}|$ and normalise it to

$$\alpha_i = \frac{\lambda_i}{|C| \binom{n}{i}}$$

so that $\alpha_0 = 1$.

Proposition 4.144. *We have*

$$\sum_{i=0}^n \alpha_i \binom{n}{i} = |C|$$

Proof. Follows from the definition of α_i and the fact that

$$\bigsqcup_{i=0}^n \{(x, y) \in C \times C \mid d(x, y) = i\} = C \times C$$

■

Proposition 4.145. *The SG matrix M_C lies in the commutant and hence can be expressed in terms of the orbital basis as*

$$M_C = n! (\alpha_0 A_0 + \alpha_1 A_1 + \dots + \alpha_n A_n)$$

Proof. We simply count the number of 1's on both sides of the equation. On the right hand side we may add up the counts because all the A_i 's have disjoint supports.

The number of 1's in A_i is simply $2^n \binom{n}{i}$. This is obtained by counting the 1's column-wise: for each one of the 2^n columns corresponding to a codeword, we may flip any i coordinates to obtain a new codeword with distance i from it.

The number of 1's in $\chi(C)\chi(C)^\top$ in the positions fixed by the 1's of A_i is given precisely by λ_i since $(\chi(C)\chi(C)^\top)(v, w) = 1_{v \in C} 1_{w \in C}$ and we restrict to only those (v, w) satisfying $d(v, w) = i$.

The result now follows by the definition of α_i . ■

We can now finally summarise our findings below.

Theorem 4.146. *If C is a binary code of length n with minimum distance d , we have*

$$|C| \leq \max \sum_{i=0}^n \binom{n}{i} x_i$$

subject to

- (i) $x_i \geq 0 \quad \forall i = 0, \dots, n$
- (ii) $x_0 = 1$ and $x_1 = x_2 = \dots = x_{d-1} = 0$
- (iii) $x_0 A_0 + \dots + x_n A_n$ being PSD

Proof. The proof readily follows from Proposition 4.145, Proposition 4.144 and the definitions of α_i . Note that the first few α_i , in particular $\alpha_1, \alpha_2, \dots, \alpha_{d-1}$ will be zero since we demand a minimum distance of at least d . ■

We are not quite done because although this may seem like an SDP, the last condition which demands positive semi-definiteness involves matrices A_i of size 2^n .

Another ray of hope: We know that a matrix is PSD if and only if all its eigenvalues are non-negative. If we were to know the eigenvalues of $\sum_{i=0}^n x_i A_i$ AND we only had $O(n)$ eigenvalues, then we can replace the third condition with the $O(n)$ conditions of these eigenvalues being non-negative which would now give us a nice linear problem.

This is indeed possible. In fact there are $n + 1$ eigenvalues. We find these in the next section.

4.3.5 Eigenvalues

Proposition 4.147. *We have the canonical isomorphism*

$$\mathbb{C}[\mathcal{B}(n)] \cong \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2 \text{ (} n \text{ times)}$$

Proof. Consider $\mathbb{C}[\mathcal{B}(n)]$. Any vector here is of length 2^n and indexed by $\mathcal{B}(n)$. The standard basis of $\mathbb{C}[\mathcal{B}(n)]$ is hence given by $\{e_X \mid X \subseteq [n]\}$.

We construct the canonical isomorphism $\phi : \mathbb{C}[\mathcal{B}(n)] \rightarrow \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2$ (n times) given by

$$e_X \mapsto a_1 \otimes \dots \otimes a_n$$

where a_i ($1 \leq i \leq n$) is given by

$$a_i = \begin{cases} \hat{1} = (0, 1)^\top \in \mathbb{C}^2 & \text{if } i \in X \\ \hat{0} = (1, 0)^\top \in \mathbb{C}^2 & \text{otherwise} \end{cases}$$

■

Lemma 4.148. *A_i acts on $a_1 \otimes \dots \otimes a_n$ in the following way. We choose any i coordinates and flip them to obtain a new tensor product. Finally add all such $\binom{n}{i}$ tensor products to get the result.*

Proof. Let us start with $e_X \in \mathbb{C}[\mathcal{B}(n)]$. We know that the action of A_i is given by $A_i e_X = \sum_{Y: d(X, Y)=i} e_Y$. Now e_X corresponds to some $a_1 \otimes \dots \otimes a_n$. Given a Y at a distance of i from X , if Y corresponds to $b_1 \otimes \dots \otimes b_n$, then it is clear that some i of the a_i have been flipped to obtain the b_i . The result follows. ■

For those seeking some formalism, when A_i acts on the tensor product space, it is realised as an operator given by

$$\sum_{\substack{S \subseteq [n] \\ |S|=i}} X_S$$

where $X_S = M_1 \otimes \cdots \otimes M_n$ with

$$M_k = \begin{cases} X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} & \text{if } k \in S \\ I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \text{otherwise} \end{cases}$$

Notation 4.149. We consider the eigenbasis of X now. This is given by

$$u = \frac{1}{\sqrt{2}}(\hat{0} + \hat{1}) ; \quad v = \frac{1}{\sqrt{2}}(\hat{0} - \hat{1})$$

Definition 4.150. Fix any $j \in \{0, 1, \dots, n\}$. Define a space $\mathbf{W}_j$ as follows.

$$W_j = \text{span}(\{u_1 \otimes \cdots \otimes u_n \mid j \text{ of the } u_i \text{ are } v \text{ and the others are } u\})$$

We shall see that this is a common eigenspace for all A_i .

Theorem 4.151. Fix $i \in \{0, 1, \dots, n\}$. Each W_j ($0 \leq j \leq n$) is an eigenspace of A_i of dimension $\binom{n}{j}$ with the corresponding eigenvalue

$$\lambda_j^{(i)} = \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k}$$

Proof. Let us take a tensor $w = u_1 \otimes \cdots \otimes u_n \in W_j$ and act A_i on it. From Lemma 4.148 we know the action of A_i . Now the key point is that u ‘flips’ to v and v ‘flips’ to $-u$. Thus for every i coordinates we choose to flip, the vector remains the same and we can pull out the sign $(-1)^{\#v \text{ flipped}}$.

Thus if J denotes the positions of w where we have a v , we have

$$A_i w = \left(\sum_{S \subseteq [n], |S|=i} (-1)^{|S \cap J|} \right) w$$

This shows that W_j is indeed an eigenspace of A_i . The dimension of W_j is simply the size of its basis which is clearly $\binom{n}{j}$. Note that the given set to be spanned is itself a basis since it is linearly independent.

We now find the exact eigenvalue $\lambda_j^{(i)}$.

$$\begin{aligned} \lambda_j^{(i)} &= \sum_{S \subseteq [n], |S|=i} (-1)^{|S \cap J|} \\ &= \sum_{k=0}^i (-1)^k |\{S \subseteq [n] : |S| = i, |S \cap J| = k\}| \\ &= \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k} \end{aligned}$$

■

Note that the given W_j are all the eigenspaces of A_i . This could follow either from observing the dimension of each W_j or simply using Theorem 4.142 to obtain a dimension of $n + 1$.

Now since by Theorem 4.142 the algebra is commutative, the A_i are all simultaneously diagonalisable and hence the W_j are in fact common eigenspaces. Thus the eigenvalues of $\sum_{i=0}^n x_i A_i$ are given by

$$\mu_j = \sum_{t=0}^n x_t \lambda_j^{(t)} \quad \forall \quad j = 0, 1, \dots, n$$

each of multiplicity $\binom{n}{j}$ corresponding to the eigenspace W_j .

4.3.6 The final result

Theorem 4.146 may be restated as the following linear programming problem.

Theorem 4.152. *The size of a binary code C of length n with minimum distance d is bounded above by*

$$\max \sum_{i=0}^n \binom{n}{i} x_i$$

where the maximum is taken over all $x_0, x_1, \dots, x_n$ such that

- (i) $x_i \geq 0 \quad \forall \quad i = 0, \dots, n$
- (ii) $x_0 = 1$ and $x_1 = \dots = x_{d-1} = 0$
- (iii) $\sum_{i=0}^n \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k} x_i \geq 0 \quad \forall \quad j = 0, 1, \dots, n$

4.4 Strong blocking sets and minimal codes from graphs

4.4.1 Asymptotically good codes

This section is a brief summary and analysis of the paper [3] on the construction of strong blocking sets using graphs.

If a linear code $C \in [n, k]_q$ has minimum distance d , we write it as $C \in [n, k, d]_q$.

Definition 4.153. A family of codes $C_i \in [n_i, k_i, d_i]_q$ of strictly increasing lengths n_i is called an **$(\mathbf{R}, \delta)$ -family of codes** if it has **rate** R and **relative distance** δ given by

$$R = \liminf_{i \rightarrow \infty} \frac{k_i}{n_i} ; \quad \delta = \liminf_{i \rightarrow \infty} \frac{d_i}{n_i}$$

Definition 4.154. A family of linear codes is said to be **asymptotically good** if both its rate and relative distance is positive, assuming they exist.

There are some good results on the existence of these codes. We cover one such result below.

Proposition 4.155. *Asymptotically good codes exists for any choice of $\delta \in [0, 1 - \frac{1}{q}]$ and $R = 1 - H_q(\delta)$ where $H_q(x)$ is the q -entropy function defined as*

$$H_q(x) = x \log_q(q-1) - x \log_q(x) - (1-x) \log_q(1-x)$$

Proof. Suppose $\delta \in [0, 1 - \frac{1}{q}]$ is fixed. Let $R = 1 - H_q(\delta)$. Pick any sequence of strictly increasing positive numbers $\{n_i\}_{i \geq 1}$. Choose $d_i = \lfloor \delta n_i \rfloor$ such that $\delta n_1 \geq 2$ so that each $d_i \geq 2$.

Recall Lemma 1.94 and Lemma 1.95. Note that by choice of d_i , we have $n_i \geq d_i \geq 2$ and hence for every k_i satisfying

$$q^{n_i - k_i} \geq V_q(n_i - 1, d_i - 2)$$

we have the existence of a code $C_i \in [n_i, k_i, d_i]_q$. Obtain such a code (the inequality is true for $k_i = 1$ and hence such a code will exist). Now for such a k_i we have

$$q^{n_i - k_i} \geq \sum_{j=0}^{d_i-2} \binom{n_i-1}{j} (q-1)^j \geq \binom{n_i-1}{d_i-2} (q-1)^{d_i-2}$$

Thus we have

$$\frac{k_i}{n_i} \leq 1 - \frac{n_i-1}{n_i} \left[\frac{1}{n_i-1} \log_q \left(\binom{n_i-1}{d_i-2} \right) \right] - \frac{d_i-2}{n_i} \log_q(q-1)$$

Thus,

$$\begin{aligned} R &\leq 1 - \left(\liminf_{i \rightarrow \infty} \frac{1}{n_i-1} \log_q \left(\binom{n_i-1}{d_i-2} \right) \right) + \delta \log_q(q-1) \\ &= 1 - H_q(\delta) \end{aligned}$$

This completes the proof since by choosing the largest possible k_i satisfying the inequality, we can take R to be exactly equal to $1 - H_q(\delta)$. Another way to see the equality is to explicitly pick $k_i = \lfloor n_i - \log_q(V_i) \rfloor$ and find the rate using that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_q(V_q(n, \lfloor \delta n \rfloor)) = H_q(\delta)$$

The limits of the log of the binomial coefficients have been calculated using Stirling's approximation. ■

This is a very weak result since we have used an extremely inefficient bound by only considering one single term from the sum formula for $V_q(n_i - 1, d_i - 2)$. An explicit and a better construction was given by Justesen [19] for the binary case. One can refer section 10.3.1 from [15] for the generalisation to all finite fields.

Proposition 4.156. *For every $0 < R < 1/2$, there is an explicit family of linear codes $C_i \in [n_i, k_i, d_i]$ of strictly increasing lengths with rate R and relative distance*

$$\delta > (1 - 2R)H_q^{-1}(1/2)$$

As a remark, we note that the choice of delta does not depend on q because $H_q^{-1}(1/2) \geq H_2^{-1}(1/2) \geq 0.11$.

4.4.2 Projective systems and their relation with codes

We now discuss our projective setting.

Definition 4.157. The *finite projective space*, $\mathbf{PG}(\mathbf{k}, \mathbf{q})$, of order k (for some integer $k \geq 0$) is defined as the following

$$PG(k, q) = (\mathbb{F}_q^{k+1} \setminus \{0\}) / \sim$$

where $v, w \in \mathbb{F}_q^{k+1} \setminus \{0\}$ are said to be equivalent, that is, $v \sim w$ if and only if v is a non-zero scalar multiple of w .

For the case of $k = 0$, we will only have a single point in the projective space. One may adopt the convention of $PG(-1, q) = \emptyset$.

Note that $PG(k, 1)$ still retains the structure of a vector space.

For this section, by a hyperplane we mean a codimension 1 space. Thus, a hyperplane is defined according to the space it is in. For instance, the hyperplanes of $PG(k, q)$ are codimension 1 spaces of the projective space. Equivalently, they are the images of the hyperplanes of $\mathbb{F}_q^{k+1}$ under the quotient map.

Throughout, we will adopt the convention of $k \geq 1$ so that we may talk of $PG(k-1, q)$, which is constructed from $\mathbb{F}_q^k$.

Definition 4.158. A multiset $M \subseteq PG(k-1, q)$ is called a *projective $[\mathbf{n}, \mathbf{k}, \mathbf{d}]_{\mathbf{q}}$ system* if M has n points as a multiset, spans the full projective space $PG(k-1, q)$ and has the property

$$d = n - \max \{H \cap M \mid H \text{ is a hyperplane in the projective space}\}$$

Essentially, the last property of d states that we cannot have too many points on one hyperplane. Of course we cannot possibly dump all the points onto one single hyperplane because they must span the projective space. The upper bound is $n - d$. Thus, $d > 1$ follows from the definition.

We now make the following definition to aid us with the subsequent theorem.

Definition 4.159. Two $[n, k, d]_q$ projective systems $M = \{p_1, \dots, p_n\}$ and $N = \{q_1, \dots, q_n\}$ are said to be **projective equivalent** if there exists an invertible matrix $A \in M_{n \times n}(\mathbb{F}_q)$ such that $[q_i] = A[p_{\sigma(i)}]$ for some permutation σ .

Equivalently, we may say $q_i = \lambda A p_{\sigma(i)}$ for some non-zero scalar $\lambda \in \mathbb{F}_q$.

Theorem 4.160. *There is a one-one correspondence between non-degenerate $[n, k, d]_q$ codes considered up to monomial equivalence and projective $[n, k, d]_q$ systems considered up to projective equivalence.*

Proof. Suppose C is a non-degenerate $[n, k, d]_q$ code. Pick some generator matrix $G_{k \times n}$ and consider its columns $g_i \in \mathbb{F}_q^k$ for $i = 1, 2, \dots, n$. We obtain the projective system $M = \{[g_1], \dots, [g_n]\}^\#$ where $[\cdot]$ denotes the equivalence class under the quotient $\sim$ and $\{\}^\#$ means that we consider the set as multiset.

Since G has rank k , the columns of G span $\mathbb{F}_q^k$ and hence M must span $PG(k-1, q)$. Now let $x \in C$ be a non-zero codeword. Obtain the unique non-zero vector $u \in \mathbb{F}_q^k$ such that $uG = x$. Thus,

$$c = (\langle u, g_1 \rangle, \dots, \langle u, g_n \rangle)$$

Thus, if H_u denotes the hyperplane of $\mathbb{F}_q^k$ with normal vector u , the weight of c is given by

$$\text{wt}((c)) = n - \left| \{g_1, \dots, g_n\}^\# \cap H_u \right|$$

Thus by definition of minimum distance,

$$d(C) = d = n - \max_{u \in \mathbb{F}_q^k \setminus \{0\}} \left| \{g_1, \dots, g_n\}^\# \cap H_u \right|$$

But observe that $\left| \{g_1, \dots, g_n\}^\# \cap H_u \right| = \left| \{[g_1], \dots, [g_n]\}^\# \cap \tilde{H}_u \right|$ where $\tilde{H}_u$ is the image of the hyperplane H_u under $\sim$, so that it is now a hyperplane of the projective space. Further, every hyperplane of the projective space arises from some non-zero $u \in \mathbb{F}_q^k$ as $\tilde{H}_u$ and hence

$$d = n - \max \{H \cap M \mid H \text{ is a hyperplane in the projective space}\}$$

Thus M is indeed a projective $[n, k, d]_q$ system. We now show that this is a well-defined map. Suppose we had two equivalent $[n, k, d]_q$ codes C and D and picked some generator matrices G and H . Then by monomial equivalence we must have

$$C = QDM$$

for some $k \times k$ invertible matrix Q over $\mathbb{F}_q$ and M is an $n \times n$ monomial matrix over $\mathbb{F}_q$. The pre-multiplication by P was done to handle the case of a specific choice of a generator matrix in accordance with Proposition 1.13.

Suppose the matrix M corresponds to the permutation $\sigma \in S_n$ followed by the scaling of the i -th coordinate by λ_i , then we can write

$$g_i = \lambda_i Q h_{\sigma(i)}$$

This shows that the projective systems obtained from G and H are equivalent.

One may write down the opposite map similarly by mapping the system to the columns of a matrix which will necessarily be of rank k because the projective system spans the projective space. Thus, it will be the generator matrix of some code which we call as the output. One needs to check the well-defined nature of the map in this case, too. Further, it is clear that the maps are inverses of each other. ■

4.4.3 Strong blocking sets and minimal codes

Definition 4.161. A multiset $M \subseteq PG(k-1, q)$ is called a **strong blocking set** of $PG(k-1, q)$ if for every hyperplane H of $PG(k-1, q)$ we have

$$\text{span}\{H \cap M\} = H$$

One notes that for a blocking set we simply require a non-empty intersection with every hyperplane. If we add that this intersection spans the hyperplane, we get a strong blocking set.

The reader may question the consequences of applying the correspondence of Theorem 4.160 to strong blocking sets. This correspondence produces a class of codes called minimal codes.

Definition 4.162. A non-zero codeword $x \in C$, for a linear code $C \in [n, k, d]_q$, is said to be minimal if there exists no y in $C \setminus \text{span}\{x\}$ with $\text{supp}(y) \subseteq \text{supp}(x)$.

Definition 4.163. A linear code $C \in [n, k, d]_q$ is said to be minimal if there exist no $x, y \in C \setminus \{0\}$ such that x is not a multiple of y and $\text{supp}(y) \subseteq \text{supp}(x)$.

Equivalently, a code C is minimal if and only if every codeword is minimal.

Theorem 4.164. Suppose C is a non-degenerate $[n, k, d]_q$ code with a generator matrix $G = [g_1 \dots g_n]_{k \times n}$ where $g_1, \dots, g_n \in \mathbb{F}_q^k$. The following are equivalent

(i) C is a minimal code.

(ii) $M = \{[g_1], \dots, [g_n]\}^\#$ is a strong blocking set in $PG(k-1, q)$.

Proof. We make the following observation. Let $Z(c)$ denote the zero set of the codeword $x \in C$, that is, $Z(c) = [n] \setminus \text{supp}(x)$. Then for every non-zero $x, y \in C$

$$\text{supp}(x) \subseteq \text{supp}(y) \iff Z(y) \subseteq Z(x) \iff M \cap H^{(y)} \subseteq M \cap H^{(x)}$$

where $H^{(x)}$ denotes the unique hyperplane H_u for $x = uG$.

Suppose C is minimal but M is not a strong blocking set of $PG(k-1, q)$. Then there exists a hyperplane H such that $K = \text{span}\{M \cap H\} \subseteq H$. Obtain a hyperplane $H' \neq H$ containing the space K of codimension at least 2. Obtain the non-zero orthogonal vectors u and u' in $\mathbb{F}_q^k$ corresponding to the hyperplanes H and H' . Thus,

$$M \cap H \subseteq \text{span}\{M \cap H\} = K \subseteq H' \implies M \cap H \subseteq M \cap H'$$

Thus if some normal vectors of H and H' produce the codewords x and x' using the generator matrix, by the observation we made earlier, we get $\text{supp}(x') \subseteq \text{supp}(x)$. Note that x and x' are obtained up to scalar multiples since there is no unique normal vector to the hyperplane. However, $H \neq H'$ implies that x and x' cannot be scalar multiples of each other. This contradicts the property of C being minimal.

Conversely, suppose M is a strong blocking set but C is not minimal. Then there exist non-zero codewords $x, y \in C$ such that $\text{supp}(x) \subseteq \text{supp}(y)$ and they are not scalar multiples of each other. By the observation, we have $M \cap H^{(y)} \subseteq M \cap H^{(x)}$ and thus

$$\text{span}\{M \cap H^{(y)}\} \subseteq \text{span}\{M \cap H^{(x)}\} \implies H^{(y)} \subseteq H^{(x)} \implies H^{(y)} = H^{(x)}$$

Thus x and y must be scalar multiples of each other which is a contradiction to what was assumed. ■

We now give some bounds on sizes of these minimal codes.

Proposition 4.165. *The size of a minimal code $C \in [n, k, d]_q$ is at least $q(k-1) + 1$.*

Proof. If $k = 1$ we are done trivially. Suppose $k \geq 2$. Let M be the SBS obtained from some generator matrix. We will bound the size of the SBS and hence it will not matter which generator matrix we pick since all of them have n columns.

Consider the multiset

$$S = \{(P, \Lambda) : P \in M, \Lambda \text{ is a hyperplane containing the point } P\}$$

We double count S . Counting hyperplanes for a fixed point we get

$$|S| = \sum_{P \in M} (\text{number of hyperplanes containing } P)$$

Since P is not zero by definition of $PG(k-1, q)$, the number of hyperplanes of $PG(k-1, q)$ containing $P \in PG(k-1, q)$ is the number of hyperplanes W of $\mathbb{F}_q^k$ which contain the subspace $\text{span}\{P\} \subseteq \mathbb{F}_q^k$. But the subspaces of $\mathbb{F}_q^k$ containing $\text{span}\{p\}$ are in one-one correspondence with subspaces of the quotient $\mathbb{F}_q^k / \text{span}\{P\}$.

Further, this correspondence preserves the codimension. The quotient has dimension $k-1$ and hence the number of codimension 1 spaces of this quotient is

$$\frac{q^{k-1} - 1}{q - 1}$$

Thus we have

$$|S| = \frac{n(q^{k-1} - 1)}{q - 1}$$

On the other hand

$$|S| = \sum_{\Lambda: \text{hyperplane of } PG(k-1, q)} |\Lambda \cap M|$$

Since $\text{span}\{\Lambda \cap M\}$ has dimension $k-1$, it must have at least $k-1$ points on it. Thus,

$$|S| \geq (k-1) \frac{q^k - 1}{q - 1}$$

Combining the two gives

$$\frac{n}{k-1} \geq q + \frac{q-1}{q^{k-1}-1} > q$$

which proves the result since we are dealing with integers. ■

This can be improved by applying the combinatorial nullstellensatz to the following.

Proposition 4.166. *The size of a minimal code $C \in [n, k, d]_q$ is at least $(k-1)(q+1)$.*

We refer the reader to [1], [7] and [17] for a proof.

This can further be improved using Delsarte's linear programming bound to the following.

Proposition 4.167. *For any prime power q there exists a constant c_q such that the size of a minimal code $C \in [n, k, d]_q$ is at least $(c_q - o(1))(k-1)(q+1)$ as $n \rightarrow \infty$.*

The proof can be found in [7] and [28].

We also have results for the upper-bound. These are the best-known upper bounds at the time the authors have written the paper.

Proposition 4.168. *The size of the smallest strong blocking set in $PG(k-1, q)$ is at most*

$$\begin{cases} \frac{2k-1}{2 - \log_2(3)} & \text{if } q = 2 \\ \frac{2k(q+1)}{4 - \log_q(q^3 - q + 1)} & \text{if } q > 2 \end{cases}$$

However this is only an existence result and has no explicit constructions. The proofs can be found in [7], [2] and [22].

4.4.4 Integrity of a graph

We come to a graph theoretic concept - the integrity of a graph.

Definition 4.169. The *integrity* of an undirected graph G with vertex set V is defined to be

$$i(G) = \min_{S \subseteq V} (|S| + \kappa(G \setminus S))$$

where $\kappa(G)$ is the size of the largest connected component of G .

We provide some bounds on this number. For this, we need to define another parameter of a graph.

Definition 4.170. For an undirected (possibly non-simple and disconnected) graph G let $\mathbf{z}(G)$ denote the largest integer z such that there are two disjoint sets of vertices in G , each of size z , with no edge between them.

Although not standardised, for our convenience, we call this number the maximum disjoint union number of the graph or MDUS for short.

Proposition 4.171. *For any undirected graph G on n vertices, we have the bounds*

$$n - 2z(G) \leq i(G) \leq n - z(G)$$

Proof. The upper bound is trivial to prove. Suppose $z(G) = z$ corresponds to the sets A and B of size z . Then choose S to be the set of all vertices other than A and B . When this set is deleted, the remaining graph $G \setminus S$ is a disjoint union of A and B and the size of the largest connected component will be bounded above by z . Thus

$$i(G) \leq |S| + z = n - 2z + z = n - z$$

The lower bound needs some casework. Suppose $z = z(G)$ and S is the set of vertices corresponding to the optimal minimum value producing $i(G)$. Let $G \setminus S$ consist of the connected components $C_1 \sqcup \cdots \sqcup C_t$. Suppose without loss of generality that $c_1 \geq \cdots \geq c_t$ where $|C_i| = c_i$. Then by definition,

$$i(G) = |S| + c_1$$

We want to show that $n - 2z \leq |S| + c_1$ or equivalently that $2z \geq c_2 + \cdots + c_t$ since $n = |S| + c_1 + \cdots + c_t$.

Note that if $t = 1$, the RHS $c_2 + \cdots + c_t$ is 0 and we are done trivially. Thus we assume that $t \geq 2$. We have the following cases.

Case 1: Suppose $c_1 \geq z + 1$.

If $c_2 + \cdots + c_t \geq z + 1$, we get a contradiction since we can choose $A = C_1$ and $B = C_2 \sqcup \cdots \sqcup C_t$ to get $z(G) \geq z + 1$ which is a contradiction. Thus,

$$c_2 + \cdots + c_t \leq z \leq 2z$$

Case 2: Suppose $c_1 \leq z$.

For the sake of contradiction suppose $c_2 + \cdots + c_t \geq 2z + 1$. Since $c_t \leq c_1 \leq z$, we must have some largest coordinate s such that $c_s + \cdots + c_t \geq z + 1$. Thus, $c_{s+1} + \cdots + c_t \leq z$. Hence

$$c_s + (c_{s+1} + \cdots + c_t) \leq c_1 + z$$

Hence,

$$2z + 1 \leq (c_2 + \cdots + c_{s-1}) + (c_s + c_{s+1} + \cdots + c_t) \leq (c_2 + \cdots + c_{s-1}) + c_1 + z$$

Thus we have

$$c_1 + c_2 + \cdots + c_{s-1} \geq z + 1$$

Picking $A = C_1 \sqcup C_{s-1}$ and $B = C_s \sqcup C_t$ we get a contradiction to the maximality of z since both A and B have size at least $z + 1$. ■

We now have important lemma in the context of spectral graph theory.

Lemma 4.172 (Expander-Mixing Lemma). *Let G be an undirected d -regular graph on n vertices. Let $e(A, B)$ denote the number of edges going between vertex sets $A, B \subseteq V(G)$ (possibly overlapping or even the same set $A = B$). Suppose the eigenvalues are $d = \lambda_1 \geq \cdots \geq \lambda_n$. Denote the second largest eigenvalue (by modulus) by $\lambda = \max\{|\lambda_2|, |\lambda_n|\}$. Then for any two sets $S, T \subseteq V(G)$,*

$$\left| e(S, T) - \frac{d}{n} |S| |T| \right| \leq \frac{\lambda}{n} \sqrt{|S| |S^c| |T| |T^c|}$$

Proof. Let $\mathbf{j} \in \mathbb{R}^n$ denote that all 1's vector. Let $\chi_S \in \mathbb{R}^n$ denote the indicator vector of a set $S \subseteq V(G)$. One observes that if A denotes the adjacency matrix

$$\chi_S^\top A \chi_T = e(S, T) = e(T, S) = \chi_T^\top A \chi_S$$

Split χ_S and χ_T into the components along and perpendicular to j to get

$$\chi_S = \frac{|S|}{n}j + \left(\chi_S - \frac{|S|}{n}j\right) = x_{\parallel} + x_{\perp}$$

$$\chi_T = \frac{|T|}{n}j + \left(\chi_T - \frac{|T|}{n}j\right) = y_{\parallel} + y_{\perp}$$

Thus we have

$$\begin{aligned} e(S, T) &= (x_{\parallel} + x_{\perp})^{\top} A (y_{\parallel} + y_{\perp}) \\ &= x_{\parallel}^{\top} A y_{\parallel} + x_{\parallel}^{\top} A y_{\perp} + x_{\perp}^{\top} A y_{\parallel} + x_{\perp}^{\top} A y_{\perp} \\ &= \frac{d}{n} |S| |T| + x_{\perp}^{\top} A y_{\perp} \end{aligned}$$

One can check the above calculation and verify that the cross-terms are both zero. Thus we get

$$\begin{aligned} \left| e(S, T) - \frac{d}{n} |S| |T| \right| &= |x_{\perp}^{\top} A y_{\perp}| \\ &\leq \lambda \|x_{\perp}\| \|y_{\perp}\| \\ &= \frac{\lambda}{n} \sqrt{|S| |S^c| |T| |T^c|} \end{aligned}$$

For finding the lengths of the vectors we have used

$$\|x_{\perp}\|^2 = \|\chi_S\|^2 - \|x_{\parallel}\|^2 = |S| - \frac{|S|^2}{n}$$

■

Corollary 4.173. *For an undirected d -regular graph on n vertices with second largest eigenvalue λ (by modulus), we have*

$$i(G) \geq \frac{d - \lambda}{d + \lambda}$$

Proof. Let $z(G)$ be as in Definition 4.170. Pick S and T of this size $z(G) = z$ such that there are no edges between these sets. Then by Lemma 4.172 we get

$$\frac{d}{n} z^2 \leq \frac{\lambda}{n} (z)(n - z) \implies z \leq \frac{\lambda n}{d + \lambda}$$

Now applying the lower bound of Proposition 4.171 we are done. ■

Since the expander-mixing lemma involves the second largest eigenvalue λ , it is natural to try and make this quantity as small as possible.

Definition 4.174. By an (n, d, λ) -**graph** we mean a simple undirected graph on n vertices which is d -regular and has eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ with $\lambda = \max\{|\lambda_2|, |\lambda_n|\}$.

Theorem 4.175 (Alon-Boppana theorem). *Let G be an (n, d, λ) -graph. Then as $n \rightarrow \infty$ we have*

$$\lambda \geq 2\sqrt{d - 1} - o(1)$$

One may find the proof in [26].

This motivated the definition (and later the classification) of Ramanujan graphs.

Definition 4.176. Let G be a connected d -regular graph with eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. If $\max\{|\lambda_i| \mid |\lambda_i| \neq d\} \leq 2\sqrt{d - 1}$, then the graph is called a **Ramanujan graph**.

The theorem below gives us the existence of graphs which are very close to being Ramanujan. This was a very important discovery in the context of Ramanujan graphs. The original paper is [23].

Theorem 4.177. *For every positive integer d and every $\varepsilon > 0$ there exists a positive integer $n_0 = n_0(d, \varepsilon)$ such that for all $n \geq n_0$ with nd being even, we can find an explicit construction of an (n, d, λ) -graph $G_{n,d}^\varepsilon$ with $\lambda \leq 2\sqrt{d-1} + \varepsilon$.*

4.4.5 Construction of strong blocking set

We begin with a very natural construction of a strong blocking set.

Proposition 4.178. *Consider any basis $\{e_1, \dots, e_k\}$ of $\mathbb{F}_q^k$ and consider all the lines $\text{span}\{e_i, e_j\}$ in the projective space. These form a tetrahedron and the collection of all points along all lines forms a strong blocking set of size*

$$\binom{k}{2}(q-1) + k$$

Proof. The size is clearly as given because every line contains $q+1$ points two of which are e_i and e_j themselves. Thus there are $q-1$ internal points and there are $\binom{k}{2}$ lines.

Let H be an arbitrary hyperplane. H cannot have all of $e_1, \dots, e_k$ due to its dimension. Suppose $e_1 \notin H$, without loss of generality. Let a normal vector to H to be u .

Consider the $k+1$ lines $\{\ell_2, \dots, \ell_k\}$ where $\ell_i = \text{span}\{e_1, e_i\}$ for all $i = 2, 3, \dots, k$. Since every line and hyperplane in the projective space must necessarily intersect, let ℓ_i meet H and Q_i for all $i = 2, 3, \dots, k$. We claim that all the Q_i are linearly independent. Suppose

$$c_2 Q_2 + \dots + c_k Q_k = 0$$

Writing $Q_j = \alpha e_1 + \beta e_j$ and imposing that Q is normal to u gives $\alpha = u_j$ and $\beta = -u_1$. Thus,

$$\sum_{j=2}^k c_j (u_j e_1) - \sum_{j=2}^k c_j u_1 e_j = 0$$

Since the e_i are all linearly independent we have $c_2 = \dots = c_k = 0$ and we are done. Thus we have shown that

$$\text{span}\{M \cap H\} = H$$

which confirms that $M = \{\text{span}\{e_i, e_j\} : i \neq j \in [k]\}$ is a strong blocking set. ■

Below we have a general way to construct strong blocking sets from a set of lines.

Definition 4.179. A set $\mathcal{L}$ of lines in $PG(k-1, q)$ is said to satisfy the **avoidance property** if there is no codimension 2 space meeting every line of $\mathcal{L}$.

Proposition 4.180. *Let $\mathcal{L}$ be a set of lines in $PG(k-1, q)$ satisfying the avoidance property. Then the union of all lines in $\mathcal{L}$ forms a strong blocking set.*

Proof. Suppose not. Let the union of the lines be denoted by $\mathcal{B}$. Then there exists a hyperplane H in $PG(k-1, q)$ such that $\mathcal{B} \cap H$ does not span H . Thus it has codimension at least 2. Let it be contained in some codimension 2 subspace K . Note that K is a codimension 1 subspace of H but a codimension 2 subspace of the projective space.

Every line $\ell \in \mathcal{L}$ meets H and hence $\ell \cap H$ is non-empty. Since $\mathcal{B} \cap H \subseteq K$, we must have $\ell \cap K$ to be non-empty. Thus we have shown that every line $\ell \in \mathcal{L}$ intersects a codimension 2 space, namely, K . Thus $\mathcal{L}$ cannot satisfy that avoidance property. ■

We now construct strong blocking sets from graphs = specifically we wish to create a system of lines which satisfies the avoidance property.

Definition 4.181. Let $M = \{p_1, \dots, p_n\}^\#$ be a multiset of points in $PG(k-1, q)$. Let G be any undirected simple graph on the vertex set M . Construct the set of lines

$$\mathcal{L}(M, G) = \mathcal{L} = \{\text{span}\{p_i, p_j\} : \{p_i, p_j\} \in E(G)\}$$

We define this to be the *set of lines from a graph*.

Note that this terminology is not used in the paper and it is for our convenience. Below is a condition on the graph

Proposition 4.182. *Let $\mathcal{L}$ be a set of lines obtained from a graph G and a multiset M of vertices as above. This collection satisfies the avoidance property if the following property is satisfied by the graph. For every $S \subseteq V(G)$, there is a connected component C of the remaining graph $G \setminus S$ such that $\text{span}\{S \cup C\} = PG(k-1, q)$.*

Proof. Suppose that the property holds and the set of lines does not satisfy the avoidance property. Then there exists some codimension 2 space K which meets every line of the system $\mathcal{L}$. Let $S = K \cap M$, that is, the points of M lying on K .

Since the property holds, we have a connected component C in $G \setminus S$, that is, points outside H such that $\text{span}\{S \cup C\} = PG(k-1, q)$. Consider all points outside K , that is $M \setminus S$ and consider the lines spanned by picking any two of these points. Call these lines ℓ_{ij} connecting points p_i and p_j lying outside K .

These lines must necessarily intersect K at some point Q_{ij} which is neither p_i nor p_j . Let Q be the collection of all these points Q_{ij} many of which may be the same point. Since Q lies in K , we must have $Q \cup S \subseteq K$.

Now fix some $p_r \in C$. Note that p_r lies outside the space K . Since every point of C is connected to p_r through some sequence of edges, we claim that $C \subseteq \text{span}\{Q \cup \{p_r\}\}$. Indeed, let p be some point of C connected via edges $e_1, \dots, e_t$. Since the ends of each edge e_j lie in C , the lines generated $\ell_1, \dots, \ell_t$ must cut K somewhere at some Q_{ij} which was already considered above and must hence lie in Q .

Now if a subspace contains both p_i and Q_{ij} , then it must also contain p_j and hence any subspace containing P_r and Q must also contain p . This proves the claim.

Finally we have

$$PG(k-1, q) = \text{span}\{S \cup C\} \subseteq \text{span}\{S \cup Q \cup \{P_r\}\} \subseteq \text{span}\{K \cup \{p_r\}\} \subseteq PG(k-1, q)$$

However, the span of K and $\{p_r\}$ must have codimension at least 1 which is a contradiction. ■

This is a good result however it is impractical since it is very difficult to choose a set M cleverly and then also build the graph cleverly so that the above property is satisfied. However we have the following result which involves projective systems, to help us achieve the condition.

Lemma 4.183. *Let M be an $[n, k, d]_q$ projective system and let G be a graph with integrity at least $n-d+1$. Then the set of lines formed from this graph with vertex set M satisfies the avoidance property.*

Proof. We verify the property from the above proposition. Let S be any arbitrary set of vertices of G . Since the integrity is at least $n-d+1$, there must be some connected component C in $G \setminus S$ with $|S| + |C| \geq n-d+1$ because of the definition of $i(G)$.

However, $d = n - \max_{H: \text{hyperplane}} |H \cap M|$, that is every hyperplane meets M in at most $n-d$ points.

Hence $S \cup C \subseteq M$ is not contained in any hyperplane of the projective space. Thus the span of $S \cup C$ must be the full space and hence the condition of Proposition 4.182 is satisfied. This proves the lemma. ■

As a remark one may note that the size of this strong blocking set created is given by $n + (q - 1) |E|$ since every line, arising from a unique edge of the graph, has $q - 1$ internal points and then we finally have the n points of M , that is, the vertices which span the $|E|$ number of lines.

Also note that picking G to be the complete graph, the integrity is always n and for the choice of M as a basis of $\mathbb{F}_q^k$, M may not be a projective system but one can check that the condition of Proposition 4.182 is satisfied and hence we retrieve the construction of Proposition 4.178.

The size can actually be made linear in qk and the below theorem, which is also the central theorem of the paper, proves this.

Theorem 4.184. *There exists an absolute constant c such that for every prime power q we have an explicit construction of a strong blocking set in $PG(k_i - 1, q)$ of size at most cqk_i for any strictly increasing sequence of positive integers k_i .*

Proof. Begin with the Justesen construction of a family of asymptotically good codes from Proposition 4.156. This gives us codes $C_i \in [n_i, k_i, d_i]_q$ with parameters $R \in (0, 1/2)$ and $\delta = 0.11(1 - 2R)$. Thus there exists some positive integer i_0 such that for all $i \geq i_0$ we have

$$\frac{d_i}{n_i} \geq \delta ; \quad \frac{k_i}{n_i} \geq R$$

Let us continue working with $i \geq i_0$ throughout.

Now recall Theorem 4.177. Since $d_i/n_i \geq \delta$ we can find an explicit family of (n_i, d_i, λ) -graphs G_i such that

$$\frac{d_i - \lambda}{d_i + \lambda} \geq 1 - \delta + \frac{1}{n_i}$$

This is because we can asymptotically ensure (by choosing a larger i_0) that

$$2\sqrt{(d_i - 1) + \varepsilon} < d_i \left(\frac{\delta - 1/n_i}{2 - \delta + 1/n_i} \right)$$

Now we apply Corollary 4.173 to get

$$i(G_i) \geq n_i - n_i\delta + 1 \geq n_i - d_i + 1$$

Now obtain projective systems M_i corresponding to the codes C_i using Theorem 4.160. Using these systems M_i and graphs G_i , applying Lemma 4.183 the lines generated will form a strong blocking set. The size of this strong blocking set is

$$n_i + (q - 1) |E(G_i)| = n_i + (q - 1) \frac{dn_i}{2} \leq \frac{dq n_i}{2} \leq \frac{d}{2R} q k_i$$

Thus the required absolute constant is $\frac{d}{2R}$ and we are done. ■

As a remark one may note that the worst case constant for the choice of $q = 2$ is 8277. This is improved with increasing values of q but can never fall below the value of 292.

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